AVP32F003x Digital Signal Processor
AVP32F003x 数字信号处理器

Data Manual
数据手册

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1 Features 特性

32-bit DSP core at 120MHz

- IEEE 754 Floating-Point Unit (FPU)
  - Support for Fast Integer Division instruction
- Trigonometric Math Unit (TMU)
  - Support for Nonlinear Proportional Integral Derivative (NLPID) control
- CRC Engine and Instructions (VCRC)
- Ten hardware breakpoints (with ERAD)

Programmable CLA co-processor at 120MHz

- Executes code independently of main CPU
- IEEE 754 single-precision floating-point instructions

On-chip memory

- 384KB of flash (ECC protected) across three independent banks
- 69KB of RAM (ECC protected)
- DCSM、JTAGLOCK and Zero-pin boot

Clock and system control

- Two internal 10-MHz oscillators
- Crystal oscillator or external clock input
- Windowed watchdog timer module
- Missing clock detection circuitry
- Dual-clock Comparator (DCC)

Power supply

- 3.3-V I/O design, 1.1-V Core supply
- Internal VREG generation allows for single-supply design
- Power on reset (POR) circuit
- Brownout reset (BOR) circuit

System peripherals

- 6-channel Direct Memory Access (DMA) controller
- 55 individually programmable multiplexed GPIO pins
- 23 digital inputs on analog pins
- 2 digital inputs/outputs on analog pins (AGPIO)

32 位 DSP 内核（120MHz 主频）

- IEEE 754 浮点单元（FPU）
  - 支持快速整数除法（FINTDIV）指令
- 三角数学单元（TMU）
  - 支持非线性比例、积分和微分控制
- 循环冗余校验引擎及指令（VCRC）
- 带 ERAD 支持的十个硬件断点

可编程的 CLA 协处理器（120MHz 主频）

- 独立于主 CPU 执行代码
- 支持 IEEE 754 单精度浮点指令

片上存储器

- 三个独立存储体，共计 384KB 带 ECC 保护的 Flash
- 69KB 带 ECC 保护的 RAM
- 双区安全、JTAG 锁定和安全引导

时钟和系统控制

- 两个 10MHz 内部振荡器
- 晶体振荡器或外部时钟输入
- 窗口看门狗定时器模块
- 时钟丢失检测电路
- 双路时钟比较器（DCC）

电源供电

- 3.3V IO 设计，1.1V 内核供电
- 内部 VREG 生成 1.1V 供电内核，支持 3.3V 单电源供电设计
- 上电复位（POR）电路
- 欠压复位（BOR）电路

系统外设

- 6 通道直接内存访问（DMA）控制器
- 55 个独立可编程多路复用通用输入/输出引脚
- 模拟引脚上支持 23 路数字输入
- 模拟引脚上支持 2 路数字输入/输出

- Enhanced Peripheral Interrupt Expansion (ePIE)
- Multiple low-power mode (LPM) support
- Embedded Real-time Analysis and Diagnostic (ERAD)
- Unique Identification (UID) number

Communications peripherals

- One Power-Management Bus (PMBUS) interface
- Two Inter-Integrated Circuit (I2C) interfaces
- One Controller Area Network (CAN/DCAN) bus port
- One Controller Area Network with Flexible Data-Rate (CANFD/MCAN) bus port
- Two Serial Peripheral Interface (SPI) ports
- Two UART-compatible Serial Communication Interface (SCI)
- Two UART-compatible Local Interconnect Network (LIN) interfaces
- Fast Serial Interface (FSI) with one transmitter and one receiver (up to 200Mbps)

Analog System

- Three 4-MSPS, 12-bit Analog-to-Digital Converters (ADCs)
  - Up to 23 external channels (includes the two GPDAC outputs)
  - Four integrated Post-Processing Blocks (PPB) per ADC
- Four windowed comparators (CMPSS) with 12-bit reference Digital-to-Analog Converters (DACs)
  - Digital glitch filters
- Two 12-bit buffered DAC outputs

Enhanced Control Peripherals

- 16 ePWM channels with eight channels that have high-resolution capability (150-ps resolution)
  - Integrated dead-band support
  - Integrated hardware trip zones (TZs)
- Three Enhanced Capture (eCAP) modules
  - High-resolution Capture (HRCAP) available on one of the three eCAP modules
- Two Enhanced Quadrature Encoder Pulse (eQEP) modules with support for CW/CCW operation modes

（AGPIO）

- 增强型外设中断扩展（ePIE）
- 支持多个低功耗模式（LPM）
- 嵌入式实时分析和诊断（ERAD）
- 唯一标识（UID）号

通信外设

- 一个 PMBUS 接口
- 两个 I2C 接口
- 一个 CAN 总线端口（兼容 CAN 2.0B）
- 一个 MCAN 总线端口（支持 CANFD）
- 两个 SPI 端口
- 两个 UART 兼容的 SCI
- 两个 UART 兼容的 LIN 接口
- 一个 FSI（带一个发送器和一个接收器，速率达 200Mbps）

模拟系统

- 三个 4MSPS 速率、12 位的模数转换器
  - 高达 23 个外部通道（包含 2 个 GPDAC 输出）
  - 每个 ADC 集成 4 个后处理块（PPB）
- 四个带 12 位 DAC 的窗口比较器（CMPSS）
  - 数字故障过滤器
- 两个 12 位的缓冲 DAC 输出

增强型控制外设

- 16 个 ePWM 通道，包含具有 150ps 分辨率的 8 个高分辨率通道
  - 集成死区支持
  - 集成硬件跳匝区（TZs）
- 三个增强型捕获模块（eCAP）
  - eCAP3 支持高分辨率捕获（HRCAP）
- 两个支持正反转工作模式的 eQEP 模块

- Eight Sigma-Delta Filter Module (SDFM) input channels (two parallel filters per channel)
  - Standard SDFM data filtering
  - Comparator filter for fast action for overvalue or undervalue condition
- Embedded Pattern Generator (EPG)
- 4 tiles Configurable Logic Block (CLB)
  - Augments existing peripheral capability
  - Supports position manager solutions
- Host Interface Controller (HIC)
  - Access to internal memory from an external host
- Background CRC (BGCRC)
  - One cycle CRC computation on 32 bits of data
- Advanced Encryption Standard (AES) accelerator
- Live Firmware Update (LFU)
  - Fast context switching from old to new firmware
  - Flash bank erase time improvements

Diagnostic features and Functional Safety

- Memory Power on Self-Test (MPOST)
- Hardware Built-in Self-Test (HWBIST)
- Systematic capability up to ASIL D and SIL 3 (TBD)
- Hardware capability up to ASIL B and SIL 2 (TBD)

Package and Temperature options

- 100-pin Quad Flatpack
- 80-pin Quad Flatpack
- 64-pin Quad Flatpack
- 48-pin Quad Flatpack
- Free-Air (T_A): –40°C to 125°C
- Junction (T_J): –40°C to 150°C

- 八个Σ-Δ滤波器模块（SDFM）输入通道（每个通道有两个并联的滤波器）
  - 支持标准的 SDFM 数据滤波
  - 比较器滤波支持高估/低估的快速动作
- 嵌入式图形生成器（EPG）
- 4 个可配置逻辑块（CLB）
  - 增强现有外设功能
  - 支持位置管理解决方案
- 主机接口控制器（HIC）
  - 可以从外部主机访问内部存储器
- 后台 CRC（BGCRC）
  - 32 位数据仅需一个 CRC 计算周期
- 高级加密标准（AES）加速器
- 实时固件更新（LFU）
  - 新旧固件快速上下文切换
  - 缩短 Flash 擦除时间

诊断特性与功能安全

- 内存开机自检（MPOST）
- 硬件内置自检（HWBIST）
- ASIL D 和 SIL 3 的系统能力（TBD）
- ASIL B 和 SIL 2 的硬件能力（TBD）

封装及温度选项

- 100 脚 QFP
- 80 脚 QFP
- 64 脚 QFP
- 48 脚 QFP
- 环境温度（T_A）：–40°C 到 125°C
- 结温（T_J）：–40°C 到 150°C

2 Applications 应用

Appliances

- Air conditioner outdoor unit

Building automation

- Door operator drive control.

Industrial machine &machine tools

- Automated sorting equipment
- Textile machine

AC inverter & VF drives

- AC drive control module
- AC drive position feedback
- AC drive power stage module

Linear motor transport systems

- Linear motor power stage

Single & multi axis servo drives

- Servo drive position feedback
- Servo drive power stage module

Speed controlled BLDC drives

- AC-input BLDC motor drive
- DC-input BLDC motor drive

Factory automation

- Robot servo drive
- Mobile robot motor control
- Position sensor

Industrial power

- Industrial AC-DC

UPS

- Three phase UPS
- Single phase online UPS

Telecom & server power

- Merchant DC/DC
- Merchant network & server PSU
- Merchant telecom rectifiers

Hybrids, electric & powertrain systems

- DC/DC converter
- Inverter & motor control
- On-board (OBC) & wireless charger

家用电器

- 空调室外机

楼宇自动化

- 门机驱动控制

工业机械与机床

- 自动分拣设备
- 纺织机

交流逆变器和变频驱动器

- 交流驱动控制模块
- 交流驱动位置反馈
- 交流驱动功率级模块

直线电机运输系统

- 直线电机功率级

单轴及多轴伺服驱动器

- 伺服驱动器位置反馈
- 伺服驱动器功率级模块

速度控制 BLDC 驱动器

- 交流输入 BLDC 电机驱动器
- 直流输入 BLDC 电机驱动器

工厂自动化

- 机器人伺服驱动器
- 移动机器人电机控制
- 位置传感器

工业电源

- 工业交流-直流

不间断电源

- 三相 UPS
- 单相在线 UPS

电信和服务器电源

- 商用 DC/DC
- 商用网络和服务器 PSU
- 商用通信电源整流器

混合动力、电动和动力总成系统

- DC/DC 转换器
- 逆变器和电机控制
- 车载充电器(OBC)和无线充电器

- Virtual engine sound system (VESS)
- Engine fan
- eTurbo/charger
- Pump
- Electric power steering (EPS)

Infotainment and cluster

- Head-up display
- Automotive head unit
- Automotive external amplifier

Body electronics & lighting

- Automotive HVAC compressor module
- DC/AC inverter
- Headlight

ADAS

- Mechanically scanning LIDAR

HEV/EV battery-management system (BMS)

- 100-Vbattery pack-passive balancing
- 12-&24-V battery pack-passive balancing
- 400-V battery pack-passive balancing
- 48-V battery pack-passive balancing

EV charging infrastructure

- AC charging (pile) station
- DC charging (pile) station
- EV charging station power module
- Wireless EV charging station

Renewable energy storage

- Energy storage power conversion system (PCS)

Solar energy

- Central inverter
- Micro inverter
- Solar power optimizer
- Solar arc protection
- Rapid shutdown
- String inverter

- 虚拟引擎声浪系统(VESS)
- 发动机风扇
- 电子涡轮/增压器
- 泵
- 电动助力转向(EPS)

信息娱乐系统和仪表组

- 抬头显示
- 汽车音响主机
- 汽车外部放大器

车身电子装置与照明

- 汽车暖通空调压缩机模块
- DC/AC 逆变器
- 头灯

ADAS

- 机械扫描激光雷达

混动/电动电池管理系统(BMS)

- 100V 电池包被动均衡
- 12V 和 24V 电池包被动均衡
- 400V 电池包被动均衡
- 48V 电池包被动均衡

电动汽车充电基础设施

- 交流充电（桩）站
- 直流充电（桩）站
- 电动汽车充电站电源模块
- 电动汽车无线充电站

可再生能源存储

- 能量存储电源转换系统(PCS)

光伏能源

- 中央逆变器
- 微型逆变器
- 光伏能源优化器
- 太阳能电弧保护
- 快速关断器
- 串式逆变器

3 Description 描述

The AVP32F003x is a member of the real-time microcontroller family of scalable, ultralow latency devices designed for efficiency in power electronics, including but not limited to: high power density, high switching frequencies, and supporting the use of GaN and SiC technologies.

These include such applications as:

- Motor drives
- Appliances
- Hybrid, electric & powertrain systems
- Solar & EV charging
- Digital power
- Body electronics & lighting
- Test & measurement

AVP32F003X 是实时微控制器系列中的一系列器件，该可扩展、超低延迟器件系列旨在提高电力电子设备的效率，包括但不限于：高功率密度、高开关频率，并支持使用 GaN 和 SiC 技术。

这些应用包括：

- 电机驱动器
- 电器
- 混合动力、电动和动力总成系统
- 太阳能和电动汽车充电
- 数字电源
- 车身电子装置与照明
- 测试与测量

The real-time control subsystem is based on 32-bit DSP core, which provides 120 MHz of signal processing performance for floating- or fixed-point code running from either on-chip flash or SRAM. The CPU is further boosted by the Floating-Point Unit (FPU), Trigonometric Math Unit (TMU), and VCRC (Cyclical Redundancy Check) extended instruction sets, speeding up common algorithms key to real-time control systems.

The CLA allows significant offloading of common tasks from the main CPU. The CLA is an independent 32-bit floating-point math accelerator that executes in parallel with the CPU. Additionally, the CLA has its own dedicated memory resources and it can directly access the key peripherals that are required in a typical control system. Support of a subset of ANSI C is standard, as are key features like hardware breakpoints and hardware task-switching.

The AVP32F003x supports up to 384KB (192KW) of flash memory divided into three 128KB (64KW) banks, which enable programming and execution in parallel. Up to 69KB (34.5KW) of on-chip SRAM is also available to supplement the flash memory.

实时控制子系统基于 32 位 A2000 DSP 内核，可针对从片上闪存或 SRAM 运行的浮点或定点代码提供 120 MHz 的信号处理性能。浮点单元 (FPU)、三角函数加速器 (TMU)和 VCRC（循环冗余校验）扩展指令集进一步增强了 A2000 CPU 的性能，从而加快了实时控制系统关键常用算法的速度。

CLA 能够将大量的常见任务从主 A2000 CPU 上卸下。CLA 是一款与 CPU 并行执行的独立 32 位浮点数学加速器。此外，CLA 自带专用存储资源，它可以直接访问典型控制系统中所需的关键外设。与硬件断点和硬件任务切换等主要特性一样，ANSI C 子集支持是标准配置。

AVP32F003X 支持高达 384KB (192KW) 的闪存，这些闪存分为三个 128KB (64KW)存储体，支持并行编程和执行。高达 69KB (34.5KW)的片上 SRAM 也可用于补充闪存。

The Live Firmware Update hardware enhancements on AVP32F003x allow fast context switching from the old

firmware to the new firmware to minimize application downtime when updating the device firmware.

High-performance analog blocks are integrated on the AVP32F003x real-time microcontroller (MCU) and are closely coupled with the processing and PWM units to provide optimal real-time signal chain performance. Sixteen PWM channels, all supporting frequency-independent resolution modes, enable control of various power stages from a 3-phase inverter to power factor correction and advanced multilevel power topologies. The inclusion of the Configurable Logic Block (CLB) allows the user to add custom logic and potentially integrate FPGA-like functions into the real-time MCU.

AVP32F003X 上的实时固件更新硬件增强允许从旧固件到新固件的快速上下文切换，以尽可能减少更新器件固件时的应用停机时间。

高性能模拟模块集成在 AVP32F003X 数字信号处理器（DSP）上，并与处理单元和 PWM 单元紧密耦合，以提供更好的实时信号链性能。16 个 PWM 通道均支持与频率无关的分辨率模式，可控制从三相逆变器到功率因数校正和高级多级电源拓扑的各种功率级。通过加入可配置逻辑块（CLB），用户可以添加自定义逻辑，还可将类似 FPGA 的功能集成到 A2000 实时 MCU 中。

Interfacing is supported through various industry-standard communication ports (such as SPI, SCI, I2C, PMBus, LIN, CAN and CAN FD) and offers multiple pin-muxing options for optimal signal placement. The Fast Serial Interface (FSI) enables up to 200Mbps of robust communications across an isolation boundary.

New to the DSP platform is the Host Interface Controller (HIC), a high-throughput interface that allows an external host to access the resources of the AVP32F003x directly.

各种业界通用通信端口（如 SPI、SCI、I2C、PMBus、LIN、CAN 和 CANFD）不仅支持连接，还提供了多个引脚复用选项，可实现出色的信号布局。快速串行接口（FSI）可跨隔离边界实现高达 200Mbps 的稳健通信。

A2000 平台新增了主机接口控制器（HIC），这是一种高吞吐量接口，允许外部主机直接访问 AVP32F003X 的资源。

3.1 Functional Block Diagram 功能框图

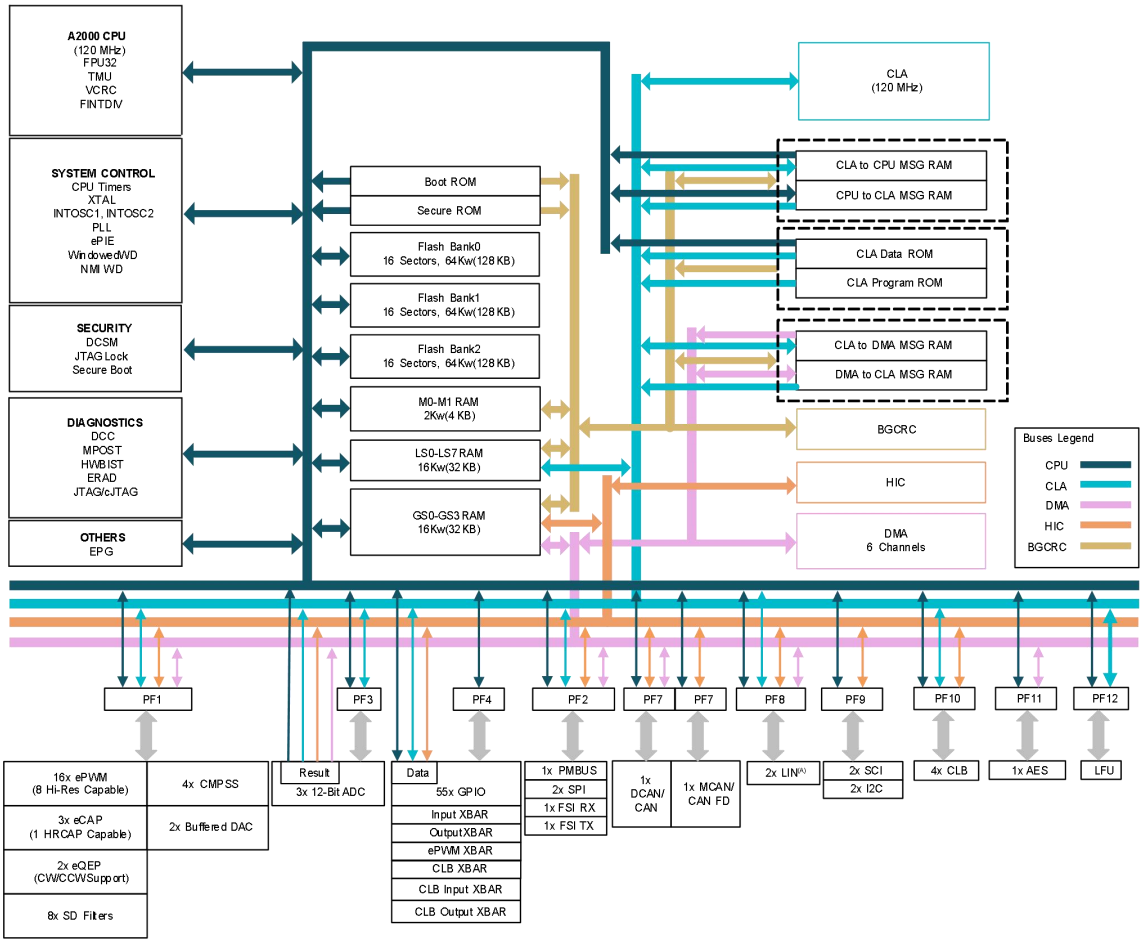


Figure 3- 1 Functional Block Diagram

4 Device Features 器件特性

Table 4-1 lists the features of the device.

Table 4-1 Device Features

FEATURE 特性		AVP32F003x AVP32F00157P
Processor and Accelerators **处理器与协处理器**		
DSP Core	Frequency (MHz) 主频	120
	FPU	Yes
	VCRC	Yes
	TMU	Yes
CLA	CLA	Yes
	CLA Frequency (MHz)	120
Memory **存储**		
Flash		384KB (192KW)
Flash Bank		3 x 128KB
RAM	Dedicated	4KB (2KW)
	Local Shared	32KB (16KW)
	Message	1KB (0.5KW)
	Global Shared	32KB (16KW)
	Total	69KB (34.5KW)
Message RAM Types		512B (256W) CPU-CLA 512B (256W) CLA-DMA
ECC		FLASH, Mx, LSx, GSx, Message RAM
Parity		ROM, CAN RAM
Security: JTAGLOCK, Zero-pin boot, Dual-zone security		Yes
System **系统**		
Configurable Logic Block (CLB)		4 Tiles on C Variants
Embedded Pattern Generator (EPG)		Yes
32-bit CPU timers		3
Advance Encryption Standard (AES)		Yes
Background CRC (BGCRC)		Yes
Live Firmware Update (LFU) Support		Yes, with enhancements and flash bank erase time improvements
Secure Boot		Yes
JTAG Lock		Yes
HWBIST		Yes
Nonmaskable Interrupt Watchdog (NMIWD) timers		1
Watchdog timers		1
Crystal oscillator/External clock input		1
Internal oscillator		2
Pins and Power Supply **引脚及电源供电**		
Internal 3.3-V to 1.1-V Voltage Regulator		Yes
GPIO pins		See Table 4-2.
AIO (analog with digital inputs)		
AGPIO (analog with digital inputs and outputs)		
Analog Peripherals **模拟外设**		
ADC 12-bit	Number of ADCs	3
	MSPS	4
	Conversion Time (ns)	250
ADC channels (single-ended, includes the two gpdac outputs)		See Table 4-2.
Temperature sensor		1
Buffered DAC		2
CMPSS (each CMPSS has two comparators and two internal DACs)		4
Control Peripherals **控制外设**		
eCAP/HRCAP modules		3 (1-eCAP3 with HRCAP capability)
ePWM/HRPWM channels		16 (8 – ePWM1 to ePWM4 with HRPWM capability)
eQEP modules		2
SDFM channels		8

FEATURE 特性	AVP32F003x AVP32F00157P
Communication Peripherals **通信外设**	
CAN (DCAN)	1
CANFD (MCAN)	1
Fast Serial Interface (FSI)	1 (1 RX and 1 TX)
I2C	2
LIN (UART-Compatible)	2
Host Interface Controller (HIC)	1
PMBus	1
SCI (UART-Compatible)	2
SPI	2
Temperature and Qualification **温度和质量**	
Free-Air temperature (T_A)	–40°C to 125°C
Junction temperature (T_J)	–40°C to 150°C

Table 4-2 lists the features of the packages.

Table 4-2 Packages Comparison

FEATURE 特性		AVP32F003x AVP32F00157P		
Flash		384KB (192KW)	256KB (128KW)	128KB (64KW)
Flash Bank		3 x 128KB	2 x 128KB	2 x 64KB
Package Options, Temperature, and Qualification **封装选项，温度和质量选项**				
S: 工业级 –40℃ to 125℃ (T_A)	QP100	AVP32F0039QP100S		
	QP80	AVP32F0039QP80S		AVP32F0033QP80S
	QP64S	AVP32F0039QP64S		
	QP48	AVP32F0039QP48S		
Q: 车规级 –40℃ to 125℃ (T_A)	QP100	AVP32F0039QP100Q		
	QP80	AVP32F0039QP80Q		
	QP80C	AVP32F00157PQP80Q		
	QP64Q	AVP32F0036QP64Q		AVP32F0033QP64Q
	QP48	AVP32F0039QP48Q		
Feature of different packages: **不同封装的特性：**				
FEATURE (PACKAGE)		GPIO pins + AGPIO pin	AIO	ADC channels
QP100		51 + 4	23	25
QP80		39 + 4	16	18
QP80C		40 + 4	16	18
QP64S		26 + 4	16	16
QP64Q		25 + 4	16	16
QP48		14 + 4	14	14

5 Pin Configuration and Functions
引脚配置和功能

5.1 Pin Diagrams
引脚框图

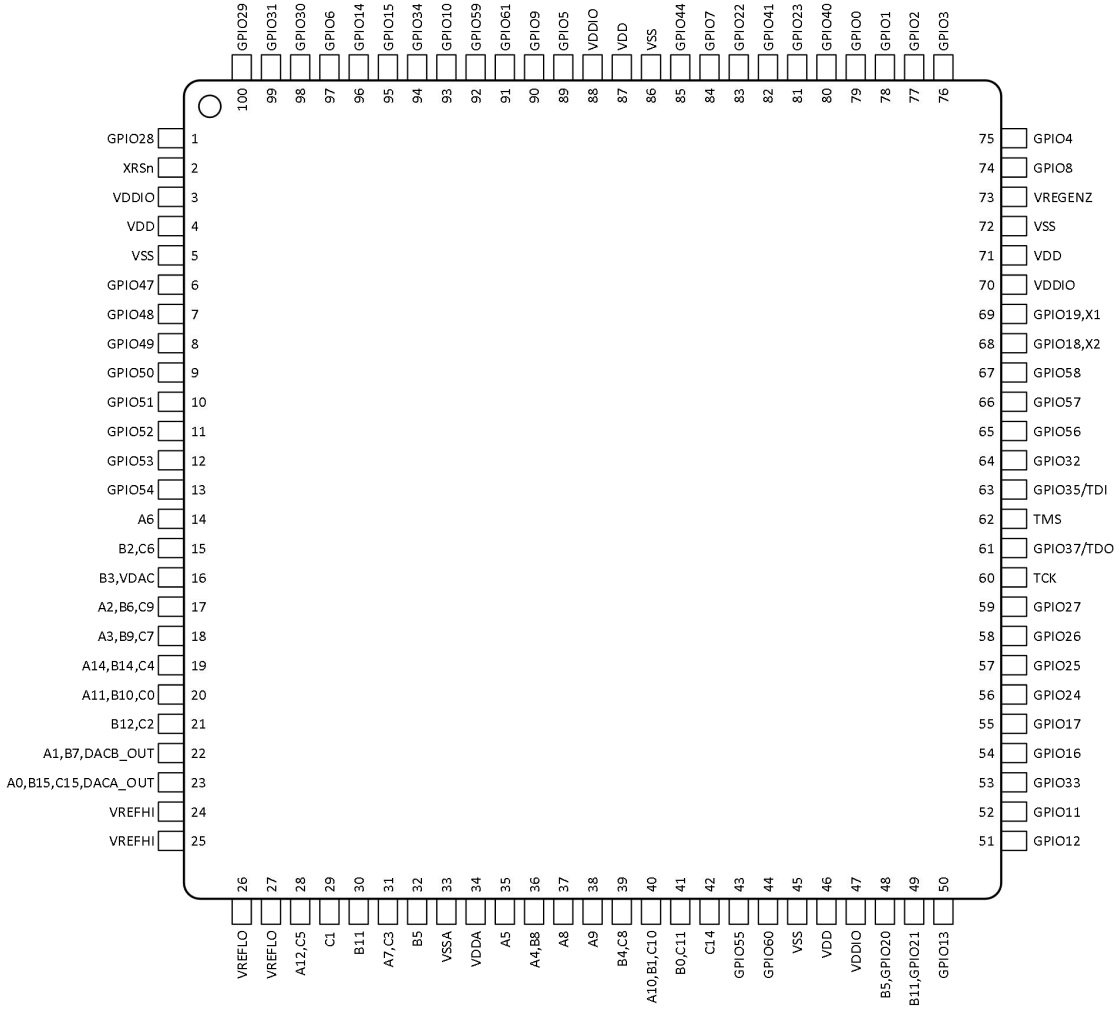


Figure 5- 1. QP100 Low-Profile Quad Flatpack (Top View)

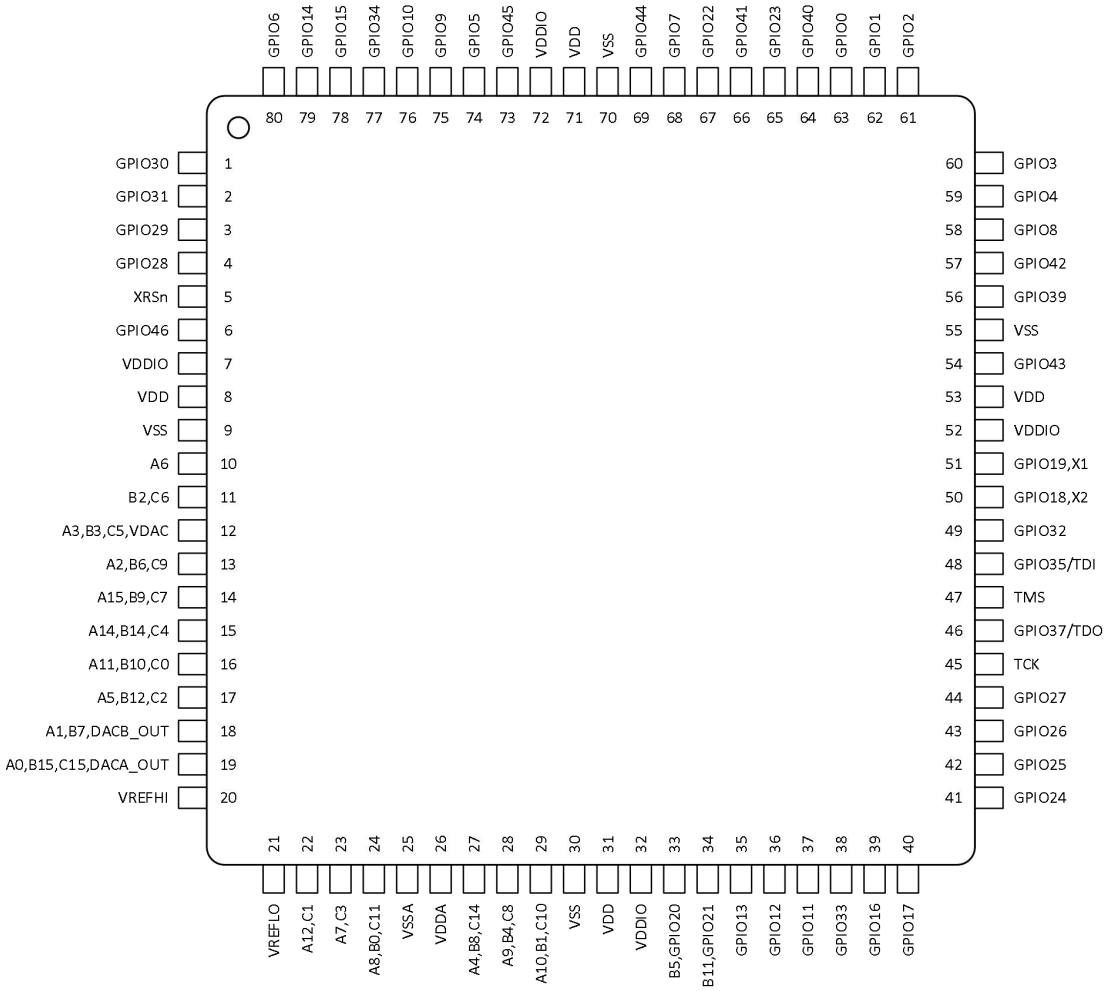


Figure 5- 2. QP80 Low-Profile Quad Flatpack (Top View)

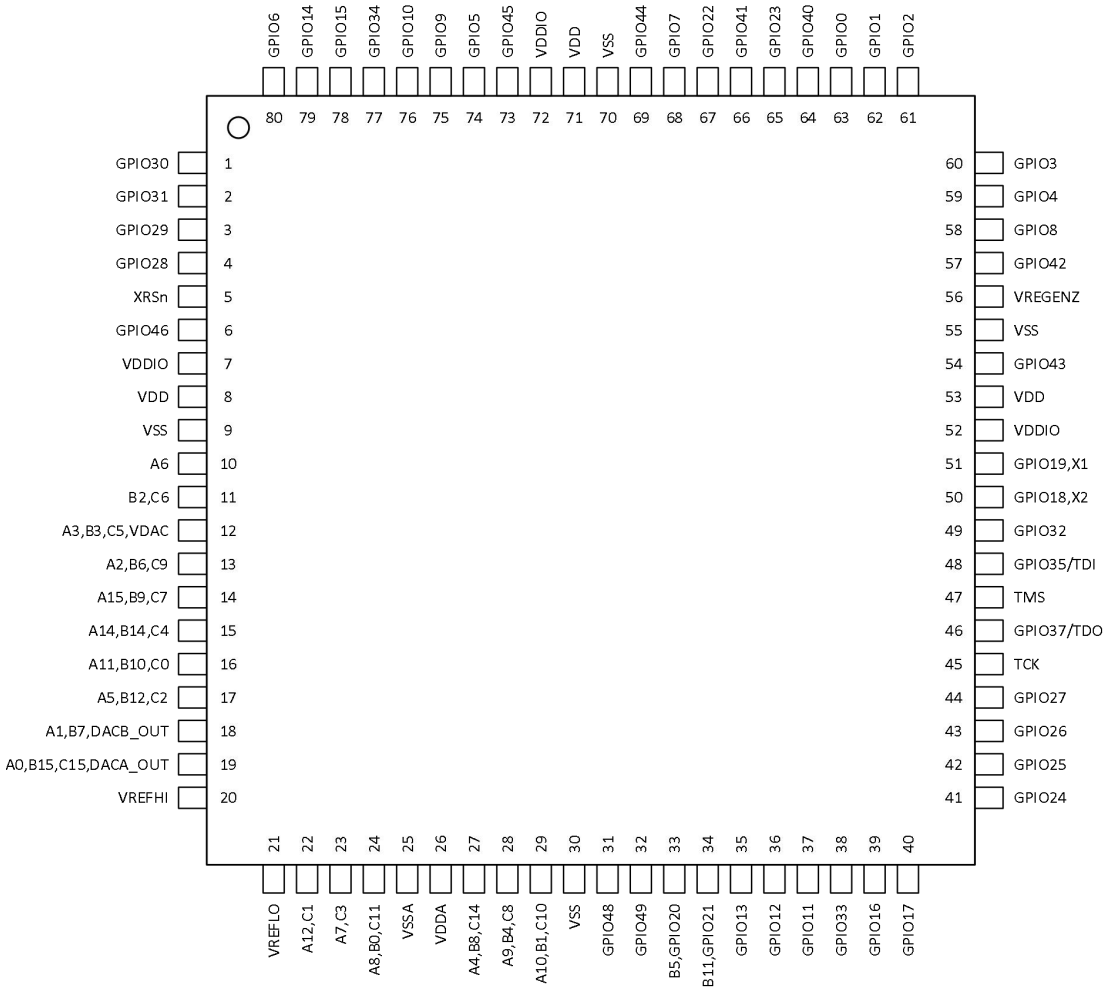


Figure 5- 3. QP80C Low-Profile Quad Flatpack (Top View)

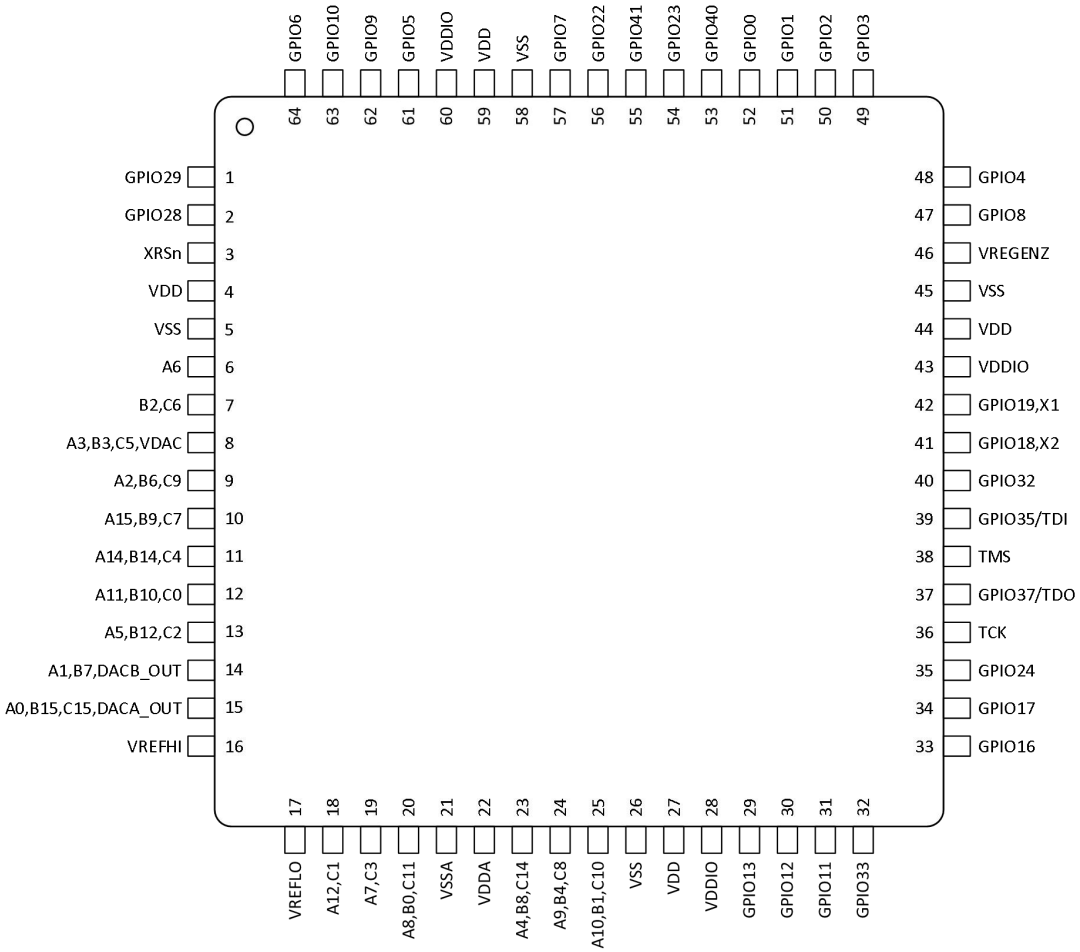


Figure 5- 4. QP64Q Low-Profile Quad Flatpack (Top View)

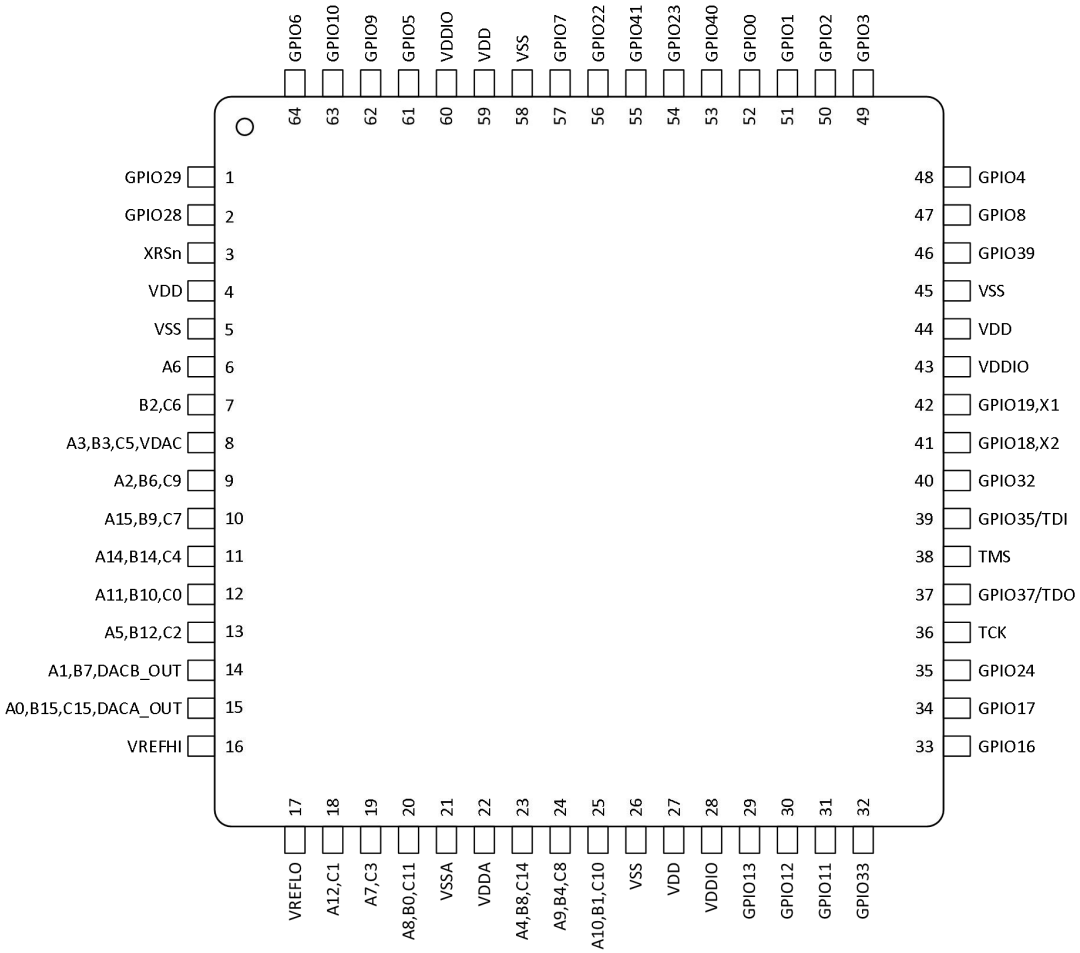


Figure 5- 5. QP64S Low-Profile Quad Flatpack (Top View)

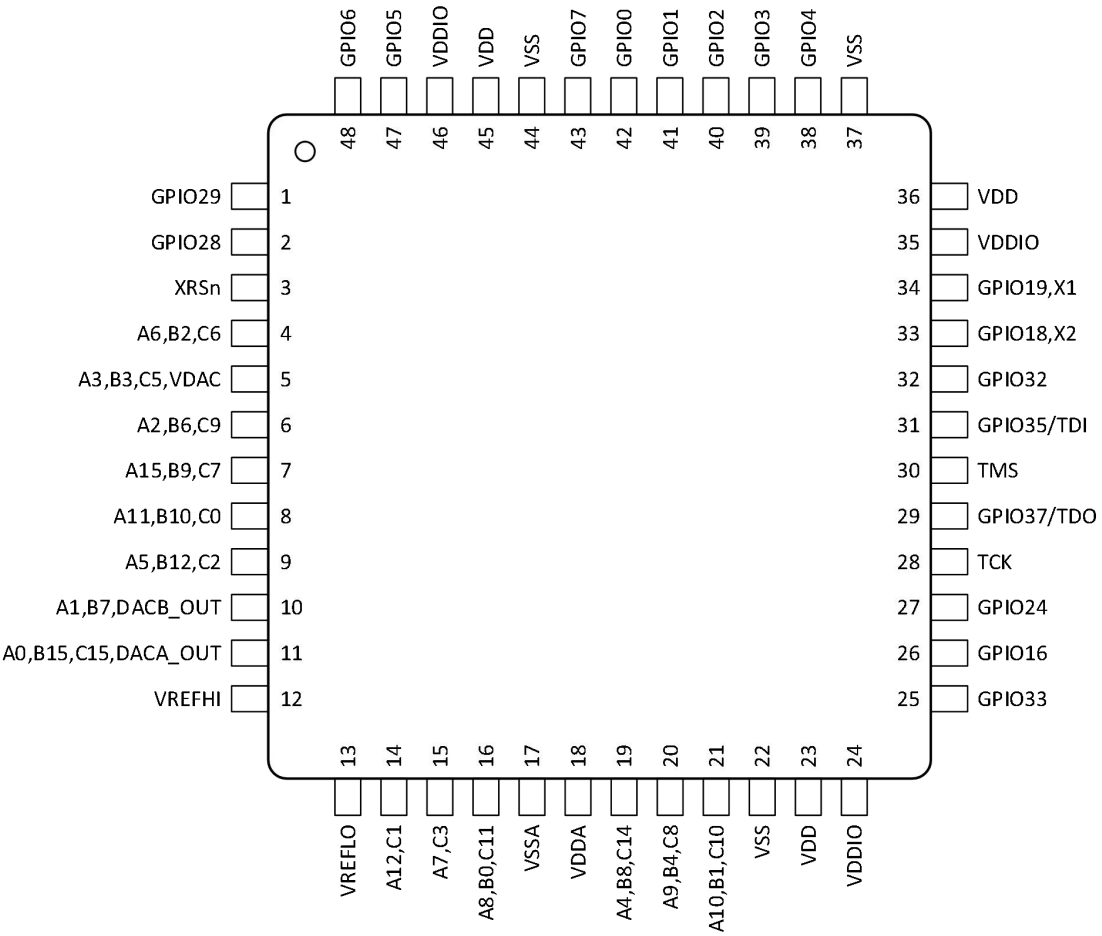


Figure 5- 6. QP48 Low-Profile Quad Flatpack (Top View)

5.2 Pin Attributes
引脚属性

Table 5-1 Pin Attributes

Signal Name	Mux Position	QP100	QP80	QP80C	QP64Q	QP64S	QP48	Pin Type	Description
Analog									
A0								I	ADC-A Input 0
B15								I	ADC-B Input 15
C15								I	ADC-C Input 15
CMP3_HP2								I	CMPSS-3 High Comparator Positive Input 2
CMP3_LP2								I	CMPSS-3 Low Comparator Positive Input 2
DACA_OUT		23	19	19	15	15	11	O	Buffered DAC-A Output.
AIO231	0, 4, 8, 12							I	Analog Pin Used For Digital Input 231 This pin also has digital mux functions which are described in the GPIO section of this table.
A1								I	ADC-A Input 1
B7								I	ADC-B Input 7
CMP1_HP4								I	CMPSS-1 High Comparator Positive Input 4
CMP1_LP4								I	CMPSS-1 Low Comparator Positive Input 4
DACB_OUT		22	18	18	14	14	10	O	Buffered DAC-B Output.
AIO232	0, 4, 8, 12							I	Analog Pin Used For Digital Input 232 This pin also has digital mux functions which are described in the GPIO section of this table.
A2								I	ADC-A Input 2
B6								I	ADC-B Input 6
C9								I	ADC-C Input 9
CMP1_HP0								I	CMPSS-1 High Comparator Positive Input 0
CMP1_LP0		17	13	13	9	9	6	I	CMPSS-1 Low Comparator Positive Input 0
AIO224	0, 4, 8, 12							I	Analog Pin Used For Digital Input 224 This pin also has digital mux functions which are described in the GPIO section of this table.
A3								I	ADC-A Input 3
CMP3_HP5								I	CMPSS-3 High Comparator Positive Input 5
CMP3_LP5		18	-	-	-	-	-	I	CMPSS-3 Low Comparator Positive Input 5
AIO229	0, 4, 8, 12							I	Analog Pin Used For Digital Input 229
A3								I	ADC-A Input 3
CMP3_HP5		-	12	12	8	8	5	I	CMPSS-3 High Comparator Positive Input 5
CMP3_LP5								I	CMPSS-3 Low Comparator Positive Input 5
A4								I	ADC-A Input 4
B8								I	ADC-B Input 8
CMP2_HP0		36	27	27	23	23	19	I	CMPSS-2 High Comparator Positive Input 0
CMP2_LP0								I	CMPSS-2 Low Comparator Positive Input 0

Signal Name	Mux Position	QP100	QP80	QP80C	QP64Q	QP64S	QP48	Pin Type	Description
AIO225	0, 4, 8, 12							I	Analog Pin Used For Digital Input 225 This pin also has digital mux functions which are described in the GPIO section of this table.
A5 CMP2_HP5 CMP2_LP5 AIO249	0, 4, 8, 12	35	-	-	-	-	-	I I I I	ADC-A Input 5 CMPSS-2 High Comparator Positive Input 5 CMPSS-2 Low Comparator Positive Input 5 Analog Pin Used For Digital Input 249
A5 CMP2_HP5 CMP2_LP5		-	17	17	13	13	9	I I I	ADC-A Input 5 CMPSS-2 High Comparator Positive Input 5 CMPSS-2 Low Comparator Positive Input 5
A6 CMP1_HP2 CMP1_LP2 AIO228	0, 4, 8, 12	14	10	10	6	6	4	I I I I	ADC-A Input 6 CMPSS-1 High Comparator Positive Input 2 CMPSS-1 Low Comparator Positive Input 2 Analog Pin Used For Digital Input 228 This pin also has digital mux functions which are described in the GPIO section of this table.
A8 CMP4_HP4 CMP4_LP4 AIO240	0, 4, 8, 12	37	-	-	-	-	-	I I I I	ADC-A Input 8 CMPSS-4 High Comparator Positive Input 4 CMPSS-4 Low Comparator Positive Input 4 Analog Pin Used For Digital Input 240 This pin also has digital mux functions which are described in the GPIO section of this table.
A8 CMP4_HP4 CMP4_LP4 AIO241	0, 4, 8, 12	-	24	24	20	20	16	I I I I	ADC-A Input 8 CMPSS-4 High Comparator Positive Input 4 CMPSS-4 Low Comparator Positive Input 4 Analog Pin Used For Digital Input 241. This pin also has digital mux functions which are described in the GPIO section of this table.
A9 CMP2_HP2 CMP2_LP2 AIO227	0, 4, 8, 12	38	28	28	24	24	20	I I I I	ADC-A Input 9 CMPSS-2 High Comparator Positive Input 2 CMPSS-2 Low Comparator Positive Input 2 Analog Pin Used For Digital Input 227 This pin also has digital mux functions which are described in the GPIO section of this table.
A10 B1 C10 CMP2_HN0 CMP2_HP3 CMP2_LN0		40	29	29	25	25	21	I I I I I I	ADC-A Input 10 ADC-B Input 1 ADC-C Input 10 CMPSS-2 High Comparator Negative Input 0 CMPSS-2 High Comparator Positive Input 3 CMPSS-2 Low Comparator Negative Input 0

Signal Name	Mux Position	QP100	QP80	QP80C	QP64Q	QP64S	QP48	Pin Type	Description
CMP2_LP3								I	CMPSS-2 Low Comparator Positive Input 3
AIO230	0, 4, 8, 12							I	Analog Pin Used For Digital Input 230 This pin also has digital mux functions which are described in the GPIO section of this table.
A11								I	ADC-A Input 11
B10								I	ADC-B Input 10
C0								I	ADC-C Input 0
CMP1_HN1								I	CMPSS-1 High Comparator Negative Input 1
CMP1_HP1								I	CMPSS-1 High Comparator Positive Input 1
CMP1_LN1		20	16	16	12	12	8	I	CMPSS-1 Low Comparator Negative Input 1
CMP1_LP1								I	CMPSS-1 Low Comparator Positive Input 1
AIO237	0, 4, 8, 12							I	Analog Pin Used For Digital Input 237 This pin also has digital mux functions which are described in the GPIO section of this table.
A12								I	ADC-A Input 12
CMP2_HN1								I	CMPSS-2 High Comparator Negative Input 1
CMP2_HP1								I	CMPSS-2 High Comparator Positive Input 1
CMP2_LN1								I	CMPSS-2 Low Comparator Negative Input 1
CMP2_LP1		28	22	22	18	18	14	I	CMPSS-2 Low Comparator Positive Input 1
AIO238	0, 4, 8, 12							I	Analog Pin Used For Digital Input 238 This pin also has digital mux functions which are described in the GPIO section of this table.
A14								I	ADC-A Input 14
B14								I	ADC-B Input 14
C4								I	ADC-C Input 4
CMP3_HP4								I	CMPSS-3 High Comparator Positive Input 4
CMP3_LP4		19	15	15	11	11	-	I	CMPSS-3 Low Comparator Positive Input 4
AIO239	0, 4, 8, 12							I	Analog Pin Used For Digital Input 239 This pin also has digital mux functions which are described in the GPIO section of this table.
A15								I	ADC-A Input 15
CMP1_HN0								I	CMPSS-1 High Comparator Negative Input 0
CMP1_HP3								I	CMPSS-1 High Comparator Positive Input 3
CMP1_LN0								I	CMPSS-1 Low Comparator Negative Input 0
CMP1_LP3		-	14	14	10	10	7	I	CMPSS-1 Low Comparator Positive Input 3
AIO233	0, 4, 8, 12							I	Analog Pin Used For Digital Input 233 This pin also has digital mux functions which are described in the GPIO section of this table.
B0		41	-	-	-	-	-	I	ADC-B Input 0
C11								I	ADC-C Input 11

Signal Name	Mux Position	QP100	QP80	QP80C	QP64Q	QP64S	QP48	Pin Type	Description
CMP2_HP4								I	CMPSS-2 High Comparator Positive Input 4
CMP2_LP4								I	CMPSS-2 Low Comparator Positive Input 4
AIO253	0, 4, 8, 12							I	Analog Pin Used For Digital Input 253
B0								I	ADC-B Input 0
C11								I	ADC-C Input 11
CMP2_HP4		-	24	24	20	20	16	I	CMPSS-2 High Comparator Positive Input 4
CMP2_LP4								I	CMPSS-2 Low Comparator Positive Input 4
B2								I	ADC-B Input 2
C6								I	ADC-C Input 6
CMP3_HP0								I	CMPSS-3 High Comparator Positive Input 0
CMP3_LP0		15	11	11	7	7	4	I	CMPSS-3 Low Comparator Positive Input 0
AIO226	0, 4, 8, 12							I	Analog Pin Used For Digital Input 226 This pin also has digital mux functions which are described in the GPIO section of this table.
B3								I	ADC-B Input 3
CMP3_HN0								I	CMPSS-3 High Comparator Negative Input 0
CMP3_HP3								I	CMPSS-3 High Comparator Positive Input 3
CMP3_LN0								I	CMPSS-3 Low Comparator Negative Input 0
CMP3_LP3		16	12	12	8	8	5	I	CMPSS-3 Low Comparator Positive Input 3
VDAC								I	Optional external reference voltage for on-chip DACs.
AIO242	0, 4, 8, 12							I	Analog Pin Used For Digital Input 242 This pin also has digital mux functions which are described in the GPIO section of this table.
B4								I	ADC-B Input 4
C8								I	ADC-C Input 8
CMP4_HP0								I	CMPSS-4 High Comparator Positive Input 0
CMP4_LP0		39	28	28	24	24	20	I	CMPSS-4 Low Comparator Positive Input 0
AIO236	0, 4, 8, 12							I	Analog Pin Used For Digital Input 236
B5								I	ADC-B Input 5
CMP1_HP5								I	CMPSS-1 High Comparator Positive Input 5
CMP1_LP5								I	CMPSS-1 Low Comparator Positive Input 5
AIO252	0, 4, 8, 12	32	-	-	-	-	-	I	Analog Pin Used For Digital Input 252 This pin also has digital mux functions which are described in the GPIO section of this table.
B5								I	ADC-B Input 5
CMP1_HP5								I	CMPSS-1 High Comparator Positive Input 5
CMP1_LP5		48	33	33	-	-	-	I	CMPSS-1 Low Comparator Positive Input 5
GPIO20	0, 4, 8, 12							I/O	General-Purpose Input Output 20
B11		30	-	-	-	-	-	I	ADC-B Input 11
CMP4_HP5								I	CMPSS-4 High Comparator Positive Input 5

Signal Name	Mux Position	QP100	QP80	QP80C	QP64Q	QP64S	QP48	Pin Type	Description
CMP4_LP5								I	CMPSS-4 Low Comparator Positive Input 5
AIO251	0, 4, 8, 12							I	Analog Pin Used For Digital Input 251
B11								I	ADC-B Input 11
CMP4_HP5								I	CMPSS-4 High Comparator Positive Input 5
CMP4_LP5		49	34	34	-	-	-	I	CMPSS-4 Low Comparator Positive Input 5
GPIO21								I/O	General-Purpose Input Output 21
C1								I	ADC-C Input 1
CMP4_HP2								I	CMPSS-4 High Comparator Positive Input 2
CMP4_LP2		29	22	22	18	18	14	I	CMPSS-4 Low Comparator Positive Input 2
AIO248	0, 4, 8, 12							I	Analog Pin Used For Digital Input 248
AIO248		29	22	22	18	18	14	I	Analog Pin Used For Digital Input 248
B12								I	ADC-B Input 12
C2								I	ADC-C Input 2
CMP3_HN1								I	CMPSS-3 High Comparator Negative Input 1
CMP3_HP1								I	CMPSS-3 High Comparator Positive Input 1
CMP3_LN1		21	17	17	13	13	9	I	CMPSS-3 Low Comparator Negative Input 1
CMP3_LP1								I	CMPSS-3 Low Comparator Positive Input 1
AIO244	0, 4, 8, 12							I	Analog Pin Used For Digital Input 244
A7								I	ADC-A Input 7
C3								I	ADC-C Input 3
CMP4_HN1								I	CMPSS-4 High Comparator Negative Input 1
CMP4_HP1								I	CMPSS-4 High Comparator Positive Input 1
CMP4_LN1		31	23	23	19	19	15	I	CMPSS-4 Low Comparator Negative Input 1
CMP4_LP1								I	CMPSS-4 Low Comparator Positive Input 1
AIO245	0, 4, 8, 12							I	Analog Pin Used For Digital Input 245
C5		28	12	12	8	8	5	I	ADC-C Input 5
B9		18	14	14	10	10	7	I	ADC-B Input 9
C7								I	ADC-C Input 7
C14								I	ADC-C Input 14
CMP4_HN0								I	CMPSS-4 High Comparator Negative Input 0
CMP4_HP3								I	CMPSS-4 High Comparator Positive Input 3
CMP4_LN0		42	-	-	-	-	-	I	CMPSS-4 Low Comparator Negative Input 0
CMP4_LP3								I	CMPSS-4 Low Comparator Positive Input 3
AIO247	0, 4, 8, 12							I	Analog Pin Used For Digital Input 247
C14								I	ADC-C Input 14
CMP4_HN0								I	CMPSS-4 High Comparator Negative Input 0
CMP4_HP3								I	CMPSS-4 High Comparator Positive Input 3
CMP4_LN0		-	27	27	23	23	19	I	CMPSS-4 Low Comparator Negative Input 0
CMP4_LP3								I	CMPSS-4 Low Comparator Positive Input 3
VREFHI		24, 25	20	20	16	16	12	I	ADC High Reference (external/internal reference mode)

Signal Name	Mux Position	QP100	QP80	QP80C	QP64Q	QP64S	QP48	Pin Type	Description
VREFLO		26, 27	21	21	17	17	13	I	ADC Low Reference
GPIO									
AIO231	0, 4, 8, 12	23	19	19	15	15	11	I	Analog Pin Used For Digital Input 231 This pin also has analog functions which are described in the GPIO section of this table.
SD1_C1	2							I	SDFM-1 Channel 1 Clock Input
HIC_BASESEL1	15							I	HIC Base address range select 1
AIO232	0, 4, 8, 12	22	18	18	14	14	10	I	Analog Pin Used For Digital Input 232 This pin also has analog functions which are described in the GPIO section of this table.
SD1_D4	2							I	SDFM-1 Channel 4 Data Input
HIC_BASESEL0	15							I	HIC Base address range select 0
AIO224	0, 4, 8, 12	17	13	13	9	9	6	I	Analog Pin Used For Digital Input 224 This pin also has analog functions which are described in the GPIO section of this table.
SD2_D3	2							I	SDFM-2 Channel 3 Data Input
HIC_A3	15							I	HIC Address 3
AIO225	0, 4, 8, 12	36	27	27	23	23	19	I	Analog Pin Used For Digital Input 225 This pin also has analog functions which are described in the GPIO section of this table.
SD2_C2	2							I	SDFM-2 Channel 2 Clock Input
HIC_NWE	15							I	HIC Data Write enable from host
AIO228	0, 4, 8, 12	14	10	10	6	6	4	I	Analog Pin Used For Digital Input 228 This pin also has analog functions which are described in the GPIO section of this table.
SD2_C1	2							I	SDFM-2 Channel 1 Clock Input
HIC_A0	15							I	HIC Address 0
AIO240	0, 4, 8, 12	37	-	-	-	-	-	I	Analog Pin Used For Digital Input 240 This pin also has analog functions which are described in the GPIO section of this table.
SD2_C1	2							I	SDFM-2 Channel 1 Clock Input
HIC_NBE1	15							I	HIC Byte enable 1
AIO241	0, 4, 8, 12	-	24	20	20	16		I	Analog Pin Used For Digital Input 241. This pin also has analog functions which are described in the GPIO section of this table.
SD2_C1	2							I	SDFM-2 Channel 1 Clock Input
HIC_NBE1	15							I	HIC Byte enable 1
AIO227	0, 4, 8, 12	38	28	28	24	24	20	I	Analog Pin Used For Digital Input 227. This pin also has analog functions which are described in the GPIO section of this table.
SD1_C3	2							I	SDFM-1 Channel 3 Clock Input
HIC_NBE0	15							I	HIC Byte enable 0

Signal Name	Mux Position	QP100	QP80	QP80C	QP64Q	QP64S	QP48	Pin Type	Description
AIO230	0, 4, 8, 12	40	29	29	25	25	21	I	Analog Pin Used For Digital Input 230. This pin also has analog functions which are described in the GPIO section of this table.
SD1_C4	2							I	SDFM-1 Channel 4 Clock Input
HIC_BASESEL2	15							I	HIC Base address range select 2
AIO237	0, 4, 8, 12	20	16	16	12	12	8	I	Analog Pin Used For Digital Input 237. This pin also has analog functions which are described in the GPIO section of this table.
SD1_D2	2							I	SDFM-1 Channel 2 Data Input
HIC_A6	15							I	HIC Address 6
AIO238	0, 4, 8, 12	28	22	22	18	18	14	I	Analog Pin Used For Digital Input 238. This pin also has analog functions which are described in the GPIO section of this table.
SD2_C3	2							I	SDFM-2 Channel 3 Clock Input
HIC_NCS	15							I	HIC Chip select input
AIO239	0, 4, 8, 12	19	15	15	11	11	-	I	Analog Pin Used For Digital Input 239. This pin also has analog functions which are described in the GPIO section of this table.
SD1_D1	2							I	SDFM-1 Channel 1 Data Input
HIC_A5	15							I	HIC Address 5
AIO233	0, 4, 8, 12	-	14	14	10	10	7	I	Analog Pin Used For Digital Input 233. This pin also has analog functions which are described in the GPIO section of this table.
SD2_D1	2							I	SDFM-2 Channel 1 Data Input
HIC_A4	15							I	HIC Address 4
AIO226	0, 4, 8, 12	15	11	11	7	7	4	I	Analog Pin Used For Digital Input 226. This pin also has analog functions which are described in the GPIO section of this table.
SD2_D4	2							I	SDFM-2 Channel 4 Data Input
HIC_A1	15							I	HIC Address 1
AIO242	0, 4, 8, 12	16	12	12	8	8	5	I	Analog Pin Used For Digital Input 242. This pin also has analog functions which are described in the ANALOG section of this table.
SD2_D2	2							I	SDFM-2 Channel 2 Data Input
HIC_A2	15							I	HIC Address 2
AIO252	0, 4, 8, 12	32	-	-	-	-	-	I	Analog Pin Used For Digital Input 252. This pin also has analog functions which are described in the ANALOG section of this table.
SD2_C4	2							I	SDFM-2 Channel 4 Clock Input

Signal Name	Mux Position	QP100	QP80	QP80C	QP64Q	QP64S	QP48	Pin Type	Description
AIO244	0, 4, 8, 12	21	17	17	13	13	9	I	Analog Pin Used For Digital Input 244. This pin also has analog functions which are described in the ANALOG section of this table.
SD1_D3	2							I	SDFM-1 Channel 3 Data Input
HIC_A7	15							I	HIC Address 7
AIO245	0, 4, 8, 12	31	23	23	19	19	15	I	Analog Pin Used For Digital Input 245. This pin also has analog functions which are described in the ANALOG section of this table.
SD1_C2	2							I	SDFM-1 Channel 2 Clock Input
HIC_NOE	15							O	HIC Output enable for data bus
GPIO0	0, 4, 8, 12	79	63	63	52	52	42	I/O	General-Purpose Input Output 0
EPWM1_A	1							O	ePWM-1 Output A
I2CA_SDA	6							I/OD	I2C-A Open-Drain Bidirectional Data
SPIA_STE	7							I/O	SPI-A Slave Transmit Enable (STE)
FSIRXA_CLK	9							I	FSIRX-A Input Clock
MCAN_RX	10							I	CAN/CAN FD Receive
CLB_OUTPUTXBAR8	11							O	CLB Output X-BAR Output 8
EQEP1_INDEX	13							I/O	eQEP-1 Index
HIC_D7	14							I/O	HIC Data 7
HIC_BASESEL1	15							I	HIC Base address range select 1
GPIO1	0, 4, 8, 12	78	62	62	51	51	41	I/O	General-Purpose Input Output 1
EPWM1_B	1							O	ePWM-1 Output B
I2CA_SCL	6							I/OD	I2C-AOpen-Drain Bidirectional Clock
SPIA_SOMI	7							I/O	SPI-A Slave Out, Master In (SOMI)
MCAN_TX	10							O	CAN/CANFD Transmit
CLB_OUTPUTXBAR7	11							O	CLB Output X-BAR Output 7
HIC_A2	13							I	HIC Address 2
FSITXA_TDM_D1	14							I	FSITX-A Time Division Multiplexed Additional Data Input
HIC_D10	15							I/O	HIC Data 10
GPIO2	0, 4, 8, 12	77	61	61	50	50	40	I/O	General-Purpose Input Output 2
EPWM2_A	1							O	ePWM-2 Output A
OUTPUTXBAR1	5							O	X-BAR Output 1
PMBUSA_SDA	6							I/OD	PMBus-A Open-Drain Bidirectional Data
SPIA_SIMO	7							I/O	SPI-A Slave In, Master Out (SIMO)
SCIA_TX	9							O	SCI-A Transmit Data
FSIRXA_D1	10							I	FSIRX-A Optional Additional Data Input
I2CB_SDA	11							I/OD	I2C-B Open-Drain Bidirectional Data
HIC_A1	13							I	HIC Address 1
CANA_TX	14							O	CAN-A Transmit
HIC_D9	15							I/O	HIC Data 9
GPIO3	0, 4, 8, 12	76	60	60	49	49	39	I/O	General-Purpose Input Output 3
EPWM2_B	1							O	ePWM-2 Output B
OUTPUTXBAR2	2, 5							O	X-BAR Output 2

Signal Name	Mux Position	QP100	QP80	QP80C	QP64Q	QP64S	QP48	Pin Type	Description
PMBUSA_SCL	6							I/OD	PMBus-A Open-Drain Bidirectional Clock
SPIA_CLK	7							I/O	SPI-A Clock
SCIA_RX	9							I	SCI-A Receive Data
FSIRXA_D0	10							I	FSIRX-A Primary Data Input
I2CB_SCL	11							I/OD	I2C-BOpen-Drain Bidirectional Clock
HIC_NOE	13							O	HIC Output enable for data bus
CANA_RX	14							I	CAN-A Receive
HIC_D4	15							I/O	HIC Data 4
GPIO4	0, 4, 8, 12							I/O	General-Purpose Input Output 4
EPWM3_A	1							O	ePWM-3 Output A
MCAN_TX	3							O	CAN/CANFD Transmit
OUTPUTXBAR3	5							O	X-BAR Output 3
CANA_TX	6							O	CAN-A Transmit
SPIB_CLK	7							I/O	SPI-B Clock
EQEP2_STROBE	9	75	59	59	48	48	38	I/O	eQEP-2 Strobe
FSIRXA_CLK	10							I	FSIRX-A Input Clock
CLB_OUTPUTXBAR6	11							O	CLB Output X-BAR Output 6
HIC_BASESEL2	13							I	HIC Base address range select 2
HIC_NWE	15							I	HIC Data Write enable from host
GPIO5	0, 4, 8, 12							I/O	General-Purpose Input Output 5
EPWM3_B	1							O	ePWM-3 Output B
OUTPUTXBAR3	3							O	X-BAR Output 3
MCAN_RX	5							I	CAN/CANFD Receive
CANA_RX	6							I	CAN-A Receive
SPIA_STE	7	89	74	74	61	61	47	I/O	SPI-A Slave Transmit Enable (STE)
FSITXA_D1	9							O	FSITX-A Optional Additional Data Output
CLB_OUTPUTXBAR5	10							O	CLB Output X-BAR Output 5
HIC_A7	13							I	HIC Address 7
HIC_D4	14							I/O	HIC Data 4
HIC_D15	15							I/O	HIC Data 15
GPIO6	0, 4, 8, 12							I/O	General-Purpose Input Output 6
EPWM4_A	1							O	ePWM-4 Output A
OUTPUTXBAR4	2							O	X-BAR Output 4
SYNCOUT	3							O	External ePWM Synchronization Pulse
EQEP1_A	5							I	eQEP-1 Input A
SPIB_SOMI	7	97	80	80	64	64	48	I/O	SPI-B Slave Out, Master In (SOMI)
FSITXA_D0	9							O	FSITX-A Primary Data Output
FSITXA_D1	11							O	FSITX-A Optional Additional Data Output
HIC_NBE1	13							I	HIC Byte enable 1
CLB_OUTPUTXBAR8	14							O	CLB Output X-BAR Output 8
HIC_D14	15							I/O	HIC Data 14
GPIO7	0, 4, 8, 12							I/O	General-Purpose Input Output 7
EPWM4_B	1							O	ePWM-4 Output B
OUTPUTXBAR5	3							O	X-BAR Output 5
EQEP1_B	5							I	eQEP-1 Input B
SPIB_SIMO	7	84	68	68	57	57	43	I/O	SPI-B Slave In, Master Out (SIMO)
FSITXA_CLK	9							O	FSITX-A Output Clock
CLB_OUTPUTXBAR2	10							O	CLB Output X-BAR Output 2
HIC_A6	13							I	HIC Address 6

Signal Name	Mux Position	QP100	QP80	QP80C	QP64Q	QP64S	QP48	Pin Type	Description
HIC_D14	15							I/O	HIC Data 14
GPIO8	0, 4, 8, 12	74	58	58	47	47	-	I/O	General-Purpose Input Output 8
EPWM5_A	1							O	ePWM-5 Output A
ADCSOCAO	3							O	ADC Start of Conversion A for External ADC
EQEP1_STROBE	5							I/O	eQEP-1 Strobe
SCIA_TX	6							O	SCI-A Transmit Data
SPIA_SIMO	7							I/O	SPI-A Slave In, Master Out (SIMO)
I2CA_SCL	9							I/OD	I2C-A Open-Drain Bidirectional Clock
FSITXA_D1	10							O	FSITX-A Optional Additional Data Output
CLB_OUTPUTXBAR5	11							O	CLB Output X-BAR Output 5
HIC_A0	13							I	HIC Address 0
FSITXA_TDM_CLK	14							I	FSITX-A Time Division Multiplexed Clock Input
HIC_D8	15							I/O	HIC Data 8
GPIO9	0, 4, 8, 12	90	75	75	62	62	-	I/O	General-Purpose Input Output 9
EPWM5_B	1							O	ePWM-5 Output B
SCIB_TX	2							O	SCI-B Transmit Data
OUTPUTXBAR6	3							O	X-BAR Output 6
EQEP1_INDEX	5							I/O	eQEP-1 Index
SCIA_RX	6							I	SCI-A Receive Data
SPIA_CLK	7							I/O	SPI-A Clock
FSITXA_D0	10							O	FSITX-A Primary Data Output
LINB_RX	11							I	LIN-B Receive
HIC_BASESEL0	13							I	HIC Base address range select 0
I2CB_SCL	14							I/OD	I2C-B Open-Drain Bidirectional Clock
HIC_NRDY	15							O	HIC Ready from device to host
GPIO10	0, 4, 8, 12	93	76	76	63	63	-	I/O	General-Purpose Input Output 10
EPWM6_A	1							O	ePWM-6 Output A
ADCSOCBO	3							O	ADC Start of Conversion B for External ADC
EQEP1_A	5							I	eQEP-1 Input A
SCIB_TX	6							O	SCI-B Transmit Data
SPIA_SOMI	7							I/O	SPI-ASlave Out, Master In (SOMI)
I2CA_SDA	9							I/OD	I2C-A Open-Drain Bidirectional Data
FSITXA_CLK	10							O	FSITX-A Output Clock
LINB_TX	11							O	LIN-B Transmit
HIC_NWE	13							I	HIC Data Write enable from host
FSITXA_TDM_D0	14							I	FSITX-A Time Division Multiplexed Data Input
CLB_OUTPUTXBAR4	15							O	CLB Output X-BAR Output 4
GPIO11	0, 4, 8, 12	52	37	37	31	31	-	I/O	General-Purpose Input Output 11
EPWM6_B	1							O	ePWM-6 Output B
OUTPUTXBAR7	3							O	X-BAR Output 7
EQEP1_B	5							I	eQEP-1 Input B
SCIB_RX	6							I	SCI-B Receive Data
SPIA_STE	7							I/O	SPI-A Slave Transmit Enable (STE)
FSIRXA_D1	9							I	FSIRX-A Optional Additional Data Input
LINB_RX	10							I	LIN-B Receive
EQEP2_A	11							I	eQEP-2 Input A
SPIA_SIMO	13							I/O	SPI-A Slave In, Master Out (SIMO)

Signal Name	Mux Position	QP100	QP80	QP80C	QP64Q	QP64S	QP48	Pin Type	Description
HIC_D6	14							I/O	HIC Data 6
HIC_NBE0	15							I	HIC Byte enable 0
GPIO12	0, 4, 8, 12							I/O	General-Purpose Input Output 12
EPWM7_A	1							O	ePWM-7 Output A
MCAN_RX	3							I	CAN/CANFD Receive
EQEP1_STROBE	5							I/O	eQEP-1 Strobe
SCIB_TX	6							O	SCI-B Transmit Data
PMBUSA_CTL	7	51	36	36	30	30	-	I/O	PMBus-A Control Signal
FSIRXA_D0	9							I	FSIRX-A Primary Data Input
B_TX	10							O	LIN-B Transmit
SPIA_CLK	11							I/O	SPI-A Clock
CANA_RX	13							I	CAN-A Receive
HIC_D13	14							I/O	HIC Data 13
HIC_INT	15							O	HIC Device interrupt
GPIO13	0, 4, 8, 12							I/O	General-Purpose Input Output 13
EPWM7_B	1							O	ePWM-7 Output B
MCAN_TX	3							O	CAN/CANFD Transmit
EQEP1_INDEX	5							I/O	eQEP-1 Index
SCIB_RX	6							I	SCI-B Receive Data
PMBUSA_ALERT	7	50	35	35	29	29	-	I/OD	PMBus-A Alert Signal
FSIRXA_CLK	9							I	FSIRX-A Input Clock
LINB_RX	10							I	LIN-B Receive
SPIA_SOMI	11							I/O	SPI-A Slave Out, Master In (SOMI)
CANA_TX	13							O	CAN-A Transmit
HIC_D11	14							I/O	HIC Data 11
HIC_D5	15							I/O	HIC Data 5
GPIO14	0, 4, 8, 12							I/O	General-Purpose Input Output 14
EPWM8_A	1							O	ePWM-8 Output A
SCIB_TX	2							O	SCI-B Transmit Data
I2CB_SDA	5							I/OD	I2C-B Open-Drain Bidirectional Data
OUTPUTXBAR3	6							O	X-BAR Output 3
PMBUSA_SDA	7	96	79	79	-	-	-	I/OD	PMBus-A Open-Drain Bidirectional Data
SPIB_CLK	9							I/O	SPI-B Clock
EQEP2_A	10							I	eQEP-2 Input A
LINB_TX	11							O	LIN-B Transmit
EPWM3_A	13							O	ePWM-3 Output A
CLB_OUTPUTXBAR7	14							O	CLB Output X-BAR Output 7
HIC_D15	15							I/O	HIC Data 15
GPIO15	0, 4, 8, 12							I/O	General-Purpose Input Output 15
EPWM8_B	1							O	ePWM-8 Output B
SCIB_RX	2							I	SCI-B Receive Data
I2CB_SCL	5							I/OD	I2C-BOpen-Drain Bidirectional Clock
OUTPUTXBAR4	6							O	Output X-BAR Output 4
PMBUSA_SCL	7	95	78	78	-	-	-	I/OD	PMBus-AOpen-Drain Bidirectional Clock
SPIB_STE	9							I/O	SPI-B Slave Transmit Enable (STE)
EQEP2_B	10							I	eQEP-2 Input B
LINB_RX	11							I	LIN-B Receive
EPWM3_B	13							O	ePWM-3 Output B
CLB_OUTPUTXBAR6	14							O	CLB Output X-BAR Output 6
HIC_D12	15							I/O	HIC Data 12
GPIO16	0, 4, 8, 12							I/O	General-Purpose Input Output 16
SPIA_SIMO	1	54	39	39	33	33	26	I/O	SPI-A Slave In, Master Out (SIMO)
OUTPUTXBAR7	3							O	X-BAR Output 7
EPWM5_A	5							O	ePWM-5 Output A
SCIA_TX	6							O	SCI-A Transmit Data

Signal Name	Mux Position	QP100	QP80	QP80C	QP64Q	QP64S	QP48	Pin Type	Description
SD1_D1	7							I	SDFM-1 Channel 1 Data Input
EQEP1_STROBE	9							I/O	eQEP-1 Strobe
PMBUSA_SCL	10							I/OD	PMBus-AOpen-Drain Bidirectional Clock
XCLKOUT	11							O	External Clock Output. This pin outputs a divided-down version of a chosen clock signal from within the device.
EQEP2_B	13							I	eQEP-2 Input B
SPIB_SOMI	14							I/O	SPI-B Slave Out, Master In (SOMI)
HIC_D1	15							I/O	HIC Data 1
GPIO17	0, 4, 8, 12	55	40	40	34	34	-	I/O	General-Purpose Input Output 17
SPIA_SOMI	1							I/O	SPI-A Slave Out, Master In (SOMI)
OUTPUTXBAR8	3							O	X-BAR Output 8
EPWM5_B	5							O	ePWM-5 Output B
SCIA_RX	6							I	SCI-A Receive Data
SD1_C1	7							I	SDFM-1 Channel 1 Clock Input
EQEP1_INDEX	9							I/O	eQEP-1 Index
PMBUSA_SDA	10							I/OD	PMBus-A Open-Drain Bidirectional Data
CANA_TX	11							O	CAN-A Transmit
HIC_D2	15							I/O	HIC Data 2
GPIO18	0, 4, 8, 12	68	50	50	41	41	33	I/O	General-Purpose Input Output 18
SPIA_CLK	1							I/O	SPI-A Clock
SCIB_TX	2							O	SCI-B Transmit Data
CANA_RX	3							I	CAN-A Receive
EPWM6_A	5							O	ePWM-6 Output A
I2CA_SCL	6							I/OD	I2C-AOpen-Drain Bidirectional Clock
SD1_D2	7							I	SDFM-1 Channel 2 Data Input
EQEP2_A	9							I	eQEP-2 Input A
PMBUSA_CTL	10							I/O	PMBus-A Control Signal - Slave Input/Master Output
XCLKOUT	11							O	External Clock Output. This pin outputs a divided-down version of a chosen clock signal from within the device.
LINB_TX	13							O	LIN-B Transmit
FSITXA_TDM_CLK	14							I	FSITX-A Time Division Multiplexed Clock Input
HIC_INT	15							O	HIC Device interrupt to host
X2	ALT							I/O	Crystal oscillator output.
GPIO19	0, 4, 8, 12	69	51	51	42	42	34	I/O	General-Purpose Input Output 19
SPIA_STE	1							I/O	SPI-A Slave Transmit Enable (STE)
SCIB_RX	2							I	SCI-B Receive Data
CANA_TX	3							O	CAN-A Transmit
EPWM6_B	5							O	ePWM-6 Output B
I2CA_SDA	6							I/OD	I2C-A Open-Drain Bidirectional Data
SD1_C2	7							I	SDFM-1 Channel 2 Clock Input
EQEP2_B	9							I	eQEP-2 Input B
PMBUSA_ALERT	10							I/OD	PMBus-A Open-Drain Bidirectional Alert Signal
CLB_OUTPUTXBAR1	11							O	CLB Output X-BAR Output 1
LINB_RX	13							I	LIN-B Receive

Signal Name	Mux Position	QP100	QP80	QP80C	QP64Q	QP64S	QP48	Pin Type	Description
FSITXA_TDM_D0	14							I	FSITX-A Time Division Multiplexed Data Input
HIC_NBE0	15							I	HIC Byte enable 0
X1	ALT							I/O	Crystal oscillator input or single-ended clock input. The device initialization software must configure this pin before the crystal oscillator is enabled. To use this oscillator, a quartz crystal circuit must be connected to X1 and X2. This pin can also be used to feed a single-ended 3.3-V level clock. See the XTAL section for usage details.
GPIO20	0, 4, 8, 12	48	33	33	-	-	-	I/O	General-Purpose Input Output 20. This pin also has analog functions which are described in the ANALOG section of this table.
EQEP1_A	1							I	eQEP-1 Input A
SPIB_SIMO	6							I/O	SPI-B Slave In, Master Out (SIMO)
SD1_D3	7							I	SDFM-1 Channel 3 Data Input
MCAN_TX	9							O	CAN/CAN FD Transmit
GPIO21	0, 4, 8, 12	49	34	34	-	-	-	I/O	General-Purpose Input Output 21 (this pin also has analog functions)
EQEP1_B	1							I	eQEP-1 Input B
SPIB_SOMI	6							I/O	SPI-B Slave Out, Master In (SOMI)
SD1_C3	7							I	SDFM-1 Channel 3 Clock Input
MCAN_RX	9							I	CAN/CAN FD Receive
GPIO22	0, 4, 8, 12	83	67	67	56	56	-	I/O	General-Purpose Input Output 22
EQEP1_STROBE	1							I/O	eQEP-1 Strobe
SCIB_TX	3							O	SCI-B Transmit Data
SPIB_CLK	6							I/O	SPI-B Clock
SD1_D4	7							I	SDFM-1 Channel 4 Data Input
LINA_TX	9							O	LIN-A Transmit
CLB_OUTPUTXBAR1	10							O	CLB Output X-BAR Output 1
LINB_TX	11							O	LIN-B Transmit
HIC_A5	13							I	HIC Address 5
EPWM4_A	14							O	ePWM-4 Output A
HIC_D13	15							I/O	HIC Data 13
GPIO23	0, 4, 8, 12	81	65	65	54	54	-	I/O	General-Purpose Input Output 23
EQEPINDEX	1							I/O	eQEP-1 Index
SCIB_RX	3							I	SCI-B Receive Data
SPIB_STE	6							I/O	SPI-B Slave Transmit Enable (STE)
SD1_C4	7							I	SDFM-1 Channel 4 Clock Input
LINA_RX	9							I	LIN-A Receive
CLB_OUTPUTXBAR3	10							O	CLB Output X-BAR Output 3
LINB_RX	11							I	LIN-B Receive
HIC_A3	13							I	HIC Address 3
EPWM4_B	14							O	ePWM-4 Output B
HIC_D11	15							I/O	HIC Data 11
GPIO24	0, 4, 8, 12	56	41	41	35	35	27	I/O	General-Purpose Input Output 24
OUTPUTXBAR1	1							O	X-BAR Output 1
EQEP2_A	2							I	eQEP-2 Input A
EPWM8_A	5							O	ePWM-8 Output A

Signal Name	Mux Position	QP100	QP80	QP80C	QP64Q	QP64S	QP48	Pin Type	Description
SPIB_SIMO	6							I/O	SPI-B Slave In, Master Out (S
SD2_D1	7							I	SDFM-2 Channel 1 Data Input
LINB_TX	9							O	LIN-B Transmit
PMBUSA_SCL	10							I/OD	PMBus-A Open-Drain Bidirectional Clock
SCIA_TX	11							O	SCI-A Transmit Data
ERRORSTS	13							O	Error Status Output. This signal requires an external pulldown.
HIC_D3	15							I/O	HIC Data 3
GPIO25	0, 4, 8, 12	57	42	42	-	-	-	I/O	General-Purpose Input Output
OUTPUTXBAR2	1							O	Output X-BAR Output 2
EQEP2_B	2							I	eQEP-2 Input B
EQEP1_A	5							I	eQEP-1 Input A
SPIB_SOMI	6							I/O	SPI-B Slave Out, Master In (SOMI)
SD2_C1	7							I	SDFM-2 Channel 1 Clock Input
FSITXA_D1	9							O	FSITX-A Optional Additional Data Output
PMBUSA_SDA	10							I/OD	PMBus-A Open-Drain Bidirectional Data
SCIA_RX	11							I	SCI-A Receive Data
HIC_BASESEL0	14							I	HIC Base address range select 0
GPIO26	0, 4, 8, 12	58	43	43	-	-	-	I/O	General-Purpose Input Output 26
OUTPUTXBAR3	1，5							O	Output X-BAR Output 3
EQEP2_INDEX	2							I/O	eQEP-2 Index
SPIB_CLK	6							I/O	SPI-B Clock
SD2_D2	7							I	SDFM-2 Channel 2 Data Input
FSITXA_D0	9							O	FSITX-A Primary Data Output
PMBUSA_CTL	10							I/O	PMBus-A Control Signal
I2CA_SDA	11							I/OD	I2C-A Open-Drain Bidirectional Data
HIC_D0	14							I/O	HIC Data 0
HIC_A1	15							I	HIC Address 1
GPIO27	0, 4, 8, 12	59	44	44	-	-	-	I/O	General-Purpose Input Output 27
OUTPUTXBAR4	1，5							O	X-BAR Output 4
EQEP2_STROBE	2							I/O	eQEP-2 Strobe
SPIB_STE	6							I/O	SPI-B Slave Transmit Enable (STE)
SD2_C2	7							I	SDFM-2 Channel 2 Clock Input
FSITXA_CLK	9							O	FSITX-A Output Clock
PMBUSA_ALERT	10							I/OD	PMBus-A Open-Drain Bidirectional Alert Signal
I2CA_SCL	11							I/OD	I2C-A Open-Drain Bidirectional Clock
HIC_D1	14							I/O	HIC Data 1
HIC_A4	15							I	HIC Address 4
GPIO28	0, 4, 8, 12	1	4	4	2	2	2	I/O	General-Purpose Input Output 28
SCIA_RX	1							I	SCI-A Receive Data
EPWM7_A	3							O	ePWM-7 Output A
OUTPUTXBAR5	5							O	X-BAR Output 5
EQEP1_A	6							I	eQEP-1 Input A
SD2_D3	7							I	SDFM-2 Channel 3 Data Input
EQEP2_STROBE	9							I/O	eQEP-2 Strobe
LINA_TX	10							O	LIN-A Transmit
SPIB_CLK	11							I/O	SPI-B Clock
ERRORSTS	13							O	Error Status Output

Signal Name	Mux Position	QP100	QP80	QP80C	QP64Q	QP64S	QP48	Pin Type	Description
I2CB_SDA	14							I/OD	I2C-B Open-Drain Bidirectional Data
HIC_NOE	15							O	HIC Output enable for data bus
GPIO29	0, 4, 8, 12							I/O	General-Purpose Input Output 29
SCIA_TX	1							O	SCI-A Transmit Data
EPWM7_B	3							O	ePWM-7 Output B
OUTPUTXBAR6	5							O	Output X-BAR Output 6
EQEP1_B	6							I	eQEP-1 Input B
SD2_C3	7							I	SDFM-2 Channel 3 Clock Input
EQEP2_INDEX	9	100	3	3	1	1	1	I/O	eQEP-2 Index
LINA_RX	10							I	LIN-A Receive
SPIB_STE	11							I/O	SPI-B Slave Transmit Enable (STE)
ERRORSTS	13							O	Error Status Output
I2CB_SCL	14							I/OD	I2C-B Open-Drain Bidirectional Clock
HIC_NCS	15							I	HIC Chip select input
AUKIN	ALT							I	Auxilary Clock Input
GPIO30	0, 4, 8, 12							I/O	General-Purpose Input Output 30
CANA_RX	1							I	CAN-A Receive
SPIB_SIMO	3							I/O	SPI-B Slave In, Master Out
OUTPUTXBAR7	5							O	Output X-BAR Output 7
EQEP1_STROBE	6	98	1	1	-	-	-	I/O	eQEP-1 Strobe
SD2_D4	7							I	SDFM-2 Channel 4 Data Input
FSIRXA_CLK	9							I	FSIRX-A Input Clock
MCAN_RX	10							I	CAN/CANFD Receive
EPWM1_A	11							O	ePWM-1 Output A
HIC_D8	14							I/O	HIC Data 8
GPIO31	0, 4, 8, 12							I/O	General-Purpose Input Output 31
CANA_TX	1							O	CAN-A Transmit
SPIB_SOMI	3							I/O	SPI-B Slave Out, Master In (SOMI)
OUTPUTXBAR8	5							O	Output X-BAR Output 8
EQEP1_INDEX	6	99	2	2	-	-	-	I/O	eQEP-1 Index
SD2_C4	7							I	SDFM-2 Channel 4 Clock Input
FSIRXA_D1	9							I	FSIRX-A Optional Additional Data Input
MCAN_TX	10							O	CAN/CAN FD Transmit
EPWM1_B	11							O	ePWM-1 Output B
HIC_D10	14							I/O	HIC Data 10
GPIO32	0, 4, 8, 12							I/O	General-Purpose Input Output 32
I2CA_SDA	1							I/OD	I2C-A Open-Drain Bidirectional Data
SPIB_CLK	3							I/O	SPI-B Clock
EPWM8_B	5							O	ePWM-8 Output B
LINA_TX	6							O	LIN-A Transmit
SD1_D2	7	64	49	49	40	40	32	I	SDFM-1 Channel 2 Data Input
FSIRXA_D0	9							I	FSIRX-A Primary Data Input
CANA_TX	10							O	CAN-A Transmit
PMBUSA_SDA	11							I/OD	PMBus-A Open-Drain Bidirectional Data
ADCSOCBO	13							O	ADC Start of Conversion B for External ADC
HIC_INT	15							O	HIC Device interrupt to host
GPIO33	0，8，12	53	38	38	32	32	25	I/O	General-Purpose Input Output 33
I2CA_SCL	1							I/OD	I2C-AOpen-Drain Bidirectional Clock

Signal Name	Mux Position	QP100	QP80	QP80C	QP64Q	QP64S	QP48	Pin Type	Description
SPIB_STE	3							I/O	SPI-B Slave Transmit Enable (STE)
OUTPUTXBAR4	5							O	X-BAR Output 4
LINA_RX	6							I	LIN-A Receive
SD1_C2	7							I	SDFM-1 Channel 2 Clock Input
FSIRXA_CLK	9							I	FSIRX-A Input Clock
CANA_RX	10							I	CAN-A Receive
EQEP2_B	11							I	eQEP-2 Input B
ADCSOCAO	13							O	ADC Start of Conversion A for External ADC
SD1_C1	14							I	SDFM-1 Channel 1 Clock Input
HIC_D0	15							I/O	HIC Data 0
GPIO34	0, 4, 8, 12	94	77	77	-	-	-	I/O	General-Purpose Input Output 34
OUTPUTXBAR1	1							O	Output X-BAR Output 1
PMBUSA_SDA	6							I/OD	PMBus-A Open-Drain Bidirectional Data
HIC_NBE1	13							I	HIC Byte enable 1
I2CB_SDA	14							I/OD	I2C-B Open-Drain Bidirectional Data
HIC_D9	15							I/O	HIC Data 9
GPIO35	0, 4, 8, 12	63	48	48	39	39	31	I/O	General-Purpose Input Output 35
SCIA_RX	1							I	SCI-A Receive Data
I2CA_SDA	3							I/OD	I2C-A Open-Drain Bidirectional Data
CANA_RX	5							I	CAN-A Receive
PMBUSA_SCL	6							I/OD	PMBus-A Open-Drain Bidirectional Clock
LINA_RX	7							I	LIN-A Receive
EQEP1_A	9							I	eQEP-1 Input A
PMBUSA_CTL	10							I/O	PMBus-A Control Signa
EPWM5_B	11							O	ePWM-5 Output B
SD2_C1	13							I	SDFM-2 Channel 1 Clock Input
HIC_NWE	14							I	HIC Data Write enable from host
TDI	15							I	JTAG Test Data Input (TDI) - TDI is the default mux selection for the pin. The internal pullup is disabled by default. The internal pullup should be enabled or an external pullup added on the board if this pin is used as JTAG TDI to avoid a floating input.
GPIO37	0, 4, 8, 12	61	46	46	37	37	29	I/O	General-Purpose Input Output 37
OUTPUTXBAR2	1							O	X-BAR Output 2
I2CA_SCL	3							I/OD	I2C-A Open-Drain Bidirectional Clock
SCIA_TX	5							O	SCI-A Transmit Data
CANA_TX	6							O	CAN-A Transmit
LINA_TX	7							O	LIN-A Transmit
EQEP1_B	9							I	eQEP-1 Input B
PMBUSA_ALERT	10							I/OD	PMBus-A Open-Drain Bidirectional Alert Signal
HIC_NRDY	14							O	HIC Ready from device to host

Signal Name	Mux Position	QP100	QP80	QP80C	QP64Q	QP64S	QP48	Pin Type	Description
TDO	15							O	JTAG Test Data Output (TDO) - TDO is the default mux selection for the pin. The internal pullup is disabled by default. The TDO function will tristate when there is no JTAG activity, leaving this pin floating; the internal pullup should be enabled or an external pullup added on the board to avoid a floating GPIO input.
GPIO39	0, 4, 8, 12	-	56	-	-	46	-	I/O	General-Purpose Input Output 39
MCAN_RX	6							I	CAN/CANFD Receive
FSIRXA_CLK	7							I	FSIRX-A Input Clock
EQEP2_INDEX	9							I/O	eQEP-2 Index
CLB_OUTPUTXBAR2	11							O	CLB Output X-BAR Output 2
SYNCOUT	13							O	External ePWM Synchronization Pulse
EQEP1_INDEX	14							I/O	eQEP-1 Index
HIC_D7	15							I/O	HIC Data 7
GPIO40	0, 4, 8, 12	80	64	64	53	53	-	I/O	General-Purpose Input Output 40
SPIB_SIMO	1							I/O	SPI-B Slave In, Master Out
EPWM2_B	5							O	ePWM-2 Output B
PMBUSA_SDA	6							I	PMBus-A Open-Drain Bidirectional Data
FSIRXA_D0	7							I	FSIRX-A Primary Data Input
SCIB_TX	9							O	SCI-B Transmit Data
EQEP1_A	10							I	eQEP-1 Input A
LINB_TX	11							O	LIN-B Transmit
HIC_NBE1	14							I	HIC Byte enable 1
HIC_D5	15							I/O	HIC Data 5
GPIO41	0, 4, 8, 12	82	66	66	55	55	-	I/O	General-Purpose Input Output 41
EPWM2_A	5							O	ePWM-2 Output A
PMBUSA_SCL	6							I/OD	PMBus-AOpen-Drain Bidirectional Clock
FSIRXA_D1	7							I	FSIRX-A Optional Additional Data Input
SCIB_RX	9							I	SCI-B Receive Data
EQEP1_B	10							I	eQEP-1 Input B
LINB_RX	11							I	LIN-B Receive
HIC_A4	13							I	HIC Address 4
SPIB_SOMI	14							I/O	SPI-B Slave Out, Master In (SOMI)
HIC_D12	15							I/O	HIC Data 12
GPIO42	0, 4, 8, 12	-	57	57	-	-	-	I/O	General-Purpose Input Output 42
LINA_RX	2							I	LIN-A Receive
OUTPUTXBAR5	3							O	X-BAR Output 5
PMBUSA_CTL	5							I/O	PMBus-A Control Signa
I2CA_SDA	6							I/OD	I2C-A Open-Drain Bidirectional Data
EQEP1_STROBE	10							I/O	eQEP-1 Strobe
CLB_OUTPUTXBAR3	11							O	CLB Output X-BAR Output 3
HIC_D2	14							I/O	HIC Data 2
HIC_A6	15							I	HIC Address 6
GPIO43	0, 4, 8, 12	-	54	54	-	-	-	I/O	General-Purpose Input Output 43
OUTPUTXBAR6	3							O	Output X-BAR Output 6
PMBUSA_ALERT	5，9							I/OD	PMBus-A Open-Drain Bidirectional Alert Signal

Signal Name	Mux Position	QP100	QP80	QP80C	QP64Q	QP64S	QP48	Pin Type	Description
I2CA_SCL	6							I/OD	I2C-AOpen-Drain Bidirectional Clock
EQEP1_INDEX	10							I/O	eQEP-1 Index
CLB_OUTPUTXBAR4	11							O	CLB Output X-BAR Output 4
SD2_D3	13							I	SDFM-2 Channel 3 Data Input
HIC_D3	14							I/O	HIC Data 3
HIC_A7	15							I	HIC Address 7
GPIO44	0, 4, 8, 12							I/O	General-Purpose Input Output 44
OUTPUTXBAR7	3							O	Output X-BAR Output 7
EQEP1_A	5							I	eQEP-1 Input A
PMBUSA_SDA	6							I/OD	PMBus-A Open-Drain Bidirectional Data
FSITXA_CLK	7							O	FSITX-A Output Clock
PMBUSA_CTL	9	85	69	69	-	-	-	I/O	PMBus-A Control Signal
CLB_OUTPUTXBAR3	10							O	CLB Output X-BAR Output 3
FSIRXA_D0	11							I	FSIRX-A Primary Data Input
HIC_D7	13							I/O	HIC Data 7
LINB_TX	14							O	LIN-B Transmit
HIC_D5	15							I/O	HIC Data 5
GPIO45	0, 4, 8, 12							I/O	General-Purpose Input Output 45
OUTPUTXBAR8	3							O	X-BAR Output 8
FSITXA_D0	7							O	FSITX-A Primary Data Output
PMBUSA_ALERT	9	-	73	73	-	-	-	I/OD	PMBus-A Open-Drain Bidirectional Alert Signal
CLB_OUTPUTXBAR4	10							O	CLB Output X-BAR Output 4
SD2_C3	13							I	SDFM-2 Channel 3 Clock Input
HIC_D6	15							I/O	HIC Data 6
GPIO46	0, 4, 8, 12							I/O	General-Purpose Input Output 46
LINA_TX	3							O	LIN-A Transmit
MCAN_TX	5							O	CAN/CANFD Transmit
FSITXA_D1	7							O	FSITX-A Optional Additional Data Output
PMBUSA_SDA	9	-	6	6	-	-	-	I/OD	PMBus-A Open-Drain Bidirectional Data
SD2_C4	13							I	SDFM-2 Channel 4 Clock Input
HIC_NWE	15							I	HIC Data Write enable from host
GPIO47	0, 4, 8, 12							I/O	General-Purpose Input Output 47
LINA_RX	3							I	LIN-A Receive
MCAN_RX	5							I	CAN/CANFD Receive
CLB_OUTPUTXBAR2	7							O	CLB Output X-BAR Output 2
PMBUSA_SCL	9	6						I/OD	PMBus-AOpen-Drain Bidirectional Clock
SD2_D4	13							I	SDFM-2 Channel 4 Data Input
FSITXA_TDM_CLK	14							I	FSITX-A Time Division Multiplexed Clock Input
HIC_A6	15							I	HIC Address 6
GPIO48	0, 4, 8, 12							I/O	General-Purpose Input Output 48
OUTPUTXBAR3	1							O	X-BAR Output 3
CANA_TX	3							O	CAN-A Transmit
SCIA_TX	6	7		31				O	SCI-A Transmit Data
SD1_D1	7							I	SDFM-1 Channel 1 Data Input
PMBUSA_SDA	9							I/OD	PMBus-A Open-Drain Bidirectional Data

Signal Name	Mux Position	QP100	QP80	QP80C	QP64Q	QP64S	QP48	Pin Type	Description
HIC_A7	15							I	HIC Address 7
GPIO49	0, 4, 8, 12	8		32				I/O	General-Purpose Input Output 49
OUTPUTXBAR4	1							O	Output X-BAR Output 4
CANA_RX	3							I	CAN-A Receive
SCIA_RX	6							I	SCI-A Receive Data
SD1_C1	7							I	SDFM-1 Channel 1 Clock Input
LINA_RX	9							I	LIN-A Receive
SD2_D1	13							I	SDFM-2 Channel 1 Data Input
FSITXA_D0	14							O	FSITX-A Primary Data Output
HIC_D2	15							I/O	HIC Data 2
GPIO50	0, 4, 8, 12	9						I/O	General-Purpose Input Output 50
EQEP1_A	1							I	eQEP-1 Input A
MCAN_TX	5							O	CAN/CANFD Transmit
SPIB_SIMO	6							I/O	SPI-B Slave In, Master Out
SD1_D2	7							I	SDFM-1 Channel 2 Data Input
I2CB_SDA	9							I/OD	I2C-B Open-Drain Bidirectional Data
SD2_D2	13							I	SDFM-2 Channel 2 Data Input
FSITXA_D1	14							O	FSITX-A Optional Additional Data Output
HIC_D3	15							I/O	HIC Data 3
GPIO51	0, 4, 8, 12	10						I/O	General-Purpose Input Output 51
EQEP1_B	1							I	eQEP-1 Input B
MCAN_RX	5							I	CAN/CANFD Receive
SPIB_SOMI	6							I/O	SPI-B Slave Out, Master In (SOMI)
SD1_C2	7							I	SDFM-1 Channel 2 Clock Input
I2CB_SCL	9							I/OD	I2C-B Open-Drain Bidirectional Clock
SD2_D3	13							I	SDFM-2 Channel 3 Data Input
FSITXA_CLK	14							O	FSITX-A Output Clock
HIC_D6	15							I/O	HIC Data 6
GPIO52	0, 4, 8, 12	11						I/O	General-Purpose Input Output 52
EQEP1_STROBE	1							I/O	eQEP-1 Strobe
CLB_OUTPUTXBAR5	5							O	CLB Output X-BAR Output 5
SPIB_CLK	6							I/O	SPI-B Clock
SD1_D3	7							I	SDFM-1 Channel 3 Data Input
SYNCOUT	9							O	External ePWM Synchronization Pulse
SD2_D4	13							I	SDFM-2 Channel 4 Data Input
FSIRXA_D0	14							I	FSIRX-A Primary Data Input
HIC_NWE	15							I	HIC Data Write enable from host
GPIO53	0, 4, 8, 12	12						I/O	General-Purpose Input Output 53
EQEP1_INDEX	1							I/O	eQEP-1 Index
CLB_OUTPUTXBAR6	5							O	CLB Output X-BAR Output 6
SPIB_STE	6							I/O	SPI-B Slave Transmit Enable (STE)
SD1_C3	7							I	SDFM-1 Channel 3 Clock Input
ADCSOCAO	9							O	ADC Start of Conversion A for External ADC
CANA_RX	10							I	CAN-A Receive

Signal Name	Mux Position	QP100	QP80	QP80C	QP64Q	QP64S	QP48	Pin Type	Description
SD1_C1	13							I	SDFM-1 Channel 1 Clock Input
FSIRXA_D1	14							I	FSIRX-A Optional Additional Data Input
GPIO54	0, 4, 8, 12							I/O	General-Purpose Input Output 54
SPIA_SIMO	1							I/O	SPI-A Slave In, Master Out (SIMO)
EQEP2_A	5							I	eQEP-2 Input A
OUTPUTXBAR2	6							O	X-BAR Output 2
SD1_D4	7							I	SDFM-1 Channel 4 Data Input
ADCSOCBO	9	13						O	ADC Start of Conversion B for External ADC
LINB_TX	10							O	LIN-B Transmit
SD1_C2	13							I	SDFM-1 Channel 2 Clock Input
FSIRXA_CLK	14							I	FSIRX-A Input Clock
FSITXA_TDM_D1	15							I	FSITX-A Time Division Multiplexed Additional Data Input
GPIO55	0，4，12							I/O	General-Purpose Input Output 55
SPIA_SOMI	1							I/O	SPI-A Slave Out, Master In (SOMI)
EQEP2_B	5							I	eQEP-2 Input B
OUTPUTXBAR3	6							O	X-BAR Output 3
SD1_C4	7	43						I	SDFM-1 Channel 4 Clock Input
ERRORSTS	9							O	Error Status Output. This signal requires an external pulldown.
LINB_RX	10							I	LIN-B Receive
SD1_C3	13							I	SDFM-1 Channel 3 Clock Input
HIC_A0	15							I	HIC Address 0
GPIO56	0, 4, 8, 12							I/O	General-Purpose Input Output 56
SPIA_CLK	1							I/O	SPI-A Clock
CLB_OUTPUTXBAR7	2							O	CLB Output X-BAR Output 7
MCAN_TX	3							O	CAN/CANFD Transmit
EQEP2_STROBE	5							I/O	eQEP-2 Strobe
SCIB_TX	6							O	SCI-B Transmit Data
SD2_D1	7	65						I	SDFM-2 Channel 1 Data Input
SPIB_SIMO	9							I/O	SPI-B Slave In, Master Out
I2CA_SDA	10							I/OD	I2C-A Open-Drain Bidirectional Data
EQEP1_A	11							I	eQEP-1 Input A
SD1_C4	13							I	SDFM-1 Channel 4 Clock Input
FSIRXA_D1	14							I	FSIRX-A Optional Additional Data Input
HIC_D6	15							I/O	HIC Data 6
GPIO57	0, 4, 8, 12							I/O	General-Purpose Input Output 57
SPIA_STE	1							I/O	SPI-A Slave Transmit Enable (STE)
CLB_OUTPUTXBAR8	2							O	CLB Output X-BAR Output 8
MCAN_RX	3							I	CAN/CANFD Receive
EQEP2_INDEX	5	66						I/O	eQEP-2 Index
SCIB_RX	6							I	SCI-B Receive Data
SD2_C1	7							I	SDFM-2 Channel 1 Clock Input
SPIB_SOMI	9							I/O	SPI-B Slave Out, Master In (SOMI)
I2CA_SCL	10							I/OD	I2C-AOpen-Drain Bidirectional Clock

Signal Name	Mux Position	QP100	QP80	QP80C	QP64Q	QP64S	QP48	Pin Type	Description
EQEP1_B	11							I	eQEP-1 Input B
FSIRXA_CLK	14							I	FSIRX-A Input Clock
HIC_D4	15							I/O	HIC Data 4
GPIO58	0, 4, 8, 12							I/O	General-Purpose Input Output 58
OUTPUTXBAR1	5							O	X-BAR Output 1
SPIB_CLK	6							I/O	SPI-B Clock
SD2_D2	7							I	SDFM-2 Channel 2 Data Input
LINA_TX	9							O	LIN-A Transmit
CANA_TX	10	67						O	CAN-A Transmit
EQEP1_STROBE	11							I/O	eQEP-1 Strobe
SD2_C2	13							I	SDFM-2 Channel 2 Clock Input
FSIRXA_D0	14							I	FSIRX-A Primary Data Input
HIC_NRDY	15							O	HIC Ready from device to host
GPIO59	0, 4, 8, 12							I/O	General-Purpose Input Output 59
OUTPUTXBAR2	5							O	X-BAR Output 2
SPIB_STE	6							I/O	SPI-B Slave Transmit Enable (STE)
SD2_C2	7							I	SDFM-2 Channel 2 Clock Input
LINA_RX	9	92						I	LIN-A Receive
CANA_RX	10							I	CAN-A Receive
EQEP1_INDEX	11							I/O	eQEP-1 Index
SD2_C3	13							I	SDFM-2 Channel 3 Clock Input
FSITXA_TDM_D1	14							I	FSITX-A Time Division Multiplexed Additional Data Input
GPIO60	0, 4, 8, 12							I/O	General-Purpose Input Output 60
MCAN_TX	3							O	CAN/CANFD Transmit
OUTPUTXBAR3	5							O	Output X-BAR Output 3
SPIB_SIMO	6	44						I/O	SPI-B Slave In, Master Out
SD2_D3	7							I	SDFM-2 Channel 3 Data Input
SD2_C4	13							I	SDFM-2 Channel 4 Clock Input
HIC_A0	15							I	HIC Address 0
GPIO61	0, 4, 8, 12							I/O	General-Purpose Input Output 61
MCAN_RX	3							I	CAN/CANFD Receive
OUTPUTXBAR4	5							O	Output X-BAR Output 4
SPIB_SOMI	6	91						I/O	SPI-B Slave Out, Master In (SOMI)
SD2_C3	7							I	SDFM-2 Channel 3 Clock Input
CANA_RX	14							I	CAN-A Receive
TEST, JTAG, AND RESET									
TCK		60	45	45	36	36	28	I	JTAG test clock with internal pullup.
TMS		62	47	47	38	38	30	I/O	JTAG test-mode select (TMS) with internal pullup. This serial control input is clocked into the TAP controller on the rising edge of TCK. This device does not have a TRSTn pin. An external pullup resistor (recommended 2.2 kΩ) on the TMS pin to VDDIO should be placed on the board to keep JTAG in reset during normal operation.

Signal Name	Mux Position	QP100	QP80	QP80C	QP64Q	QP64S	QP48	Pin Type	Description
XRSn		2	5	5	3	3	3	I/OD	Device Reset (in) and Watchdog Reset (out). During a power-on condition, this pin is driven low by the device. An external circuit may also drive this pin to assert a device reset. This pin is also driven low by the MCU when a watchdog reset occurs. During watchdog reset, the XRSn pin is driven low for the watchdog reset duration of 512 OSCCLK cycles. A resistor between 2.2 kΩ and 10 kΩ should be placed between XRSn and VDDIO. If a capacitor is placed between XRSn and VSS for noise filtering, it should be 100 nF or smaller. These values will allow the watchdog to properly drive the XRSn pin to VOL within 512 OSCCLK cycles when the watchdog reset is asserted. This pin is an open-drain output with an internal pullup. If this pin is driven by an external device, it should be done using an open-drain device.
POWER AND GROUND									
VDD		4, 46, 71, 87	8, 31, 53, 71	8, 53, 71	4, 27, 44, 59	4, 27, 44, 59	23, 36, 45		1.1V Digital Logic Power Pins. See the Power Management Module (PMM) section for usage details.
VDDA		34	26	26	22	22	18		3.3V Analog Power Pins. Place a minimum 2.2-µF decoupling capacitor on each pin. See the Power Management Module (PMM) section for usage details.
VDDIO		3, 47, 70, 88	7, 32, 52, 72	7, 52, 72	28, 43, 60	28, 43, 60	24, 35, 46		3.3V Digital I/O Power Pins. See the Power Management Module (PMM) section for usage details.
VREGENZ		73		56	46			I	Internal voltage regulator disable with internal pulldown. Tie low to VSS to enable internal VREG. Tie high to VDDIO to use an external supply. See the Power Management Module (PMM) section for usage details.
VSS		5, 45, 72, 86	9, 30, 55, 70	9, 30, 55, 70	5, 26, 45, 58	5, 26, 45, 58	22, 37, 44		Digital Ground
VSSA		33	25	25	21	21	17		Analog Ground

5.3 Signal Descriptions
信号描述

5.3.1 Analog Signals
模拟信号

Table 5-2. Analog Signals

SIGNALNAME	PINTYPE	DESCRIPTION	QP100	QP80/QP80C	QP64S/QP64Q	QP48
A0	I	ADC-A Input 0	23	19	15	11
A1	I	ADC-A Input 1	22	18	14	10
A2	I	ADC-A Input 2	17	13	9	6
A3	I	ADC-A Input 3	18	12	8	5
A4	I	ADC-A Input 4	36	27	23	19
A5	I	ADC-A Input 5	35	17	13	9
A6	I	ADC-A Input 6	14	10	6	4
A7	I	ADC-A Input 7	31	23	19	15
A8	I	ADC-A Input 8	37	24	20	16
A9	I	ADC-A Input 9	38	28	24	20
A10	I	ADC-A Input 10	40	29	25	21
A11	I	ADC-A Input 11	20	16	12	8
A12	I	ADC-A Input 12	28	22	18	14
A14	I	ADC-A Input 14	19	15	11	
A15	I	ADC-A Input 15		14	10	7
AIO224	I	Analog Pin Used For Digital Input 224	17	13	9	6
AIO225	I	Analog Pin Used For Digital Input 225	36	27	23	19
AIO226	I	Analog Pin Used For Digital Input 226	15	11	7	4
AIO227	I	Analog Pin Used For Digital Input 227	38	28	24	20
AIO228	I	Analog Pin Used For Digital Input 228	14	10	6	4
AIO229	I	Analog Pin Used For Digital Input 229	18			
AIO230	I	Analog Pin Used For Digital Input 230	40	29	25	21
AIO231	I	Analog Pin Used For Digital Input 231	23	19	15	11
AIO232	I	Analog Pin Used For Digital Input 232	22	18	14	10
AIO233	I	Analog Pin Used For Digital Input 233		14	10	7
AIO236	I	Analog Pin Used For Digital Input 236	39	28	24	20
AIO237	I	Analog Pin Used For Digital Input 237	20	16	12	8
AIO238	I	Analog Pin Used For Digital Input 238	28	22	18	14
AIO239	I	Analog Pin Used For Digital Input 239	19	15	11	
AIO240	I	Analog Pin Used For Digital Input 240	37			
AIO241	I	Analog Pin Used For Digital Input 241		24	20	16
AIO242	I	Analog Pin Used For Digital Input 242	16	12	8	5
AIO244	I	Analog Pin Used For Digital Input 244	21	17	13	9
AIO245	I	Analog Pin Used For Digital Input 245	31	23	19	15
AIO247	I	Analog Pin Used For Digital Input 247	42			
AIO248	I	Analog Pin Used For Digital Input 248	29	22	18	14
AIO249	I	Analog Pin Used For Digital Input 249	35			
AIO251	I	Analog Pin Used For Digital Input 251	30			

SIGNALNAME	PINTYPE	DESCRIPTION	QP100	QP80/QP80C	QP64S/QP64Q	QP48
AIO252	I	Analog Pin Used For Digital Input 252	32			
AIO253	I	Analog Pin Used For Digital Input 253	41			
B0	I	ADC-B Input 0	41	24	20	16
B1	I	ADC-B Input 1	40	29	25	21
B2	I	ADC-B Input 2	15	11	7	4
B3	I	ADC-B Input 3	16	12	8	5
B4	I	ADC-B Input 4	39	28	24	20
B5	I	ADC-B Input 5	32，48	33		
B6	I	ADC-B Input 6	17	13	9	6
B7	I	ADC-B Input 7	22	18	14	10
B8	I	ADC-B Input 8	36	27	23	19
B9	I	ADC-B Input 9	18	14	10	7
B10	I	ADC-B Input 10	20	16	12	8
B11	I	ADC-B Input 11	30, 49	34		
B12	I	ADC-B Input 12	21	17	13	9
B14	I	ADC-B Input 14	19	15	11	
B15	I	ADC-B Input 15	23	19	15	11
C0	I	ADC-C Input 0	20	16	12	8
C1	I	ADC-C Input 1	29	22	18	14
C2	I	ADC-C Input 2	21	17	13	9
C3	I	ADC-C Input 3	31	23	19	15
C4	I	ADC-C Input 4	19	15	11	
C5	I	ADC-C Input 5	28	12	8	5
C6	I	ADC-C Input 6	15	11	7	4
C7	I	ADC-C Input 7	18	14	10	7
C8	I	ADC-C Input 8	39	28	24	20
C9	I	ADC-C Input 9	17	13	9	6
C10	I	ADC-C Input 10	40	29	25	21
C11	I	ADC-C Input 11	41	24	20	16
C14	I	ADC-C Input 14	42	27	23	19
C15	I	ADC-C Input 15	23	19	15	11
CMP1_HN0	I	CMPSS-1 High Comparator Negative Input 0		14	10	7
CMP1_HN1	I	CMPSS-1 High Comparator Negative Input 1	20	16	12	8
CMP1_HP0	I	CMPSS-1 High Comparator Positive Input 0	17	13	9	6
CMP1_HP1	I	CMPSS-1 High Comparator Positive Input 1	20	16	12	8
CMP1_HP2	I	CMPSS-1 High Comparator Positive Input 2	14	10	6	4
CMP1_HP3	I	CMPSS-1 High Comparator Positive Input 3		14	10	7
CMP1_HP4	I	CMPSS-1 High Comparator Positive Input 4	22	18	14	10
CMP1_HP5	I	CMPSS-1 High Comparator Positive Input 5	32，48	33		
CMP1_LN0	I	CMPSS-1 Low Comparator Negative Input 0		14	10	7
CMP1_LN1	I	CMPSS-1 Low Comparator Negative Input 1	20	16	12	8
CMP1_LP0	I	CMPSS-1 Low Comparator Positive Input 0	17	13	9	6
CMP1_LP1	I	CMPSS-1 Low Comparator Positive Input 1	20	16	12	8
CMP1_LP2	I	CMPSS-1 Low Comparator Positive Input 2	14	10	6	4
CMP1_LP3	I	CMPSS-1 Low Comparator Positive Input 3		14	10	7

SIGNALNAME	PINTYPE	DESCRIPTION	QP100	QP80/QP80C	QP64S/QP64Q	QP48
CMP1_LP4	I	CMPSS-1 Low Comparator Positive Input 4	22	18	14	10
CMP1_LP5	I	CMPSS-1 Low Comparator Positive Input 5	32，48	33		
CMP2_HN0	I	CMPSS-2 High Comparator Negative Input 0	40	29	25	21
CMP2_HN1	I	CMPSS-2 High Comparator Negative Input 1	28	22	18	14
CMP2_HP0	I	CMPSS-2 High Comparator Positive Input 0	36	27	23	19
CMP2_HP1	I	CMPSS-2 High Comparator Positive Input 1	28	22	18	14
CMP2_HP2	I	CMPSS-2 High Comparator Positive Input 2	38	28	24	20
CMP2_HP3	I	CMPSS-2 High Comparator Positive Input 3	40	29	25	21
CMP2_HP4	I	CMPSS-2 High Comparator Positive Input 4	41	24	20	16
CMP2_HP5	I	CMPSS-2 High Comparator Positive Input 5	35	17	13	9
CMP2_LN0	I	CMPSS-2 Low Comparator Negative Input 0	40	29	25	21
CMP2_LN1	I	CMPSS-2 Low Comparator Negative Input 1	28	22	18	14
CMP2_LP0	I	CMPSS-2 Low Comparator Positive Input 0	36	27	23	19
CMP2_LP1	I	CMPSS-2 Low Comparator Positive Input 1	28	22	18	14
CMP2_LP2	I	CMPSS-2 Low Comparator Positive Input 2	38	28	24	20
CMP2_LP3	I	CMPSS-2 Low Comparator Positive Input 3	40	29	25	21
CMP2_LP4	I	CMPSS-2 Low Comparator Positive Input 4	41	24	20	16
CMP2_LP5	I	CMPSS-2 Low Comparator Positive Input 5	35	17	13	9
CMP3_HN0	I	CMPSS-3 High Comparator Negative Input 0	16	12	8	5
CMP3_HN1	I	CMPSS-3 High Comparator Negative Input 1	21	17	13	9
CMP3_HP0	I	CMPSS-3 High Comparator Positive Input 0	15	11	7	4
CMP3_HP1	I	CMPSS-3 High Comparator Positive Input 1	21	17	13	9
CMP3_HP2	I	CMPSS-3 High Comparator Positive Input 2	23	19	15	11
CMP3_HP3	I	CMPSS-3 High Comparator Positive Input 3	16	12	8	5
CMP3_HP4	I	CMPSS-3 High Comparator Positive Input 4	19	15	11	
CMP3_HP5	I	CMPSS-3 High Comparator Positive Input 5	18	12	8	5
CMP3_LN0	I	CMPSS-3 Low Comparator Negative Input 0	16	12	8	5
CMP3_LN1	I	CMPSS-3 Low Comparator Negative Input 1	21	17	13	9
CMP3_LP0	I	CMPSS-3 Low Comparator Positive Input 0	15	11	7	4
CMP3_LP1	I	CMPSS-3 Low Comparator Positive Input 1	21	17	13	9
CMP3_LP2	I	CMPSS-3 Low Comparator Positive Input 2	23	19	15	11
CMP3_LP3	I	CMPSS-3 Low Comparator Positive Input 3	16	12	8	5
CMP3_LP4	I	CMPSS-3 Low Comparator Positive Input 4	19	15	11	
CMP3_LP5	I	CMPSS-3 Low Comparator Positive Input 5	18	12	8	5
CMP4_HN0	I	CMPSS-4 High Comparator Negative Input 0	42	27	23	19
CMP4_HN1	I	CMPSS-4 High Comparator Negative Input 1	31	23	19	15
CMP4_HP0	I	CMPSS-4 High Comparator Positive Input 0	39	28	24	20
CMP4_HP1	I	CMPSS-4 High Comparator Positive Input 1	31	23	19	15
CMP4_HP2	I	CMPSS-4 High Comparator Positive Input 2	29	22	18	14
CMP4_HP3	I	CMPSS-4 High Comparator Positive Input 3	42	27	23	19
CMP4_HP4	I	CMPSS-4 High Comparator Positive Input 4	37	24	20	16
CMP4_HP5	I	CMPSS-4 High Comparator Positive Input 5	30, 49	34		
CMP4_LN0	I	CMPSS-4 Low Comparator Negative Input 0	42	27	23	19
CMP4_LN1	I	CMPSS-4 Low Comparator Negative Input 1	31	23	19	15
CMP4_LP0	I	CMPSS-4 Low Comparator Positive Input 0	39	28	24	20

SIGNALNAME	PINTYPE	DESCRIPTION	QP100	QP80/QP80C	QP64S/QP64Q	QP48
CMP4_LP1	I	CMPSS-4 Low Comparator Positive Input 1	31	23	19	15
CMP4_LP2	I	CMPSS-4 Low Comparator Positive Input 2	29	22	18	14
CMP4_LP3	I	CMPSS-4 Low Comparator Positive Input 3	42	27	23	19
CMP4_LP4	I	CMPSS-4 Low Comparator Positive Input 4	37	24	20	16
CMP4_LP5	I	CMPSS-4 Low Comparator Positive Input 5	30, 49	34		
DACA_OUT	O	Buffered DAC-A Output.	23	19	15	11
DACB_OUT	O	Buffered DAC-B Output.	22	18	14	10
VDAC	I	Optional external reference voltage for on-chip DACs.	16	12	8	5
VREFHI	I	ADC High Reference. In external reference mode, externally drive the high reference voltage onto this pin. In internal reference mode, a voltage is driven onto this pin by the device. In either mode, place at least a 2.2-µF capacitor on this pin. This capacitor should be placed as close to the device as possible between the VREFHI and VREFLO pins.	24, 25	20	16	12
VREFLO	I	ADC Low Reference	26, 27	21	17	13

5.3.2Digital Signals
数字信号

Table 5-3. Digital Signals

SIGNAL NAME	PIN TYPE	DESCRIPTION	GPIO	QP100	QP80	QP80C	QP64S	QP64Q	QP48
ADCSOCAO	O	ADC Start of Conversion A for External ADC	8, 33, 53	12, 53, 74	38, 58	38, 58	32, 47	32, 47	25
ADCSOCBO	O	ADC Start of Conversion B for External ADC	10, 32, 54	13, 64, 93	49, 76	49, 76	40, 63	40, 63	32
AUXCLKIN	I	Auxiliary Clock Input	29	100	3	3	1	1	1
CANA_RX	I	CAN-A Receive	3, 5, 12, 18, 30, 33, 35, 49, 53, 59, 61	8, 12, 51, 53, 63, 68, 76, 89, 91, 92, 98	1, 36, 38, 48, 50, 60, 74	1, 32, 36, 38, 48, 50, 60, 74	30, 32, 39, 41, 49, 61	30, 32, 39, 41, 49, 61	25, 31, 33, 39, 47
CANA_TX	O	CAN-A Transmit	2, 4, 13, 17, 19, 31, 32, 37, 48, 58	7, 50, 55, 61, 64, 67, 69, 75, 77, 99	2, 35, 40, 46, 49, 51, 59, 61	2, 31, 35, 40, 46, 49, 51, 59, 61	29, 34, 37, 40, 42, 48, 50	29, 34, 37, 40, 42, 48, 50	29, 32, 34, 38, 40
CLB_OUTPUTXBAR1	O	CLB Output X-BAR Output 1	19, 22	69, 83	51, 67	51, 67	42, 56	42, 56	34
CLB_OUTPUTXBAR2	O	CLB Output X-BAR Output 2	7, 39, 47	6, 84	56, 68	68	46, 57	57	43
CLB_OUTPUTXBAR3	O	CLB Output X-BAR Output 3	23, 42, 44	81, 85	57, 65, 69	57, 65, 69	54	54	
CLB_OUTPUTXBAR4	O	CLB Output X-BAR Output 4	10, 43, 45	93	54, 73, 76	57, 65, 69	63	63	
CLB_OUTPUTXBAR5	O	CLB Output X-BAR Output 5	5, 8, 52	11, 74, 89	58, 74	58, 74	47, 61	47, 61	47
CLB_OUTPUTXBAR6	O	CLB Output X-BAR Output 6	4, 15, 53	12, 75, 95	59, 78	59, 78	48	48	38
CLB_OUTPUTXBAR7	O	CLB Output X-BAR Output 7	1, 14, 56	65, 78, 96	62, 79	62, 79	51	51	41
CLB_OUTPUTXBAR8	O	CLB Output X-BAR Output 8	0, 6, 57	66, 79, 97	63, 80	63, 80	52, 64	52, 64	42, 48
EPWM1_A	O	ePWM-1 Output A	0, 30	79, 98	1, 63	1, 63	52	52	42
EPWM1_B	O	ePWM-1 Output B	1, 31	78, 99	2, 62	2, 62	51	51	41
EPWM2_A	O	ePWM-2 Output A	2, 41	77, 82	61, 66	61, 66	50, 55	50, 55	40
EPWM2_B	O	ePWM-2 Output B	3, 40	76, 80	60, 64	60, 64	49, 53	49, 53	39
EPWM3_A	O	ePWM-3 Output A	4, 14	75, 96	59, 79	59, 79	48	48	38
EPWM3_B	O	ePWM-3 Output B	5, 15	89, 95	74, 78	74, 78	61	61	47
EPWM4_A	O	ePWM-4 Output A	6, 22	83, 97	67, 80	67, 80	56, 64	56, 64	48
EPWM4_B	O	ePWM-4 Output B	7, 23	81, 84	65, 68	65, 68	54, 57	54, 57	43
EPWM5_A	O	ePWM-5 Output A	8, 16	54, 74	39, 58	39, 58	33, 47	33, 47	26
EPWM5_B	O	ePWM-5 Output B	9, 17, 35	55, 63, 90	40, 48, 75	40, 48, 75	34, 39, 62	34, 39, 62	31
EPWM6_A	O	ePWM-6 Output A	10, 18	68, 93	50, 76	50, 76	41, 63	41, 63	33
EPWM6_B	O	ePWM-6 Output B	11, 19	52, 69	37, 51	37, 51	31, 42	31, 42	34
EPWM7_A	O	ePWM-7 Output A	12, 28	1, 51	4, 36	4, 36	2, 30	2, 30	2
EPWM7_B	O	ePWM-7 Output B	13, 29	50, 100	3, 35	3, 35	1, 29	1, 29	1
EPWM8_A	O	ePWM-8 Output A	14, 24	56, 96	41, 79	41, 79	35	35	27
EPWM8_B	O	ePWM-8 Output B	15, 32	64, 95	49, 78	49, 78	40	40	32
EQEP1_A	I	eQEP-1 Input A	6, 10, 20, 25, 28, 35, 40, 44, 50, 56	1, 9, 48, 57, 63, 65, 80, 85, 93, 97	4, 33, 42, 48, 64, 69, 76, 80	4, 33, 42, 48, 64, 69, 76, 80	2, 39, 53, 63, 64	2, 39, 53, 63, 64	2, 31, 48
EQEP1_B	I	eQEP-1 Input B	7, 11, 21, 29, 37, 41, 51, 57	10, 49, 52, 61, 66, 82, 84, 100	3, 34, 37, 46, 66, 68	3, 34, 37, 46, 66, 68	1, 31, 37, 55, 57	1, 31, 37, 55, 57	1, 29, 43
EQEP1_INDEX	I/O	eQEP-1 Index	0, 9, 13, 17, 23, 31, 39, 43, 53, 59	12, 50, 55, 79, 81, 90, 92, 99	2, 35, 40, 54, 56, 63, 65, 75	2, 35, 40, 54, 63, 65, 75	29, 34, 46, 52, 54, 62	29, 34, 52, 54, 62	42
EQEP1_STROBE	I/O	eQEP-1 Strobe	8, 12, 16, 22, 30, 42, 52, 58	11, 51, 54, 67, 74, 83, 98	1, 36, 39, 57, 58, 67	1, 36, 39, 57, 58, 67	30, 33, 47, 56	30, 33, 47, 56	26

SIGNAL NAME	PIN TYPE	DESCRIPTION	GPIO	QP100	QP80	QP80C	QP64S	QP64Q	QP48
EQEP2_A	I	eQEP-2 Input A	11, 14, 18,24, 54	13, 52, 56, 68, 96	37, 41, 50, 79	37, 41, 50, 79	31, 35, 41	31, 35, 41	27, 33
EQEP2_B	I	eQEP-2 Input B	15, 16, 19,25, 33, 55	43, 53, 54, 57, 69, 95	38, 39, 42, 51, 78	38, 39, 42, 51, 78	32, 33, 42	32, 33, 42	25,26, 34
EQEP2_INDEX	I/O	eQEP-2 Index	26, 29, 39, 57	58, 66, 100	3, 43, 56	3, 43	1, 46	1	1
EQEP2_STROBE	I/O	eQEP-2 Strobe	4, 27, 28, 56	1, 59, 65, 75	4, 44, 59	4, 44, 59	2, 48	2, 48	2, 38
ERRORSTS	O	Error Status Output. This signal requires an external pulldown.	24, 28, 29, 55	1, 43, 56, 100	3, 4, 41	3, 4, 41	1, 2, 35	1, 2, 35	1, 2,27
FSIRXA_CLK	I	FSIRX-A Input Clock	0, 4, 13, 30,33, 39, 54, 57	13, 50, 53, 66, 75, 79, 98	1, 35, 38, 56, 59, 63	1, 35, 38, 59, 63	29, 32, 46, 48, 52	29, 32, 48, 52	25,38, 42
FSIRXA_D0	I	FSIRX-A Primary Data Input	3, 12, 32, 40,44, 52, 58	11, 51, 64, 67, 76, 80, 85	36, 49, 60,64, 69	36, 49, 60, 64, 69	30, 40, 49, 53	30, 40, 49, 53	32, 39
FSIRXA_D1	I	FSIRX-A Optional Additional Data Input	2, 11, 31, 41, 53, 56	12, 52, 65, 77, 82, 99	2, 37, 61, 66	2, 37, 61, 66	31, 50, 55	31, 50, 55	40
FSITXA_CLK	O	FSITX-A Output Clock	7, 10, 27, 44, 51	10, 59, 84, 85, 93	44, 68, 69, 76	44, 68, 69, 76	57, 63	57, 63	43
FSITXA_D0	O	FSITX-A Primary Data Output	6, 9, 26, 45, 49	8, 58, 90, 97	43, 73, 75, 80	32, 43, 73, 75, 80	62, 64	62, 64	48
FSITXA_D1	O	FSITX-A Optional Additional Data Output	5, 6, 8, 25, 46, 50	9, 57, 74, 89, 97	6, 42, 58, 74, 80	6, 42, 58, 74, 80	47, 61, 64	47, 61, 64	47, 48
FSITXA_TDM_CLK	I	FSITX-A Time Division Multiplexed Clock Input	8, 18, 47	6, 68, 74	50, 58	50, 58	41, 47	41, 47	33
FSITXA_TDM_D0	I	FSITX-A Time Division Multiplexed Data Input	10, 19	69, 93	51, 76	51, 76	42, 63	42, 63	34
FSITXA_TDM_D1	I	FSITX-A Time Division Multiplexed Additional Data Input	1, 54, 59	13, 78, 92	62	62	51	51	41
GPIO0	I/O	General-Purpose Input Output 0	0	79	63	63	52	52	42
GPIO1	I/O	General-Purpose Input Output 1	1	78	62	62	51	51	41
GPIO2	I/O	General-Purpose Input Output 2	2	77	61	61	50	50	40
GPIO3	I/O	General-Purpose Input Output 3	3	76	60	60	49	49	39
GPIO4	I/O	General-Purpose Input Output 4	4	75	59	59	48	48	38
GPIO5	I/O	General-Purpose Input Output 5	5	89	74	74	61	61	47
GPIO6	I/O	General-Purpose Input Output 6	6	97	80	80	64	64	48
GPIO7	I/O	General-Purpose Input Output 7	7	84	68	68	57	57	43
GPIO8	I/O	General-Purpose Input Output 8	8	74	58	58	47	47	
GPIO9	I/O	General-Purpose Input Output 9	9	90	75	75	62	62	
GPIO10	I/O	General-Purpose Input Output 10	10	93	76	76	63	63	
GPIO11	I/O	General-Purpose Input Output 11	11	52	37	37	31	31	
GPIO12	I/O	General-Purpose Input Output 12	12	51	36	36	30	30	
GPIO13	I/O	General-Purpose Input Output 13	13	50	35	35	29	29	
GPIO14	I/O	General-Purpose Input Output 14	14	96	79	79			

SIGNAL NAME	PIN TYPE	DESCRIPTION	GPIO	QP100	QP80	QP80C	QP64S	QP64Q	QP48
GPIO15	I/O	General-Purpose Input Output 15	15	95	78	78			
GPIO16	I/O	General-Purpose Input Output 16	16	54	39	39	33	33	26
GPIO17	I/O	General-Purpose Input Output 17	17	55	40	40	34	34	
GPIO18	I/O	General-Purpose Input Output 18	18	68	50	50	41	41	33
GPIO19	I/O	General-Purpose Input Output 19	19	69	51	51	42	42	34
GPIO20	I/O	General-Purpose Input Output 20	20	48	33	33			
GPIO21	I/O	General-Purpose Input Output 21	21	49	34	34			
GPIO22	I/O	General-Purpose Input Output 22	22	83	67	67	56	56	
GPIO23	I/O	General-Purpose Input Output 23	23	81	65	65	54	54	
GPIO24	I/O	General-Purpose Input Output 24	24	56	41	41	35	35	27
GPIO25	I/O	General-Purpose Input Output 25	25	57	42	42			
GPIO26	I/O	General-Purpose Input Output 26	26	58	43	43			
GPIO27	I/O	General-Purpose Input Output 27	27	59	44	44			
GPIO28	I/O	General-Purpose Input Output 28	28	1	4	4	2	2	2
GPIO29	I/O	General-Purpose Input Output 29	29	100	3	3	1	1	1
GPIO30	I/O	General-Purpose Input Output 30	30	98	1	1			
GPIO31	I/O	General-Purpose Input Output 31	31	99	2	2			
GPIO32	I/O	General-Purpose Input Output 32	32	64	49	49	40	40	32
GPIO33	I/O	General-Purpose Input Output 33	33	53	38	38	32	32	25
GPIO34	I/O	General-Purpose Input Output 34	34	94	77	77			
GPIO35	I/O	General-Purpose Input Output 35	35	63	48	48	39	39	31
GPIO37	I/O	General-Purpose Input Output 37	37	61	46	46	37	37	29
GPIO39	I/O	General-Purpose Input Output 39	39		56	-	46	-	
GPIO40	I/O	General-Purpose Input Output 40	40	80	64	64	53	53	
GPIO41	I/O	General-Purpose Input Output 41	41	82	66	66	55	55	
GPIO42	I/O	General-Purpose Input Output 42	42		57	57			
GPIO43	I/O	General-Purpose Input Output 43	43		54	54			
GPIO44	I/O	General-Purpose Input Output 44	44	85	69	69			
GPIO45	I/O	General-Purpose Input Output 45	45		73	73			
GPIO46	I/O	General-Purpose Input Output 46	46		6	6			
GPIO47	I/O	General-Purpose Input Output 47	47	6					
GPIO48	I/O	General-Purpose Input Output 48	48	7		31			
GPIO49	I/O	General-Purpose Input Output 49	49	8		32			
GPIO50	I/O	General-Purpose Input Output 50	50	9					
GPIO51	I/O	General-Purpose Input Output 51	51	10					
GPIO52	I/O	General-Purpose	52	11					

SIGNAL NAME	PIN TYPE	DESCRIPTION	GPIO	QP100	QP80	QP80C	QP64S	QP64Q	QP48
		Input Output 52							
GPIO53	I/O	General-Purpose Input Output 53	53	12					
GPIO54	I/O	General-Purpose Input Output 54	54	13					
GPIO55	I/O	General-Purpose Input Output 55	55	43					
GPIO56	I/O	General-Purpose Input Output 56	56	65					
GPIO57	I/O	General-Purpose Input Output 57	57	66					
GPIO58	I/O	General-Purpose Input Output 58	58	67					
GPIO59	I/O	General-Purpose Input Output 59	59	92					
GPIO60	I/O	General-Purpose Input Output 60	60	44					
GPIO61	I/O	General-Purpose Input Output 61	61	91					
HIC_A0	I	HIC Address 0	8, 55, 60	14, 43, 44, 74	10, 58	10, 58	6, 47	6, 47	4
HIC_A1	I	HIC Address 1	2, 26	15, 58, 77	11, 43, 61	11, 43, 61	7, 50	7, 50	4, 40
HIC_A2	I	HIC Address 2	1	16, 78	12, 62	12, 62	8, 51	8, 51	5, 41
HIC_A3	I	HIC Address 3	23	17, 81	13, 65	13, 65	9, 54	9, 54	6
HIC_A4	I	HIC Address 4	27, 41	59, 82	14, 44, 66	14, 44, 66	10, 55	10, 55	7
HIC_A5	I	HIC Address 5	22	19, 83	15, 67	15, 67	11, 56	11, 56	
HIC_A6	I	HIC Address 6	7, 42, 47	6, 20, 84	16, 57, 68	16, 57, 68	12, 57	12, 57	8, 43
HIC_A7	I	HIC Address 7	5, 43, 48	7, 21, 89	17, 54, 74	17, 54, 74, 31	13, 61	13, 61	9, 47
HIC_BASESEL0	I	HIC Base address range select 0	9, 25	22, 57, 90	18, 42, 75	18, 42, 75	14, 62	14, 62	10
HIC_BASESEL1	I	HIC Base address range select 1	0	23, 79	19, 63	19, 63	15, 52	15, 52	11, 42
HIC_BASESEL2	I	HIC Base address range select 2	4	40, 75	29, 59	29, 59	25, 48	25, 48	21, 38
HIC_D0	I/O	HIC Data 0	26, 33	53, 58	38, 43	38, 43	32	32	25
HIC_D1	I/O	HIC Data 1	16, 27	54, 59	39, 44	39, 44	33	33	26
HIC_D2	I/O	HIC Data 2	17, 42, 49	8, 55	40, 57	32, 40, 57	34	34	
HIC_D3	I/O	HIC Data 3	24, 43, 50	9, 56	41, 54	41, 54	35	35	27
HIC_D4	I/O	HIC Data 4	3, 5, 57	66, 76, 89	60, 74	60, 74	49, 61	49, 61	39, 47
HIC_D5	I/O	HIC Data 5	13, 40, 44	50, 80, 85	35, 64, 69	35, 64, 69	29, 53	29, 53	
HIC_D6	I/O	HIC Data 6	11, 45, 51, 56	10, 52, 65	37, 73	37, 73	31	31	
HIC_D7	I/O	HIC Data 7	0, 39, 44	79, 85	56, 63, 69	63, 69	46, 52	52	42
HIC_D8	I/O	HIC Data 8	8, 30	74, 98	1, 58	1, 58	47	47	
HIC_D9	I/O	HIC Data 9	2, 34	77, 94	61, 77	61, 77	50	50	40
HIC_D10	I/O	HIC Data 10	1, 31	78, 99	2, 62	2, 62	51	51	41
HIC_D11	I/O	HIC Data 11	13, 23	50, 81	35, 65	35, 65	29, 54	29, 54	
HIC_D12	I/O	HIC Data 12	15, 41	82, 95	66, 78	66, 78	55	55	
HIC_D13	I/O	HIC Data 13	12, 22	51, 83	36, 67	36, 67	30, 56	30, 56	
HIC_D14	I/O	HIC Data 14	6, 7	84, 97	68, 80	68, 80	57, 64	57, 64	43, 48
HIC_D15	I/O	HIC Data 15	5, 14	89, 96	74, 79	74, 79	61	61	47
HIC_INT	O	HIC Device interrupt to host	12, 18, 32	51, 64, 68	36, 49, 50	36, 49, 50	30, 40, 41	30, 40, 41	32, 33
HIC_NBE0	I	HIC Byte enable 0	11, 19	38, 52, 69	28, 37, 51	28, 37, 51	24, 31, 42	24, 31, 42	20, 34
HIC_NBE1	I	HIC Byte enable 1	6, 34, 40	37, 80, 94, 97	24, 64, 77, 80	24, 64, 77, 80	20, 53, 64	20, 53, 64	16, 48
HIC_NCS	I	HIC Chip select input	29	28, 100	3, 22	3, 22	1, 18	1, 18	1, 14

SIGNAL NAME	PIN TYPE	DESCRIPTION	GPIO	QP100	QP80	QP80C	QP64S	QP64Q	QP48
HIC_NOE	O	HIC Output enable for data bus	3, 28	1, 31, 76	4, 23, 60	4, 23, 60	2, 19, 49	2, 19, 49	2, 15, 39
HIC_NRDY	O	HIC Ready from device to host	9, 37, 58	61, 67, 90	46, 75	46, 75	37, 62	37, 62	29
HIC_NWE	I	HIC Data Write enable from host	4, 10, 35, 46, 52	11, 36, 63, 75, 93	6, 27, 48, 59, 76	6, 27, 48, 59, 76	23, 39, 48, 63	23, 39, 48, 63	19, 31, 38
I2CA_SCL	I/OD	I2C-A Open-Drain Bidirectional Clock	1, 8, 18, 27, 33, 37, 43, 57	53, 59, 61, 66, 68, 74, 78	38, 44, 46, 50, 54, 58, 62	38, 44, 46, 50, 54, 58, 62	32, 37, 41, 47, 51	32, 37, 41, 47, 51	25, 29, 33, 41
I2CA_SDA	I/OD	I2C-A Open-Drain Bidirectional Data	0, 10, 19, 26, 32, 35, 42, 56	58, 63, 64, 65, 69, 79, 93	43, 48, 49, 51, 57, 63, 76	43, 48, 49, 51, 57, 63, 76	39, 40, 42, 52, 63	39, 40, 42, 52, 63	31, 32, 34, 42
I2CB_SCL	I/OD	I2C-B Open-Drain Bidirectional Clock	3, 9, 15, 29, 51	10, 76, 90, 95, 100	3, 60, 75, 78	3, 60, 75, 78	1, 49, 62	1, 49, 62	1, 39
I2CB_SDA	I/OD	I2C-B Open-Drain Bidirectional Data	2, 14, 28, 34, 50	1, 9, 77, 94, 96	4, 61, 77, 79	4, 61, 77, 79	2, 50	2, 50	2, 40
LINA_RX	I	LIN-A Receive	23, 29, 33, 35, 42, 47, 49, 59	6, 8, 53, 63, 81, 92, 100	3, 38, 48, 57, 65	3, 32, 38, 48, 57, 65	1, 32, 39, 54	1, 32, 39, 54	1, 25, 31
LINA_TX	O	LIN-A Transmit	22, 28, 32, 37, 46, 58	1, 61, 64, 67, 83	4, 6, 46, 49, 67	4, 6, 46, 49, 67	2, 37, 40, 56	2, 37, 40, 56	2, 29, 32
LINB_RX	I	LIN-B Receive	9, 11, 13, 15, 19, 23, 41, 55	43, 50, 52, 69, 81, 82, 90, 95	35, 37, 51, 65, 66, 75, 78	35, 37, 51, 65, 66, 75, 78	29, 31, 42, 54, 55, 62	29, 31, 42, 54, 55, 62	34
LINB_TX	O	LIN-B Transmit	10, 12, 14, 18, 22, 24, 40, 44, 54	13, 51, 56, 68, 80, 83, 85, 93, 96	36, 41, 50, 64, 67, 69, 76, 79	36, 41, 50, 64, 67, 69, 76, 79	30, 35, 41, 53, 56, 63	30, 35, 41, 53, 56, 63	27, 33
MCAN_RX	I	CAN/CAN FD Receive	0, 5, 12, 21, 30, 39, 47, 51, 57, 61	6, 10, 49, 51, 66, 79, 89, 91, 98	1, 34, 36, 56, 63, 74	1, 34, 36, 63, 74	30, 46, 52, 61	30, 52, 61	42, 47
MCAN_TX	O	CAN/CAN FD Transmit	1, 4, 13, 20, 31, 46, 50, 56, 60	9, 44, 48, 50, 65, 75, 78, 99	2, 6, 33, 35, 59, 62	2, 6, 33, 35, 59, 62	29, 48, 51	29, 48, 51	38, 41
OUTPUTXBAR1	O	Output X-BAR Output 1	2, 24, 34, 58	56, 67, 77, 94	41, 61, 77	41, 61, 77	35, 50	35, 50	27, 40
OUTPUTXBAR2	O	Output X-BAR Output 2	3, 25, 37, 54, 59	13, 57, 61, 76, 92	42, 46, 60	42, 46, 60	37, 49	37, 49	29, 39
OUTPUTXBAR3	O	Output X-BAR Output 3	4, 5, 14, 26, 48, 55, 60	7, 43, 44, 58, 75, 89, 96	43, 59, 74, 79	31, 43, 59, 74, 79	48, 61	48, 61	38, 47
OUTPUTXBAR4	O	Output X-BAR Output 4	6, 15, 27, 33, 49, 61	8, 53, 59, 91, 95, 97	38, 44, 78, 80	32, 38, 44, 78, 80	32, 64	32, 64	25, 48
OUTPUTXBAR5	O	Output X-BAR Output 5	7, 28, 42	1, 84	4, 57, 68	4, 57, 68	2, 57	2, 57	2, 43
OUTPUTXBAR6	O	Output X-BAR Output 6	9, 29, 43	90, 100	3, 54, 75	3, 54, 75	1, 62	1, 62	1
OUTPUTXBAR7	O	Output X-BAR Output 7	11, 16, 30, 44	52, 54, 85, 98	1, 37, 39, 69	1, 37, 39, 69	31, 33	31, 33	26
OUTPUTXBAR8	O	Output X-BAR Output 8	17, 31, 45	55, 99	2, 40, 73	2, 40, 73	34	34	
PMBUSA_ALERT	I/OD	PMBus-A Open-Drain Bidirectional Alert Signal	13, 19, 27, 37, 43, 45	50, 59, 61, 69	35, 44, 46, 51, 54, 73	35, 44, 46, 51, 54, 73	29, 37, 42	29, 37, 42	29, 34
PMBUSA_CTL	I/O	PMBus-A Control Signal - Slave Input/Master Output	12, 18, 26, 35, 42, 44	51, 58, 63, 68, 85	36, 43, 48, 50, 57, 69	36, 43, 48, 50, 57, 69	30, 39, 41	30, 39, 41	31, 33
PMBUSA_SCL	I/OD	PMBus-A Open-Drain Bidirectional Clock	3, 15, 16, 24, 35, 41, 47	6, 54, 56, 63, 76, 82, 95	39, 41, 48, 60, 66, 78	39, 41, 48, 60, 66, 78	33, 35, 39, 49, 55	33, 35, 39, 49, 55	26, 27, 31, 39
PMBUSA_SDA	I/OD	PMBus-A Open-Drain Bidirectional Data	2, 14, 17, 25, 32, 34, 40, 44, 46, 48	7, 55, 57, 64, 77, 80, 85, 94, 96	6, 40, 42, 49, 61, 64, 69, 77, 79	6, 31, 40, 42, 49, 61, 64, 69, 77, 79	34, 40, 50, 53	34, 40, 50, 53	32, 40
SCIA_RX	I	SCI-A Receive Data	3, 9, 17, 25, 28, 35, 49	1, 8, 55, 57, 63, 76, 90	4, 40, 42, 48, 60, 75	4, 32, 40, 42, 48, 60, 75	2, 34, 39, 49, 62	2, 34, 39, 49, 62	2, 31, 39

SIGNAL NAME	PIN TYPE	DESCRIPTION	GPIO	QP100	QP80	QP80C	QP64S	QP64Q	QP48
SCIA_TX	O	SCI-A Transmit Data	2, 8, 16, 24, 29, 37, 48	7, 54, 56, 61, 74, 77, 100	3, 39, 41, 46, 58, 61	3, 31, 39, 41, 46, 58, 61	1, 33, 35, 37, 47, 50	1, 33, 35, 37, 47, 50	1, 26, 27, 29, 40
SCIB_RX	I	SCI-B Receive Data	11, 13, 15, 19, 23, 41, 57	50, 52, 66, 69, 81, 82, 95	35, 37, 51, 65, 66, 78	35, 37, 51, 65, 66, 78	29, 31, 42, 54, 55	29, 31, 42, 54, 55	34
SCIB_TX	O	SCI-B Transmit Data	9, 10, 12, 14, 18, 22, 40, 56	51, 65, 68, 80, 83, 90, 93, 96	36, 50, 64, 67, 75, 76, 79	36, 50, 64, 67, 75, 76, 79	30, 41, 53, 56, 62, 63	30, 41, 53, 56, 62, 63	33
SD1_C1	I	SDFM-1 Channel 1 Clock Input	17, 33, 49, 53	8, 12, 23, 53, 55	19, 38, 40	19, 32, 38, 40	15, 32, 34	15, 32, 34	11, 25
SD1_C2	I	SDFM-1 Channel 2 Clock Input	19, 33, 51, 54	10, 13, 31, 53, 69	23, 38, 51	23, 38, 51	19, 32, 42	19, 32, 42	15, 25, 34
SD1_C3	I	SDFM-1 Channel 3 Clock Input	21, 53, 55	12, 38, 43, 49	28, 34	28, 34	24	24	20
SD1_C4	I	SDFM-1 Channel 4 Clock Input	23, 55, 56	40, 43, 65, 81	29, 65	29, 65	25, 54	25, 54	21
SD1_D1	I	SDFM-1 Channel 1 Data Input	16, 48	7, 19, 54	15, 39	15, 31, 39	11, 33	11, 33	26
SD1_D2	I	SDFM-1 Channel 2 Data Input	18, 32, 50	9, 20, 64, 68	16, 49, 50	16, 49, 50	12, 40, 41	12, 40, 41	8, 32, 33
SD1_D3	I	SDFM-1 Channel 3 Data Input	20, 52	11, 21, 48	17, 33	17, 33	13	13	9
SD1_D4	I	SDFM-1 Channel 4 Data Input	22, 54	13, 22, 83	18, 67	18, 67	14, 56	14, 56	10
SD2_C1	I	SDFM-2 Channel 1 Clock Input	25, 35, 57	14, 37, 57, 63, 66	10, 24, 42, 48	10, 24, 42, 48	6, 20, 39	6, 20, 39	4, 16, 31
SD2_C2	I	SDFM-2 Channel 2 Clock Input	27, 58, 59	36, 59, 67, 92	27, 44	27, 44	23	23	19
SD2_C3	I	SDFM-2 Channel 3 Clock Input	29, 45, 59, 61	28, 91, 92, 100	3, 22, 73	3, 22, 73	1, 18	1, 18	1, 14
SD2_C4	I	SDFM-2 Channel 4 Clock Input	31, 46, 60	32, 44, 99	2，6	2，6			
SD2_D1	I	SDFM-2 Channel 1 Data Input	24, 49, 56	8, 56, 65	14, 41	14, 32, 41	10, 35	10, 35	7, 27
SD2_D2	I	SDFM-2 Channel 2 Data Input	26, 50, 58	9, 16, 58, 67	12, 43	12, 43	8	8	5
SD2_D3	I	SDFM-2 Channel 3 Data Input	28, 43, 51, 60	1, 10, 17, 44	4, 13, 54	4, 13, 54	2, 9	2, 9	2，6
SD2_D4	I	SDFM-2 Channel 4 Data Input	30, 47, 52	6, 11, 15, 98	1, 11	1, 11	7	7	4
SPIA_CLK	I/O	SPI-A Clock	3, 9, 12, 18, 56	51, 65, 68, 76, 90	36, 50, 60, 75	36, 50, 60, 75	30, 41, 49, 62	30, 41, 49, 62	33, 39
SPIA_SIMO	I/O	SPI-A Slave In, Master Out (SIMO)	2, 8, 11, 16, 54	13, 52, 54, 74, 77	37, 39, 58, 61	37, 39, 58, 61	31, 33, 47, 50	31, 33, 47, 50	26, 40
SPIA_SOMI	I/O	SPI-A Slave Out, Master In (SOMI)	1, 10, 13, 17, 55	43, 50, 55, 78, 93	35, 40, 62, 76	35, 40, 62, 76	29, 34, 51, 63	29, 34, 51, 63	41
SPIA_STE	I/O	SPI-A Slave Transmit Enable (STE)	0, 5, 11, 19, 57	52, 66, 69, 79, 89	37, 51, 63, 74	37, 51, 63, 74	31, 42, 52, 61	31, 42, 52, 61	34, 42, 47
SPIB_CLK	I/O	SPI-B Clock	4, 14, 22, 26, 28, 32, 52, 58	1, 11, 58, 64, 67, 75, 83, 96	4, 43, 49, 59, 67, 79	4, 43, 49, 59, 67, 79	2, 40, 48, 56	2, 40, 48, 56	2, 32, 38
SPIB_SIMO	I/O	SPI-B Slave In, Master Out (SIMO)	7, 20, 24, 30, 40, 50, 56, 60	9, 44, 48, 56, 65, 80, 84, 98	1, 33, 41, 64, 68	1, 33, 41, 64, 68	35, 53, 57	35, 53, 57	27, 43
SPIB_SOMI	I/O	SPI-B Slave Out, Master In (SOMI)	6, 16, 21, 25, 31, 41, 51, 57, 61	10, 49, 54, 57, 66, 82, 91, 97, 99	2, 34, 39, 42, 66, 80	2, 34, 39, 42, 66, 80	33, 55, 64	33, 55, 64	26, 48
SPIB_STE	I/O	SPI-B Slave Transmit Enable (STE)	15, 23, 27, 29, 33, 53, 59	12, 53, 59, 81, 92, 95, 100	3, 38, 44, 65, 78	3, 38, 44, 65, 78	1, 32, 54	1, 32, 54	1, 25

SIGNAL NAME	PIN TYPE	DESCRIPTION	GPIO	QP100	QP80	QP80C	QP64S	QP64Q	QP48
SYNCOUT	O	External ePWM Synchronization Pulse	6, 39, 52	11, 97	56, 80	80	46, 64	64	48
TDI	I	JTAG Test Data Input (TDI) - TDI is the default mux selection for the pin. The internal pullup is disabled by default. The internal pullup should be enabled or an external pullup added on the board if this pin is used as JTAG TDI to avoid a floating input.	35	63	48	48	39	39	31
TDO	O	JTAG Test Data Output (TDO) - TDO is the default mux selection for the pin. The internal pullup is disabled by default. The TDO function will tristate when there is no JTAG activity, leaving this pin floating; the internal pullup should be enabled or an external pullup added on the board to avoid a floating GPIO input.	37	61	46	46	37	37	29
X1	I/O	Crystal oscillator input or single-ended clock input. The device initialization software must configure this pin before the crystal oscillator is enabled. To use this oscillator, a quartz crystal circuit must be connected to X1 and X2. This pin can also be used to feed a single-ended 3.3-V level clock. See the XTAL section for usage details.	19	69	51	51	42	42	34
X2	I/O	Crystal oscillator output.	18	68	50	50	41	41	33
XCLKOUT	O	External Clock Output. This pin outputs a divided-down version of a chosen clock signal from within the device.	16, 18	54, 68	39, 50	39, 50	33, 41	33, 41	26, 33

5.3.3Power and Ground
电源和地

Table 5-4. Power and Ground

SIGNAL NAME	PIN TYPE	DESCRIPTION	QP100	QP80	QP80C	QP64S	QP64Q	QP48
VDD		1.1V Digital Logic Power Pins. See the Power Management Module (PMM) section for usage details.	4, 46, 71, 87	8, 31, 53, 71	8, 53, 71	4, 27, 44, 59	4, 27, 44, 59	23, 36, 45
VDDA		3.3-V Analog Power Pins. Place a minimum 2.2-µF decoupling capacitor on each pin. See the Power Management Module (PMM) section for usage details.	34	26	26	22	22	18
VDDIO		3.3-V Digital I/O Power Pins. See the Power Management Module (PMM) section for usage details.	3, 47, 70, 88	7, 32, 52, 72	7, 52, 72	28, 43, 60	28, 43, 60	24, 35, 46
VREGENZ	I	Internal voltage regulator disable with internal pulldown. Tie low to VSS to enable internal VREG. Tie high to VDDIO to use an external supply. See the Power Management Module (PMM) section for usage details.	73	-	56	-	46	-
VSS		Digital Ground	5, 45, 72, 86	9, 30, 55, 70	9, 30, 55, 70	5, 26, 45, 58	5, 26, 45, 58	22, 37, 44
VSSA		Analog Ground	33	25	25	21	21	17

5.3.4Test, JTAG, and Reset
测试、JTAG 和复位

Table 5-5. Test, JTAG, and Reset

SIGNAL NAME	PIN TYPE	DESCRIPTION	QP100	QP80	QP80C	QP64S	QP64Q	QP48
TCK	I	JTAG test clock with internal pullup.	60	45	45	36	36	28
TMS	I/O	JTAG test-mode select (TMS) with internal pullup. This serial control input is clocked into the TAP controller on the rising edge of TCK. This device does not have a TRSTn pin. An external pullup resistor (recommended 2.2 kΩ) on the TMS pin to VDDIO should be placed on the board to keep JTAG in reset during normal operation.	62	47	47	38	38	30
XRSn	I/OD	Device Reset (in) and Watchdog Reset (out). During a power-on condition, this pin is driven low by the device. An external circuit may also drive this pin to assert a device reset. This pin is also driven low by the MCU when a watchdog reset occurs. During watchdog reset, the XRSn pin is driven low for the watchdog reset duration of 512 OSCCLK cycles. A resistor between 2.2 kΩ and 10 kΩ should be placed between XRSn and VDDIO. If a capacitor is placed between XRSn and VSS for noise filtering, it should be 100 nF or smaller. These values will allow the watchdog to properly drive the XRSn pin to VOL within 512 OSCCLK cycles when the watchdog reset is asserted. This pin is an open-drain output with an internal pullup. If this pin is driven by an external device, it should be done using an open- drain device.	2	5	5	3	3	3

5.4 Pin Multiplexing
引脚复用

5.4.1 GPIO Muxed Pins GPIO
多路复用引脚

Table 5-6 lists the GPIO muxed pins. The default mode for each GPIO pin is the GPIO function, except GPIO35 and GPIO37, which default to TDI and TDO, respectively. Secondary functions can be selected by setting both the GPyGMUXn.GPIOz and GPyMUXn.GPIOz register bits. The GPyGMUXn register should be configured before the GPyMUXn to avoid transient pulses on GPIOs from alternate mux selections. Columns that are not shown and blank cells are reserved GPIO Mux settings. GPIO ALT functions cannot be configured with the GPyMUXn and GPyGMUXn registers. These are special functions that need to be configured from the module.

表 5-6 列出了 GPIO 多路复用引脚。每个 GPIO 引脚的默认模式都是 GPIO 功能，但 GPIO35 和 GPIO37 除外，这两个引脚的默认模式分别为 TDI 和 TDO。可以通过设置 GPyGMUXn.GPIOz 和 GPyMUXn.GPIOz 寄存器位来选择辅助功能。GPyGMUXn 寄存器应在 GPyMUXn 之前配置，以避免交替的多路复用器选择对 GPIO 产生瞬时脉冲。未显示的列和空白单元格是保留的 GPIO 多路复用器设置。GPIOALT 功能不能通过 GPyMUXn 和 GPyGMUXn 寄存器进行配置。这些是需要从模块进行配置的特殊功能。

Note

GPIO36 and GPIO38 do not exist on this device. GPIO62 to GPIO63 exist but are not pinned out on any packages. Boot ROM enables pullups on GPIO62 to GPIO63. For more details, see Section 5.5.
此器件上不存在 GPIO36 和 GPIO38。GPIO62 至 GPIO63 存在，但在任何封装上都没有引脚输出。引导 ROM 在 GPIO62 至 GPIO63 上启用上拉电阻。如需了解更多详情，请参阅第 5.5 节。

5.4.1.1 GPIO Muxed Pins

Table 5-6 GPIO 多路复用引脚

0, 4, 8, 12	1	2	3	5	6	7	9	10	11	13	14	15	ALT
GPIO0	EPWM1_A				I2CA_SDA	SPIA_STE	FSIRXA_CLK	MCAN_RX	CLB_OUTPUTXBAR8	EQEP1_INDEX	HIC_D7	HIC_BASESEL1	
GPIO1	EPWM1_B				I2CA_SCL	SPIA_SOMI		MCAN_TX	CLB_OUTPUTXBAR7	HIC_A2	FSITXA_TDM_D1	HIC_D10	
GPIO2	EPWM2_A			OUTPUTXBAR1	PMBUSA_SDA	SPIA_SIMO	SCIA_TX	FSIRXA_D1	I2CB_SDA	HIC_A1	CANA_TX	HIC_D9	
GPIO3	EPWM2_B	OUTPUTXBAR2		OUTPUTXBAR2	PMBUSA_SCL	SPIA_CLK	SCIA_RX	FSIRXA_D0	I2CB_SCL	HIC_NOE	CANA_RX	HIC_D4	
GPIO4	EPWM3_A		MCAN_TX	OUTPUTXBAR3	CANA_TX	SPIB_CLK	EQEP2_STROBE	FSIRXA_CLK	CLB_OUTPUTXBAR6	HIC_BASESEL2		HIC_NWE	
GPIO5	EPWM3_B		OUTPUTXBAR3	MCAN_RX	CANA_RX	SPIA_STE	FSITXA_D1	CLB_OUTPUTXBAR5		HIC_A7	HIC_D4	HIC_D15	
GPIO6	EPWM4_A	OUTPUTXBAR4	SYNCOUT	EQEP1_A		SPIB_SOMI	FSITXA_D0		FSITXA_D1	HIC_NBE1	CLB_OUTPUTXBAR8	HIC_D14	
GPIO7	EPWM4_B		OUTPUTXBAR5	EQEP1_B		SPIB_SIMO	FSITXA_CLK	CLB_OUTPUTXBAR2		HIC_A6		HIC_D14	
GPIO8	EPWM5_A		ADCSOCAO	EQEP1_STROBE	SCIA_TX	SPIA_SIMO	I2CA_SCL	FSITXA_D1	CLB_OUTPUTXBAR5	HIC_A0	FSITXA_TDM_CLK	HIC_D8	
GPIO9	EPWM5_B	SCIB_TX	OUTPUTXBAR6	EQEP1_INDEX	SCIA_RX	SPIA_CLK		FSITXA_D0	LINB_RX	HIC_BASESEL0	I2CB_SCL	HIC_NRDY	
GPIO10	EPWM6_A		ADCSOCBO	EQEP1_A	SCIB_TX	SPIA_SOMI	I2CA_SDA	FSITXA_CLK	LINB_TX	HIC_NWE	FSITXA_TDM_D0	CLB_OUTPUTXBAR4	
GPIO11	EPWM6_B		OUTPUTXBAR7	EQEP1_B	SCIB_RX	SPIA_STE	FSIRXA_D1	LINB_RX	EQEP2_A	SPIA_SIMO	HIC_D6	HIC_NBE0	
GPIO12	EPWM7_A		MCAN_RX	EQEP1_STROBE	SCIB_TX	PMBUSA_CTL	FSIRXA_D0	LINB_TX	SPIA_CLK	CANA_RX	HIC_D13	HIC_INT	
GPIO13	EPWM7_B		MCAN_TX	EQEP1_INDEX	SCIB_RX	PMBUSA_ALERT	FSIRXA_CLK	LINB_RX	SPIA_SOMI	CANA_TX	HIC_D11	HIC_D5	
GPIO14	EPWM8_A	SCIB_TX		I2CB_SDA	OUTPUTXBAR3	PMBUSA_SDA	SPIB_CLK	EQEP2_A	LINB_TX	EPWM3_A	CLB_OUTPUTXBAR7	HIC_D15	
GPIO15	EPWM8_B	SCIB_RX		I2CB_SCL	OUTPUTXBAR4	PMBUSA_SCL	SPIB_STE	EQEP2_B	LINB_RX	EPWM3_B	CLB_OUTPUTXBAR6	HIC_D12	
GPIO16	SPIA_SIMO		OUTPUTXBAR7	EPWM5_A	SCIA_TX	SD1_D1	EQEP1_STROBE	PMBUSA_SCL	XCLKOUT	EQEP2_B	SPIB_SOMI	HIC_D1	
GPIO17	SPIA_SOMI		OUTPUTXBAR8	EPWM5_B	SCIA_RX	SD1_C1	EQEP1_INDEX	PMBUSA_SDA	CANA_TX			HIC_D2	
GPIO18	SPIA_CLK	SCIB_TX	CANA_RX	EPWM6_A	I2CA_SCL	SD1_D2	EQEP2_A	PMBUSA_CTL	XCLKOUT	LINB_TX	FSITXA_TDM_CLK	HIC_INT	X2
GPIO19	SPIA_STE	SCIB_RX	CANA_TX	EPWM6_B	I2CA_SDA	SD1_C2	EQEP2_B	PMBUSA_ALERT	CLB_OUTPUTXBAR1	LINB_RX	FSITXA_TDM_D0	HIC_NBE0	X1
GPIO20	EQEP1_A				SPIB_SIMO	SD1_D3	MCAN_TX						
GPIO21	EQEP1_B				SPIB_SOMI	SD1_C3	MCAN_RX						
GPIO22	EQEP1_STROBE		SCIB_TX		SPIB_CLK	SD1_D4	LINA_TX	CLB_OUTPUTXBAR1	LINB_TX	HIC_A5	EPWM4_A	HIC_D13	

0, 4, 8, 12	1	2	3	5	6	7	9	10	11	13	14	15	ALT
GPIO23	EQEP1_INDEX		SCIB_RX		SPIB_STE	SD1_C4	LINA_RX	CLB_OUTPUTXBAR3	LINB_RX	HIC_A3	EPWM4_B	HIC_D11	
GPIO24	OUTPUTXBAR1	EQEP2_A		EPWM8_A	SPIB_SIMO	SD2_D1	LINB_TX	PMBUSA_SCL	SCIA_TX	ERRORSTS		HIC_D3	
GPIO25	OUTPUTXBAR2	EQEP2_B		EQEP1_A	SPIB_SOMI	SD2_C1	FSITXA_D1	PMBUSA_SDA	SCIA_RX		HIC_BASESEL0		
GPIO26	OUTPUTXBAR3	EQEP2_INDEX		OUTPUTXBAR3	SPIB_CLK	SD2_D2	FSITXA_D0	PMBUSA_CTL	I2CA_SDA		HIC_D0	HIC_A1	
GPIO27	OUTPUTXBAR4	EQEP2_STROBE		OUTPUTXBAR4	SPIB_STE	SD2_C2	FSITXA_CLK	PMBUSA_ALERT	I2CA_SCL		HIC_D1	HIC_A4	
GPIO28	SCIA_RX		EPWM7_A	OUTPUTXBAR5	EQEP1_A	SD2_D3	EQEP2_STROBE	LINA_TX	SPIB_CLK	ERRORSTS	I2CB_SDA	HIC_NOE	
GPIO29	SCIA_TX		EPWM7_B	OUTPUTXBAR6	EQEP1_B	SD2_C3	EQEP2_INDEX	LINA_RX	SPIB_STE	ERRORSTS	I2CB_SCL	HIC_NCS	AUXCLKIN
GPIO30	CANA_RX		SPIB_SIMO	OUTPUTXBAR7	EQEP1_STROBE	SD2_D4	FSIRXA_CLK	MCAN_RX	EPWM1_A		HIC_D8		
GPIO31	CANA_TX		SPIB_SOMI	OUTPUTXBAR8	EQEP1_INDEX	SD2_C4	FSIRXA_D1	MCAN_TX	EPWM1_B		HIC_D10		
GPIO32	I2CA_SDA		SPIB_CLK	EPWM8_B	LINA_TX	SD1_D2	FSIRXA_D0	CANA_TX	PMBUSA_SDA	ADCSOCBO		HIC_INT	
GPIO33	I2CA_SCL		SPIB_STE	OUTPUTXBAR4	LINA_RX	SD1_C2	FSIRXA_CLK	CANA_RX	EQEP2_B	ADCSOCAO	SD1_C1	HIC_D0	
GPIO34	OUTPUTXBAR1				PMBUSA_SDA					HIC_NBE1	I2CB_SDA	HIC_D9	
GPIO35	SCIA_RX		I2CA_SDA	CANA_RX	PMBUSA_SCL	LINA_RX	EQEP1_A	PMBUSA_CTL	EPWM5_B	SD2_C1	HIC_NWE	TDI	
GPIO37	OUTPUTXBAR2		I2CA_SCL	SCIA_TX	CANA_TX	LINA_TX	EQEP1_B	PMBUSA_ALERT			HIC_NRDY	TDO	
GPIO39					MCAN_RX	FSIRXA_CLK	EQEP2_INDEX		CLB_OUTPUTXBAR2	SYNCOUT	EQEP1_INDEX	HIC_D7	
GPIO40	SPIB_SIMO			EPWM2_B	PMBUSA_SDA	FSIRXA_D0	SCIB_TX	EQEP1_A	LINB_TX		HIC_NBE1	HIC_D5	
GPIO41				EPWM2_A	PMBUSA_SCL	FSIRXA_D1	SCIB_RX	EQEP1_B	LINB_RX	HIC_A4	SPIB_SOMI	HIC_D12	
GPIO42		LINA_RX	OUTPUTXBAR5	PMBUSA_CTL	I2CA_SDA			EQEP1_STROBE	CLB_OUTPUTXBAR3		HIC_D2	HIC_A6	
GPIO43			OUTPUTXBAR6	PMBUSA_ALERT	I2CA_SCL		PMBUSA_ALERT	EQEP1_INDEX	CLB_OUTPUTXBAR4	SD2_D3	HIC_D3	HIC_A7	
GPIO44			OUTPUTXBAR7	EQEP1_A	PMBUSA_SDA	FSITXA_CLK	PMBUSA_CTL	CLB_OUTPUTXBAR3	FSIRXA_D0	HIC_D7	LINB_TX	HIC_D5	
GPIO45			OUTPUTXBAR8			FSITXA_D0	PMBUSA_ALERT	CLB_OUTPUTXBAR4		SD2_C3		HIC_D6	
GPIO46			LINA_TX	MCAN_TX		FSITXA_D1	PMBUSA_SDA			SD2_C4		HIC_NWE	
GPIO47			LINA_RX	MCAN_RX		CLB_OUTPUTXBAR2	PMBUSA_SCL			SD2_D4	FSITXA_TDM_CLK	HIC_A6	
GPIO48	OUTPUTXBAR3		CANA_TX		SCIA_TX	SD1_D1	PMBUSA_SDA					HIC_A7	
GPIO49	OUTPUTXBAR4		CANA_RX		SCIA_RX	SD1_C1	LINA_RX			SD2_D1	FSITXA_D0	HIC_D2	

0, 4, 8, 12	1	2	3	5	6	7	9	10	11	13	14	15	ALT
GPIO50	EQEP1_A			MCAN_TX	SPIB_SIMO	SD1_D2	I2CB_SDA			SD2_D2	FSITXA_D1	HIC_D3	
GPIO51	EQEP1_B			MCAN_RX	SPIB_SOMI	SD1_C2	I2CB_SCL			SD2_D3	FSITXA_CLK	HIC_D6	
GPIO52	EQEP1_STROBE			CLB_OUTPUTXBAR5	SPIB_CLK	SD1_D3	SYNCOUT			SD2_D4	FSIRXA_D0	HIC_NWE	
GPIO53	EQEP1_INDEX			CLB_OUTPUTXBAR6	SPIB_STE	SD1_C3	ADCSOCAO	CANA_RX		SD1_C1	FSIRXA_D1		
GPIO54	SPIA_SIMO			EQEP2_A	OUTPUTXBAR2	SD1_D4	ADCSOCBO	LINB_TX		SD1_C2	FSIRXA_CLK	FSITXA_TDM_D1	
GPIO55	SPIA_SOMI			EQEP2_B	OUTPUTXBAR3	SD1_C4	ERRORSTS	LINB_RX		SD1_C3		HIC_A0	
GPIO56	SPIA_CLK	CLB_OUTPUTXBAR7	MCAN_TX	EQEP2_STROBE	SCIB_TX	SD2_D1	SPIB_SIMO	I2CA_SDA	EQEP1_A	SD1_C4	FSIRXA_D1	HIC_D6	
GPIO57	SPIA_STE	CLB_OUTPUTXBAR8	MCAN_RX	EQEP2_INDEX	SCIB_RX	SD2_C1	SPIB_SOMI	I2CA_SCL	EQEP1_B		FSIRXA_CLK	HIC_D4	
GPIO58				OUTPUTXBAR1	SPIB_CLK	SD2_D2	LINA_TX	CANA_TX	EQEP1_STROBE	SD2_C2	FSIRXA_D0	HIC_NRDY	
GPIO59				OUTPUTXBAR2	SPIB_STE	SD2_C2	LINA_RX	CANA_RX	EQEP1_INDEX	SD2_C3	FSITXA_TDM_D1		
GPIO60			MCAN_TX	OUTPUTXBAR3	SPIB_SIMO	SD2_D3				SD2_C4		HIC_A0	
GPIO61			MCAN_RX	OUTPUTXBAR4	SPIB_SOMI	SD2_C3					CANA_RX		
AIO224		SD2_D3										HIC_A3	
AIO225		SD2_C2										HIC_NWE	
AIO226		SD2_D4										HIC_A1	
AIO227		SD1_C3										HIC_NBE0	
AIO228		SD2_C1										HIC_A0	
AIO229													
AIO230		SD1_C4										HIC_BASESEL2	
AIO231		SD1_C1										HIC_BASESEL1	
AIO232		SD1_D4										HIC_BASESEL0	
AIO233		SD2_D1										HIC_A4	
AIO236													
AIO237		SD1_D2										HIC_A6	
AIO238		SD2_C3										HIC_NCS	
AIO239		SD1_D1										HIC_A5	
AIO240		SD2_C1										HIC_NBE1	
AIO241		SD2_C1										HIC_NBE1	
AIO242		SD2_D2										HIC_A2	
AIO244		SD1_D3										HIC_A7	
AIO245		SD1_C2										HIC_NOE	
AIO247													
AIO248													

0, 4, 8, 12	1	2	3	5	6	7	9	10	11	13	14	15	ALT
AIO249													
AIO251													
AIO252		SD2_C4											
AIO253													

5.4.2Digital Inputs on ADC Pins (AIOs)
ADC 引脚上的数字输入(AIO)

PIOs on port H (GPIO224–GPIO253) are multiplexed with analog pins. These are also referred to as AIOs. These pins can only function in input mode. By default, these pins will function as analog pins and the GPIOs are in a high-Z state. The GPHAMSEL register is used to configure these pins for digital or analog operation.

端口 H 上的 GPIO(GPIO224–GPIO253)与模拟引脚进行多路复用。这也称为 AIO。这些引脚只能在输入模式下工作。默认情况下,这些引脚将用作模拟引脚,并且 GPIO 处于高阻抗状态。GPHAMSEL 寄存器用于配置这些引脚的数字或模拟操作。

Note

If digital signals with sharp edges (high dv/dt) are connected to the AIOs, cross-talk can occur with adjacent analog signals. The user should therefore limit the edge rate of signals connected to AIOs if adjacent channels are being used for analog functions.

如果将具有尖锐边缘（高 dv/dt）的数字信号连接到 AIO，则相邻的模拟信号可能会发生串扰。因此，如果相邻通道用于模拟功能，用户应限制连接到 AIO 的信号的边沿速率。

5.4.3Digital Inputs and Outputs on ADC Pins (AGPIOs)
ADC 引脚上的数字输入和输出(AGPIO)

Some GPIOs on this device are multiplexed with analog pins. These are also referred to as AGPIOs. Unlike AIOs, AGPIOs have full input and output capability. This device has two GPIOs (GPIO20, GPIO21) that offer this feature on the 100-Pin PZ and 80-Pin PN packages.

100-Pin PZ: On this package, there are dedicated pins for B5 (pin 32) and B11 (pin 30) which respectively also have AIO252 and AIO251 functionality. In addition, GPIO20 (pin 48) and GPIO21 (pin 49) are also available as B5 and B11 respectively. Since B5 and B11 are dedicated pins on this package, it is recommended to use them instead of the ones on GPIO20/21.

80-Pin PN: On this package, GPIO20 (pin 33) and GPIO21 (pin 34) are also available as B5 and B11 respectively. There are no dedicated pin for B5 and B11.

By default the AGPIOs are not connected and have to be configured. Table 5-7 truth table shows how to configure the AGPIOs using B5 (pin 32) and GPIO20 (pin 48) on the 100-Pin PZ as an example.

该器件上的某些 GPIO 与模拟引脚进行多路复用。这些也称为 AGPIO。与 AIO 不同，AGPIO 具有完整的输入和输出能力。该器件具有两个 GPIO（GPIO20、GPIO21），可在 100 引脚 QP 和 80 引脚 QP 封装上提供此特性。

100 引脚 QP：这种封装上具有用于 B5（引脚 32）和 B11（引脚 30）的专用引脚，这些引脚分别还具有 AIO252 和 AIO251 功能。此外，GPIO20（引脚 48）和 GPIO21（引脚 49）也可分别用作 B5 和 B11。由于 B5 和 B11 是这种封装上的专用引脚，因此建议使用它们代替 GPIO20/21 上的引脚。

80 引脚 QP：在这种封装上，GPIO20（引脚 33）和 GPIO21（引脚 34）也可分别用作 B5 和 B11。B5 和 B11 没有专用引脚。

默认情况下，AGPIO 未连接，必须进行配置。表 5-7 真值表以 100 引脚 QP 上的 B5（引脚 32）和 GPIO20（引脚 48）为例展示了如何配置 AGPIO。

Table 5-7 AGPIO Configuration

AGPIOCTRLA.bit.GPIO20	GPAAMSEL.bit.GPIO20	GPHAMSEL.bit.GPIO252	B5 CONNECTED TO			GPIO20 CONNECTED TO		
			ADC	GPIO20	AIO252	ADC	GPIO20	AIO252
0	0	1	Yes	-	-	-	Yes	-
0	1	1	Yes	-	-	-	-	-
1	0	1	Yes	-	-	-	Yes	-
1	1	1	-	-	-	Yes	-	-
0	0	0	Yes	-	Yes	-	Yes	-
0	1	0	Yes	-	Yes	-	-	-
1	0	0	Yes	-	Yes	-	Yes	-
1	1	0	-	-	Yes	Yes	-	-

Note

If digital signals with sharp edges (high dv/dt) are connected to the AGPIOs, cross-talk can occur with adjacent analog signals. The user should therefore limit the edge rate of signals connected to AGPIOs if adjacent channels are being used for analog functions.

如果将具有尖锐边缘（高 dv/dt）的数字信号连接到 AGPIO，则相邻的模拟信号可能会发生串扰。因此，如果相邻通道用于模拟功能，用户应限制连接到 AGPIO 的信号的边沿速率。

5.4.4GPIO Input X-BAR

GPIO 输入 X-BAR

The Input X-BAR is used to route signals from a GPIO to many different IP blocks such as the ADCs, eCAPs, ePWMs, and external interrupts (see Figure 5- 7). Table 5-8 lists the input X-BAR destinations.

输入 X-BAR 用于将信号从 GPIO 路由到许多不同的 IP 块，例如 ADC、eCAP、ePWM 和外部中断（请参阅图5-7）。表 5-8 列出了输入 X-BAR 目标。

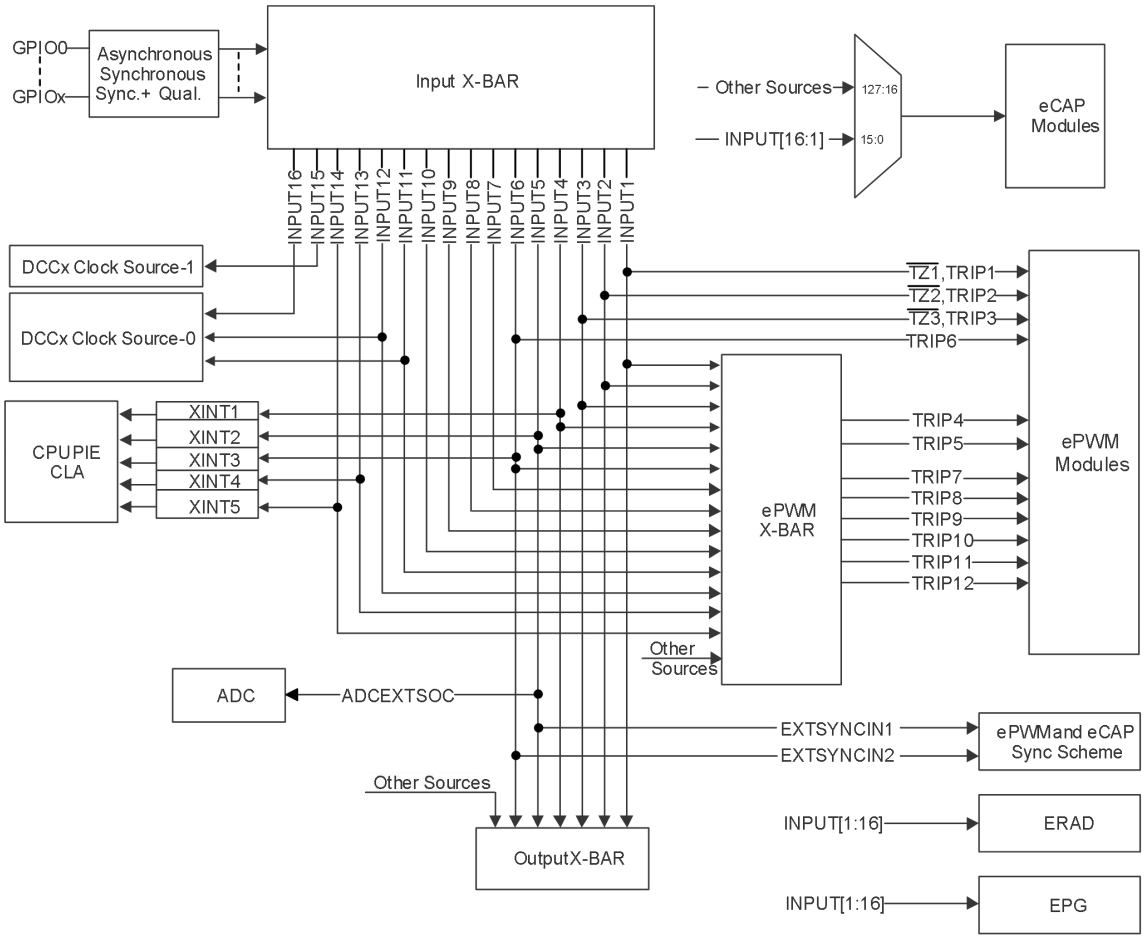


Figure 5- 7 Input X-BAR

Table 5-8 Input X-BAR Destinations

INPUT	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
ECAP / HRCAP	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
EPWM X-BAR	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes		
CLB X-BAR	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes		
OUTPUT X-BAR	Yes	Yes	Yes	Yes	Yes	Yes										
CPU XINT				XINT1	XINT2	XINT3							XINT4	XINT5		
EPWM TRIP	TZ1, TRIP1	TZ2, TRIP2	TZ3, TRIP3			TRIP6										
ADC START OF CONVERSION					ADCEX TSOC											

INPUT	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
EPWM / ECAP SYNC					EXTSYNCIN1	EXTSYNCIN2										
DCCx											CLK0	CLK0			CLK1	CLK0
EPG													EPG1IN1	EPG1IN2	EPG1IN3	EPG1IN4
ERAD	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

5.4.5 GPIO Output X-BAR, CLB X-BAR, CLB Output X-BAR, and ePWM X-BAR

GPIO 输出 X-BAR、CLBX-BAR、CLB 输出 X-BAR 和 ePWMX-BAR

The Output X-BAR has eight outputs that can be selected on the GPIO mux as OUTPUTXBARx. The CLB X-BAR has eight outputs that are connected to the CLB global mux as AUXSIGx. The CLB Output X-BAR has eight outputs that can be selected on the GPIO mux as CLB_OUTPUTXBARx. The ePWM X-BAR has eight outputs that are connected to the TRIPx inputs of the ePWM. The sources for the Output X-BAR, CLB X-BAR, CLB Output X-BAR, and ePWM X-BAR are shown in Figure 5- 8.

输出交叉开关（Output X - BAR）有八个输出端，可在通用输入输出（GPIO）复用器上选择作为 OUTPUTXBARx。可配置逻辑块交叉开关（CLB X - BAR）有八个输出端，这些输出端作为辅助信号 AUXSIGx 连接到可配置逻辑块（CLB）全局复用器。可配置逻辑块输出交叉开关（CLB Output X - BAR）有八个输出端，可在 GPIO 复用器上选择作为 CLB_OUTPUTXBARx。增强型脉宽调制交叉开关（ePWM X - BAR）有八个输出端，这些输出端连接到增强型脉宽调制器（ePWM）的 TRIPx 输入端。输出交叉开关、CLB 交叉开关、CLB 输出交叉开关和 ePWM 交叉开关的信号源如图 5-8 所示。

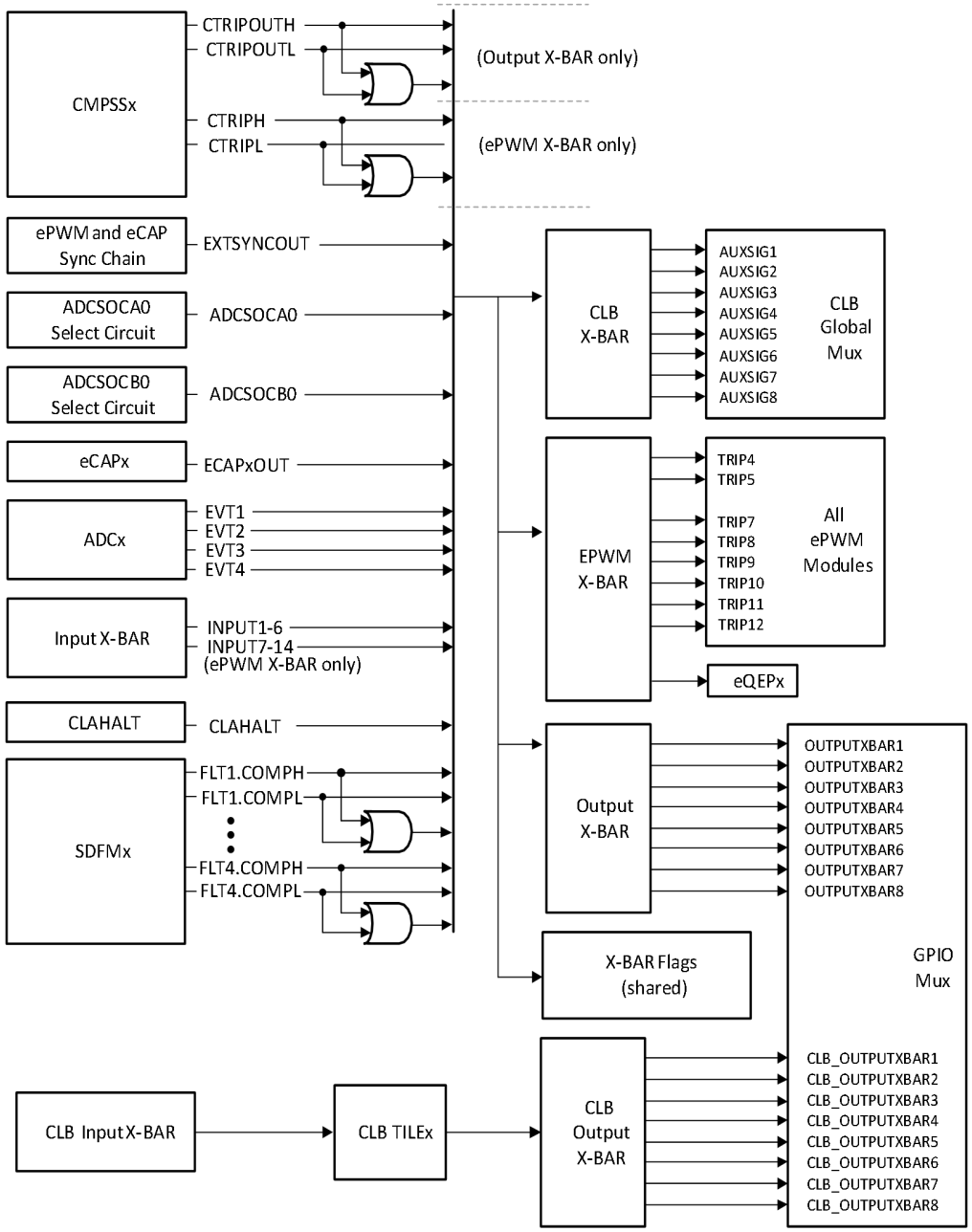


Figure 5- 8 Output X-BAR, CLB X-BAR, CLB Output X-BAR, and ePWM X-BAR Sources

5.5 Pins With Internal Pullup and Pulldown
带有内部上拉和下拉的引脚

Some pins on the device have internal pullups or pulldowns. Table 5-9 lists the pull direction and when it is active. The pullups on GPIO pins are disabled by default and can be enabled through software. To avoid any floating unbonded inputs, the Boot ROM will enable internal pullups on GPIO pins that are not bonded out in a particular package. Other pins noted in Table 5-9 with pullups and pulldowns are always on and cannot be disabled.

设备上的一些引脚具有内部上拉或下拉电阻。表 5-9 列出了上拉或下拉方向以及激活条件。GPIO 引脚的上拉电阻默认是禁用的，可以通过软件启用。为避免出现任何浮空的未绑定输入，启动 ROM 将为特定封装中未引出的 GPIO 引脚启用内部上拉电阻。表 5-9 中注明具有上拉和下拉电阻的其他引脚始终处于开启状态，无法禁用。

Table 5-9 Pins with Internal Pullup and Pulldown

PIN	RESET (XRSn=0)	DEVICE BOOT	APPLICATION
GPIOx	Pullup disabled	Pullup disabled(1)	Application defined
GPIO35/TDI	Pullup disabled		Application defined
GPIO37/TDO	Pullup disabled		Application defined
AGPIOx	Pullup disabled	Pullup disabled	Application defined
TCK	Pullup active		
TMS	Pullup active		
XRSn	Pullup active		
Other pins (include AIOs)	No pullup or pulldown present		

(1) Pins not bonded out in a given package will have the internal pullups enabled by the Boot ROM.
给定封装中未绑定的引脚将具有由引导 ROM 启用内部上拉。

5.6 Connections for Unused Pins
未使用引脚的连接

For applications that do not need to use all functions of the device, Table 5-10 lists acceptable conditioning for any unused pins. When multiple options are listed in Table 5-10, any option is acceptable. Pins not listed in Table 5-10 must be connected according to Section 5.

对于不需要使用设备所有功能的应用，Table 5-10 列出了任何未使用引脚的可接受处理方式。当 Table 5-10 中列出多个选项时，任何一个选项都是可以接受的。Table 5-10 中未列出的引脚必须按照第 5 节的要求进行连接。

Table 5-10 Connections for Unused Pins

SIGNAL NAME	ACCEPTABLE PRACTICE
ANALOG	
VREFHI	Tie to VDDA (applies only if ADC is not used in the application)
VREFLO	Tie to VSSA
Analog input pins with DACx_OUT	● No Connect ● Tie to VSSA through 4.7-kΩ or larger resistor
Analog input pins (except DACx_OUT)	● No Connect ● Tie to VSSA ● Tie to VSSA through resistor
Analog input pins (shared with GPIOs)(1)	● No connection (digital input mode with internal pullup enabled) ● No connection (digital output mode with internal pullup disabled) ● Pullup or pulldown resistor (any value resistor, digital input mode, and with internal pullup disabled)
DIGITAL	
GPIOx	● No connection (input mode with internal pullup enabled) ● No connection (output mode with internal pullup disabled) ● Pullup or pulldown resistor (any value resistor, input mode, and with internal pullup disabled)
GPIO35/TDI	When TDI mux option is selected (default), the GPIO is in Input mode. ● Internal pullup enabled ● External pullup resistor
GPIO37/TDO	When TDO mux option is selected (default), the GPIO is in Output mode only during JTAG activity; otherwise, it is in a tri-state condition. The pin must be biased to avoid extra current on the input buffer. ● Internal pullup enabled ● External pullup resistor
TCK	● No Connect ● Pullup resistor
TMS	Pullup resistor
GPIO19/X1	Turn XTAL off and: ● Input mode with internal pullup enabled. ● Input mode with external pullup or pulldown resistor. ● Output mode with internal pullup disabled.
GPIO18/X2	Turn XTAL off and: ● Input mode with internal pullup enabled. ● Input mode with external pullup or pulldown resistor. ● Output mode with internal pullup disabled.
POWER AND GROUND	
VDD	All VDD pins must be connected per Section 5.3. Pins should not be used to bias any external circuits.
VDDA	If a dedicated analog supply is not used, tie to VDDIO.
VDDIO	All VDDIO pins must be connected per Section 5.3.
VSS	All VSS pins must be connected to board ground.
VSSA	If an analog ground is not used, tie to VSS.

(1) AGPIO pins share analog and digital functionality. The actions here only apply if these pins are also not being used for analog functions.

AGPIO 引脚共享模拟和数字功能。此处的操作仅适用于这些引脚也不用于模拟功能的情况。

6 Specifications
规格参数

6.1 Absolute Maximum Ratings
最大额定值

over operating free-air temperature range (unless otherwise noted)(1)(2)
在自然通风条件下的工作温度范围内（除非另有说明）(1)(2)

		MIN	MAX	Unit
Supply voltage	VDDIO with respect to VSS	-0.3	4.6	V
	VDDA with respect to VSS	-0.3	4.6	
	VDD with respect to VSS	-0.3	1.5	
Input voltage	V_{IN} (3.3V)	-0.3	4.6	V
Output voltage	V_O	-0.3	4.6	V
Input clamp current	Digital/analog input (per pin)，I_{IK}（V_{IN} < VSS/VSSA or V_{IN} > VDDIO/ VDDA）(4)	-20	20	mA
	Total for all inputs, $I_{IKTOTAL}$ （V_{IN} < VSS/VSSA or V_{IN} > VDDIO/VDDA）	-20	20	
Output current	Digital output (per pin), I_{OUT}	-20	20	mA
Free-Air temperature	T_A	-40	125	℃
Operating junction temperature	T_J	-40	150	℃
Storage temperature(3)	Tstg	–65	150	℃

(1) Operation outside the Absolute Maximum Ratings may cause permanent device damage. Absolute Maximum Ratings do not imply functional operation of the device at these or any other conditions beyond those listed under Recommended Operating Conditions. If used outside the Recommended Operating Conditions but within the Absolute Maximum Ratings, the device may not be fully functional, and this may affect device reliability, functionality, performance, and shorten the device lifetime.
超出绝对最大额定值运行可能会对器件造成永久损坏。绝对最大额定值并不表示器件在这些条件下或在建议运行条件以外的任何其他条件下能够正常运行。如果超出建议工作条件但在绝对最大额定值范围内使用，器件可能不会完全正常运行，这可能影响器件的可靠性、功能和性能，并缩短器件寿命。

(2) All voltage values are with respect to VSS, unless otherwise noted.
除非另有说明，否则所有电压值均以 VSS 为基准。

(3) Long-term high-temperature storage or extended use at maximum temperature conditions may result in a reduction of overall device life.
长期高温贮存或在最大温度条件下超期使用可能会导致器件总体使用寿命缩短。

(4) Continuous clamp current per pin is ±2 mA. Do not operate in this condition continuously as VDDIO/VDDA voltage may internally rise and impact other electrical specifications.
每个引脚的连续钳位电流为±2mA。请勿在此条件下连续工作，因为 V_{DDIO}/V_{DDA} 电压可能会在内部上升并影响其他电气规格。

6.2 ESD Ratings – Automotive

			VALUE	UNIT
AVP32F003X in 100-pin QP package				
$V_{(ESD)}$ Electrostatic discharge	Human body model (HBM), per AEC Q100-002(1)	All pins	±2000	V
	Charged device model (CDM), per AEC Q100-011	All pins	±500	
		Corner pins on 100-pin QP: 1, 25, 26, 50, 51, 75, 76, 100	±750	
AVP32F003X in 80-pin QP package				
$V_{(ESD)}$ Electrostatic discharge	Human body model (HBM), per AEC Q100-002(1)	All pins	±2000	V
	Charged device model (CDM), per AEC Q100-011	All pins	±500	
		Corner pins on 80-pin QP: 1, 20, 21, 40, 41, 60, 61, 80	±750	
AVP32F003X in 64-pin QP package				
$V_{(ESD)}$ Electrostatic discharge	Human body model (HBM), per AEC Q100-002(1)	All pins	±2000	V
	Charged device model (CDM), per AEC Q100-011	All pins	±500	
		Corner pins on 64-pin QP: 1, 16, 17, 32, 33, 48, 49, 64	±750	
AVP32F003X in 48-pin QP package				
$V_{(ESD)}$ Electrostatic discharge	Human body model (HBM), per AEC Q100-002(1)	All pins	±2000	V
	Charged device model (CDM), per AEC Q100-011	All pins	±500	
		Corner pins on 48-pin QP: 1, 12, 13, 24, 25, 36, 37, 48	±750	

(1) AEC Q100-002 indicates HBM stressing is done in accordance with the ANSI/ESDA/JEDEC JS-001 specification.
AEC Q100-002 指示应当按照 ANSI/ESDA/JEDEC JS-001 规范执行 HBM 应力测试。

6.3 ESD Ratings – Commercial

			VALUE	UNIT
AVP32F003X in 100-pin QP package				
$V_{(ESD)}$ Electrostatic discharge (ESD)	Human-body model (HBM), per ANSI/ESDA/JEDEC JS-001(1)		±2000	V
	Charged-device model (CDM), per ANSI/ESDA/JEDEC JS-002(2)	All pins	±500	
		Corner pins on 100-pin QP: 1, 25, 26, 50, 51, 75, 76, 100	±750	
AVP32F003X in 80-pin QP package				
$V_{(ESD)}$ Electrostatic discharge (ESD)	Human-body model (HBM), per ANSI/ESDA/JEDEC JS-001(1)		±2000	V
	Charged-device model (CDM), per ANSI/ESDA/JEDEC JS-002(2)	All pins	±500	
		Corner pins on 80-pin QP: 1, 20, 21, 40, 41, 60, 61, 80	±750	
AVP32F003X in 64-pin QP package				
$V_{(ESD)}$ Electrostatic discharge (ESD)	Human-body model (HBM), per ANSI/ESDA/JEDEC JS-001(1)		±2000	V
	Charged-device model (CDM), per ANSI/ESDA/JEDEC JS-002(2)	All pins	±500	
		Corner pins on 64-pin QP: 1, 16, 17, 32, 33, 48, 49, 64	±750	
AVP32F003X in 48-pin QP package				
$V_{(ESD)}$ Electrostatic discharge (ESD)	Human-body model (HBM), per ANSI/ESDA/JEDEC JS-001(1)		±2000	V
	Charged-device model (CDM), per ANSI/ESDA/JEDEC JS-002(2)	All pins	±500	
		Corner pins on 48-pin QP: 1, 12, 13, 24, 25, 36, 37, 48	±750	

(1) JEDEC document JEP155 states that 500-V HBM allows safe manufacturing with a standard ESD control process.

JEDEC 文档 JEP155 指出：500V HBM 时能够在标准 ESD 控制流程下安全生产。

(2) JEDEC document JEP157 states that 250-V CDM allows safe manufacturing with a standard ESD control process.

JEDEC 文档 JEP157 指出：250V CDM 能够在标准 ESD 控制流程下安全生产。

6.4 Recommended Operating Conditions
推荐工作条件

		MIN	NOM	MAX	UNIT
Device supply voltage, VDDIO and VDDA	Internal BOR enabled(3)	VBOR-VDDIO(MAX)+VBOR-VDDIO-GB(2)	3.3	3.63	V
	Internal BOR disabled	2.8	3.3	3.63	
Device supply voltage, VDD		1.04	1.1	1.21	V
Device ground, VSS			0		V
Analog ground, VSSA			0		V
SRSUPPLY	Supply ramp rate(4)				
VIN	Digital input voltage	VSS–0.3		VDDIO+0.3	V
	Analog input voltage	VSSA–0.3		VDDA+0.3	V
Junction temperature, T_J (1)		-40		150	°C
Free-Air temperature, T_A		-40		125	°C

(1) Operation above T_J = 105°C for extended duration will reduce the lifetime of the device.
在T_J=105°C 以上的温度下长时间运行将缩短器件的使用寿命。

(2) See the *Power Management Module (PMM)* section.
请参阅“电源管理模块(PMM)”部分。

(3) Internal BOR is enabled by default.
默认情况下会启用内部 BOR。

(4) See the *Power Management Module Operating Conditions* table.
请参阅“电源管理模块运行条件”表。

6.5 Power Consumption Summary
功耗汇总

Current values listed in this section are representative for the test conditions given and not the absolute maximum possible. The actual device currents in an application will vary with application code and pin configurations. Section 6.5.1 lists the system current consumption values. Section 6.5.2 lists the system current consumption with VREG disabled.

本小节中列出的电流值仅代表给定的测试条件下的值，而不是可能的绝对最大值。应用中的实际器件电流将随应用代码和引脚配置的不同而变化。第 6.5.1 节列出了系统流耗值。第 6.5.2 节列出了禁用 VREG 时的系统流耗。

6.5.1 System Current Consumption
系统流耗

over operating free-air temperature range (unless otherwise noted).
TYP : V_{nom}, 30°C

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
OPERATING MODE						
I_{DDIO}	VDDIO current consumption during operational usage	This is an estimation of current for a typical heavily loaded application. Actual currents will vary depending on system		70	97	mA

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
I_{DDA}	VDDA current consumption during operational usage	activity, I/O electrical loading and switching frequency. This includes Core supply current with Internal Vreg Enabled. - CPU is running from RAM - Flash is powered up - X1/X2 crystal is powered up - PLL is enabled, SYSCLK=Max Device frequency - Analog modules are powered up - Outputs are static without DC Load - Inputs are static high or low		5	12	mA
IDLE MODE						
I_{DDIO}	VDDIO current consumption while device is in Idle mode	- CPU is in IDLE mode - Flash is powered down - PLL is Enabled, SYSCLK=Max Device Frequency, CPUCLK is gated - X1/X2 crystal is powered up - Analog Modules are powered down - Outputs are static without DC Load - Inputs are static high or low		25	50	mA
I_{DDA}	VDDA current consumption while device is in Idle mode			0.2	7.1	mA
STANDBY MODE						
I_{DDIO}	VDDIO current consumption while device is in Standby mode	- CPU is in STANDBY mode - Flash is powered down - PLL is Enabled, SYSCLK & CPUCLK are gated - X1/X2 crystal is powered down - Analog Modules are powered down - Outputs are static without DC Load - Inputs are static high or low		8.3	32.8	mA
I_{DDA}	VDDA current consumption while device is in Standby mode			0.9	6	mA
HALT MODE						
I_{DDIO}	VDDIO current consumption while device is in Halt mode	- CPU is in HALT mode - Flash is powered down - PLL is Disabled, SYSCLK & CPUCLK are gated - X1/X2 crystal is powered down - Analog Modules are powered down - Outputs are static without DC Load - Inputs are static high or low		6.7	30.8	mA
I_{DDA}	VDDA current consumption while device is in Halt mode			1	6	mA
Flash Erase/Program						
I_{DDIO}	VDDIO current consumption during Erase/Program cycle(1)	- CPU is running from RAM - Flash going through continuous Program/Erase operation - PLL is enabled, SYSCLK at 120 MHz.		45	106	mA

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
I_{DDA}	VDDA current consumption during Erase/Program cycle	- Peripheral clocks are turned OFF. - X1/X2 crystal is powered up - Analog is powered down - Outputs are static without DC Load - Inputs are static high or low		0.3	2.5	mA
RESET MODE						
I_{DDIO}	VDDIO current consumption while reset is active(2)			10.3		mA
I_{DDA}	VDDA current consumption while reset is active(2)			0.8		mA

(1) Brownout events during flash programming can corrupt flash data and permanently lock the device. Programming environments using alternate power sources (such as a USB programmer) must be capable of supplying the rated current for the device and other system components with sufficient margin to avoid supply brownout conditions.
闪存编程期间的欠压事件可能会损坏闪存数据并永久锁定器件。使用备用电源（例如 USB 编程器）的编程环境必须能够为器件和其他系统组件提供额定电流，并留有足够的裕度以避免电源欠压情况。

(2) This is the current consumption while reset is active, that is XRSn is low.
这是复位激活时的流耗，即 XRSn 为低电平。

6.5.2System Current Consumption - VREG Disable - External Supply
系统流耗-禁用 VREG-外部电源

over operating free-air temperature range (unless otherwise noted).
TYP： V_{nom}, 30°C

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
OPERATING MODE						
I_{DD}	VDD current consumption during operational usage	This is an estimation of current for a typical heavily loaded application. Actual currents will vary depending on system activity, I/O electrical loading and switching frequency. - CPU is running from RAM - Flash is powered up - X1/X2 crystal is powered up - PLL is enabled, SYSCLK=Max Device frequency - Analog modules are powered up - Outputs are static without DC Load - Inputs are static high or low		60	97	mA
I_{DDIO}	VDDIO current consumption during operational usage			11.6	12.8	mA
I_{DDA}	VDDA current consumption during operational usage			1.5	7.4	mA
IDLE MODE						
I_{DD}	VDD current consumption while device is in Idle mode	- CPU is in IDLE mode - Flash is powered down - PLL is Enabled, SYSCLK=Max Device Frequency, CPUCLK is gated - X1/X2 crystal is powered up - Analog Modules are powered down - Outputs are static without DC Load - Inputs are static high or low		18.5	51	mA
I_{DDIO}	VDDIO current consumption while device is in Idle mode			6.8	7.6	mA
I_{DDA}	VDDA current consumption while device is in Idle mode			1	5.3	mA
STANDBY MODE						
I_{DD}	VDD current consumption while device is in Standby mode	- CPU is in STANDBY mode - Flash is powered down - PLL is Enabled, SYSCLK & CPUCLK are gated - X1/X2 crystal is powered down - Analog Modules are powered down - Outputs are static without DC Load - Inputs are static high or low		1.9	33	mA
I_{DDIO}	VDDIO current consumption while device is in Standby mode			6.2	7	mA
I_{DDA}	VDDA current consumption while device is in Standby mode			0.9	5.1	mA
HALT MODE						
I_{DD}	VDD current consumption while device is in Halt mode	- CPU is in HALT mode - Flash is powered down - PLL is Disabled, SYSCLK & CPUCLK are gated - X1/X2 crystal is powered down - Analog Modules are powered down - Outputs are static without DC Load - Inputs are static high or low		1.1	32	mA
I_{DDIO}	VDDIO current consumption while device is in Halt mode			5.3	6.2	mA
I_{DDA}	VDDA current consumption while device is in Halt mode			0.9	4.8	mA
FLASH ERASE/PROGRAM						
I_{DD}	VDD Current consumption during Erase/Program cycle(1)	- CPU is running from RAM - Flash going through		41	60.5	mA

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
I_{DDIO}	VDDIO Current consumption during Erase/Program cycle(1)	continuous Program/Erase operation - PLL is enabled, SYSCLK at 120 MHz. - Peripheral clocks are turned OFF. - X1/X2 crystal is powered up - Analog is powered down - Outputs are static without DC Load - Inputs are static high or low		10	15	mA
I_{DDA}	VDDA Current consumption during Erase/Program cycle			0.3	2.5	mA
RESET MODE						
I_{DD}	VDD current consumption while reset is active(2)			4.0		mA
I_{DDIO}	VDDIO current consumption while reset is active(2)			6.0		mA
I_{DDA}	VDDA current consumption while reset is active(2)			1.0		mA

(1) Brownout events during flash programming can corrupt flash data and permanently lock the device. Programming environments using alternate power sources (such as a USB programmer) must be capable of supplying the rated current for the device and other system components with sufficient margin to avoid supply brownout conditions.
闪存编程期间的欠压事件可能会损坏闪存数据并永久锁定器件。使用备用电源（例如 USB 编程器）的编程环境必须能够为器件和其他系统组件提供额定电流，并留有足够的裕度以避免电源欠压情况。

(2) This is the current consumption while reset is active, that is XRSn is low.
这是复位激活时的流耗，即 XRSn 为低电平。

6.5.3Operating Mode Test Description
工作模式测试说明

Section 6.5.1 and Section 6.5.5.1 list the current consumption values for the operational mode of the device. The operational mode provides an estimation of what an application might encounter. The test condition for these measurements has the following properties:

- Code is executing from RAM.
- FLASH is read and kept in active state.
- No external components are driven by I/O pins.
- All peripherals have clocks enabled.
- The CPU is actively executing code.
- All analog peripherals are powered up. ADCs and DACs are periodically converting.

第 6.5.1 节和第 6.5.5.1 节列出了器件工作模式下的流耗值。工作模式下将估算应用可能遇到的流耗。这些测量的测试条件具有以下属性：

- 从 RAM 执行代码。
- 闪存被读取，并保持激活状态。
- I/O 引脚不驱动任何外部元件。
- 所有外设都启用了时钟。
- CPU 正在主动执行代码。
- 所有模拟外设均已通电。ADC 和 DAC 会定期转换。

6.5.4Current Consumption Graphs 流耗图

The below graphs show a typical representation of the relationship between frequency, temperature, supply(VREG Enabled), and current consumption on the device. Actual results will vary based on the system implementation and conditions.
Figure 6-1 shows the typical operating current profile across temperature and core supply voltage. Figure 6-2 shows the typical idle current profile across temperature and core supply voltage. Figure 6-3 shows the typical standby current profile across temperature and core supply voltage. Figure 6-4 shows the typical halt current profile across temperature and core supply voltage.

图示为器件温度、电源（VREG 启用）与电流消耗之间的关系。实际结果将因系统实现情况和具体条件而异。

图 6-1 所示为整个温度和内核电源电压范围内的典型工作电流曲线。图 6-2 所示为整个温度和内核电源电压下的典型空闲电流曲线。图 6-3 所示为整个温度和内核电源电压范围内的典型待机电流曲线。图 6-4 所示为整个温度和内核电源电压范围内的典型停机电流曲线。

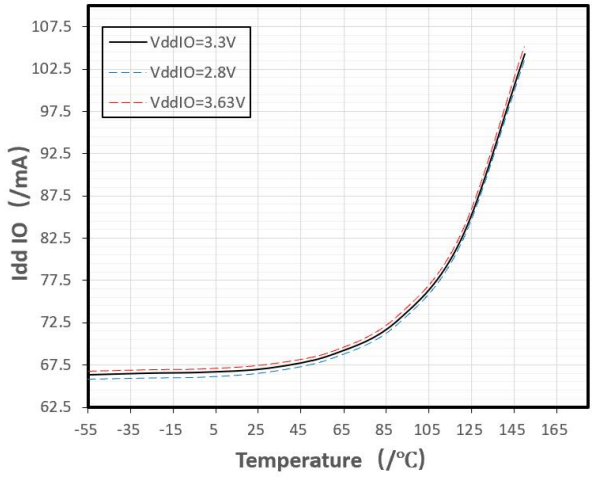


Figure 6-1 Operating Current Versus Temperature

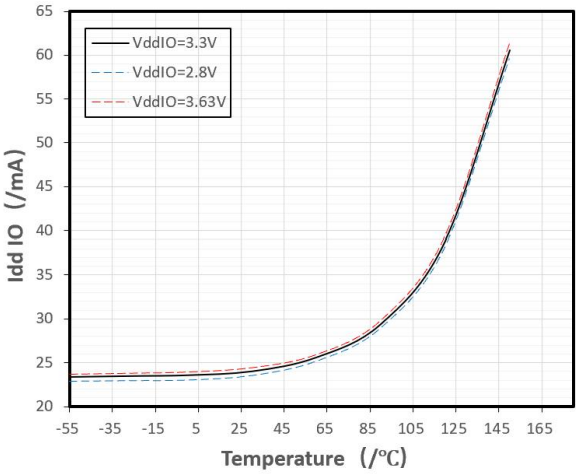


Figure 6-2 Current Versus Temperature - Idle Mode

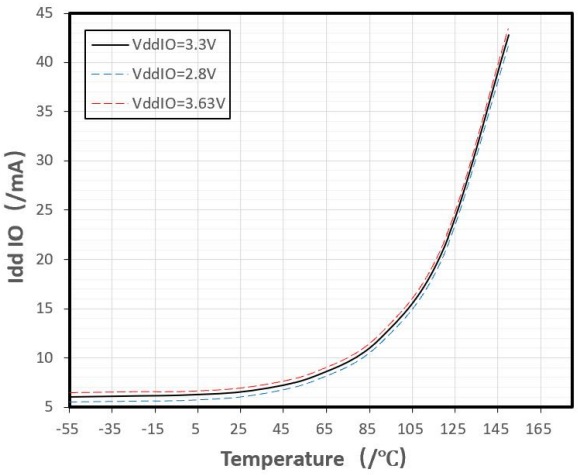


Figure 6- 3 Current Versus Temperature – Standby Mode

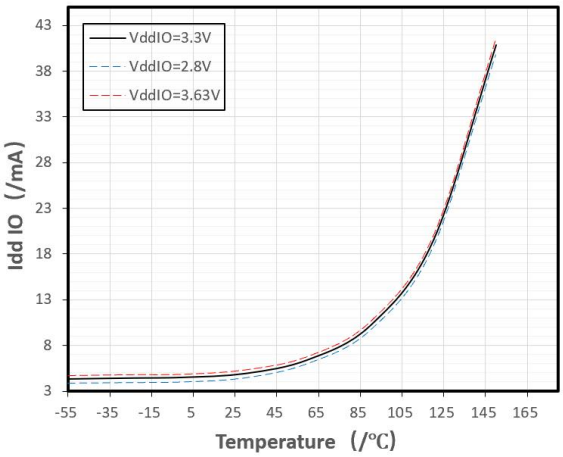


Figure 6- 4 Current Versus Temperature – Halt Mode

6.5.5Reducing Current Consumption

AVP32F003X devices provide some methods to reduce the device current consumption:

- One of the two low-power modes—IDLE or STANDBY—could be entered during idle periods in the application.
- The flash module may be powered down if the code is run from RAM.
- Disable the pullups on pins that assume an output function.
- Each peripheral has an individual clock-enable bit (PCLKCRx). Reduced current consumption may be achieved by turning off the clock to any peripheral that is not used in a given application. Section 6.5.4.1 lists the typical current reduction that may be achieved by disabling the clocks using the PCLKCRx register.
- To realize the lowest VDDA current consumption in an LPM, see the Analog-to-Digital Converter (ADC) chapter of the *AVP32F003X Technical Reference Manual* to ensure each module is powered down as well.

AVP32F003X 器件提供了一些降低器件流耗的方法：

- 在应用的空闲期间，可以进入两种低功耗模式中的任何一种：空闲或待机。
- 如果代码从 RAM 中运行，闪存模块可能会断电。
- 禁用假定具有输出功能的引脚上的上拉电阻。
- 每个外设都有一个单独的时钟使能位(PCLKCRx)。通过关闭给定应用中未使用的任何外设的时钟，可以减少流耗。第 6.5.4.1 节列出了通过使用 PCLKCRx 寄存器禁用时钟可以实现的电流降低量典型值。
- 为了在 LPM 模式下实现更低的 VDDA 流耗，请参阅 AVP32F003X 实时微控制器技术参考手册中的“模数转换器（ADC）”一章，确保每个模块也断电。

6.5.5.1 Typical Current Reduction per Disabled Peripheral
每个禁用外设的典型电流降低

PERIPHERAL	I_{DD} CURRENT REDUCTION (mA)
ADC(1)	0.80
CLA	0.40
CLABGCRC	0.21
CLB	1.60
CMPSS(1)	0.45
CPUBGCRC	0.28
CPU TIMER	0.04
GPDAC	0.21
DCAN	1.17
DCC	0.10
DMA	0.60
eCAP1 and eCAP2	0.09
eCAP3(2)	0.31
ePWM1 to ePWM4(3)	1.05
ePWM5 to ePWM8	0.74
ERAD	1.89
eQEP	0.09
FSIRX	0.34
FSITX	0.28
HIC	0.17
I2C	0.27
LIN	0.40
MCAN(CANFD)	1.36
PMBUS	0.34
SCI	0.26
SDFM	1.79
SPI	0.08

(1) This current represents the current drawn by the digital portion of the each module.
此电流代表了每个模块的数字部分汲取的电流。

(2) eCAP3 can also be configured as HRCAP.
eCAP3 也可以配置为 HRCAP。

(3) ePWM1 to ePWM4 can also be configured as HRPWM.
ePWM1 到ePWM4 也可以配置为 HRPWM。

6.6 Electrical Characteristics
电气特性

over recommended operating conditions (unless otherwise noted)
在建议运行条件下测得（除非另有说明）

PARAMETER			TEST CONDITIONS	MIN	TYP	MAX	UNIT
Digital and analog IO 数字和模拟 IO							
V_{OH}	High-level output voltage		I_{OH}=I_{OH}MIN	VDDIO*0.8			V
			I_{OH}=–100uA	VDDIO–0.2			
V_{OL}	Low-level output voltage		I_{OL}=I_{OL}MAX			0.4	V
			I_{OL}=100uA			0.2	
I_{OH}	High-level output source current for all output pins			–4			mA
I_{OL}	Low-level output sink current for all output pins					4	mA
R_{OH}	High-level output impedance for all output pins		V_{OH}=VDDIO–0.4V	50	66	96	Ω
R_{OL}	Low-level output impedance for all output pins		V_{OL}=0.4V	48	60	84	Ω
V_{IH}	High-level input voltage			2.0			V
V_{IL}	Low-level input voltage					0.8	V
$V_{HYSTERESIS}$	Input hysteresis			125			mV
$I_{PULLDOWN}$	Input current	Pins with pulldown	VDDIO=3.3V V_{IN}=VDDIO		120		µA
I_{PULLUP}	Input current	Digital inputs with pullup enabled(1)	VDDIO=3.3V V_{IN}=0V		160		µA
$R_{PULLDOWN}$	**Weak pull-down resistor**			22	31	62	kΩ
R_{PULLUP}	**Weak pull-up resistor**			19	30	54	kΩ
I_{LEAK}	Pin leakage	Digital inputs	Pullups and outputs disabled 0V≤V_{IN}≤VDDIO			0.1	µA
		Analog pins (except ADCINB3/VDAC)	Analog drivers disabled 0V≤V_{IN}≤VDDA			0.1	
		ADCINB3/VDAC			0.2	4.4	
C_I	Input capacitance	Digital inputs			2		pF
		Analog pins(2)					
VREG, POR and BOR							
VREG, POR, BOR(3)							

(1) See Pins With Internal Pullup and Pulldown table for a list of pins with a pullup or pulldown.
有关带有上拉或下拉的引脚列表，请参阅“带有内部上拉和下拉的引脚”表。

(2) The analog pins are specified separately; see the Per-Channel Parasitic Capacitance tables that are in the ADC Input Model section.
模拟引脚是单独指定的；请参阅“ADC 输入模型”部分中的“每通道寄生电容”表。

(3) See the *Power Management Module (PMM)* section.
请参阅“电源管理模块（PMM）”部分。

6.7 Thermal Resistance Characteristics for 100-pin QP Package
100 引脚 QP 封装的热阻特性

		°C/W(1)	AIR FLOW (lfm)(2)
$R\Theta_{JC}$	Junction-to-case thermal resistance	10.60	N/A
$R\Theta_{JB}$	Junction-to-board thermal resistance	22.42	N/A
$R\Theta_{JA}$（High k PCB）	Junction-to-free air thermal resistance	39.56	0
$R\Theta_{JMA}$	Junction-to-moving air thermal resistance	31.65	150
		30.16	250
		28.99	500
Psi_{JT}	Junction-to-package top	0.36	0
		0.40	150
		0.52	250
		0.65	500
Psi_{JB}	Junction-to-board	21.61	0
		20.89	150
		20.48	250
		20.20	500

(1) These values are based on a JEDEC-defined 2S2P system (with the exception of the Theta JC [$R\Theta_{JC}$] value, which is based on a JEDEC-defined 1S0P system) and will change based on environment as well as application. For more information, see these EIA/JEDEC standards:
以上值基于 JEDEC 定义的 2S2P 系统（基于 JEDEC 定义的 1S0P 系统的 ThetaJC[$R\Theta_{JC}$]值除外），将随环境和应用的变化而更改。有关更多信息，请参阅以下 EIA/JEDEC 标准：

- JESD51-2, Integrated Circuits Thermal Test Method Environmental Conditions - Natural Convection (Still Air)
  JESD51-2，集成电路热测试方法环境条件-自然对流（静止空气）
- JESD51-3, *Low Effective Thermal Conductivity Test Board for Leaded Surface Mount Packages*
  JESD51-3，导电性表面安装封装低有效热导率测试板
- JESD51-7, *High Effective Thermal Conductivity Test Board for Leaded Surface Mount Packages*
  JESD51-7，导电性表面安装封装高有效热导率测试板
- JESD51-9, *Test Boards for Area Array Surface Mount Package Thermal Measurements*
  JESD51-9，用于区域阵列表面安装封装热测量的测试板

(2) lfm = linear feet per minute
lfm=线性英尺/分钟

6.8 Thermal Resistance Characteristics for 80-pin QP Package
80 引脚 QP 封装的热阻特性

		°C/W(1)	AIR FLOW (lfm)(2)
$R\Theta_{JC}$	Junction-to-case thermal resistance	14.2	N/A
$R\Theta_{JB}$	Junction-to-board thermal resistance	21.9	N/A
$R\Theta_{JA}$（High k PCB）	Junction-to-free air thermal resistance	49.9	0
		38.3	150
		36.7	250
		34.4	500
Psi_{JT}	Junction-to-package top	0.8	0
		1.18	150
		1.34	250

		°C/W(1)	AIR FLOW (lfm)(2)
		1.62	500
Psi_{JB}	Junction-to-board	21.6	0
		20.7	150
		20.5	250
		20.1	500

(1) These values are based on a JEDEC-defined 2S2P system (with the exception of the Theta JC [$R\Theta_{JC}$] value, which is based on a JEDEC-defined 1S0P system) and will change based on environment as well as application. For more information, see these EIA/JEDEC standards:

以上值基于 JEDEC 定义的 2S2P 系统（基于 JEDEC 定义的 1S0P 系统的 ThetaJC[$R\Theta_{JC}$]值除外），将随环境和应用的变化而更改。有关更多信息，请参阅以下 EIA/JEDEC 标准：

- JESD51-2, Integrated Circuits Thermal Test Method Environmental Conditions - Natural Convection (Still Air)
  JESD51-2，集成电路热测试方法环境条件-自然对流（静止空气）
- JESD51-3, *Low Effective Thermal Conductivity Test Board for Leaded Surface Mount Packages*
  JESD51-3，导电性表面安装封装低有效热导率测试板
- JESD51-7, *High Effective Thermal Conductivity Test Board for Leaded Surface Mount Packages*
  JESD51-7，导电性表面安装封装高有效热导率测试板
- JESD51-9, *Test Boards for Area Array Surface Mount Package Thermal Measurements*
  JESD51-9，用于区域阵列表面安装封装热测量的测试板

(2) lfm = linear feet per minute
lfm=线性英尺/分钟

6.9 Thermal Resistance Characteristics for 64-pin QP Package
64 引脚 QP 封装的热阻特性

		°C/W(1)	AIR FLOW (lfm)(2)
$R\Theta_{JC}$	Junction-to-case thermal resistance	12.4	N/A
$R\Theta_{JB}$	Junction-to-board thermal resistance	23.5	N/A
$R\Theta_{JA}$（High k PCB）	Junction-to-free air thermal resistance	44.15	0
$R\Theta_{JMA}$	Junction-to-moving air thermal resistance	36.4	150
		35.0	250
		33.1	500
Psi_{JT}	Junction-to-package top	0.48	0
		0.69	150
		0.85	250
		1.09	500
Psi_{JB}	Junction-to-board	22.30	0
		22.34	150
		22.22	250
		22.02	500

(1) These values are based on a JEDEC-defined 2S2P system (with the exception of the Theta JC [$R\Theta_{JC}$] value, which is based on a JEDEC-defined 1S0P system) and will change based on environment as well as application. For more information, see these EIA/JEDEC standards:

以上值基于 JEDEC 定义的 2S2P 系统（基于 JEDEC 定义的 1S0P 系统的 ThetaJC[$R\Theta_{JC}$]值除外），将随环境和应用的变化而更改。有关更多信息，请参阅以下 EIA/JEDEC 标准：

- JESD51-2, Integrated Circuits Thermal Test Method Environmental Conditions - Natural Convection (Still Air)
  JESD51-2，集成电路热测试方法环境条件-自然对流（静止空气）
- JESD51-3, *Low Effective Thermal Conductivity Test Board for Leaded Surface Mount Packages*
  JESD51-3，导电性表面安装封装低有效热导率测试板
- JESD51-7, *High Effective Thermal Conductivity Test Board for Leaded Surface Mount Packages*

JESD51-7，导电性表面安装封装高有效热导率测试板

- JESD51-9, *Test Boards for Area Array Surface Mount Package Thermal Measurements*
  JESD51-9，用于区域阵列表面安装封装热测量的测试板

(2) lfm = linear feet per minute
lfm=线性英尺/分钟

6.10 Thermal Resistance Characteristics for 48-pin QP Package
48 引脚 QP 封装的热阻特性

		°C/W(1)	AIR FLOW (lfm)(2)
$R\Theta_{JC}$	Junction-to-case thermal resistance	16.2	N/A
$R\Theta_{JB}$	Junction-to-board thermal resistance	22.3	N/A
$R\Theta_{JA}$（High k PCB）	Junction-to-free air thermal resistance	56.7	0
		50.4	150
		48.2	250
		45	500
Psi_{JT}	Junction-to-package top	0.7	0
		0.94	150
		1.1	250
		1.38	500
Psi_{JB}	Junction-to-board	22	0
		28.7	150
		28.4	250
		28	500

(1) These values are based on a JEDEC-defined 2S2P system (with the exception of the Theta JC [$R\Theta_{JC}$] value, which is based on a JEDEC-defined 1S0P system) and will change based on environment as well as application. For more information, see these EIA/JEDEC standards:
以上值基于 JEDEC 定义的 2S2P 系统（基于 JEDEC 定义的 1S0P 系统的 ThetaJC[$R\Theta_{JC}$]值除外），将随环境和应用的变化而更改。有关更多信息，请参阅以下 EIA/JEDEC 标准：

- JESD51-2, Integrated Circuits Thermal Test Method Environmental Conditions - Natural Convection (Still Air)
  JESD51-2，集成电路热测试方法环境条件-自然对流（静止空气）
- JESD51-3, *Low Effective Thermal Conductivity Test Board for Leaded Surface Mount Packages*
  JESD51-3，导电性表面安装封装低有效热导率测试板
- JESD51-7, *High Effective Thermal Conductivity Test Board for Leaded Surface Mount Packages*
  JESD51-7，导电性表面安装封装高有效热导率测试板
- JESD51-9, *Test Boards for Area Array Surface Mount Package Thermal Measurements*
  JESD51-9，用于区域阵列表面安装封装热测量的测试板

(2) lfm = linear feet per minute
lfm=线性英尺/分钟

6.11 Thermal Design Considerations
散热设计注意事项

Based on the end application design and operational profile, the I_{DD} and I_{DDIO} currents could vary. Systems that exceed the recommended maximum power dissipation in the end product may require additional thermal enhancements. Ambient temperature (T_A) varies with the end application and product design. The critical factor that affects reliability and functionality is T_J, the junction temperature, not the ambient temperature. Hence, care should be taken to keep T_J within the specified limits. T_{case} should be measured to

estimate the operating junction temperature T_J.

根据最终应用设计和运行情况，I_{DD} 和 I_{DDIO} 电流可能有所不同。最终产品中超过建议最大功率耗散的系统可能需要额外的散热增强措施。环境温度(T_A)随最终应用和产品设计的不同而异。影响可靠性和功能性的关键参数是结温 T_J，而不是环境温度。因此，应该注意将 T_J 保持在指定限值内。应该测量 T_{case} 以评估工作结温 T_J。

6.12 System 系统

6.12.1 Power Management Module (PMM) 电源管理模块(PMM)

6.12.1.1 Introduction 引言

The Power Management Module (PMM) handles all the power management functions required for device operation.

电源管理模块（PMM）可以处理运行器件时所需的所有电源管理功能。

6.12.1.2 Overview 概述

The block diagram of the PMM is shown in Figure 6-5. As can be seen, the PMM comprises of various subcomponents, which are described in the subsequent sections.

在图 6-5 中给出了 PMM 的框图。可以看出，PMM 包含多个子组件，这些子组件将在后续章节中进行介绍。

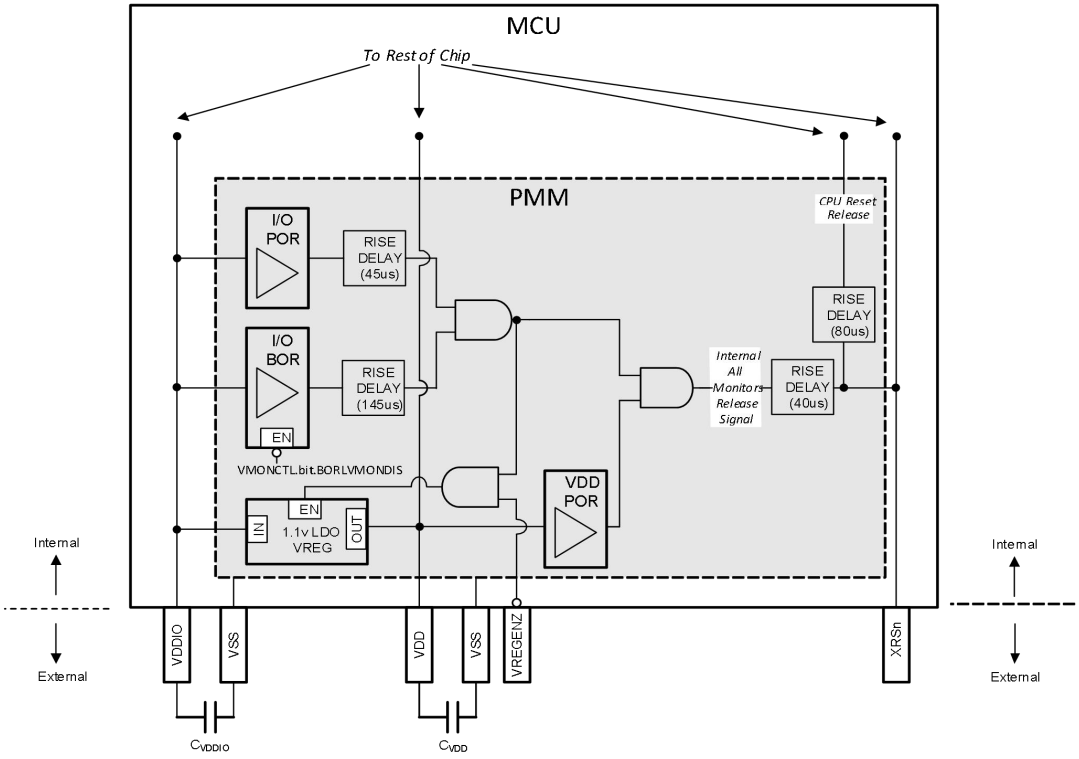


Figure 6- 5 PMM Block Diagram

6.12.1.2.1 Power Rail Monitors
电源轨监测器

The PMM has voltage monitors on the supply rails that release the XRSn signal high once the voltages cross the set threshold during power up. They also function to trip the XRSn signal low if any of the voltages drop below the programmed levels. The various voltage monitors are described in subsequent sections.

PMM 在电源轨上有电压监测器，一旦电压在上电期间超过设定阈值，XRSn 信号便会释放为高电平。如果任何电压降至编程的电平以下，这些电压监测器还可以使 XRSn 信号跳闸为低电平。后续各节将介绍各种电压监测器。

Note

Not all the voltage monitors are supported for device operation in an application after boot up. In the case where a voltage monitor is not supported, an external supervisor is recommended if the device needs supply voltage monitoring while the application is running.

启动后，并非所有电压监测器都支持在应用中运行器件。在不支持电压监测器的情况下，如果器件在应用运行过程中需要监测电源电压，则建议使用外部监控器。

The three voltage monitors (I/O POR, I/O BOR, VDD POR) all have to release their respective outputs before the device begins operation (that is, XRSn goes high). However, if any of the voltage monitors trips, XRSn is driven low. The I/Os are held in high impedance when any of the voltage monitors trip.

三个电压监测器（I/OPOR、I/OBOR、VDDPOR）都必须在器件开始运行（即XRSn 变为高电平）之前释放各自的输出。但如果任何电压监测器跳闸，XRSn 将被驱动为低电平。当任何电压监测器跳闸时，I/O 保持高阻抗。

6.12.1.2.1.1 I/O POR (Power-On Reset) Monitor
I/O POR（上电复位）监测器

The I/O POR monitor supervises the VDDIO rail. During power up, this is the first monitor to release (that is, first to untrip) on VDDIO.

I/OPOR 监测器会监控 VDDIO 电源轨。在上电期间，这是第一个在 VDDIO 上释放的监测器（即第一个要解除跳闸的监测器）。

6.12.1.2.1.2 I/O BOR (Brown-Out Reset) Monitor
I/O BOR（欠压复位）监测器

The I/O BOR monitor also supervises the VDDIO rail. During power up, this is the second monitor to release (that is, second to untrip) on VDDIO. This monitor has a tighter tolerance compared to the I/O POR.

Any drop in voltage below the recommended operating voltages will trip the I/O BOR and reset the device but this can be disabled by setting VMONCTL.bit.BORLVMONDIS to 1. The I/O BOR can only be disabled after the device has fully booted up. If the I/O BOR is disabled, the I/O POR will reset the device for voltage drops.

Note

The level at which the I/O POR trips is well below the minimum recommended voltage for VDDIO, and therefore should not be used for device supervision.

I/O BOR 监测器还会监控 VDDIO 电源轨。在上电期间，这是第二个在 VDDIO 上释放的监测器（即第二个要解除跳闸的监测器）。与 I/O POR 相比，该监测器具有更严格的容差。

只要电压降至低于建议工作电压，都会导致 I/O BOR 跳闸并复位器件，但可以通过将 VMONCTL.bit.BORLVMONDIS 设置为 1 来禁用该功能。只有在器件完全启动后，才能禁用 I/O BOR。如果 I/O BOR 被禁用，I/O POR 将在电压下降时复位器件。

备注

I/O POR 跳闸的电平远低于 VDDIO 的最小建议电压，因此不应用于器件监控。

Figure 6- 6 shows the operating region of the I/O BOR.

图 6-6 所示为 I/O BOR 的工作区域。

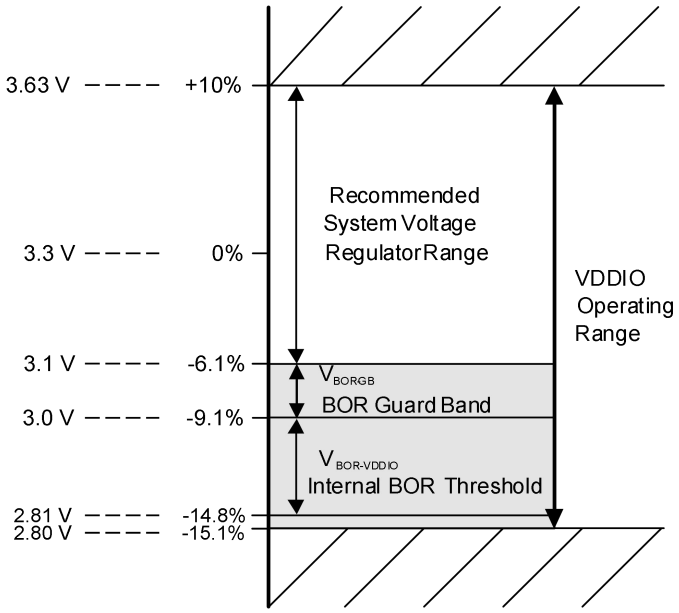


Figure 6- 6 I/O BOR Operating Range

6.12.1.2.1.3 VDD POR (Power-On Reset) Monitor
VDD POR（上电复位）监测器

The VDD POR monitor supervises the VDD rail. During power up, this monitor releases (that is, untrips) once the voltage crosses the programmed trip level on VDD.

VDDPOR 监测器可监控 VDD 电源轨。在上电期间，一旦电压超过 VDD 上编程的跳闸电平，该监测器就会释放（即解除跳闸）。

Note

VDD POR is programmed at a level below the minimum recommended voltage for VDD, and therefore it should not be relied upon for VDD supervision if that is required in the application.

VDDPOR 编程为低于 VDD 最小建议电压的水平，因此，如果应用中需要 VDD 监控，不应该依赖 VDDPOR 来进行此监控。

6.12.1.2.2 External Supervisor Usage
外部监控器使用情况

VDDIO Monitoring: The I/O BOR is supported for application use, so an external supervisor is not

required to monitor the I/O rail.

VDD Monitoring: The VDD POR is not supported for application use. If VDD monitoring is required by the application, an external supervisor should be used to monitor the VDD rail.

VDDIO 监控：I/OBOR 支持应用使用，因此无需外部监控器来监控 I/O 电源轨。

VDD 监控：VDDPOR 不支持应用使用。如果应用需要 VDD 监控，则应使用外部监控器监控 VDD 电源轨。

Note

The use of an external supervisor with the internal VREG is not supported. If VDD monitoring is required by the application, a package with a VREGENZ pin must be used to power VDD externally.

不支持将外部监控器与内部 VREG 一起使用。如果应用需要 VDD 监控，则必须使用具有 VREGENZ 引脚的封装从外部为 VDD 供电。

6.12.1.2.3 Delay Blocks
延迟块

The delay blocks in the path of the voltage monitors work together to delay the release time between the voltage monitors and XRSn. These delays ensure that the voltages are stable when XRSn releases in external VREG mode. The delay blocks are only active during power up (that is, when VDDIO and VDD are ramping up).
The delay blocks contribute to the minimum slew rates specified in Power Management Module Electrical Data and Timing for the power rails.

电压监控器路径中的延迟块协同工作，以延迟电压监控器和 XRSn 之间的释放时间。当 XRSn 在外部VREG 模式下释放时，这些延迟可确保电压稳定。延迟块仅在上电期间（即，当 VDDIO 和 VDD 斜升时）有效。

延迟块有助于实现“电源管理模块电气数据和时序”章节中所指定的电源轨最小压摆率。

Note

The delay numbers specified in the block diagram are typical numbers.

方框图中指定的延迟数字是典型值。

6.12.1.2.4 Internal 1.1V LDO Voltage Regulator (VREG)
内部 1.1V LDO 稳压器(VREG)

The internal VREG is supplied by the VDDIO rail and can generate the 1.1 V required to power the VDD pins. It is enabled by tying the VREGENZ pin low. Although the internal VREG eliminates the need to use an external supply for VDD, decoupling capacitors are still required on the VDD pins for VREG stability and transients. See VDD Decoupling for details.

内部 VREG 由 VDDIO 电源轨供电，并可以生成为 VDD 引脚供电所需的 1.1V 电压。启用它的方法是将 VREGENZ 引脚绑定为低电平。虽然有了内部 VREG 后便不再需要为 VDD 使用外部电源，但 VDD 引脚上仍需要去耦电容器以确保 VREG 稳定性和避免瞬变。详细信息，请参阅“VDD 去耦”章节。

6.12.1.2.5 VREGENZ

The VREGENZ (VREG disable) pin controls the state of the internal VREG. To enable the internal VREG, the VREGENZ pin should be tied low. For applications supplying VDD externally (external VREG), the internal VREG should be disabled by tying the VREGENZ pin high.

VREGENZ（VREG 禁用）引脚可控制内部 VREG 的状态。要启用内部 VREG，VREGENZ 引脚应连接至低电平。对于从外部为 VDD 供电（外部 VREG）的应用，应通过将 VREGENZ 引脚连接至高电平来禁用内部 VREG。

Note

Not all device packages have VREGENZ pinned out. For packages without VREGENZ, external VREG mode is not supported.

并非所有器件封装都具有 VREGENZ 引脚输出。对于没有 VREGENZ 的封装，不支持外部 VREG 模式。

6.12.1.3 External Components
外部元件

6.12.1.3.1 Decoupling Capacitors
去耦电容器

VDDIO and VDD require decoupling capacitors for correct operation. The requirements are outlined in subsequent sections.

VDDIO 和 VDD 需要去耦电容器才能正常运行。后续章节将概述这些要求。

6.12.1.3.1.1 VDDIO Decoupling
VDDIO 去耦

A minimum amount of decoupling capacitance should be placed on VDDIO. See the CVDDIO parameter in Power Management Module Electrical Data and Timing. The actual amount of decoupling capacitance to use is a requirement of the power supply driving VDDIO. Either of the configurations outlined below is acceptable:

- Configuration 1: Place a decoupling capacitor on each VDDIO pin per the CVDDIO parameter.
- Configuration 2: Install a single decoupling capacitor that is the equivalent of CVDDIO * VDDIO pins.

应在 VDDIO 上放置最小的去耦电容值。请参阅 C_{VDDIO} 参数（位于“电源管理模块电气数据和时序”中）。实际使用的去耦电容值取决于驱动 VDDIO 的电源。可接受以下任一配置：

- **配置 1：**根据 C_{VDDIO} 参数在每个 VDDIO 引脚上放置一个去耦电容器。
- **配置 2：**安装等效于 C_{VDDIO}*VDDIO 引脚的单个去耦电容器。

Note

It is critical to have the decoupling capacitor or capacitors close to the device pins.

将去耦电容器（一个或多个）靠近器件引脚放置至关重要。

6.12.1.4 Power Sequencing
电源时序

6.12.1.4.1 Supply Pins Ganging
电源引脚联动

It is strongly recommended that all 3.3-V rails be tied together and supplied from a single source. This list includes:

- VDDIO
- VDDA

In addition, no power pin should be left unconnected.
In external VREG mode, the VDD pins should be tied together and supplied from a single source.
In internal VREG mode, tying the VDD pins together is optional as long as each VDD pin has a capacitor on it. See VDD Decoupling for VDD decoupling configurations.
The analog modules on the device have fairly high PSRR; therefore, in most cases, noise on VDDA will have to exceed the recommended operating conditions of the supply rails before the analog modules see performance degradation. Therefore, supplying VDDA separately typically offers minimal benefits. Nevertheless, for the purposes of noise improvement, placing a pi filter between VDDIO and VDDA is acceptable.

强烈建议将所有 3.3V 电源轨连接在一起并由单个电源供电。该列表包括：

- VDDIO
- VDDA

此外，任何电源引脚都不应悬空。

在外部VREG 模式下，VDD 引脚应连接在一起并由单个电源供电。

在内部VREG 模式下，将VDD 引脚连接在一起是可选操作，只要每个 VDD 引脚上都有一个电容器即可。请参阅 VDD 去耦，了解VDD 去耦配置。

器件上的模拟模块具有相当高的 PSRR；因此，在大多数情况下，VDDA 上的噪声必须超过电源轨的建议工作条件之后，模拟模块才会出现性能下降。因此，单独为 VDDA 供电带来的好处通常微乎其微。然而，为了改善噪声，一种可接受的做法是在 VDDIO 和VDDA 之间放置一个π型滤波器。

Note

All the supply pins per rail are tied together internally. For example, all VDDIO pins are tied together internally, all VDD pins are tied together internally, and so forth.

每个电源轨的所有电源引脚在内部连接在一起。例如，所有 VDDIO 引脚在内部连接在一起，所有 VDD 引脚在内部连接在一起，以此类推。

6.12.1.4.2 Signal Pins Power Sequence
信号引脚电源序列

Before powering the device, no voltage larger than 0.3 V above VDDIO or 0.3 V below VSS should be applied to any digital pin; and no voltage larger than 0.3 V above VDDA or 0.3 V below VSSA should be applied to any analog pin (including VREFHI and VDAC). Simply, the signal pins should only be driven after XRSn goes high, provided all the 3.3-V rails are tied together. This sequencing is still required even if VDDIO and VDDA are not tied together.

CAUTION

If the above sequence is violated, device malfunction and possibly damage can occur as current will flow through unintended parasitic paths in the device.

在为器件供电之前，不得对任何数字引脚施加比 VDDIO 高0.3V 以上的电压或比 VSS 低0.3V 以上的电压，也不得对任何模拟引脚（包括 VREFHI 和 VDAC）施加比 77VDDA 高 0.3V 以上的电压或比 VSSA 低 0.3V 以上的电压。简单地说，只应在 XRSn 变为高电平后驱动信号引脚，且前提是所有 3.3V 电源轨连接在一起。即使 VDDIO 和 VDDA 未连接在一起，仍需要进行此时序控制。

CAUTION

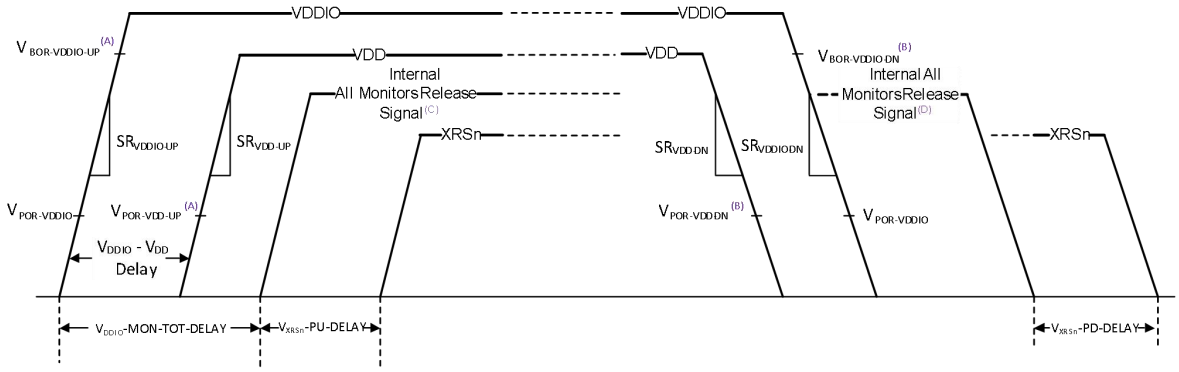
如果违反上述序列，则可能会发生器件故障，甚至可能造成损坏，因为电流将流经器件中的意外寄生路径。

6.12.1.4.3 Supply Pins Power Sequence
电源引脚电源序列

6.12.1.4.3.1 External VREG/VDD Mode Sequence
外部 VREG/VDD 模式序列

Figure 6-7 depicts the power sequencing requirements for external VREG mode. The values for all the parameters indicated can be found in Power Management Module Electrical Data and Timing.

图6-7 显示了外部 VREG 模式的电源时序控制要求。所有参数的值可在“电源管理模块电气数据和时序”章节中找到。



A. 该跳闸点是 XRSn 释放之前的跳闸点。请参阅“电源管理模块特性”表。
This trip point is the trip point before XRSn releases. See the Power Management Module Characteristics table.

B. 该跳闸点是 XRSn 释放之后的跳闸点。请参阅“电源管理模块特性”表。
This trip point is the trip point after XRSn releases. See the Power Management Module Characteristics table.

C. 上电期间，所有监控器释放信号在所有 POR 和BOR 监控器释放后变为高电平。请参阅 PMM 方框图。
During power up, the All Monitors Release Signal goes high after all POR and BOR monitors are released. See the PMM Block Diagram.

D. 在断电期间，如果任何 POR 或BOR 监测器跳闸，所有监测器释放信号都会变为低电平。请参阅 PMM 方框图。
During power down, the All Monitors Release Signal goes low if any of the POR or BOR monitors are tripped. See the PMM Block Diagram.

Figure 6-7 External VREG Power Up Sequence

- **For Power Up:**
  1. VDDIO (that is, the 3.3-V rail) should come up first with the minimum slew rate specified.
  2. VDD (that is, the 1.1-V rail) should come up next with the minimum slew rate specified.
  3. The time delta between the VDDIO rail coming up and when the VDD rail can come up is also specified.
  4. After the times specified by VDDIO-MON-TOT-DELAY and VXRSN-PD-DELAY, XRSn will be released and the device starts the boot-up sequence.

  There is an additional delay between XRSn releasing (that is, going high) and the boot-up sequence starting.
  5. The I/O BOR monitor has different release points during power up and power down.
  6. During power up, both VDDIO and VDD rails have to be up before XRSn releases.
- **在上电期间：**
  1. VDDIO（即3.3V 电源轨）应首先以指定的最小压摆率出现。
  2. VDD（即1.1V 电源轨）应其次以指定的最小压摆率出现。
  3. 还指定了 VDDIO 电源轨和 VDD 电源轨出现时间之间的时间差。
  4. 在 $V_{DDIO-MON-TOT-DELAY}$ 和 $V_{XRSN-PD-DELAY}$ 指定的时间之后，XRSn 将被释放，并且器件会开始启动序列。

     在XRSn 释放（即变为高电平）和启动序列之间还有额外的延迟。
  5. I/OBOR 监测器在上电和断电期间具有不同的释放点。
  6. 在上电期间，VDDIO 和VDD 电源轨都必须在 XRSn 释放之前启动。
- **For Power Down:**
  1. There is no requirement between VDDIO and VDD on which should power down first; however, there is a minimum slew rate specification.
  2. The I/O BOR monitor has different release points during power up and power down.
  3. Any of the POR or BOR monitors that trips during power down will cause XRSn to go low after $V_{XRSN-PD-DELAY}$.
- **在掉电期间：**
  1. 对 VDDIO 和 VDD 中哪个应先断电没有要求；但是，有最小压摆率规格。
  2. I/OBOR 监测器在上电和断电期间具有不同的释放点。
  3. 在断电期间跳闸的任何 POR 或 BOR 监测器都会导致 XRSn 在 VXRSN-PD-DELAY 之后变为低电平。

Note

The *All Monitors Release Signal* is an internal signal.

“所有监控器释放信号”是一个内部信号。

Note

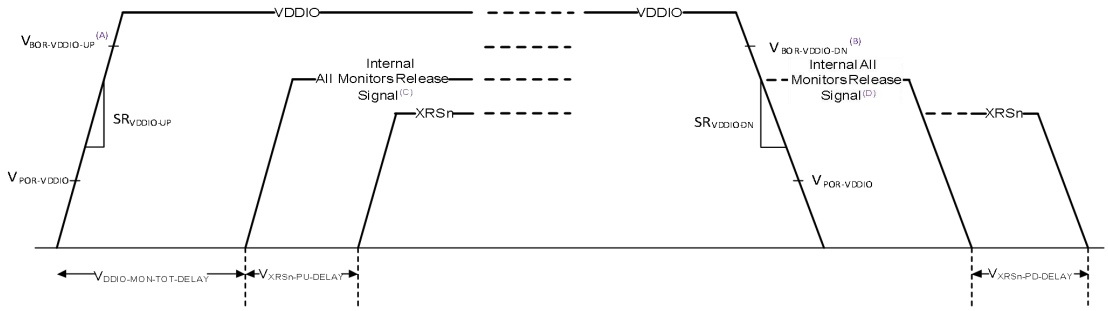
If there is an external circuit driving XRSn (for example, a supervisor), the boot-up sequence does not start until the XRSn pin is released by all internal and external sources.

如果有一个驱动 XRSn 的外部电路（例如，监测器），在所有内部和外部源释放 XRSn 引脚之前，启动序列不会启动。

6.12.1.4.3.2 内部 VREG/VDD 模式序列

Figure 6-8 depicts the power sequencing requirements for internal VREG mode. The values for all the parameters indicated can be found in Power Management Module Electrical Data and Timing.

图6-8 显示了内部 VREG 模式的电源时序控制要求。所有所示参数的值均可在电源管理模块电气数据和时序中找到。



A. 该跳闸点是 XRSn 释放之前的跳闸点。请参阅“电源管理模块特性”表。
This trip point is the trip point before XRSn releases. See the Power Management Module Characteristics table.

B. 该跳闸点是 XRSn 释放之后的跳闸点。请参阅“电源管理模块特性”表。
This trip point is the trip point after XRSn releases. See the Power Management Module Characteristics table.

C. 上电期间，所有监测器释放信号在所有 POR 和BOR 监控器释放后变为高电平。请参阅 PMM 方框图。
During power up, the All Monitors Release Signal goes high after all POR and BOR monitors are released. See the PMM Block Diagram.

D. 在断电期间，如果任何 POR 或BOR 监测器跳闸，所有监测器释放信号都会变为低电平。请参阅 PMM 方框图。
During power down, the All Monitors Release Signal goes low if any of the POR or BOR monitors are tripped. See the PMM Block Diagram.

Figure 6-8 Internal VREG Power Up Sequence

- **For Power Up:**
  1. VDDIO (that is, the 3.3-V rail) should come up with the minimum slew rate specified.
  2. The Internal VREG powers up after the I/O monitors (I/O POR and I/O BOR) are released.
  3. After the times specified by VDDIO-MON-TOT-DELAY and VXRSN-PU-DELAY, XRSn will be released and the device starts the boot-up sequence.
  4. There is an additional delay between XRSn releasing (that is, going high) and the boot-up sequence starting.
  5. The I/O BOR monitor has different release points during power up and power down.
- **For Power Down:**
  1. The only requirement on VDDIO during power down is the slew rate.
  2. The I/O BOR monitor has different release points during power up and power down.
  3. The I/O BOR tripping will cause XRSn to go low after VXRSN-PD-DELAY and also power down the Internal VREG.

- **在上电期间：**
  1. VDDIO（即 3.3V 电源轨）应提供指定的最小压摆率。
  2. 在释放 I/O 监测器（I/OPOR 和 I/OBOR）后，内部 VREG 将上电。
  3. 在 VDDIO-MON-TOT-DELAY 和 VXRSN-PU-DELAY 指定的时间过后，XRSn 将被释放，并且器件的启动序列将开始。
  4. 在 XRSn 释放（即变为高电平）和启动序列开始之间有额外的延迟。
  5. I/OBOR 监测器在上电和断电期间具有不同的释放点。

- **在断电期间：**

1. 在断电期间对 VDDIO 的唯一要求是压摆率。
2. I/OBOR 监测器在上电和断电期间具有不同的释放点。
3. I/OBOR 跳闸将导致 XRSn 在 VXRSN-PD-DELAY 之后变为低电平，并使内部 VREG 断电。

备注

The *All Monitors Release Signal* is an internal signal.
所有监控器释放信号是一个内部信号。

备注

If there is an external circuit driving XRSn (for example, a supervisor), the boot-up sequence does not start until the XRSn pin is released by all internal and external sources.
如果有一个驱动 XRSn 的外部电路（例如，监控器），在所有内部和外部源释放 XRSn 引脚之前，启动序列不会开始。

6.12.1.4.3.3 Supply Sequencing Summary and Effects of Violations 电源时序摘要和违规影响

The acceptable power-up sequence for the rails is summarized below. "Power up" here means the rail in question has reached the minimum recommended operating voltage.

电源轨的可接受上电序列汇总如下。此处的“上电”表示相关电源轨已达到建议的最低工作电压。

CAUTION

Non-acceptable sequences will lead to reliability concerns and possibly damage.
不可接受的序列会导致可靠性问题并可能导致损坏。

For simplicity, it is recommended that all 3.3-V rails be tied together, and to follow the descriptions in Supply Pins Power Sequence.

为简单起见，建议将所有 3.3V 电源轨连接在一起，并按照电源引脚电源序列中的说明进行操作。

Table 6-1 External VREG Sequence Summary

CASE	RAILS POWER-UP ORDER			ACCEPTABLE
	VDDIO	VDDA	VDD	
A	1	2	3	Yes
B	1	3	2	Yes
C	2	1	3	-
D	2	3	1	-
E	3	2	1	-
F	3	1	2	-
G	1	1	2	Yes
H	2	2	1	-

Table 6-2 Internal VREG Sequence Summary

CASE	RAILS POWER-UP ORDER		ACCEPTABLE
	VDDIO	VDDA	
A	1	2	Yes
B	2	1	-
C	1	1	Yes

Note

The analog modules on the device should only be powered after VDDA has reached the minimum recommended operating voltage.

应当仅在 VDDA 达到建议的最低工作电压后才为器件上的模拟模块供电。

6.12.1.4.3.4 Supply Slew Rate
电源压摆率

VDDIO has a minimum slew rate requirement. If the minimum slew rate is not met, XRSn might toggle a few times until VDDIO crosses the I/O BOR region.

Note

The toggling on XRSn has no adverse effect on the device as boot only starts once XRSn is steadily high. However if XRSn from the device is used to gate the reset signal of other ICs, then the slew rate requirement should be met to prevent this toggling.

VDD has a minimum slew rate requirement in external VREG mode. If the minimum slew rate is not met, the device can release from reset and start booting before VDD has reached the minimum operating voltage, which can result in the device not functioning correctly.

Note

If the minimum slew rate cannot be met, a supervisor must be used on VDD to keep XRSn low until VDD crosses the minimum operating voltage to ensure correct device functionality.

VDDIO 有最低压摆率要求。如果不满足最低压摆率要求，XRSn 可能会切换几次，直到VDDIO越过I/OBOR 区域。

备注

XRSn 上的切换操作对器件没有不利影响，因为只有当 XRSn 稳定为高电平时才会开始引导。但是，如果使用器件的 XRSn 来选通其他 IC 的复位信号，则应满足压摆率要求以防止这种切换。

VDD 在外部 VREG 模式下具有最低压摆率要求。如果不满足最低压摆率要求，器件可能会在 VDD 达到最低工作电压之前解除复位并开始引导，这可能导致器件无法正常工作。

备注

如果无法满足最低压摆率要求，必须在 VDD 上使用监控器来保持 XRSn 为低电平，直到 VDD 超过最低工作电压，以此确保器件正常工作。

6.12.1.5 Power Management Module Electrical Data and Timing 电源管理模块电气数据和时序

6.12.1.5.1 Power Management Module Operating Conditions 电源管理模块运行条件

over recommended operating conditions (unless otherwise noted)
在建议运行条件下测得（除非另有说明）

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
General 通用						
C_{VDDIO}[1][2]	VDDIO Capacitance Per Pin[7]		0.1			µF
C_{VDDA}[1][2]	VDDA Capacitance Per Pin[7]		2.2			µF
$SR_{VDDIO-UP}$[3]	Supply Ramp Up Rate of 3.3V Rail (VDDIO)		8		100	mV/µs
$SR_{VDDIO-DN}$[3]	Supply Ramp Down Rate of 3.3V Rail (VDDIO)		20		100	mV/µs
$V_{BOR-VDDIO-GB}$[5]	VDDIO Brown Out Reset Voltage Guardband			0.1		V
External VREG 外部 VREG						
C_{VDD} Total[1][4]	Total VDD Capacitance[7]		10			µF
SR_{VDD-UP}[3]	Supply Ramp Up Rate of 1.1V Rail (VDD)		3.5		100	mV/µs
SR_{VDD-DN}[3]	Supply Ramp Down Rate of 1.1V Rail (VDD)		10		100	mV/µs
V_{DDIO}-V_{DD} Delay[6]	Ramp Delay Between VDDIO and VDD		0		No Restrictions	µs
Internal VREG 内部 VREG						
C_{VDD} TOTAL[4]	Total VDD Capacitance[7]		10		26.8	µF

(1) 去耦电容的确切值取决于为这些引脚供电的系统电压调节解决方案。
The exact value of the decoupling capacitance depends on the system voltage regulation solution that is supplying these pins.
(2) 建议将3.3V 电压轨（VDDIO、VDDA）连接在一起并由单电源供电。

It is recommended to tie the 3.3V rails (VDDIO, VDDA) together and supply them from a single source.

(3) 请参阅*电源压摆率*一节。电源斜坡速率高于最大值会触发片上 ESD 保护。
See the *Supply Slew Rate* section. Supply ramp rate faster than the maximum can trigger the on-chip ESD protection.

(4) 请参阅*电源管理模块(PMM)*一节，了解总去耦电容的可能配置。
See the *Power Management Module (PMM)* section on possible configurations for the total decoupling capacitance.

(5) 建议使用 $V_{BOR-VDDIO-GB}$，避免由于正常电源噪声或 3.3VVDDIO 系统稳压器上的负载瞬态事件而导致 BOR-VDDIO 复位。要防止在器件正常运行期间激活 BOR-VDDIO，良好的系统稳压器设计和去耦电容（符合系统稳压器规格）非常重要。$V_{BOR-VDDIO-GB}$ 的值是一个系统级设计注意事项；此处列出的电压是许多应用的典型值。
It is recommended to use $V_{BOR-VDDIO-GB}$ to avoid BOR-VDDIO resets due to normal supply noise or load-transient events on the 3.3-V VDDIO system regulator. Good system regulator design and decoupling capacitance (following the system regulator specifications) are important to prevent activation of the BOR-VDDIO during normal device operation. The value of $V_{BOR-VDDIO-GB}$ is a system-level design consideration; the voltage listed here is typical for many applications.

(6) 3.3V 电源轨斜升时和 1.1V 电源轨斜升时之间的延迟。请参阅 *VREG 序列摘要*表，了解允许的电源斜坡序列。
Delay between when the 3.3-V rail ramps up and when the 1.1-V rail ramps up. See the *VREG Sequence Summary* table for the allowable supply ramp sequences.

(7) 最大电容器容差应为 20%。
Max capacitor tolerance should be 20%.

6.12.1.5.2 Power Management Module Characteristics
电源管理模块特征

over recommended operating conditions (unless otherwise noted)
在建议运行条件下测得（除非另有说明）

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
V_{VREG}	Internal Voltage Regulator Output		0.96	1.1	1.32	V
$V_{VREG-PU}$	Internal Voltage Regulator Power Up Time				350	µs
$V_{VREG-INRUSH}$(5)	Internal Voltage Regulator Inrush Current			650		mA
$V_{POR-VDDIO}$	VDDIO Power on Reset Voltage	Before and After XRSn Release		2.3		V
$V_{BOR-VDDIO-UP}$(1)	VDDIO Brown Out Reset Voltage on Ramp Up	Before XRSn Release		2.7		V
$V_{BOR-VDDIO-DN}$(1)	VDDIO Brown Out Reset Voltage on Ramp Down	After XRSn Release	2.81		3.0	V
$V_{POR-VDD-UP}$(2)	VDD Power on Reset Voltage on Ramp Up	Before XRSn Release		0.95		V
$V_{POR-VDD-DN}$(2)	VDD Power on Reset Voltage on Ramp Down	After XRSn Release		0.95		V
$V_{XRSn-PU-DELAY}$(3)	XRSn Release Delay after Supplies are Ramped Up During Power Up	This is the final delay		40		µs
$V_{XRSn-PD-DELAY}$(4)	XRSn Trip Delay after Supplies are Ramped Down During Power Down			2		µs
$V_{DDIO-MON-TOT-DELAY}$	Total Delays in Path of VDDIO Monitors (POR, BOR)			145		µs
$V_{XRSn-MON-RELEASE-DELAY}$	XRSn Release Delay after a VDD POR Event	Supplies Within Operating Range		40		µs
	XRSn Release Delay after a VDDIO BOR			140		µs
	XRSn Release Delay after a VDDIO POR Event			185		µs

(1) 请参阅电源电压图。
See the *Supply Voltages* figure.

(2) $V_{POR-VDD}$ 不受支持，它设置为在低于建议运行条件的电平跳闸。如果需要监测 VDD，则需要一个外部监控器。
$V_{POR-VDD}$ is not supported and it is set to trip at a level below the recommended operating conditions. If monitoring of VDD is needed, an external supervisor is required.

(3) 电源在越过相应电源轨的最低建议运行条件后被视为完全斜升。在该延迟生效之前，需要释放所有 POR 和 BOR 监测器。RC 网络延迟将与该延迟相加。
Supplies are considered fully ramped up after they cross the minimum recommended operating conditions for the respective rail. All POR and BOR monitors need to be released before this delay takes effect. RC network delay will add to this.

(4) 断电时，任何跳闸的 POR 或 BOR 监测器都会立即使 XRSn 跳闸。该延迟是任何 POR、BOR 监测器跳闸和 XRSn 变为低电平之间的时间。该延迟是变量，取决于电源的斜降速率。RC 网络延迟将与该延迟相加。
On power down, any of the POR or BOR monitors that trips will immediately trip XRSn. This delay is the time between any of the POR, BOR monitors tripping and XRSn going low. It is variable and depends on the ramp down rate of the supply. RC network delay will add to this.

(5) 这是内部 VREG 导通时在 VDDIO 电源轨上汲取的瞬态电流。因此，当 VREG 导通时，VDDIO 电源轨上可能会出现一些压降，这可能导致 VREG 逐步斜升。这不会对器件产生不利影响，但如果需要，可以通过在 VDDIO 上使用足够的去耦电容器或选择能够提供此瞬态电流的 LDO 或直流/直流稳压器来降低影响。
This is the transient current drawn on the VDDIO rail when the internal VREG turns on. Due to this, there might be some voltage drops on the VDDIO rail when the VREG turns on which could cause the VREG to ramp up in steps. There is no detriment to the device from this but the effect can be reduced if desired by using sufficient decoupling capacitors on VDDIO or picking an LDO/DC-DC that can supply this transient current.

Supply Voltages

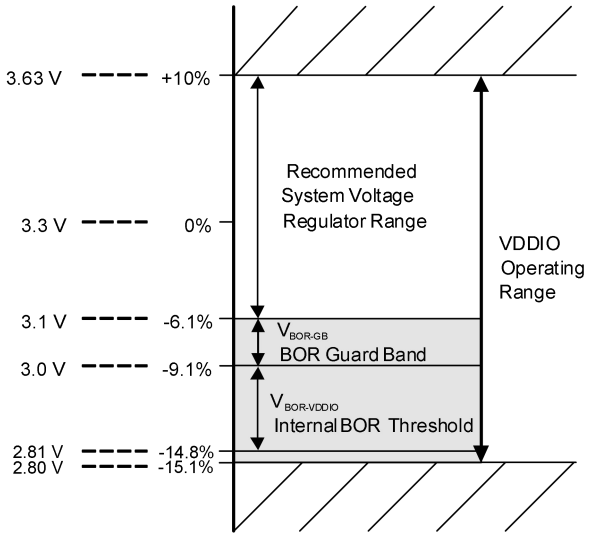


Figure 6- 9 Supply Voltages

6.12.2 Reset Timing
复位时序

XRSn is the device reset pin. It functions as an input and open-drain output. The device has a built-in power-on reset (POR) and brown-out reset (BOR) monitors. During power up, the monitor circuits keep the XRSn pin low. For more details, see the Power Management Module (PMM) section. A watchdog or NMI watchdog reset will also drive the pin low. An external open-drain circuit may drive the pin to assert a device reset.

A resistor with a value from 2.2 kΩ to 10 kΩ should be placed between XRSn and VDDIO. A capacitor should be placed between XRSn and VSS for noise filtering, it should be 100 nF or smaller. These values will allow the watchdog to properly drive the XRSn pin to VOL within 512 OSCCLK cycles when the watchdog reset is asserted. Figure 6- 10 shows the recommended reset circuit.

XRSn 是器件复位引脚。它用作输入和漏极开路输出。该器件内置上电复位(POR)和欠压复位(BOR)

监控器。在上电期间，监控器电路会将 XRSn 引脚保持为低电平。有关更多详细信息，请参阅“电源管理模块(PMM)”部分。看门狗或 NMI 看门狗复位也会驱动引脚至低电平。外部开漏电路可以驱动该引脚，从而使器件复位生效。

应在 XRSn 和 VDDIO 之间放置一个阻值为 2.2kΩ至10kΩ的电阻。应在 XRSn 和 VSS 之间放置一个电容器进行噪声滤除；电容应为 100nF 或更小。当看门狗复位生效时，这些值允许看门狗在 512 个 OSCCLK 周期内正确地将 XRSn 引脚驱动至 V_{OL}。图 6-10 显示了推荐的复位电路。

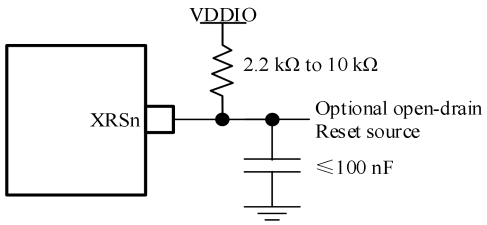


Figure 6-10 Reset Circuit

6.12.2.1 Reset Sources
复位源

The Reset Signals table summarizes the various reset signals and their effect on the device.
“复位信号”表总结了各种复位信号及其对器件的影响。

Table 6-3 Reset Signals

Reset Source	CPU Core Reset (A2000, FPU, TMU)	Peripherals Reset	JTAG / Debug Logic Reset	IOs	$\overline{XRS}$ Output
POR	Yes	Yes	Yes	Hi-Z	Yes
BOR	Yes	Yes	Yes	Hi-Z	Yes
$\overline{XRS}$ Pin	Yes	Yes	No	Hi-Z	-
$\overline{WDRS}$	Yes	Yes	No	Hi-Z	Yes
$\overline{NMIWDRS}$	Yes	Yes	No	Hi-Z	Yes
$\overline{SYSRS}$ (Debugger Reset)	Yes	Yes	No	Hi-Z	No
$\overline{SCCRESET}$	Yes	Yes	No	Hi-Z	No
SIMRESET.$\overline{XRS}$	Yes	Yes	No	Hi-Z	Yes
SIMRESET.$\overline{CPU1RS}$	Yes	Yes	No	Hi-Z	No
$\overline{HWBISTRS}$	Yes	No	No	No	No

The parameter $t_{h(boot\text{-}mode)}$ must account for a reset initiated from any of these sources.
参数 $t_{h(boot\text{-}mode)}$ 必须考虑从这些来源启动的复位。

CAUTION

Some reset sources are internally driven by the device. Some of these sources will drive XRSn low, use this to disable any other devices driving the boot pins. The SCCRESET and debugger reset sources do not drive XRSn; therefore, the pins used for boot mode should not be actively driven by other devices in the system. The boot configuration has a provision for changing the boot pins in OTP.

有些复位源由器件内部驱动。其中一些源会将 XRSn 驱动为低电平，用于禁用驱动引导引脚的任何其他器件。SCCRESET 和调试器复位源不会驱动 XRSn；因此，用于引导模式的引脚不应由系统中的其他器件主动驱动。引导配置规定可更改 OTP 中的引导引脚。

6.12.2.2 Reset Electrical Data and Timing
复位电气数据和时序

6.12.2.2.1 Reset - XRSn - Timing Requirements
复位-XRSn-时序要求

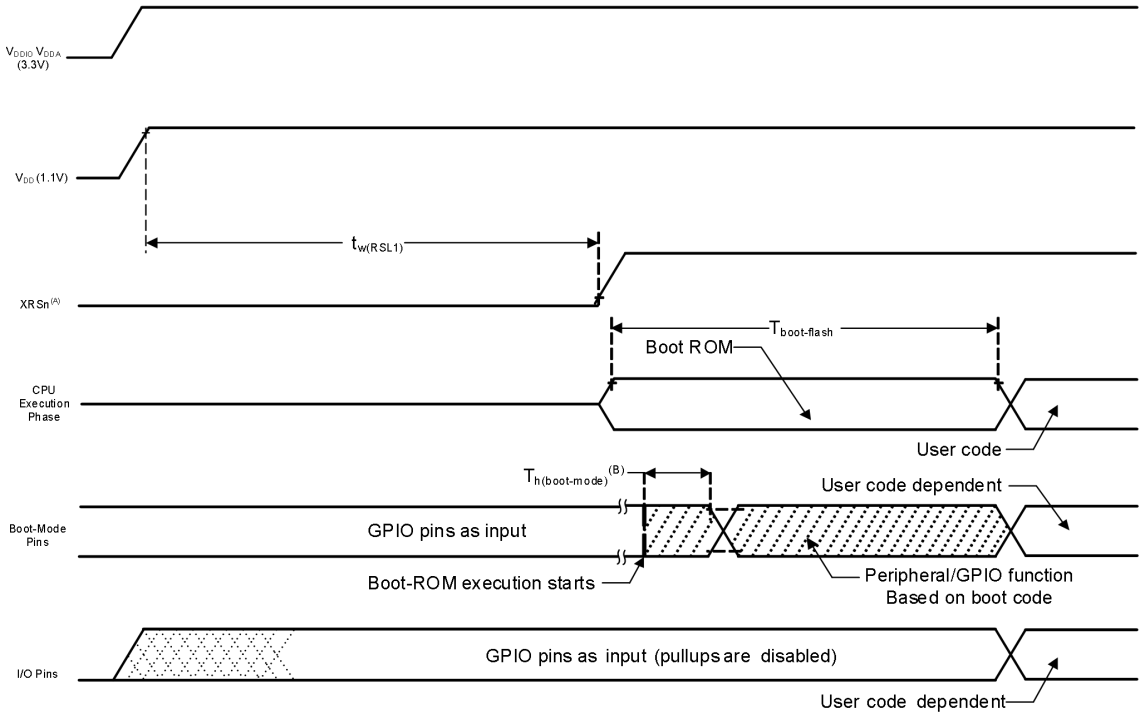
		MIN	MAX	UNIT
$t_{h(boot-mode)}$	Hold time for boot-mode pins	1.5		ms
$t_{w(RSL2)}$	Pulse duration, XRSn low on warm reset	3.2		µs

6.12.2.2.2 Reset - XRSn - Switching Characteristics
复位-XRSn-开关特性

over recommended operating conditions (unless otherwise noted)
在建议运行条件下测得（除非另有说明）

PARAMETER		MIN	TYP	MAX	UNIT
$t_{w(RSL1)}$	Pulse duration, XRSn driven low by device after supplies are stable		100		µs
$t_{w(WDRS)}$	Pulse duration, reset pulse generated by watchdog		$512t_{c(OSCCLK)}$		cycles
$t_{boot-flash}$	Boot-ROM execution time to first instruction fetch in flash			1.2	ms

6.12.2.2.3 Reset Timing Diagrams
复位时序图

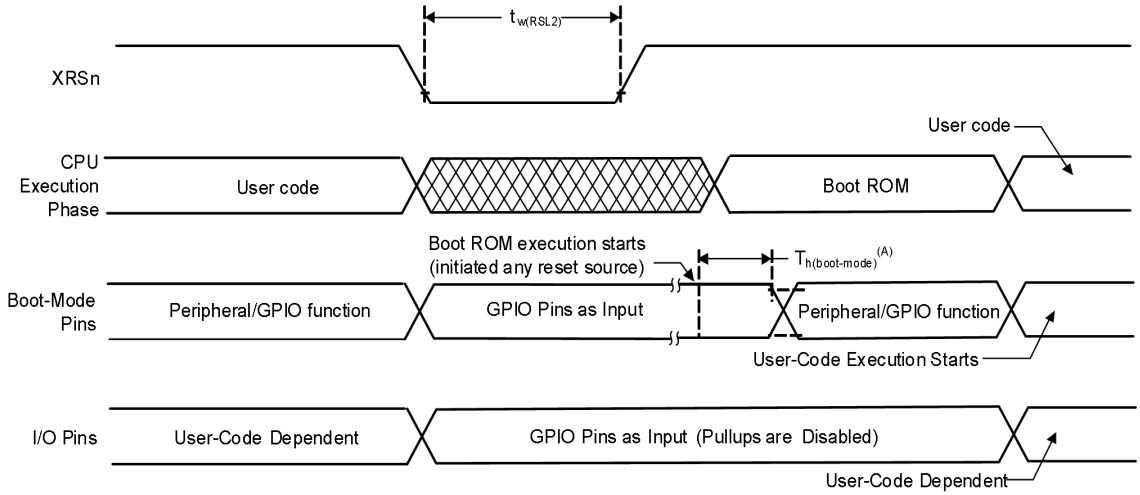


A. XRSn 引脚可以由监控器或外部上拉电阻从外部驱动，请参阅“引脚属性”表。片上监控器将保持该引脚为低电平，直到电源处于有效范围内。
The XRSn pin can be driven externally by a supervisor or an external pullup resistor, see the Pin Attributes table. On-chip monitors will hold this pin low until the supplies are in a valid range.

B. 从任何源（参阅“复位源”部分）复位后，引导ROM 代码将对引导模式引脚进行采样。基于引导模式引脚的状态，引导代码分支到目的内存或者引导代码功能。如果引导 ROM 代码在上电条件后（在调试程序环境中）执行，则引导代码执行时间基于当前的 SYSCLK 速度。SYSCLK 将基于用户环境，可以启用或不启用 PLL。

After reset from any source (see the Reset Sources section), the boot ROM code samples Boot Mode pins. Based on the status of the Boot Mode pin, the boot code branches to destination memory or boot code function. If boot ROM code executes after power-on conditions (in debugger environment), the boot code execution time is based on the current SYSCLK speed. The SYSCLK will be based on user environment and could be with or without PLL enabled.

Figure 6- 11 Power-on Reset



A. 从任何源（参阅“复位源”部分）复位后，引导ROM 代码将对引导模式引脚进行采样。基于引导模式引脚的状态，引导代码分支到目的内存或者引导代码功能。如果引导 ROM 代码在上电条件后（在调试程序环境中）执行，则引导代码执行时间基于当前的 SYSCLK 速度。SYSCLK 将基于用户环境，可以启用或不启用 PLL。

After reset from any source (see the Reset Sources section), the Boot ROM code samples BOOT Mode pins. Based on the status of the Boot Mode pin, the boot code branches to destination memory or boot code function. If Boot ROM code executes after power-on conditions (in debugger environment), the Boot code execution time is based on the current SYSCLK speed. The SYSCLK will be based on user environment and could be with or without PLL enabled.

Figure 6- 12 Warm Reset

6.12.3 Clock Specifications
时钟规格

6.12.3.1 Clock Sources
时钟源

Table 6-4 Possible Reference Clock Sources

CLOCK SOURCE	DESCRIPTION
INTOSC1	Internal oscillator 1. Zero-pin overhead 10-MHz internal oscillator.
INTOSC2(1)	Internal oscillator 2. Zero-pin overhead 10-MHz internal oscillator.
X1(XTAL)	External crystal or resonator connected between the X1 and X2 pins or single-ended clock connected to the X1 pin.

(1) 复位时，内部振荡器 2(INTOSC2)为PLL(OSCCLK)的默认时钟源。

On reset, internal oscillator 2 (INTOSC2) is the default clock source for the PLL (OSCCLK).

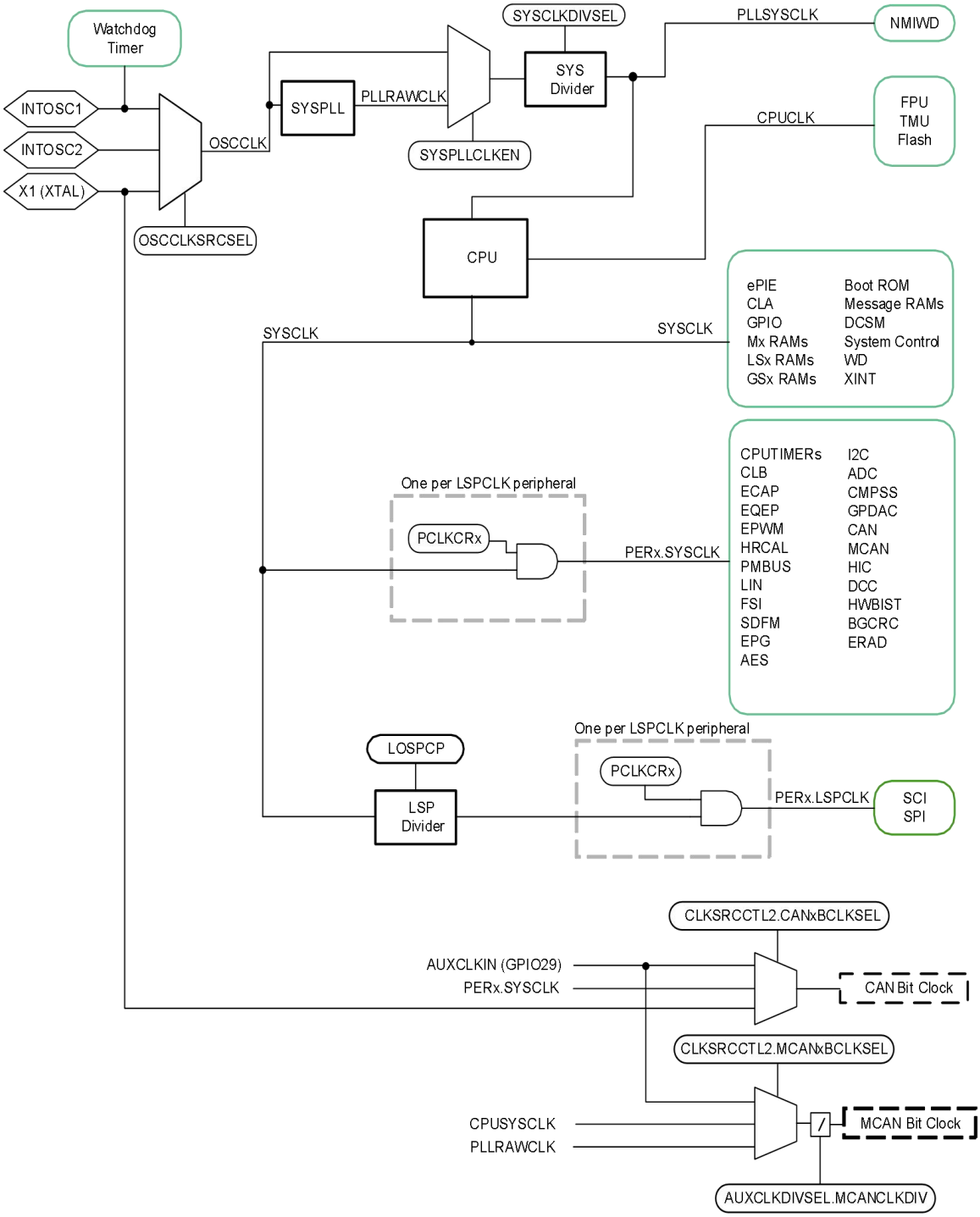


Figure 6- 13 Clocking System

SYSPLL

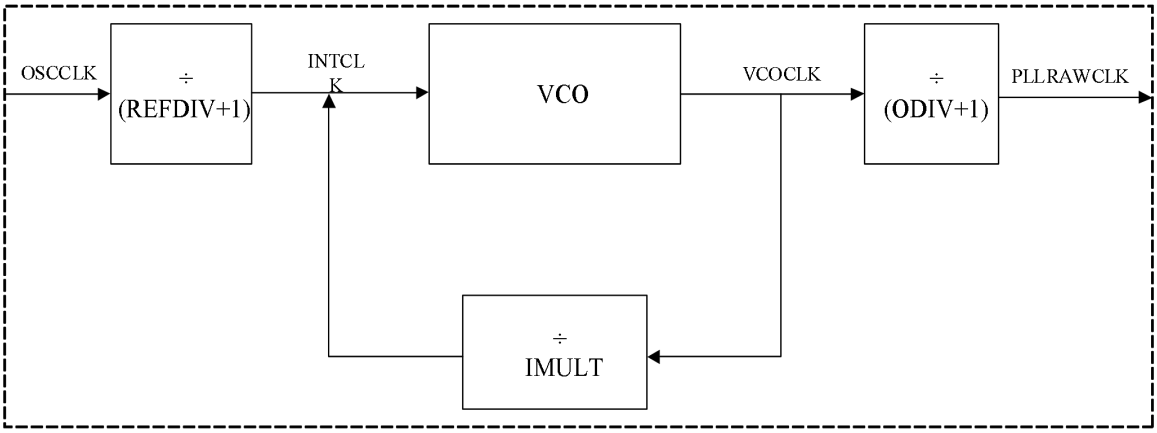


Figure 6- 14 System PLL

In Figure 6- 14,

$$f_{PLLRAWCLK} = \frac{f_{OSCCLK}}{(REFDIV + 1)} \times \frac{IMULT}{(ODIV + 1)}$$

6.12.3.2 Clock Frequencies, Requirements, and Characteristics 时钟频率、要求和特性

This section provides the frequencies and timing requirements of the input clocks, PLL lock times, frequencies of the internal clocks, and the frequency and switching characteristics of the output clock.

本节提供了输入时钟的频率和时序要求、PLL 锁定时间、内部时钟频率以及输出时钟的频率和开关特性。

6.12.3.2.1 Input Clock Frequency and Timing Requirements, PLL Lock Times 输入时钟频率和时序要求，PLL 锁定时间

6.12.3.2.1.1 Input Clock Frequency 输入时钟频率

		MIN	MAX	UNIT
$f_{(XTAL)}$	Frequency, X1/X2, from external crystal or resonator	10	20	MHz
$f_{(X1)}$	Frequency, X1, from external oscillator	10	25	MHz
$f_{(AUXI)}$	Frequency, AUXCLKIN, from external oscillator	10	60	MHz

6.12.3.2.1.2 XTAL Oscillator Characteristics XTAL 振荡器特征

over recommended operating conditions (unless otherwise noted)
在建议运行条件下测得（除非另有说明）

PARAMETER		MIN	TYP	MAX	UNIT
X1 V_{IL}	Valid low-level input voltage (Comparator)	–0.3		0.3*VDDIO	V
X1 V_{IH}	Valid high-level input voltage (Comparator)	0.7*VDDIO		VDDIO+0.3	V

6.12.3.2.1.3 X1 Input Level Characteristics When Using an External Clock Source - Not a Crystal
使用外部时钟源（非晶体）时的 X1 输入电平特性

over recommended operating conditions (unless otherwise noted)
在建议运行条件下测得（除非另有说明）

PARAMETER		MIN	MAX	UNIT
X1 V_{IL}	Valid low-level input voltage (Buffer)	–0.3	0.3*VDDIO	V
X1 V_{IH}	Valid high-level input voltage (Buffer)	0.7*VDDIO	VDDIO+0.3	V

6.12.3.2.1.4 X1 Timing Requirements
X1 时序要求

		MIN	MAX	UNIT
$t_{f(X1)}$	Fall time, X1		6	ns
$t_{r(X1)}$	Rise time, X1		6	ns
$t_{w(X1L)}$	Pulse duration, X1 low as a percentage of $t_{c(X1)}$	45%	55%	
$t_{w(X1H)}$	Pulse duration, X1 high as a percentage of $t_{c(X1)}$	45%	55%	

6.12.3.2.1.5 AUXCLKIN Timing Requirements
AUXCLKIN 时序要求

		MIN	MAX	UNIT
$t_{f(AUXI)}$	Fall time, AUXCLKIN		6	ns
$t_{r(AUXI)}$	Rise time, AUXCLKIN		6	ns
$t_{w(AUXL)}$	Pulse duration, AUXCLKIN low as a percentage of $t_{c(XCI)}$	45%	55%	
$t_{w(AUXH)}$	Pulse duration, AUXCLKIN high as a percentage of $t_{c(XCI)}$	45%	55%	

6.12.3.2.1.6 APLL Characteristics APLL 特性

over recommended operating conditions (unless otherwise noted)
在自然通风条件下的工作温度范围内测得（除非另有说明）

PARAMETER	MIN	TYP	MAX	UNIT
PLL Lock time				
SYS PLL Lock Time(1)			5μs+(1024*(REFDIV+1)*$t_{c(OSCCLK)}$)	μs

(1) 此处的PLL 锁定时间定义了 PLL 启用SYSPLLCTL1[PLLENA]=1)后PLL 锁定所需的典型时间。此处未考虑使用双路钟比较器(DCC）验证PLL 时钟的额外时间。
The PLL lock time here defines the typical time that takes for the PLL to lock once PLL is enabled (SYSPLLCTL1[PLLENA]=1). Additional time to verify the PLL clock using Dual Clock Comparator (DCC) is not accounted here.

6.12.3.2.1.7 XCLKOUT Switching Characteristics - PLL Bypassed or Enabled
XCLKOUT 开关特性-旁路或启用 PLL

over recommended operating conditions (unless otherwise noted)
在建议运行条件下测得（除非另有说明）

PARAMETER(1)		MIN	MAX	UNIT
$t_{f(XCO)}$	Fall time, XCLKOUT		5	ns
$t_{r(XCO)}$	Rise time, XCLKOUT		5	ns
$t_{w(XCOL)}$	Pulse duration, XCLKOUT low	H–2(2)	H+2(2)	ns
$t_{w(XCOH)}$	Pulse duration, XCLKOUT high	H–2(2)	H+2(2)	ns
$f_{(XCO)}$	Frequency, XCLKOUT		60	MHz

(1) 假定这些参数有 40pF 的负载。
A load of 40 pF is assumed for these parameters.
(2) $H=0.5t_{c(XCO)}$

6.12.3.2.1.8 Internal Clock Frequencies
内部时钟频率

		MIN	TYP	MAX	UNIT
$f_{(SYSCLK)}$	Frequency, device (system) clock	2		120	MHz
$t_{c(SYSCLK)}$	Period, device (system) clock	8.33		500	ns
$f_{(INTCLK)}$	Frequency, system PLL going into VCO (after REFDIV)	2		20	MHz
$f_{(VCOCLK)}$	Frequency, system PLL VCO (before ODIV)	220		600	MHz
$f_{(PLLRAWCLK)}$	Frequency, system PLL output (before SYSCLK divider)	6		240	MHz
$f_{(PLL)}$	Frequency, PLLSYSCLK	2		120	MHz
$f_{(PLL_LIMP)}$	Frequency, PLL Limp Frequency (1)		45/(ODIV+1)		MHz
$f_{(LSP)}$	Frequency, LSPCLK	2		120	MHz
$t_{c(LSPCLK)}$	Period, LSPCLK	8.33		500	ns
$f_{(OSCCLK)}$	Frequency, OSCCLK (INTOSC1 or INTOSC2 or XTAL or X1)	See respective clock			MHz
$f_{(EPWM)}$	Frequency, EPWMCLK			120	MHz
$f_{(HRPWM)}$	Frequency, HRPWMCLK	60		120	MHz

(1) OSCCLK 无效时的 PLL 输出频率（OSCCLK 丢失导致 PLL 变为跛行模式）。
PLL output frequency when OSCCLK is dead (Loss of OSCCLK causes PLL to Limp).

6.12.3.3 Input Clocks and PLLs
输入时钟和 PLL

In addition to the internal 0-pin oscillators, three types of external clock sources are supported:

- A single-ended 3.3-V external clock. The clock signal should be connected to X1, as shown in Figure 6- 15, with the XTALCR.SE bit set to 1.
- An external crystal. The crystal should be connected across X1 and X2 with its load capacitors connected to VSS as shown in Figure 6- 16.
- An external resonator. The resonator should be connected across X1 and X2 with its ground connected to VSS as shown in Figure 6- 17.

除了内部 0 引脚振荡器外，还支持三种类型的外部时钟源：

- 单端 3.3V 外部时钟。时钟信号应连接到 X1（如图 6-15 所示），且 XTALCR.SE 位设置为 1。
- 外部晶体。如图 6-16 所示，晶体应连接在 X1 和 X2 之间，其负载电容器连接至 VSS。
- 外部谐振器。如图 6-17 所示，谐振器应连接在 X1 和 X2 之间，且其接地端连接至 VSS。

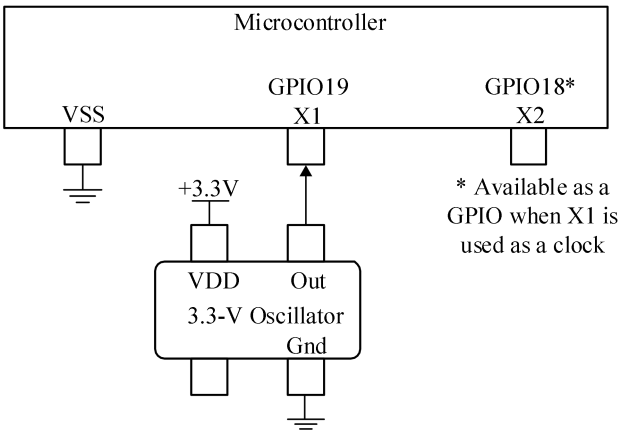


Figure 6- 15 Single-ended 3.3-V External Clock

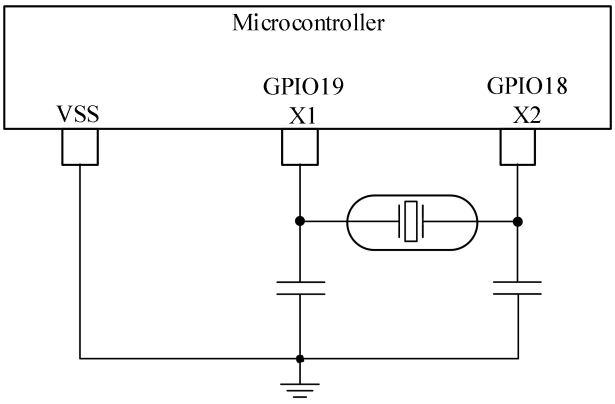


Figure 6- 16 External Crystal

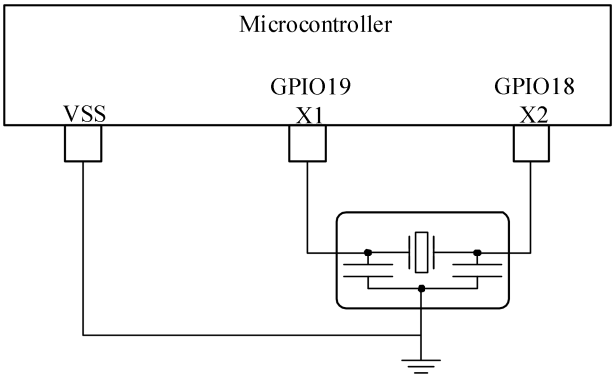


Figure 6- 17 External Resonator

6.12.3.4 XTAL Oscillator
XTAL 振荡器

6.12.3.4.1 Introduction 引言

The XTAL oscillator in this device is an embedded electrical oscillator that, when paired with a compatible crystal, can generate the system clock required by the device.

该器件中的 XTAL 振荡器是一种嵌入式电子振荡器，当与兼容的晶体配对使用时，可生成器件所需的系统时钟。

6.12.3.4.2 Overview 概述

The following sections describe the components of the electrical oscillator and crystal.

以下几节将介绍电振荡器和晶体的元件。

6.12.3.4.2.1 Electrical Oscillator 电子振荡器

The electrical oscillator in this device is a Pierce oscillator design. It is a positive feedback inverter circuit that requires a tuning circuit in order to oscillate. When this oscillator is paired with a compatible crystal, a tank circuit is formed. This tank circuit oscillates at the fundamental frequency of the crystal component. On this device, the oscillator is designed to operate in parallel resonance mode due to the shunt capacitor (C0) and required load capacitors (CL). Figure 6-18 illustrates the components of the electrical oscillator and the tank circuit.

该器件中的电子振荡器是皮尔斯振荡器设计。它是一个正反馈逆变器电路，需要一个调优电路才能振荡。当这个振荡器与一个兼容的晶体配对时，会形成振荡电路。该振荡电路在晶体元件的基频处振荡。在该器件上，由于分流电容器(C0)和所需的负载电容器(CL)，振荡器被设计成在并联谐振模式下运行。图 6-18 所示为电子振荡器和振荡电路的元件。

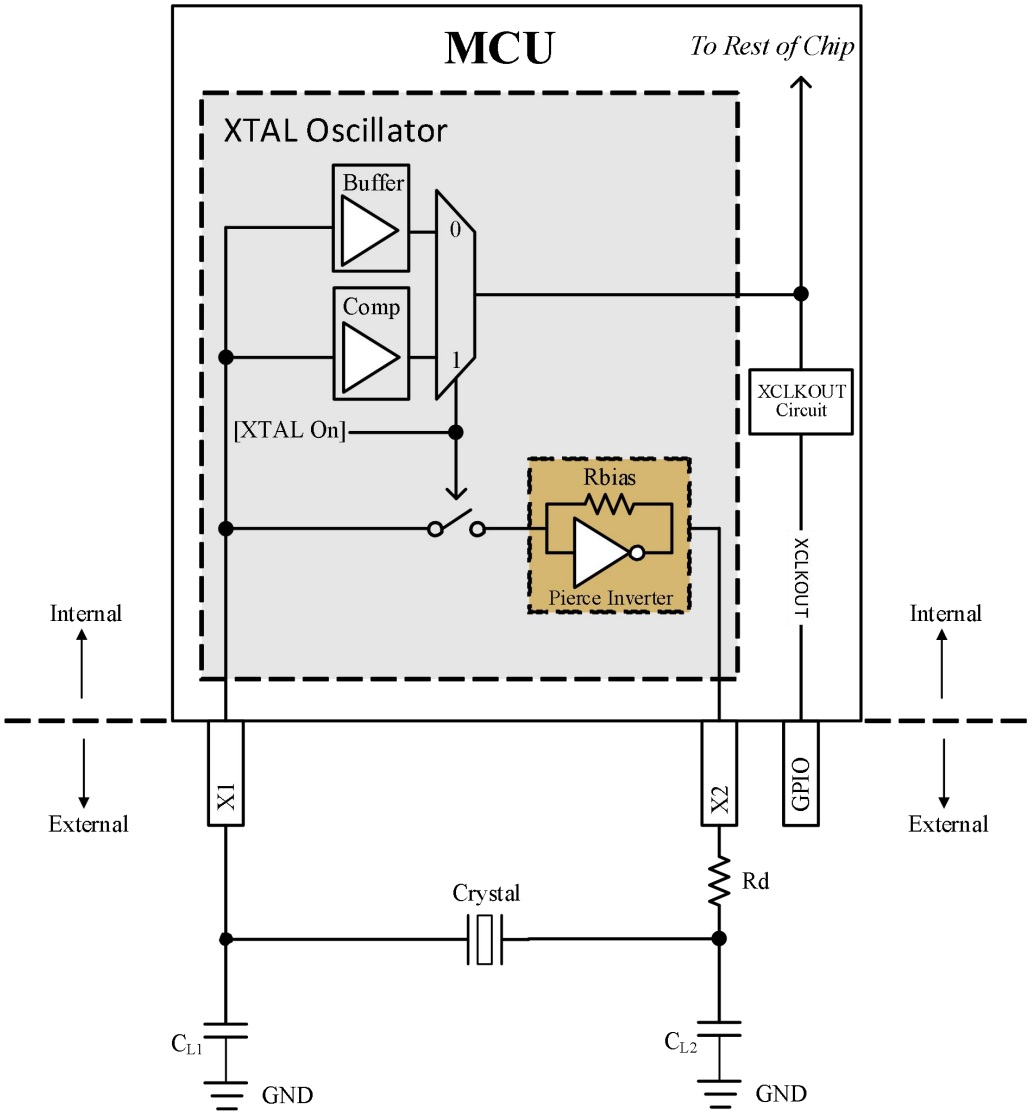


Figure 6- 18 Electrical Oscillator Block Diagram

6.12.3.4.2.1.1 Modes of Operation
运行模式

The electrical oscillator in this device has two modes of operation: crystal mode and single-ended mode.
此器件中的电振荡器有两种工作模式：晶体模式和单端模式。

6.12.3.4.2.1.1.1 Crystal Mode of Operation
晶体的工作模式

In the crystal mode of operation, a quartz crystal with load capacitors has to be connected to X1 and X2. This mode of operation is engaged when [XTAL On] = 1, which is achieved by setting XTALCR.OSCOFF = 0 and XTALCR.SE = 0. There is an internal bias resistor for the feedback loop so an external one should not be used. Adding an external bias resistor will create a parallel resistance with the internal Rbias, moving the bias point of operation and possibly leading to clipped waveforms, out-of-specification duty cycle, and reduction in the effective negative resistance.

In this mode of operation, the resultant clock on X1 is passed through a comparator (Comp) to the rest of the chip. The clock on X1 needs to meet the VIH and VIL of the comparator. See the XTAL Oscillator

Characteristics table for the VIH and VIL requirements of the comparator.

在晶体工作模式下，必须将带有负载电容器的石英晶体连接到 X1 和X2。

当[XTALOn]=1 时，会启用此工作模式，这是通过设置 XTALCR.OSCOFF=0 和XTALCR.SE=0 来实现的。反馈环路有一个内部偏置电阻器，因此不应使用外部偏置电阻器。添加外部偏置电阻器会产生与内部 Rbias 并联的电阻，从而移动工作偏置点并可能导致波形削波、占空比超出规格以及有效负电阻降低。

在此工作模式下，X1 上的结果时钟通过比较器(Comp)传递到芯片的其余部分。X1 上的时钟需要满足比较器的 VIH 和VIL。有关比较器的 VIH 和VIL 要求，请参阅*XTAL 振荡器特性*表。

6.12.3.4.2.1.1.2 Single-Ended Mode of Operation
单端工作模式

In the single-ended mode of operation, a clock signal is connected to X1 with X2 left unconnected. A quartz crystal should not be used in this mode.
This mode is enabled when [XTAL On] = 0, which can be achieved by setting XTALCR.OSCOFF = 1 and XTALCR.SE = 1.
In this mode of operation, the clock on X1 is passed through a buffer (Buffer) to the rest of the chip. See the *X1 Input Level Characteristics When Using an External Clock Source (Not a Crystal)* table for the input requirements of the buffer.

在单端工作模式下，一个时钟信号连接至 X1，而 X2 悬空。在此模式下不应使用石英晶体。

当[XTALOn]=0 时会启用此模式，这可通过设置 XTALCR.OSCOFF=1 和 XTALCR.SE=1 来实现。

在此工作模式下，X1 上的时钟通过一个缓冲器(Buffer)传递到芯片的其余部分。有关缓冲器的输入要求，请参阅使用外部时钟源（非晶体）时的 *X1* 输入电平特征表。

6.12.3.4.2.1.2 XTAL Output on XCLKOUT
XCLKOUT 上的 XTAL 输出

The output of the electrical oscillator that is fed to the rest of the chip can be brought out on XCLKOUT for observation by configuring the CLKSRCCTL3.XCLKOUTSEL and XCLKOUTDIVSEL.XCLKOUTDIV registers. See the GPIO Muxed Pins table for a list of GPIOs that XCLKOUT comes out on.

通过配置 CLKSRCCTL3.XCLKOUTSEL 和XCLKOUTDIVSEL.XCLKOUTDIV 寄存器，可以将馈入芯片其余部分的电振荡器输出引出到 XCLKOUT 上以供观察。如需查看输出 XCLKOUT 的 GPIO 的列表，请参阅“GPIO 多路复用引脚”表。

6.12.3.4.2.2 Quartz Crystal
石英晶体

Electrically, a quartz crystal can be represented by an LCR (Inductor-Capacitor-Resistor) circuit. However, unlike an LCR circuit, crystals have very high Q due to the low motional resistance and are also very underdamped. Components of the crystal are shown in Figure 6- 19 and explained below.

石英晶体可以由 LCR（电感-电容-电阻）电路进行电气表示。然而，与LCR 电路不同，晶体由于低动态电阻而具有非常高的 Q 值，并且阻尼也非常低。晶体元件如图 6-19 所示，并在下文中有相应说明。

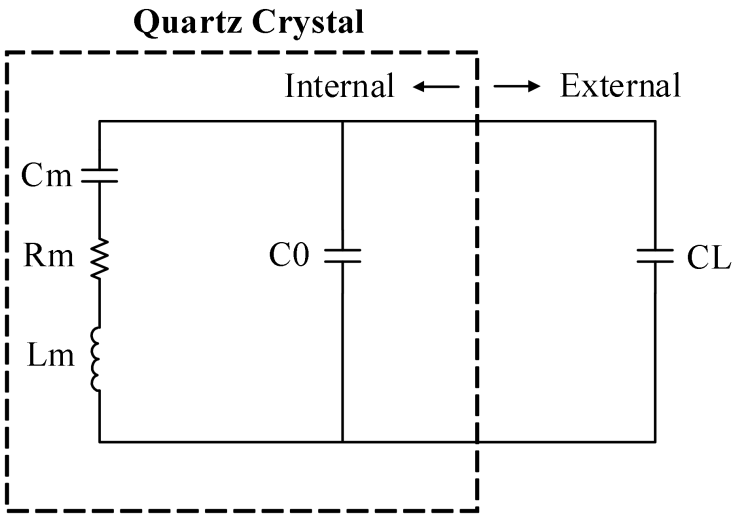


Figure 6- 19 Crystal Electrical Representation

Cm (Motional capacitance): Denotes the elasticity of the crystal.

Rm (Motional resistance): Denotes the resistive losses within the crystal. This is not the ESR of the crystal but can be approximated as such depending on the values of the other crystal components.

Lm (Motional inductance): Denotes the vibrating mass of the crystal.
C0 (Shunt capacitance): The capacitance formed from the two crystal electrodes and stray package capacitance.

CL (Load capacitance): This is the effective capacitance seen by the crystal at its electrodes. It is external to the crystal. The frequency ppm specified in the crystal data sheet is usually tied to the CL parameter.

Cm（动态电容）： 表示晶体的弹性。

Rm（动态电阻）： 表示晶体内的电阻损耗。这不是晶体的 ESR，但可以根据其他晶体元件的值进行近似计算。

Lm（动态电感）： 表示晶体的振动质量。

C0（并联电容）： 由两个晶体电极和杂散封装电容形成的电容。

CL（负载电容）： 这是晶体在其电极处看到的有效电容。它位于晶体外部。晶体数据表中指明的频率 ppm 通常与 CL 参数相关联。

Note that most crystal manufacturers specify CL as the effective capacitance seen at the crystal pins, while some crystal manufacturers specify CL as the capacitance on just one of the crystal pins. Check with the crystal manufacturer for how the CL is specified in order to use the correct values in calculations.

From Figure 6- 18, CL1 and CL2 are in series; so, to find the equivalent total capacitance seen by the crystal, the capacitance series formula has to be applied which simply evaluates to [CL1]/2 if CL1 = CL2.
It is recommended that a stray PCB capacitance be added to this value. 3 pF to 5 pF are reasonable estimates, but the actual value will depend on the PCB in question.

请注意，大多数晶体制造商将 CL 指定为晶体引脚上的有效电容，而一些晶体制造商将 CL 指定为仅其中一个晶体引脚上的电容。请与晶体制造商核实 CL 的指定值，以便在计算中使用正确的值。

根据图 6-18，CL1 和 CL2 是串联的；因此，要找到晶体看到的等效总电容，必须应用电容串联公式：如果 CL1=CL2，只需计算[CL1]/2 即可。

建议将杂散 PCB 电容与该值相加。合理的估算值为 3pF 至 5pF，但实际值将取决于相关的 PCB。

Note that the load capacitance is a requirement of both the electrical oscillator and crystal. The value chosen has to satisfy both the electrical oscillator and the crystal.

The effect of CL on the crystal is frequency-pulling. If the effective load capacitance is lower than the target, the crystal frequency will increase and vice-versa. However, the effect of frequency-pulling is usually very minimal and typically results in less than 10-ppm variation from the nominal frequency.

请注意，电振荡器和晶体都需要负载电容。所选的值必须同时满足电振荡器和晶振的要求。

CL 对晶体的影响是频率牵引。如果有效负载电容低于目标值，晶体频率将增加，反之亦然。然而，频率牵引的影响通常非常小，通常会导致与标称频率相差不到 10ppm。

6.12.3.4.2.3 GPIO 工作模式
GPIO Modes of Operation

On this device, X1 and X2 can be used as GPIO19 and GPIO18, respectively, depending on the operating mode of the XTAL.

在此器件上，X1 和 X2 分别可用作 GPIO19 和 GPIO18，具体取决于 XTAL 的工作模式。

6.12.3.4.3 Functional Operation
正常运行

6.12.3.4.3.1 ESR – Effective Series Resistance
ESR–有效串联电阻

Effective Series Resistance is the resistive load the crystal presents to the electrical oscillator at resonance. The higher the ESR, the lower the Q, and less likely the crystal will start up or maintain oscillation. The relationship between ESR and the crystal components is indicated below.

有效串联电阻是晶体在谐振时提供给电子振荡器的电阻负载。ESR 越高，Q 越低，晶体启动或保持振荡的可能性就越小。ESR 和晶体元件之间的关系如下所示。

$$ESR = Rm * (1 + \frac{C0}{CL})^2 \quad (1)$$

Note that ESR is not the same as motional resistance of the crystal, but can be approximated as such if the effective load capacitance is much greater than the shunt capacitance.

请注意，ESR 与晶体的动态电阻不同，但如果有效负载电容远高于分流电容、则可以按此近似计算。

6.12.3.4.3.2 Rneg – Negative Resistance
Rneg-负电阻

Negative resistance is the impedance presented by the electrical oscillator to the crystal. It is the amount of energy the electrical oscillator must supply to the crystal to overcome the losses incurred during oscillation. Rneg depicts a circuit that provides rather than consume energy and can also be viewed as the overall gain of the circuit.

The generally accepted practice is to have Rneg > 3x ESR to 5x ESR to ensure the crystal starts up under all conditions. Note that it takes slightly more energy to start up the crystal than it does to sustain oscillation; therefore, if it can be ensured that the negative resistance requirement is met at start-up, then oscillation sustenance will not be an issue.

Figure 6-20 and Figure 6-21 show the variation between negative resistance and the crystal components for this device. As can be seen from the graphs, the crystal shunt capacitance (C0) and effective load capacitance (CL) greatly influence the negative resistance of the electrical oscillator. Note that these are typical graphs; so, refer to Table 6-5 for minimum and maximum values for design considerations.

负电阻是电振荡器向晶体呈现的阻抗。这是电振荡器为了克服振荡期间产生的损耗而必须为晶体提供的能量。Rneg 描述了一个提供而不是消耗能量的电路，也可以看作是电路的总体增益。

为确保晶体在所有条件下都能启动，普遍接受的做法是让 Rneg>3 倍ESR 至5 倍ESR。请注意，启动晶体所需的能量略大于维持振荡所需的能量；因此，如果能够确保在启动时满足负电阻要求，则维持振荡将不是问题。

图 6-20 和图 6-21 所示为该器件的负电阻与晶体元件之间的差异。从图中可以看出，晶体并联电容(C0)和有效负载电容(CL)对电振荡器的负电阻有极大影响。请注意，这些是典型图；因此，请参阅表 6-5 了解设计中需要注意的最小值和最大值。

6.12.3.4.3.3 Start-up Time
启动时间

Start-up time is an important consideration when selecting the components of the crystal circuit. As mentioned in the Rneg – Negative Resistance section, for reliable start-up across all conditions, it is recommended that the Rneg > 3x ESR to 5x ESR of the crystal.
Crystal ESR and the dampening resistor (Rd) greatly affect the start-up time. The higher the two values, the longer the crystal takes to start up. Longer start-up times are usually a sign that the crystal and components are not a correct match.
Refer to Crystal Oscillator Specifications for the typical start-up times. Note that the numbers specified here are typical numbers provided for guidance only. Actual start-up time depends heavily on the crystal in question and the external components.

在选择晶体电路的元件时，启动时间是一个重要的考虑因素。如“Rneg-负电阻”一节所述，为了在所有条件下实现可靠启动，建议晶体的 Rneg>3 倍 ESR 至 5 倍 ESR。

晶体 ESR 和阻尼电阻(Rd)会极大地影响启动时间。这两个值越高，晶体启动所需的时间就越长。较长的启动时间通常表明晶体和元件未正确匹配。

如需了解典型的启动时间，请参阅晶体振荡器规格。请注意，此处指定的数字是仅供参考的典型数字。实际启动时间在很大程度上取决于所涉及的晶体和外部元件。

6.12.3.4.3.3.1 X1/X2 Precondition
X1/X2 前提条件

On this device, the GPIO19/18 alternate functionality on X1/X2 can be used to speed up the start-up time of the crystal if needed. This functionality is achieved by preconditioning the load capacitors CL1 and CL2 to a known state before the XTAL is turned on.

在该器件上，X1/X2 上的GPIO19/18 备选功能可用于在需要时缩短晶体的启动时间。此功能是通过将负载电容器 CL1 和CL2 预调节到 XTAL 开启前的已知状态来实现的。

6.12.3.4.3.4 DL – Drive Level
DL–驱动电平

Drive level refers to how much power is provided by the electrical oscillator and dissipated by the crystal. The maximum drive level specified in the crystal manufacturer’s data sheet is usually the maximum the crystal can dissipate without damage or significant reduction in operating life. On the other hand, the drive level specified by the electrical oscillator is the maximum power it can provide. The actual power provided by the electrical oscillator is not necessarily the maximum power and depends on the crystal and board components.
For cases where the actual drive level from the electrical oscillator exceeds the maximum drive level specification of the crystal, a dampening resistor (Rd) should be installed to limit the current and reduce the power dissipated by the crystal. Note that Rd reduces the circuit gain; and therefore, the actual value

to use should be evaluated to make sure all other conditions for start-up and sustained oscillation are met.

驱动电平是指电子振荡器提供以及晶体耗散的功率。晶体制造商数据表中指定的最大驱动电平通常是晶体在不损坏或显著缩短使用寿命的情况下可以耗散的最大驱动电平。另一方面，电子振荡器指定的驱动电平是它可以提供的最大功率。电子振荡器提供的实际功率不一定是最大功率，具体取决于晶体和电路板元件。

如果电子振荡器的实际驱动电平超过晶体的最大驱动电平规格，则应安装阻尼电阻器(Rd)以限制电流并降低晶体的功率耗散。请注意，Rd 会降低电路增益；因此，应评估要使用的实际值，以确保满足启动和持续振荡的所有其他条件。

6.12.3.4.4 How to Choose a Crystal
如何选择晶体

Using Crystal Oscillator Specifications as a reference:

1. Pick a crystal frequency (for example, 20 MHz).
2. Check that the ESR of the crystal <=50 Ω per specifications for 20 MHz.
3. Check that the load capacitance requirement of the crystal manufacturer is within 6 pF and 12 pF per specifications for 20 MHz.
   - As mentioned, CL1 and CL2 are in series; so, provided CL1 = CL2, effective load capacitance CL = [CL1]/2.
   - Adding board parasitics to this results in CL = [CL1]/2 + Cstray
4. Check that the maximum drive level of the crystal >= 1 mW. If this requirement is not met, a dampening resistor Rd can be used. Refer to DL – Drive Level on other points to consider when using Rd.

请参考“晶体振荡器规格”一节：

1. 选择一个晶体频率（例如，20MHz）。
2. 确认晶体的 ESR<=50Ω，符合 20MHz 的规格。
3. 确认晶体制造商的负载电容要求位于 6pF 和 12pF 之间，符合 20MHz 的规格。
   - 如前所述，CL1 和 CL2 是串联的；因此，如果 CL1=CL2，则有效负载电容 CL=[CL1]/2。
   - 在此基础上加上电路板寄生效应会得到 CL=[CL1]/2+杂散电容
4. 确认晶体的最大驱动电平>=1mW。如果不满足此要求，则可以使用阻尼电阻 Rd。请参阅 DL-驱动电平，了解使用 Rd 时要考虑的其他要点。

6.12.3.4.5 Testing 测试

It is recommended that the user have the crystal manufacturer completely characterize the crystal with their board to ensure the crystal always starts up and maintains oscillation.
Below is a brief overview of some measurements that can be performed:
Due to how sensitive the crystal circuit is to capacitance, it is recommended that scope probes not be connected to X1 and X2. If scope probes must be used to monitor X1/X2, an active probe with <1-pF capacitance should be used.

Frequency

1. Bring out the XTAL on XCLKOUT.
2. Measure this frequency as the crystal frequency.

Negative Resistance

1. Bring out the XTAL on XCLKOUT.

2. Place a potentiometer in series with the crystal between the load capacitors.
3. Increase the resistance of the potentiometer until the clock on XCLKOUT stops.
4. This resistance plus the crystal's actual ESR is the negative resistance of the electrical oscillator.

Start-Up Time

1. Turn off the XTAL.
2. Bring out the XTAL on XCLKOUT.
3. Turn on the XTAL and measure how long it takes the clock on XCLKOUT to stay within 45% and 55% duty cycle.

建议用户让晶体制造商使用其电路板对晶体进行完整表征，以确保晶体始终启动并保持振荡。下面简要概述了可执行的一些测量：

由于晶体电路对电容非常敏感，建议不要将示波器探针连接到 X1 和 X2。如果必须使用示波器探针来监测 X1/X2，则应使用电容小于 1pF 的有源探针。

频率

1. 在 XCLKOUT 上引出 XTAL。
2. 测量该频率作为晶体频率。

负电阻

1. 在 XCLKOUT 上引出 XTAL。
2. 在负载电容器之间放置一个与晶体串联的电位器。
3. 增加电位器的电阻，直到 XCLKOUT 上的时钟停止。
4. 该电阻加上晶体的实际 ESR 就是电振荡器的负电阻。

启动时间

1. 关闭 XTAL。
2. 在 XCLKOUT 上引出 XTAL。
3. 开启 XTAL 并测量 XCLKOUT 上的时钟保持在 45%和 55%占空比范围内所需的时间。

6.12.3.4.6 Common Problems and Debug Tips
常见问题和调试提示

Crystal Fails to Start Up

- Go through the How to Choose a Crystal section and make sure there are no violations.

Crystal Takes a Long Time to Start Up

- If a dampening resistor Rd is installed, it is too high.
- If no dampening resistor is installed, either the crystal ESR is too high or the overall circuit gain is too low due to high load capacitance.

晶体无法启动

- 浏览如何选择晶体部分，确保没有违规。

晶体需要很长时间才能启动

- 如果安装了阻尼电阻 Rd，则其过高。
- 如果未安装阻尼电阻，则晶体 ESR 过高或由于高负载电容而导致总电路增益过低。

6.12.3.4.7 Crystal Oscillator Specifications
晶体振荡器规格

6.12.3.4.7.1 Crystal Oscillator Parameters
晶体振荡器参数

		MIN	MAX	UNIT
CL1, CL2	Load capacitance	12	24	pF
C0	Crystal shunt capacitance		7	pF

6.12.3.4.7.2 Crystal Equivalent Series Resistance (ESR) Requirements
晶振等效串联电阻(ESR)要求

For the Crystal Equivalent Series Resistance (ESR) Requirements table:

1. Crystal shunt capacitance (C0) should be less than or equal to 7 pF.
2. ESR = Negative Resistance/3

对于晶振等效串联电阻（ESR）需求表：

1. 晶振并联电容（C0）应小于或等于 7pF。
2. ESR=负电阻/3

Table 6-5 Crystal Equivalent Series Resistance (ESR) Requirements

CRYSTAL FREQUENCY (MHz)	MAXIMUM ESR (Ω) (CL1=CL2=12pF)	MAXIMUM ESR (Ω) (CL1=CL2=24pF)
10	55	110
12	50	95
14	50	90
16	45	75
18	45	65
20	45	50

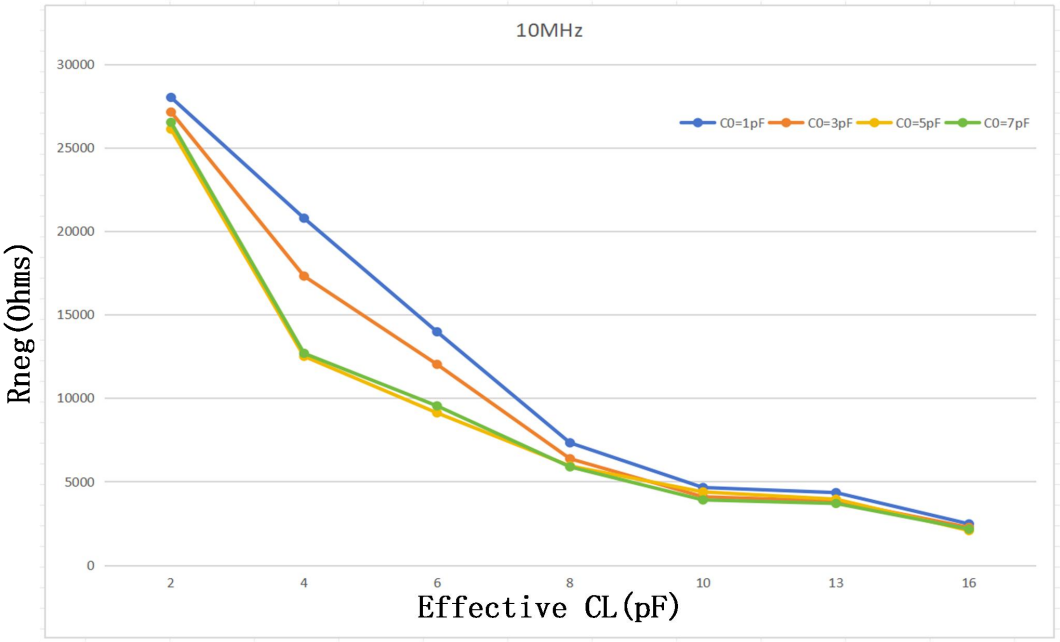


Figure 6- 20 Negative Resistance Variation at 10 MHz

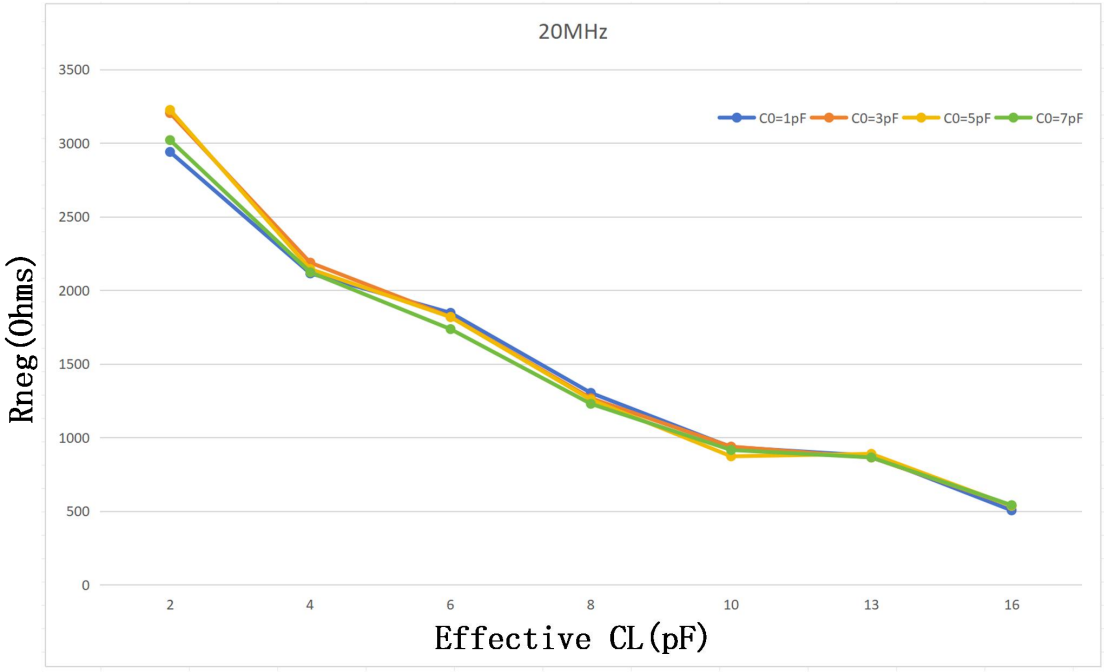


Figure 6- 21 Negative Resistance Variation at 20 MHz

6.12.3.4.7.3 Crystal Oscillator Electrical Characteristics
晶体振荡器电气特性

over recommended operating conditions (unless otherwise noted)
在推荐的工作条件下（除非另有说明）

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
Start-up time(1)	f=10MHz	ESR MAX =110Ω CL1=CL2=24pF C0=7pF		4		ms
	f=20MHz	ESR MAX =50Ω CL1=CL2=24pF C0=7pF		2		ms
Crystal drive level (DL)					1	mW

(1) Start-up time is dependent on the crystal and tank circuit components. It is recommended that the crystal vendor characterize the application with the chosen crystal.
启动时间取决于晶体和振荡电路元件。建议晶体供应商使用所选晶体来表征应用。

6.12.3.5 Internal Oscillators
内部振荡器

To reduce production board costs and application development time, all AVP32F003X devices contain two independent internal oscillators, referred to as INTOSC1 and INTOSC2. By default, INTOSC2 is set as the source for the system reference clock (OSCCLK) and INTOSC1 is set as the backup clock source.

为了减少电路板生产成本和缩短应用开发时间，AVP32F003X 器件包含两个独立的内部振荡器，称为 INTOSC1 和 INTOSC2。默认情况下，INTOSC2 设置为系统参考时钟(OSCCLK)源，INTOSC1 设置为备用时钟源。

6.12.3.5.1 INTOSC Characteristics
INTOSC 特性

over recommended operating conditions (unless otherwise noted)
在建议运行条件下测得（除非另有说明）

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
f_{INTOSC}	Frequency, INTOSC1 and INTOSC2(1)	-40°C to 125°C	9.82(–1.8%)	10	10.1(1.0%)	MHz
		–30°C to 90°C	9.86(–1.4%)	10	10.1(1.0%)	
		–10°C to 85°C	9.90(–1.0%)	10	10.1(1.0%)	
$f_{INTOSC-STABILITY}$	Frequency stability at room temperature	30°C, Nominal VDD		±0.1		%
$t_{INT0SC-ST}$	Start-up and settling time				20	µs

(1) INTOSC frequency may shift due to the thermal and mechanical stress of solder reflow. A post-reflow bake can restore the unit to its original datasheet performance.
由于回流焊的热应力和机械应力，INTOSC 频率可能会发生偏移。回流焊后烘烤可以使器件恢复到原始数据表性能。

6.12.4 Flash Parameters 闪存参数

Table 6-6 lists the minimum required Flash wait states with different clock sources and frequencies. Wait state is the value set in register FRDCNTL[RWAIT].

表 6-6 列出了不同时钟源和频率下所需的最低闪存等待状态。等待状态是寄存器 FRDCNTL[RWAIT]中设置的值。

Table 6-6 Minimum Required Flash Wait States with Different Clock Sources and Frequencies

CPUCLK（MHz）	EXTERNAL OSCILLATOR OR CRYSTAL		INTOSC1 OR INTOSC2	
	NORMAL OPERATION	BANK OR PUMP SLEEP(1)	NORMAL OPERATION	BANK OR PUMP SLEEP(1)
116<CPUCLK≤120	5		5	6
100<CPUCLK≤116				5
97<CPUCLK≤100	4		4	5
80<CPUCLK≤97				4
77<CPUCLK≤80	3		3	4
60<CPUCLK≤77				3
58<CPUCLK≤60	2		2	3
40<CPUCLK≤58				2
38<CPUCLK≤40	1		1	2
20<CPUCLK≤38				1
19<CPUCLK≤20	0		0	1
CPUCLK≤19				0

(1) Flash SLEEP operations require an extra wait state when using INTOSC as the clock source for the frequency ranges indicated. Any wait state FRDCNTL[RWAIT] change must be made before beginning a SLEEP mode operation. This setting impacts both flash banks.
当使用 INTOSC 作为所示频率范围的时钟源时，闪存睡眠操作需要一个额外的等待状态。任何等待状态 FRDCNTL[RWAIT]更改都必须在开始睡眠模式操作之前进行。此设置对两个闪存存储体都有影响。

AVP32F003X devices have an improved 128-bit prefetch buffer that provides high flash code execution efficiency across wait states. Figure 6-22 and Figure 6-23 illustrate typical efficiency across wait-state settings compared to previous-generation devices with a 64-bit prefetch buffer. Wait-state execution efficiency with a prefetch buffer will depend on how many branches are present in application software. Two examples of linear code and if-then-else code are provided.

AVP32F003X 器件具有经改进的 128 位预取缓冲器，可在不同等待状态下提供更高的闪存代码执行效率。图 6-22 和图 6-23 展示了该系列器件与采用 64 位预取缓冲器的上一代器件在不同等待状态设置下的典型效率比较情况。使用预取缓冲器时的等待状态执行效率将取决于应用软件中存在的分支数量。此处提供了线性代码和 if-then-else 代码的两个示例。

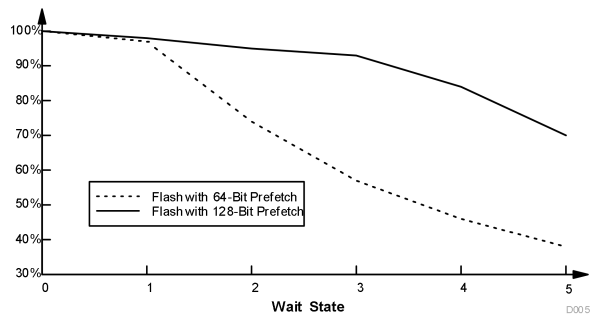


Figure 6-22 Application Code With Heavy 32-Bit Floating-Point Math Instructions

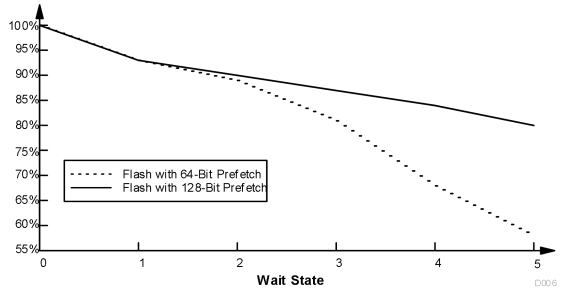


Figure 6- 23 Application Code With 16-Bit If-Else Instructions

Section 6.12.4.1 lists the Flash parameters.

Note

The Main Array flash programming must be aligned to 64-bit address boundaries and each 64-bit word may only be programmed once per write/erase cycle.

Note

It is important to provide the correct sector mask for the bank erase command. If the mask is mistakenly chosen to erase an inaccessible sector (belongs to another security zone), the bank erase command will continue attempting to erase the sector endlessly and the FSM will never exit (since erase will not succeed). To avoid such a situation, user must take care to provide the correct mask. However, given that there is a chance of choosing an incorrect mask, it is recommended to initialize the max allowed erase pulses to zero after the max number of pulses are issued by the FSM for the bank erase operation. This will ensure that the FSM will end the bank erase command after trying to erase the inaccessible sector up to the max allowed erase pulses.

The Example_EraseBanks() function in the flash API usage example depicts the implementation of this sequence (content of the while loop waiting for the FSM to complete the bank erase command). Users must use this code as-is irrespective of whether or not security is used by the application to also ensure that the FSM exits from bank erase operations in case of an erase-failure.

第 6.12.4.1 节列出了闪存参数。

备注

主阵列闪存编程必须与 64 位地址边界对齐，并且每个 64 位字在每个写/擦除周期只能编程一次。

备注

为存储体擦除命令提供正确的扇区掩码非常重要。如果错误地选择掩码来擦除无法访问的扇区（属于另一个安全区域），存储体擦除命令将继续尝试不停地擦除扇区并且 FSM 永远不会退出（因为擦除不会成功）。为避免这种情况，用户必须注意提供正确的掩码。但是，鉴于有可能选择不正确的掩码，建议在 FSM 为进行存储体擦除操作发出最大脉冲数后，将允许的最大擦除脉冲数初始化为零。这将确保 FSM 在尝试擦除不可访问的扇区到允许的最大擦除脉冲后结束存储体擦除命令。

闪存 API 使用示例中的 Example_EraseBanks()函数描述了该序列的实现 (等待 FSM 完成存储体擦除命令的 while 循环的内容）。此外还为了确保 FSM 在擦除失败时退出存储体擦除操作，无论应用是否使用安全性，用户都必须按原样使用此代码。

6.12.4.1 Flash Parameters
闪存参数

PARAMETER			MIN	TYP	MAX	UNIT
Program Time(1)		128 data bits+16 ECC bits		150	300	μs
		8KB (Sector)		50	100	ms
Sector Erase Time(2)(3)	<25 cycles	8KB (Sector)		15	56	ms
	1k cycles			26	133	ms
	2k cycles			31	226	ms
	20k cycles			123	1026	ms
Bank Erase Time(2)(3)	<25 cycles	128KB (Bank)		21	78	ms
	1k cycles			35	183	ms
	2k cycles			42	310	ms
	20k cycles			169	1410	ms
N_{wec} Write/Erase Cycles per Sector					20000	cycles
N_{wec} Write/Erase Cycles for Entire Flash (Combined for all Sectors)					100000	cycles
$T_{retention}$ Data retention duration (T_J=85oC)			10			years

(1) Program time is at the maximum device frequency. Program time includes overhead of the flash state machine but does not include the time to transfer the following into RAM:
- Code that uses flash API to program the flash
- Flash API itself
- Flash data to be programmed

In other words, the time indicated in this table is applicable after all the required code/data is available in the device RAM, ready for programming. The transfer time will significantly vary depending on the speed of the JTAG debug probe used.
Program time calculation is based on programming 144 bits at a time at the specified operating frequency. Program time includes Program verify by the CPU. The program time does not degrade with write/erase (W/E) cycling, but the erase time does.
Erase time includes Erase verify by the CPU and does not involve any data transfer.

编程时间是最大器件频率下的值。编程时间包括闪存状态机的开销，但不包括将以下内容传输到 RAM 的时间：
- 使用闪存 API 对闪存进行编程的代码
- 闪存 API 本身
- 要编程的闪存数据

换言之，此表中显示的时间是指器件 RAM 中的所有必需代码/数据都变为可用状态并准备好进行编程之后的相应时间。所用 JTAG 调试探头的速度对传输时间有显著影响。

编程时间的计算以在指定的工作频率下一次编程 144 位为基础。编程时间包含 CPU 对编程的验证。写入/擦除(W/E)循环不会缩短编程时间，但会缩短擦除时间。擦除时间包括由 CPU 对编程的验证，不涉及任何数据传输。

(2) Erase time includes Erase verify by the CPU.
擦除时间包含 CPU 对擦除的验证。

(3) The on-chip flash memory is in an erased state when the device is shipped. As such, erasing the flash memory is not required prior to programming, when programming the device for the first time. However, the erase operation is needed on all subsequent programming operations.
当器件出货时，片上闪存存储器处于一个被擦除状态。这样，当首次编辑器件时，在编程前无需擦除闪存存储器。然而，对于所有随后的编程操作，需要执行擦除操作。

6.12.5 RAM and ROM Parameters
RAM 和 ROM 参数

All volatile memory (RAM and ROM) on the AVP32F003X device is 0 Wait-state for both reads and writes, meaning the memory operates at the same speed as SYSCLK. RAM Block Properties summarizes the characteristics of the different RAM instances on the device. ROM Properties summarizes the aspects of the ROM instances on the device

AVP32F003X 器件上所有易失性存储器（RAM 和 ROM）的读取和写入均为 0 等待状态，这意味着存储器以与 SYSCLK 相同的速度运行。RAM 块属性总结了器件上不同 RAM 实例的特征。ROM 属性总结了器件上 ROM 实例的各个方面。

RAM 块属性

RAM TYPE	SIZE EACH	FETCH TIME (CYCLES)	READ TIME (CYCLES)	WRITE TIME (CYCLES)	SUPPORTED BUS WIDTHS (BITS)	HOST ACCESS LIST	WAIT STATES	BURST ACCESS SUPPORT
M0	2KB	2	2	1	16/32	A2000	0	No
M1								
LS RAM (LS0–LS7)	4KB					A2000/CLA		
GS RAM (GS0–GS3)	8KB					A2000/DMA/HIC		
CLA to CPU Message RAM	256B					A2000/CLA		
CPU to CLA Message RAM								
CLA to DMA Message RAM						CLA/DMA		
DMA to CLA Message RAM								

ROM 属性

ROM TYPE	SIZE EACH	FETCH TIME (CYCLES)	READ TIME (CYCLES)	SUPPORTED BUS WIDTHS (BITS)	HOST ACCESS LIST	WAIT STATES	BURST ACCESS SUPPORT
Boot ROM	64KB	2	2	16/32	A2000	0	No
Secure ROM	48KB						
CLA Data ROM	8KB				A2000/CLA		
CLA Program ROM	96KB				CLA		

6.12.6 Emulation/JTAG
仿真/JTAG

The JTAG (IEEE Standard 1149.1-1990 Standard Test Access Port and Boundary Scan Architecture) port

has four dedicated pins: TMS, TDI, TDO, and TCK. The cJTAG (IEEE Standard 1149.7-2009 for Reduced-Pin and Enhanced-Functionality Test Access Port and Boundary-Scan Architecture) port is a compact JTAG interface requiring only two pins (TMS and TCK), which allows other device functionality to be muxed to the traditional GPIO35 (TDI) and GPIO37 (TDO) pins.

JTAG(IEEE 标准1149.1-1990 标准测试访问端口和边界扫描架构)端口有四个专用引脚:TMS、TDI、TDO 和 TCK。cJTAG（针对简化引脚和增强功能测试访问端口以及边界扫描架构的 IEEE 标准 1149.7-2009）端口，是一个只需要两个引脚（TMS 和 TCK）的紧凑型 JTAG 接口，此接口可实现多路复用为传统 GPIO35(TDI)和 GPIO37(TDO)引脚的其他器件功能。

Typically, no buffers are needed on the JTAG signals when the distance between the MCU target and the JTAG header is smaller than 6 inches (15.24 cm), and no other devices are present on the JTAG chain. Otherwise, each signal should be buffered. Additionally, for most JTAG debug probe operations at 10 MHz, no series resistors are needed on the JTAG signals. However, if high emulation speeds are expected (35 MHz or so), 22-Ω resistors should be placed in series on each JTAG signal.

通常情况下，当MCU 目标和JTAG 接头之间的距离小于 6 英寸(15.24cm)，并且JTAG 链上没有其他器件时，JTAG 信号上不需要缓冲器。否则，每个信号都应被缓冲。此外，对于大多数 10MHz 下的JTAG 调试探针操作，JTAG 信号上不需要串联电阻器。但是，如果需要高仿真速度（35MHz 左右），则应在每个 JTAG 信号上串联 22Ω电阻。

The PD (Power Detect) pin of the JTAG debug probe header should be connected to the board's 3.3-V supply. Header GND pins should be connected to board ground. TDIS (Cable Disconnect Sense) should also be connected to board ground. The JTAG clock should be looped from the header TCK output pin back to the RTCK input pin of the header (to sense clock continuity by the JTAG debug probe). This MCU does not support the EMU0 and EMU1 signals that are present on 14-pin and 20-pin emulation headers. These signals should always be pulled up at the emulation header through a pair of board pullup resistors ranging from 2.2 kΩ to 4.7 kΩ (depending on the drive strength of the debugger ports). Typically, a 2.2-kΩ value is used.

JTAG 调试探针头的 PD（电源检测）引脚应连接到电路板 3.3V 电源。接头 GND 引脚应连接至电路板接地。TDIS（电缆断开感应）也应连接至电路板接地。JTAG 时钟应从接头 TCK 输出引脚环回到接头的 RTCK 输入引脚（以通过 JTAG 调试探针检测时钟连续性）。此MCU 不支持 14 引脚和 20 引脚仿真接头上的 EMU0 和EMU1 信号。这些信号应始终通过一对 2.2kΩ至 4.7kΩ（取决于调试器端口的驱动强度）的板载上拉电阻在仿真接头处上拉。通常使用 2.2kΩ的阻值。

Header pin $\overline{\text{RESET}}$ is an open-drain output from the JTAG debug probe header that enables board components to be reset through JTAG debug probe commands (available only through the 20-pin header). Figure 6- 24 shows how the 14-pin JTAG header connects to the MCU's JTAG port signals. Figure 6- 25 shows how to connect to the 20-pin JTAG header. The 20-pin JTAG header pins EMU2, EMU3, and EMU4 are not used and should be grounded.

接头引脚$\overline{\text{RESET}}$是 JTAG 调试探针接头的开漏输出，通过 JTAG 调试探针命令使电路板元件复位（仅通过 20 引脚接头可用）。图 6-24 显示了如何将 14 引脚 JTAG 接头连接到 MCU 的 JTAG 端口信号图 6-25 显示了如何连接到 20 引脚 JTAG 接头。20 引脚 JTAG 接头引脚 EMU2、EMU3 和 EMU4 未使用，应接地。

Note

JTAG Test Data Input (TDI) is the default mux selection for the pin. The internal pullup is disabled by default. If this pin is used as JTAG TDI, the internal pullup should be enabled or an external pullup added on the board to avoid a floating input. In the cJTAG option, this pin can be used as GPIO.

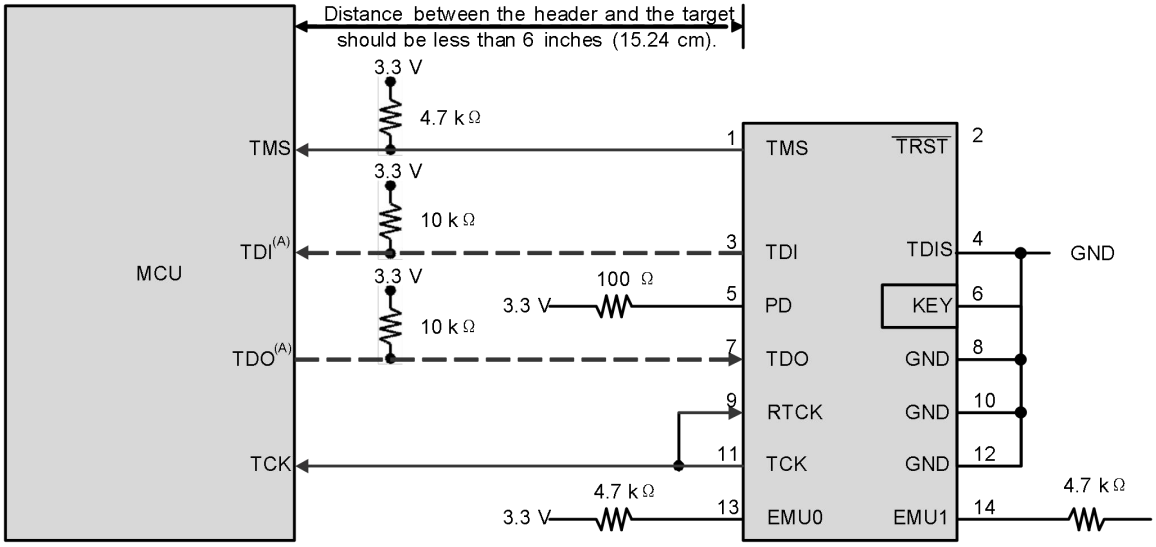
JTAG Test Data Output (TDO) is the default mux selection for the pin. The internal pullup is disabled by default. The TDO function will be in a tri-state condition when there is no JTAG activity, leaving

this pin floating. The internal pullup should be enabled or an external pullup added on the board to avoid a floating GPIO input. In the cJTAG option, this pin can be used as GPIO.

备注

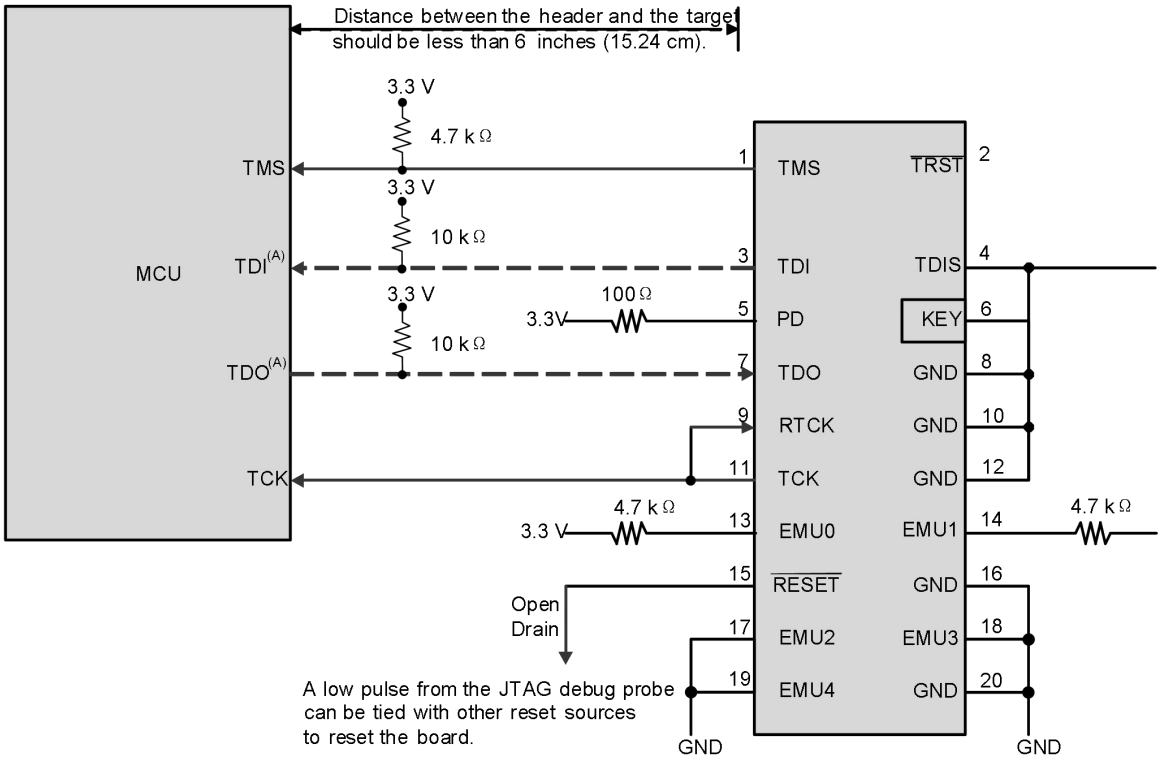
JTAG 测试数据输入(TDI)是引脚的默认多路复用器选择。默认情况下，内部上拉处于禁用状态。如果此引脚被用作 JTAGTDI，应该启用内部上拉电阻器或在电路板上增加一个外部上拉电阻器来避免悬空输入。在 cJTAG 选项中，此引脚可用作 GPIO。

JTAG 测试数据输出(TDO)是引脚的默认多路复用器选择。默认情况下，内部上拉处于禁用状态。当没有 JTAG 活动时，TDO 函数将处于三态条件，使此引脚悬空。应启用内部上拉或在电路板上添加外部上拉，以避免 GPIO 输入悬空。在 cJTAG 选项中，此引脚可用作 GPIO。



A. TDI and TDO connections are not required for cJTAG option and these pins can be used as GPIOs instead.
cJTAG 选项不需要 TDI 和TDO 连接，这些引脚可用作 GPIO。

Figure 6- 24 Connecting to the 14-Pin JTAG Header



A. TDI and TDO connections are not required for cJTAG option and these pins can be used as GPIOs instead.
cJTAG 选项不需要 TDI 和TDO 连接，这些引脚可用作 GPIO。

Figure 6- 25 Connecting to the 20-Pin JTAG Header

6.12.6.1 JTAG Electrical Data and Timing
JTAG 电气数据和时序

6.12.6.1.1 JTAG Timing Requirements
JTAG 时序要求

NO.			MIN	MAX	UNIT
1	$t_{c(TCK)}$	Cycle time, TCK	66.66		ns
1a	$t_{w(TCKH)}$	Pulse duration, TCK high (40% of t_c)	26.66		ns
1b	$t_{w(TCKL)}$	Pulse duration, TCK low (40% of t_c)	26.66		ns
3	$t_{su(TDI-TCKH)}$	Input setup time, TDI valid to TCK high	7		ns
	$t_{su(TMS-TCKH)}$	Input setup time, TMS valid to TCK high	7		
4	$t_{h(TCKH-TDI)}$	Input hold time, TDI valid from TCK high	7		ns
	$t_{h(TCKH-TMS)}$	Input hold time, TMS valid from TCK high	7		

6.12.6.1.2 JTAG Switching Characteristics
JTAG 开关特征

over recommended operating conditions (unless otherwise noted)
在推荐的工作条件下（除非另有说明）

NO.	PARAMETER		MIN	MAX	UNIT
2	$t_{d(TCKL-TDO)}$	Delay time, TCK low to TDO valid	6	20	ns

6.12.6.1.3 JTAG Timing Diagram
JTAG 时序图

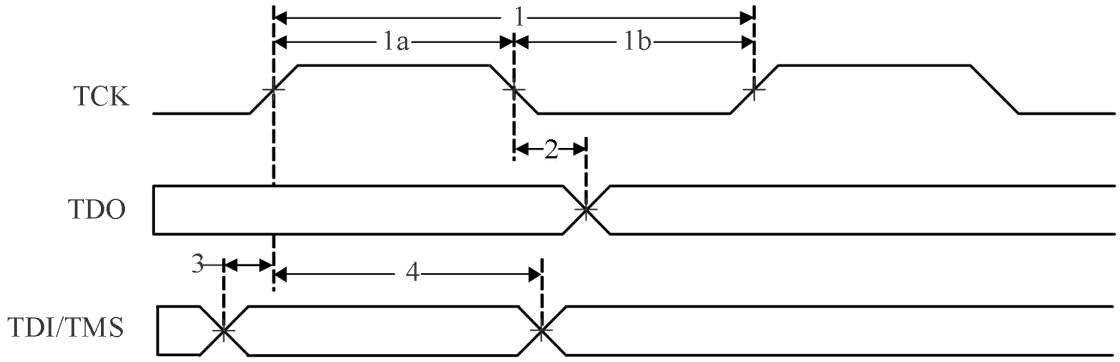


Figure 6- 26 JTAG Timing

6.12.6.2 cJTAG Electrical Data and Timing
cJTAG 电气数据和时序

6.12.6.2.1 cJTAG Timing Requirements
cJTAG 时序要求

NO.			MIN	MAX	UNIT
1	$t_{c(TCK)}$	Cycle time, TCK	100		ns
1a	$t_{w(TCKH)}$	Pulse duration, TCK high (40% of t_c)	40		ns
1b	$t_{w(TCKL)}$	Pulse duration, TCK low (40% of t_c)	40		ns
3	$t_{su(TMS-TCKH)}$	Input setup time, TMS valid to TCK high	7		ns
	$t_{su(TMS-TCKL)}$	Input setup time, TMS valid to TCK low	7		ns
4	$t_{h(TCKH-TMS)}$	Input hold time, TMS valid from TCK high	2		ns
	$t_{h(TCKL-TMS)}$	Input hold time, TMS valid from TCK low	2		ns

6.12.6.2.2 cJTAG Switching Characteristics
cJTAG 开关特性

over recommended operating conditions (unless otherwise noted)
在建议运行条件下测得（除非另有说明）

NO.	PARAMETER		MIN	MAX	UNIT
2	$t_{d(TCKL-TMS)}$	Delay time, TCK low to TMS valid	5	20	ns
5	$t_{dis(TCKH-TMS)}$	Delay time, TCK high to TMS disable		20	ns

6.12.6.2.3 cJTAG Timing Diagram
cJTAG 时序图

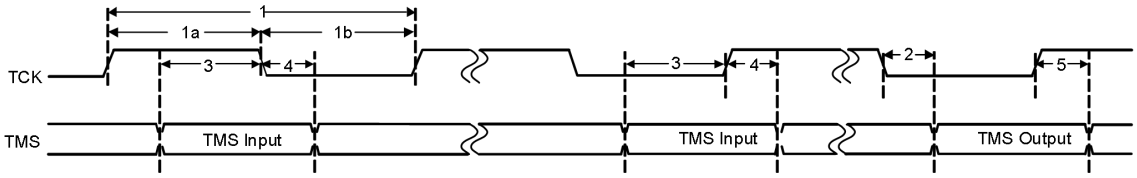


Figure 6- 27 cJTAG Timing

6.12.7 GPIO Electrical Data and Timing
GPIO 电气数据和时序

The peripheral signals are multiplexed with general-purpose input/output (GPIO) signals. On reset, GPIO pins are configured as inputs. For specific inputs, the user can also select the number of input qualification cycles to filter unwanted noise glitches.
The GPIO module contains an Output X-BAR which allows an assortment of internal signals to be routed to a GPIO in the GPIO mux positions denoted as OUTPUTXBARx. The GPIO module also contains an Input X-BAR which is used to route signals from any GPIO input to different IP blocks such as the ADCs, eCAPs, ePWMs, and external interrupts.

外设信号与通用输入/输出(GPIO)信号多路复用。复位时，GPIO 引脚配置为输入。对于特定的输入，用户还能选择输入滤波周期的数量来滤除不必要的噪声干扰。

GPIO 模块包含输出 X-BAR，其允许将各种内部信号路由到 GPIO 多路复用器位置中的 GPIO 上，并表示为 OUTPUTXBARx。GPIO 模块还包含输入 X-BAR，用于将来自任何 GPIO 输入的信号路由到不同的 IP 块，例如 ADC、eCAP、ePWM 和外部中断。

6.12.7.1 GPIO – Output Timing
GPIO-输出时序

6.12.7.1.1 General-Purpose Output Switching Characteristics
通用输出开关特征

over recommended operating conditions (unless otherwise noted)
在推荐的工作条件下（除非另有说明）

PARAMETER			MIN	MAX	UNIT
$t_{r(GPO)}$	Rise time, GPIO switching low to high	All GPIOs		8(1)	ns
$t_{f(GPO)}$	Fall time, GPIO switching high to low	All GPIOs		8(1)	ns
t_{fGPO}	Toggling frequency, GPIO pins			60	MHz

(1) Rise time and fall time vary with load. These values assume a 20-pF load.
上升时间和下降时间随负载而变化。这些值假定负载为 20pF。

6.12.7.1.2 General-Purpose Output Timing Diagram
通用输出时序图

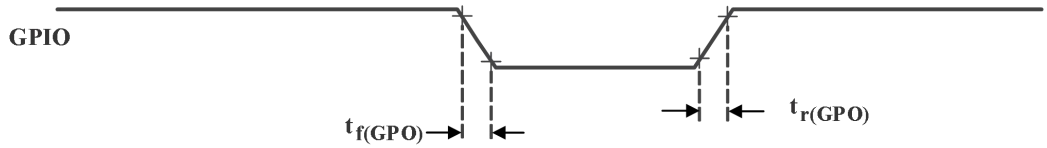


Figure 6- 28 General-Purpose Output Timing

6.12.7.2 GPIO – Input Timing
GPIO-输入时序

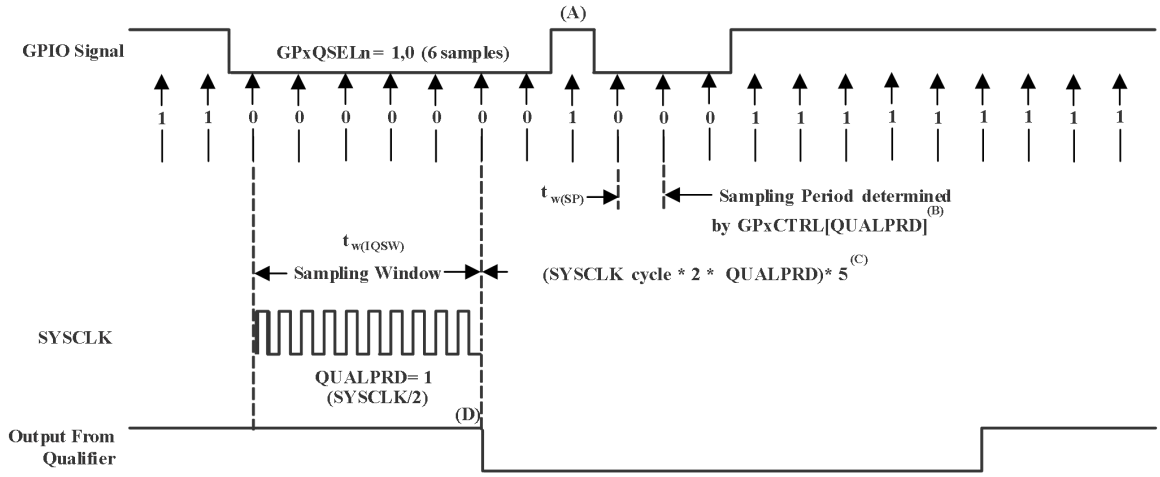
6.12.7.2.1 General-Purpose Input Timing Requirements
通用输入时序要求

PARAMETER			MIN	MAX	UNIT
$t_{w(SP)}$	Sampling period	QUALPRD=0	$1t_{c(SYSCLK)}$		cycles
		QUALPRD≠0	$2t_{c(SYSCLK)}$*QUALPRD		cycles
$t_{w(IQSW)}$	Input qualifier sampling window		$t_{w(SP)}$*(n(1)-1)		cycles
$t_{w(GPI)}$(2)	Pulse duration, GPIO low/high	Synchronous mode	$2t_{c(SYSCLK)}$		cycles
		With input qualifier	$t_{w(IQSW)}+t_{w(SP)}+1t_{c(SYSCLK)}$		cycles

(1) "n" represents the number of qualification samples as defined by GPxQSELn register.
"n"代表由 GPxQSELn 寄存器定义的滤波采样的数量。

(2) For $t_{w(GPI)}$, pulse width is measured from V_{IL} to V_{IL} for an active low signal and V_{IH} to V_{IH} for an active high signal.
对于 $t_{w(GPI)}$, 对低电平有效信号在 V_{IL} 至 V_{IL} 之间测量脉宽，而高电平有效信号，在 V_{IH} 至 V_{IH} 之间测量脉宽。

6.12.7.2.2 Sampling Mode
采样模式



A. This glitch will be ignored by the input qualifier. The QUALPRD bit field specifies the qualification sampling period. It can vary from 00 to 0xFF. If QUALPRD = 00, then the sampling period is 1 SYSCLK cycle. For any other value "n", the qualification sampling period in 2n SYSCLK cycles (that is, at every 2n SYSCLK cycles, the GPIO pin will be sampled).
输入滤波将忽略此短时脉冲波干扰。QUALPRD 位字段指定了滤波采样周期。该位字段可以在 00 至 0xFF 之间变化。如果 QUALPRD=00，那么采样周期为 1 个SYSCLK 周期。对于任何其他的"n"值，滤波采样周期为 2nSYSCLK 周期（也就是说，在每2n 个 SYSCLK 周期上，GPIO 引脚将被采样）。

B. The qualification period selected through the GPxCTRL register applies to groups of eight GPIO pins.
通过GPxCTRL 寄存器选择的滤波周期会应用于包含 8 个GPIO 引脚的组。

C. The qualification block can take either three or six samples. The GPxQSELn Register selects which sample mode is used.
此滤波块可取 3 个或者 6 个采样。GPxQSELn 寄存器选择使用哪种采样模式。

D. In the example shown, for the qualifier to detect the change, the input should be stable for 10 SYSCLK cycles or greater. In other words, the inputs should be stable for (5 × QUALPRD × 2) SYSCLK cycles. This would ensure 5 sampling periods for detection to occur. Because external signals are driven asynchronously, an 13-SYSCLK-wide pulse ensures reliable recognition.
在所示的示例中，为了使滤波检测到变化，输入应该在 10 个 SYSCLK 周期或者更长周期内保持稳定。换句话说，输入应该在(5×QUALPRD×2)个SYSCLK 周期内保持稳定。这将确保有 5 个采样周期用于检测。由于外部信号是异步驱动

的，因此13SYSCLK 宽的脉冲确保了可靠的识别。

Figure 6- 29 Sampling Mode

6.12.7.3 Sampling Window Width for Input Signals
输入信号的采样窗口宽度

The following section summarizes the sampling window width for input signals for various input qualifier configurations. Sampling frequency denotes how often a signal is sampled with respect to SYSCLK.
Sampling frequency = SYSCLK/(2 × QUALPRD), if QUALPRD ≠ 0
Sampling frequency = SYSCLK, if QUALPRD = 0
Sampling period = SYSCLK cycle × 2 × QUALPRD, if QUALPRD ≠ 0
In the previous equations, SYSCLK cycle indicates the time period of SYSCLK. Sampling period = SYSCLK cycle, if QUALPRD = 0
In a given sampling window, either 3 or 6 samples of the input signal are taken to determine the validity of the signal. This is determined by the value written to GPxQSELn register.

下述小节总结了各种输入滤波配置下的输入信号的采样窗口宽度。采样频率表明相对于 SYSCLK 对信号进行采样的频率。

如果 QUALPRD≠0，采样频率=SYSCLK/(2×QUALPRD)

如果 QUALPRD=0，则采样频率=SYSCLK

如果 QUALPRD≠0，则采样周期=SYSCLK 周期×2×QUALPRD

在上面的等式中，SYSCLK 周期表示 SYSCLK 的时间周期。如果 QUALPRD=0，则采样周期 =SYSCLK 周期

在给定的采样窗口中，采取输入信号的 3 个或者 6 个采样来确定信号的有效性。这是由写入到 GPxQSELn 寄存器的值确定的。

Case 1:

Qualification using 3 samples
Sampling window width = (SYSCLK cycle × 2 × QUALPRD) × 2, if QUALPRD ≠ 0
Sampling window width = (SYSCLK cycle) × 2, if QUALPRD = 0

Case 2:

Qualification using 6 samples
Sampling window width = (SYSCLK cycle × 2 × QUALPRD) × 5, if QUALPRD ≠ 0
Sampling window width = (SYSCLK cycle) × 5, if QUALPRD = 0

情况 1:

使用 3 个采样进行滤波

如果 QUALPRD≠0，则采样窗口宽度=（SYSCLK 周期 x2xQUALPRD）×2

如果 QUALPRD=0，则采样窗口宽度=（SYSCLK 周期）x2

情况 2:

使用 6 个采样进行滤波

如果 QUALPRD≠0，则采样窗口宽度=（SYSCLK 周期 x2xQUALPRD）×5

如果 QUALPRD=0，则采样窗口宽度=（SYSCLK 周期）x5

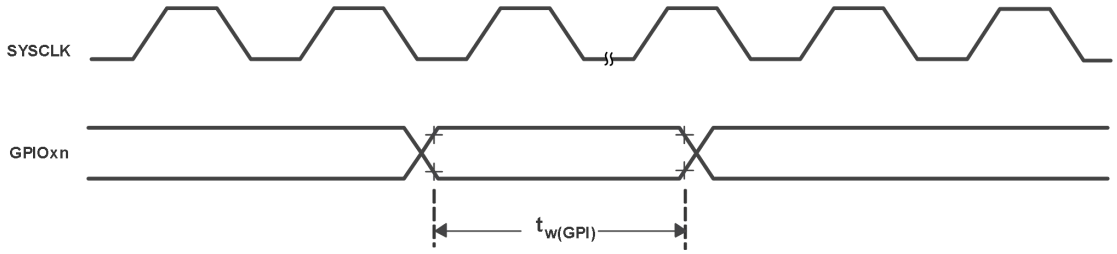


Figure 6-30 General-Purpose Input Timing

6.12.8 Interrupts 中断

The A2000 CPU has fourteen peripheral interrupt lines. Two of them (INT13 and INT14) are connected directly to CPU timers 1 and 2, respectively. The remaining twelve are connected to peripheral interrupt signals through the enhanced Peripheral Interrupt Expansion (ePIE) module. The ePIE multiplexes up to sixteen peripheral interrupts into each CPU interrupt line. It also expands the vector table to allow each interrupt to have its own ISR. This allows the CPU to support a large number of peripherals.
An interrupt path is divided into three stages—the peripheral, the ePIE, and the CPU. Each stage has its own enable and flag registers. This system allows the CPU to handle one interrupt while others are pending, implement and prioritize nested interrupts in software, and disable interrupts during certain critical tasks.
Figure 6-31 shows the interrupt architecture for this device.

A2000CPU 有14 条外设中断线路，其中两条（INT13 和INT14）分别直接连接到 CPU 定时器 1 和2。其余 12 条通过增强型外设中断扩展(ePIE)模块连接到外设中断信号。ePIE 将多达 16 个外设中断多路复用到每条 CPU 中断线路中。它还会扩展矢量表以允许每个中断都有自己的 ISR。这使得 CPU 能够支持大量外设。

中断路径分为三级：外设、ePIE 和CPU。每一级都有其自身的使能和标志寄存器。该系统允许 CPU 处理一个中断并让其他中断挂起，在软件中实施嵌套中断并确定其优先级，以及在某些关键任务期间禁用中断。

图 6-31 所示为该器件的中断架构。

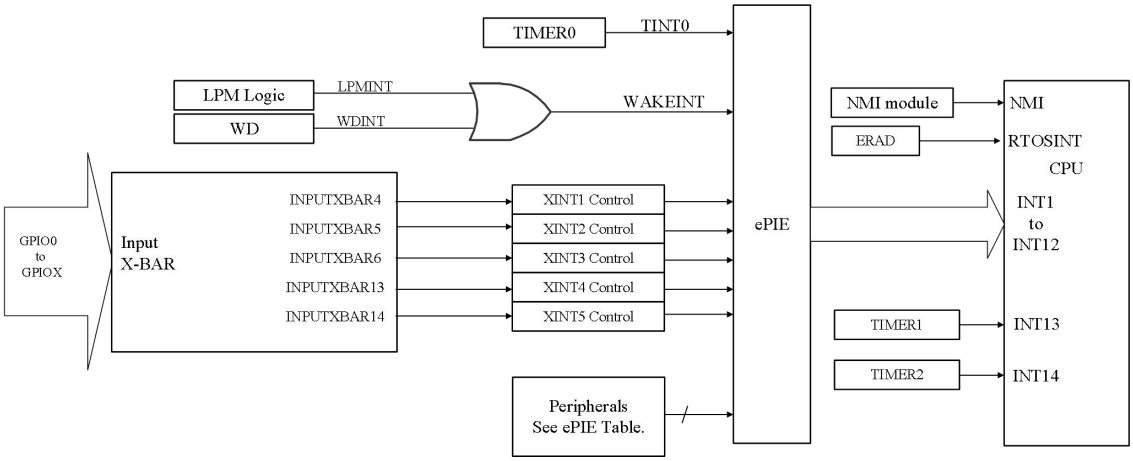


Figure 6-31 Device Interrupt Architecture

6.12.8.1 External Interrupt (XINT) Electrical Data and Timing
外部中断(XINT)电气数据和时序

For an explanation of the input qualifier parameters, see the General-Purpose Input Timing Requirements table.

有关输入滤波参数的说明，请参阅“通用输入时序要求”表。

6.12.8.1.1 External Interrupt Timing Requirements
外部中断时序要求

			MIN	MAX	UNIT
$t_{w(INT)}$	Pulse duration, INT input low/high	Synchronous	$2t_{c(SYSCLK)}$		cycles
		With qualifier	$t_{w(IQSW)}+t_{w(SP)}+1t_{c(SYSCLK)}$		cycles

6.12.8.1.2 External Interrupt Switching Characteristics
外部中断开关特性

over recommended operating conditions (unless otherwise noted)

在建议运行条件下测得（除非另有说明）

PARAMETER		MIN	MAX	UNIT
$t_{d(INT)}$	Delay time, INT low/high to interrupt-vector fetch(1)	$t_{w(IQSW)}+14t_{c(SYSCLK)}$	$t_{w(IQSW)}+t_{w(SP)}+14t_{c(SYSCLK)}$	cycles

(1) This assumes that the ISR is in a single-cycle memory.
这是假设 ISR 是在单周期存储器中。

6.12.8.1.3 External Interrupt Timing
外部中断时序

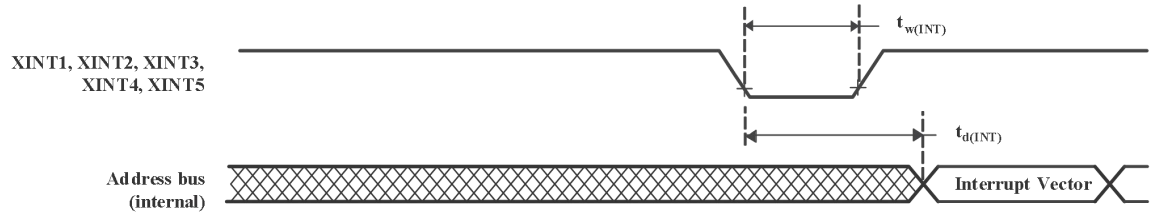


Figure 6- 32 External Interrupt Timing

6.12.9 Low-Power Modes
低功率模式

This device has HALT, IDLE and STANDBY as clock-gating low-power modes.

该器件具有停机、空闲和待机三种时钟门控低功耗模式。

6.12.9.1 Clock-Gating Low-Power Modes
时钟门控低功耗模式

IDLE and HALT modes on this device are similar to those on other A2000 devices. Table 6-7 describes the effect on the system when any of the clock-gating low-power modes are entered.

该器件上的空闲和停机模式与其他 A2000 器件上的类似。表 6-7 描述了进入任何一种时钟门控低功耗模式时对系统的影响。

Table 6-7 Effect of Clock-Gating Low-Power Modes on the Device

MODULES/ CLOCK DOMAIN	IDLE	STANDBY	HALT
SYSCLK	Active	Gated	Gated
CPUCLK	Gated	Gated	Gated
Clock to modules connected to PERx.SYSCLK	Active	Gated	Gated
WDCLK	Active	Active	Gated if CLKSRCCTL1.WDHALTI = 0
PLL	Powered	Powered	Software must power down PLL before entering HALT.
INTOSC1	Powered	Powered	Powered down if CLKSRCCTL1.WDHALTI = 0
INTOSC2	Powered	Powered	Powered down if CLKSRCCTL1.WDHALTI = 0
Flash(1)	Powered	Powered	Powered
XTAL(2)	Powered	Powered	Powered

(1) The Flash module is not powered down by hardware in any LPM. It may be powered down using software if required by the application. For more information, see the Flash and OTP Memory section of the System Control chapter in the *AVP32F003X Technical Reference Manual*.
在任意一种低功耗模式下，闪存模块不会由硬件断电。如果应用需要，可使用软件将其断电。有关更多信息，请参阅《AVP32F003X 技术参考手册》中“系统控制”一章的“闪存和 OTP 存储器”一节。

(2) The XTAL is not powered down by hardware in any LPM. It may be powered down by software setting the XTALCR.OSCOFF bit to 1. This can be done at any time during the application if the XTAL is not required.
在任意一种低功耗模式下，XTAL 不会由硬件断电。它可以通过软件将 XTALCR.OSCOFF 位设置为 1 来断电。如果不需要 XTAL，可以在应用期间的任何时间完成此操作。

6.12.9.2 Low-Power Mode Wake-up Timing 低功耗模式唤醒时序

For an explanation of the input qualifier parameters, see the General-Purpose Input Timing Requirements table.
有关输入滤波参数的说明，请参阅“通用输入时序要求”表。

6.12.9.2.1 IDLE Mode Timing Requirements
空闲模式时序要求

PARAMETER			MIN	MAX	UNIT
$t_{w(WAKE)}$	Pulse duration, external wake-up signal	Without input qualifier	$2t_{c(SYSCLK)}$		cycles
		With input qualifier	$2t_{c(SYSCLK)}+t_{w(IQSW)}$		

6.12.9.2.2 IDLE Mode Switching Characteristics
空闲模式开关特性

over recommended operating conditions (unless otherwise noted)
在推荐的运行条件下测得（除非另有说明）

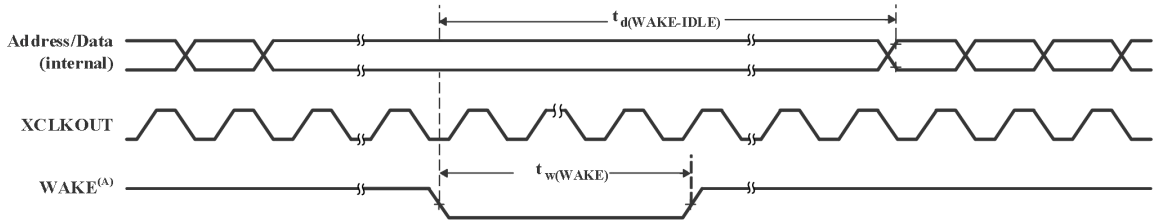
PARAMETER			TEST CONDITIONS	MIN	MAX	UNIT
$t_{d(WAKE-IDLE)}$	Delay time, external wake signal to program execution resume(1)	From Flash (active state)	Without input qualifier		$40t_{c(SYSCLK)}$	cycles
			With input qualifier		$40t_{c(SYSCLK)}+t_{w(WAKE)}$	cycles
		From Flash (sleep state)	Without input qualifier		$9316t_{c(SYSCLK)}$(2)	cycles
			With input qualifier		$9316t_{c(SYSCLK)}$(2)$+t_{w(WAKE)}$	cycles
		From RAM	Without input		$25t_{c(SYSCLK)}$	cycles

			qualifier			
			With input qualifier		$25t_{c(SYSCLK)}+t_{w(WAKE)}$	cycles

(1) This is the time taken to begin execution of the instruction that immediately follows the IDLE instruction. Execution of an ISR (triggered by the wake-up signal) involves additional latency.
这个时间是在 IDLE 指令之后立即开始指令执行的时间。ISR（由唤醒信号触发）的执行涉及额外延迟。

(2) This value is based on the flash power-up time, which is a function of the SYSCLK frequency, flash wait states (RWAIT), and FPAC1[PSLEEP].
该值基于闪存上电时间，其是SYSCLK 频率、闪存等待状态(RWAIT)和FPAC1[PSLEEP]的函数。

6.12.9.2.3 IDLE Entry and Exit Timing Diagram
进入和退出空闲模式时序图



A. WAKE can be any enabled interrupt, WDINT or XRSn. After the IDLE instruction is executed, a delay of five OSCCLK cycles (minimum) is needed before the wake-up signal could be asserted.
WAKE 可以是任何启用的中断、WDINT或XRSn。IDLE 指令执行后，在唤醒信号生效前需要 5 个OSCCLK 周期（最少）的延迟。

Figure 6- 33 IDLE Entry and Exit Timing Diagram

6.12.9.2.4 STANDBY Mode Timing Requirements
待机模式时序要求

PARAMETER			MIN	MAX	UNIT
$t_{w(WAKE-INT)}$	Pulse duration, external wake-up signal	QUALSTDBY=0|$2t_{c(OSCCLK)}$	$3t_{c(OSCCLK)}$		Cycles
		QUALSTDBY>0| $(2+QUALSTDBY)t_{c(OSCCLK)}$ (1)	$(2+QUALSTDBY)*t_{c(OSCCLK)}$		

(1) QUALSTDBY 是LPMCR 寄存器中的一个 6 位字段。
QUALSTDBY is a 6-bit field in the LPMCR register.

6.12.9.2.5 STANDBY Mode Switching Characteristics
待机模式开关特征

over recommended operating conditions (unless otherwise noted)
在推荐的工作条件下（除非另有说明）

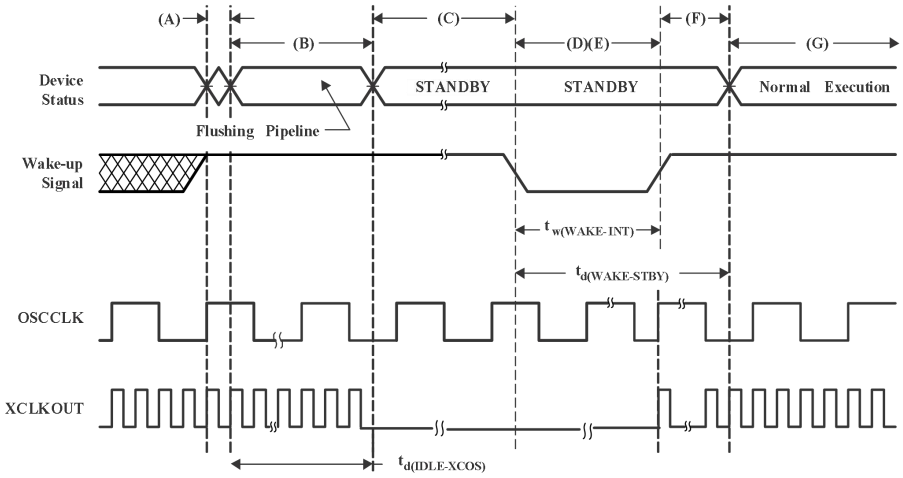
PARAMETER		TEST CONDITION	MIN	MAX	UNIT
$t_{d(IDLE-XCOS)}$	Delay time, IDLE instruction executed to XCLKOUT stop			$16t_{c(INTOSC1)}$	cycles
$t_{d(WAKE-STBY)}$	Delay time, external wake signal to program execution resume(1)	Wakeup from flash (Flash module in active state)		$175t_{c(SYSCLK)}+t_{w(WAKE-INT)}$	cycles
$t_{d(WAKE-STBY)}$		Wakeup from flash (Flash module in sleep state)		$9316t_{c(SYSCLK)}$(2)$+t_{w(WAKE-INT)}$	cycles
$t_{d(WAKE-STBY)}$		Wakeup from RAM		$3t_{c(OSC)}+15t_{c(SYSCLK)}+t_{w(WAKE-INT)}$	cycles

(1) This is the time taken to begin execution of the instruction that immediately follows the IDLE instruction. Execution of an ISR

(triggered by the wake-up signal) involves additional latency.
这个时间是在 IDLE 指令之后立即开始指令执行的时间。ISR（由唤醒信号触发）的执行涉及额外延迟。

(2) This value is based on the flash power-up time, which is a function of the SYSCLK frequency, flash wait states (RWAIT), and FPAC1[PSLEEP].
该值基于闪存上电时间，其是SYSCLK 频率、闪存等待状态(RWAIT)和FPAC1[PSLEEP]的函数。

6.12.9.2.6 STANDBY Entry and Exit Timing Diagram
进入和退出待机模式时序图



A. IDLE instruction is executed to put the device into STANDBY mode.
执行空闲指令将器件置于待机模式。

B. The LPM block responds to the STANDBY signal, SYSCLK is held for a maximum 16 INTOSC1 clock cycles before being turned off. This delay enables the CPU pipeline and any other pending operations to flush properly.
LPM 块响应待机信号，SYSCLK 在关闭之前最多保持 16 个INTOSC1 时钟周期。此延迟使 CPU 流水线和其他待处理的操作能够正确刷新。

C. Clock to the peripherals are turned off. However, the PLL and watchdog are not shut down. The device is now in STANDBY mode. After the IDLE instruction is executed, a delay of five OSCCLK cycles (minimum) is needed before the wake-up signal could be asserted.
外设的时钟被关闭。然而，PLL 和看门狗并未关闭。此器件现在处于待机模式。IDLE 指令执行后，在唤醒信号生效前需要 5 个OSCCLK 周期（最小值）的延迟。

D. The external wake-up signal is driven active.
外部唤醒信号被驱动为有效。

E. The wake-up signal fed to a GPIO pin to wake up the device must meet the minimum pulse width requirement. Furthermore, this signal must be free of glitches. If a noisy signal is fed to a GPIO pin, the wakeup behavior of the device will not be deterministic and the device may not exit low-power mode for subsequent wakeup pulses.
馈送到GPIO 引脚的唤醒信号必须符合最小脉冲宽度要求。此外，此信号必须免受短时脉时波干扰。如果噪声信号馈送到 GPIO 引脚，器件的唤醒行为将不确定，并且对于后续的唤醒脉冲，器件可能不会退出低功耗模式。

F. After a latency period, the STANDBY mode is exited.
经过一个延迟周期后，退出待机模式。

G. Normal execution resumes. The device will respond to the interrupt (if enabled).
正常执行重新开始。器件将响应中断（如果启用）。

Figure 6- 34 STANDBY Entry and Exit Timing Diagram

6.12.9.2.7 HALT Mode Timing Requirements
停机模式时序要求

PARAMETER		MIN	MAX	UNIT
$t_{w(WAKE-GPIO)}$	Pulse duration, GPIO wake-up signal(1)	$t_{oscst}+2t_{c(OSCCLK)}$		cycles
$t_{w(WAKE-XRS)}$	Pulse duration, $\overline{XRS}$ wake-up signal(1)	$t_{oscst}+8t_{c(OSCCLK)}$		cycles

(1) For applications using X1/X2 for OSCCLK, the user must characterize their specific oscillator start-up time as it is dependent on circuit/layout external to the device. See *Crystal Oscillator (XTAL)* section for more information. For applications using INTOSC1 or INTOSC2 for OSCCLK, see the Internal Oscillators section for t_{oscst}. Oscillator start-up time does not apply to applications using a single-ended crystal on the X1 pin, as it is powered externally to the device.
对于将 X1/X2 用于 OSCCLK 的应用，用户必须表征其特定的振荡器启动时间，因为它取决于器件外部的电路/布局。要了解更多信息，请参阅晶体振荡器(XTAL)部分。对于使用 INTOSC1 或 INTOSC2 作为 OSCCLK 的应用，请参阅“内部振荡器”部分，了解 t_{oscst}。振荡器启动时间不适用于在 X1 引脚上使用单端晶振的应用，因为它由器件外部供电。

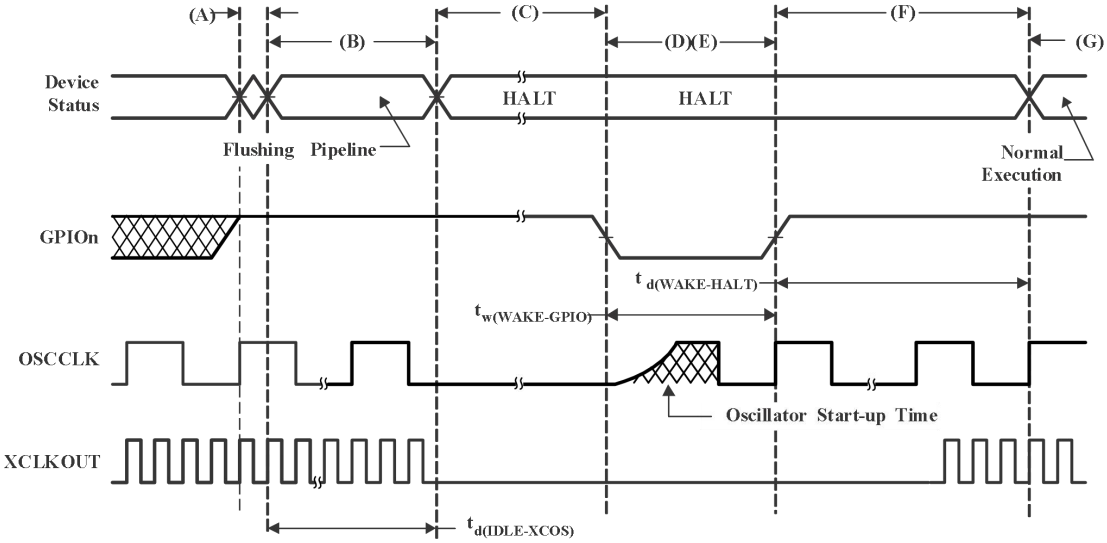
6.12.9.2.8 HALT Mode Switching Characteristics
停机模式开关特征

over recommended operating conditions (unless otherwise noted)
在推荐的工作条件下（除非另有说明）

PARAMETER		MIN	MAX	UNIT
$t_{d(IDLE-XCOS)}$	Delay time, IDLE instruction executed to XCLKOUT stop		$16t_{c(INTOSC1)}$	cycles
$t_{d(WAKE-HALT)}$	Delay time, external wake signal end to CPU1 program execution resume			cycles
	Wakeup from Flash - Flash module in active state		$75t_{c(OSCCLK)}$	
	Wakeup from Flash - Flash module in sleep state		$9316t_{c(SYSCLK)}+75t_{c(OSCCLK)}$ (1)	
	Wakeup from RAM		$75t_{c(OSCCLK)}$	

(1) This value is based on the flash power-up time, which is a function of the SYSCLK frequency, flash wait states (RWAIT), and FPAC1[PSLEEP].
该值基于闪存上电时间，其是 SYSCLK 频率、闪存等待状态(RWAIT)和 FPAC1[PSLEEP]的函数。

6.12.9.2.9 HALT Entry and Exit Timing Diagram
进入和退出停机模式时序图



A. IDLE instruction is executed to put the device into HALT mode.
执行IDLE 指令以将器件置于 HALT 模式。

B. The LPM block responds to the HALT signal, SYSCLK is held for a maximum 16 INTOSC1 clock cycles before being turned off. This delay enables the CPU pipeline and any other pending operations to flush properly.
LPM 块响应HALT 信号，SYSCLK 在关闭之前最多保持 16 个INTOSC1 时钟周期。此延迟使 CPU 流水线和其他待处理的操作能够正确刷新。

C. Clocks to the peripherals are turned off and the PLL is shut down. If a quartz crystal or ceramic resonator is used as the clock source, the internal oscillator is shut down as well. The device is now in HALT mode and consumes very little power. It is possible to keep the zero-pin internal oscillators (INTOSC1 and INTOSC2) and the watchdog alive in HALT MODE. This is done by writing 1 to CLKSRCCTL1.WDHALTI. After the IDLE instruction is executed, a delay of five OSCCLK cycles (minimum) is needed before the wake-up signal could be asserted.
到外设的时钟被关闭并且 PLL 被关断。如果一个石英晶振或者陶瓷谐振器被用作时钟源，内部振荡器也被关断。器件现在处于停机模式，并且功耗非常低。可以在停机模式中保持零引脚内部振荡器（INTOSC1 和INTOSC2）以及看门狗处于活动中。为实现这一点，需要向CLKSRCCTL1.WDHALTI 写入 1。IDLE 指令执行后，在唤醒信号生效前需要 5 个OSCCLK 周期（最少）的延迟。

D. When the GPIOn pin (used to bring the device out of HALT) is driven low, the oscillator is turned on and the oscillator wake-up sequence is initiated. The GPIO pin should be driven high only after the oscillator has stabilized. This enables the provision of a clean clock signal during the PLL lock sequence. Because the falling edge of the GPIO pin asynchronously begins the wake-up procedure, care should be taken to maintain a low noise environment before entering and during HALT mode.
当GPIOn 引脚（用于使器件脱离 HALT 模式）被驱动为低电平时，振荡器被打开并且振荡器唤醒序列被启动。只有当振荡器稳定时，GPIO 才应被驱动为高电平。这使得在 PLL 锁序列期间提供洁净的时钟信号。由于 GPIO 引脚的下降沿会以异步方式开始唤醒过程，因此在进入停机模式之前和在此模式期间，应该注意保持低噪声环境。

E. The wake-up signal fed to a GPIO pin to wake up the device must meet the minimum pulse width requirement. Furthermore, this signal must be free of glitches. If a noisy signal is fed to a GPIO pin, the wake-up behavior of the device will not be deterministic and the device may not exit low-power mode for subsequent wake-up pulses.
馈送到GPIO 引脚的唤醒信号必须符合最小脉冲宽度要求。此外，此信号不能有毛刺。如果噪声信号馈送到 GPIO 引脚，器件的唤醒行为将是不确定的并且在随后的唤醒脉冲中器件可能不会退出低功耗模式。

F. When CLKIN to the core is enabled, the device will respond to the interrupt (if enabled), after some latency. The HALT mode is now exited.
当内核的CLKIN 已启用时，器件将在一些延迟后响应中断（如果已启用）。现在退出停机模式。

G. Normal operation resumes.
恢复正常运行。

H. The user must relock the PLL upon HALT wakeup to ensure a stable PLL lock.
用户必须在停机唤醒时重新锁定 PLL，以确保稳定的 PLL 锁定。

Figure 6-35 HALT Entry and Exit Timing Diagram

6.13 Analog Peripherals
模拟外设

The analog subsystem module is described in this section.
The analog modules on this device include the ADC, temperature sensor, CMPSS, and buffered

DAC. The analog subsystem has the following features:

- Flexible voltage references
  - The ADCs are referenced to VREFHIx and VSSA pins
    - VREFHIx pin voltage can be driven in externally or can be generated by an internal bandgap voltage reference
    - The internal voltage reference range can be selected to be 0 V to 3.3 V or 0 V to 2.5 V
  - The buffered DACs are referenced to VREFHIx and VSSA
    - Alternately, these DACs can be referenced to the VDAC pin and VSSA
  - The comparator DACs are referenced to VDDA and VSSA
    - Alternately, these DACs can be referenced to the VDAC pin and VSSA
- Flexible pin usage
  - Buffered DAC outputs, comparator subsystem inputs, and digital inputs (AIOs)/outputs (AGPIOs) are multiplexed with ADC inputs
  - Internal connection to VREFLO on all ADCs for offset self-calibration

Figure 6-36 shows the Analog Subsystem Block Diagram for the 100-pin PZ LQFP.

Figure 6-37 shows the Analog Subsystem Block Diagram for the 80-pin PN LQFP.

Figure 6-38 shows the Analog Subsystem Block Diagram for the 64-pin PM LQFP.

Figure 6-39 shows the Analog Subsystem Block Diagram for the 48-pin PT LQFP.

Figure 6-40 shows the analog group connections. Section 6.13.1 lists the analog pins and internal connections. Section 6.13.2 lists descriptions of analog signals.

本节介绍了模拟子系统模块。

该器件上的模拟模块包括 ADC、温度传感器、CMPSS 和缓冲 DAC。

模拟子系统具有以下特性：

- 灵活的电压基准
  - ADC 以VREFHIx 和VSSA 引脚为基准
- VREFHIx 引脚电压可由外部驱动或由内部带隙电压基准生成
- 内部电压基准范围可选为 0V 至 3.3V 或 0V 至 2.5V
  - 缓冲 DAC 以 VREFHIx 和 VSSA 为基准
- 或者，这些DAC 能以VDAC 引脚和VSSA 为基准
  - 比较器 DAC 以 VDDA 和 VSSA 为基准
- 或者，这些 DAC 能以 VDAC 引脚和 VSSA 为基准
- 灵活地使用引脚
  - 缓冲 DAC 输出、比较器子系统输入和数字输入(AIO)/输出(AGPIO)与 ADC 输入进行多路复用
  - 所有 ADC 上的 VREFLO 的内部连接用于失调电压自我校准

图 6-36 显示了 100 引脚 LQFP 的模拟子系统方框图。

图 6-37 展示了 80 引脚 LQFP 的模拟子系统方框图。

图 6-38 显示了 64 引脚 LQFP 的模拟子系统方框图。

图 6-39 展示了 48 引脚 LQFP 的模拟子系统方框图。

图 6-40 显示了模拟组连接。第 6.13.1 节列出了模拟引脚和内部连接。第 6.13.2 节列出了模拟信号的说明。

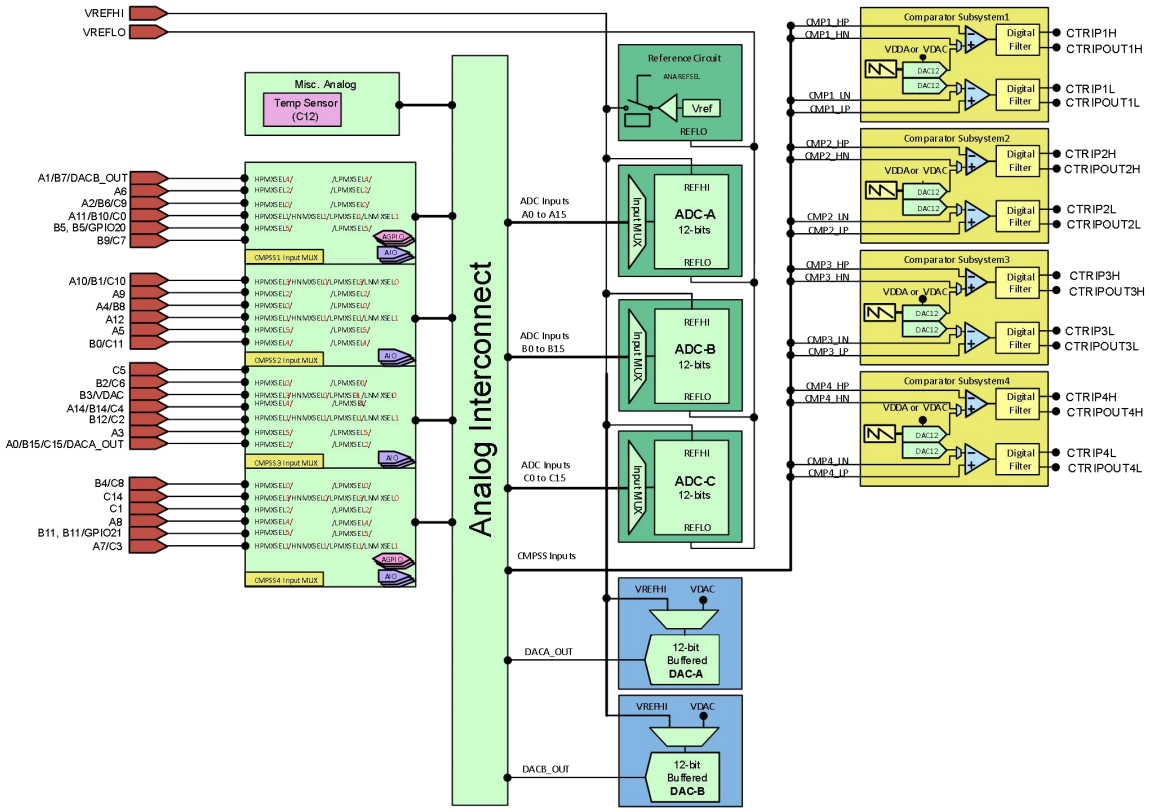


Figure 6- 36 Analog Subsystem Block Diagram (100-Pin LQFP)

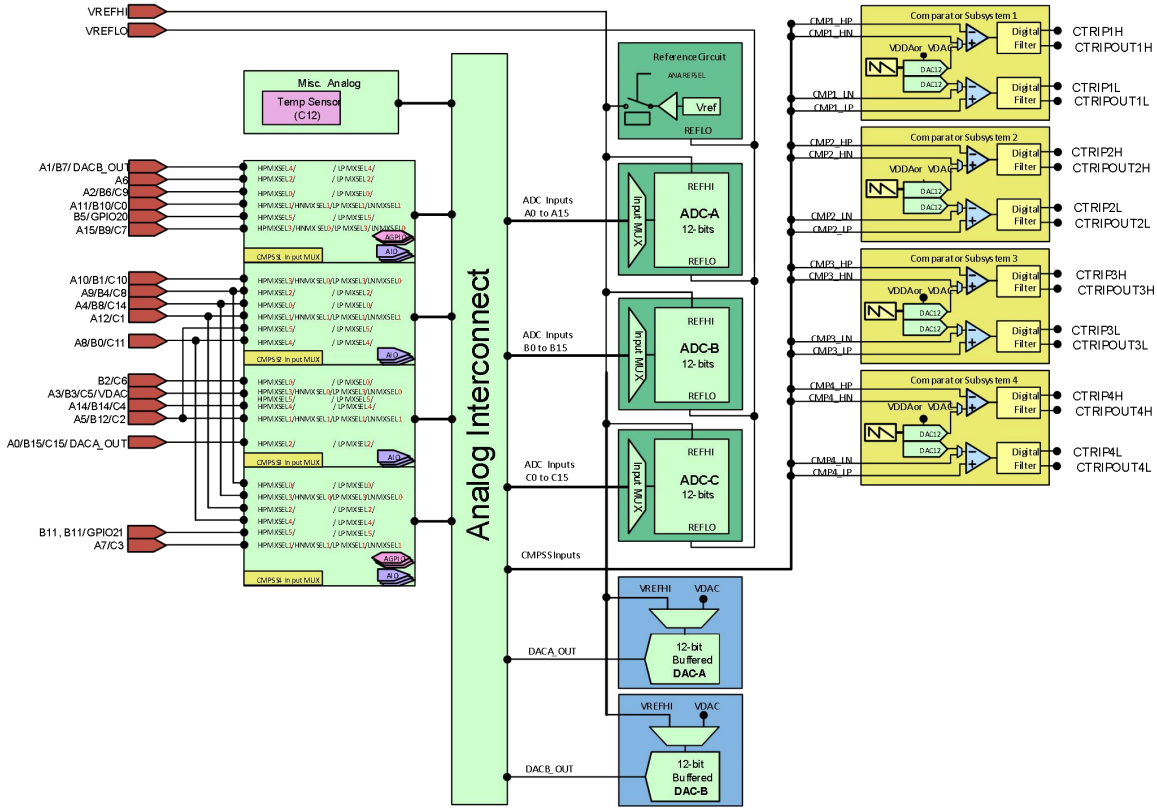


Figure 6- 37 Analog Subsystem Block Diagram (80-Pin LQFP)

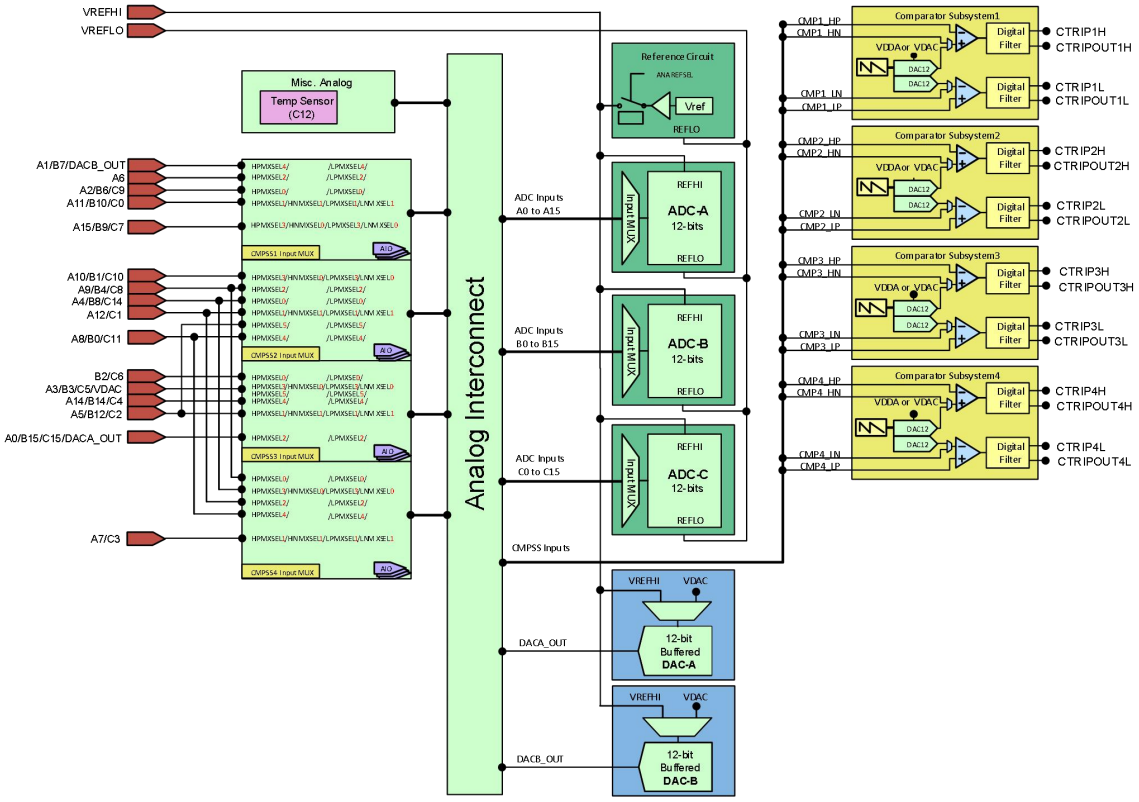


Figure 6- 38 Analog Subsystem Block Diagram (64-Pin LQFP)

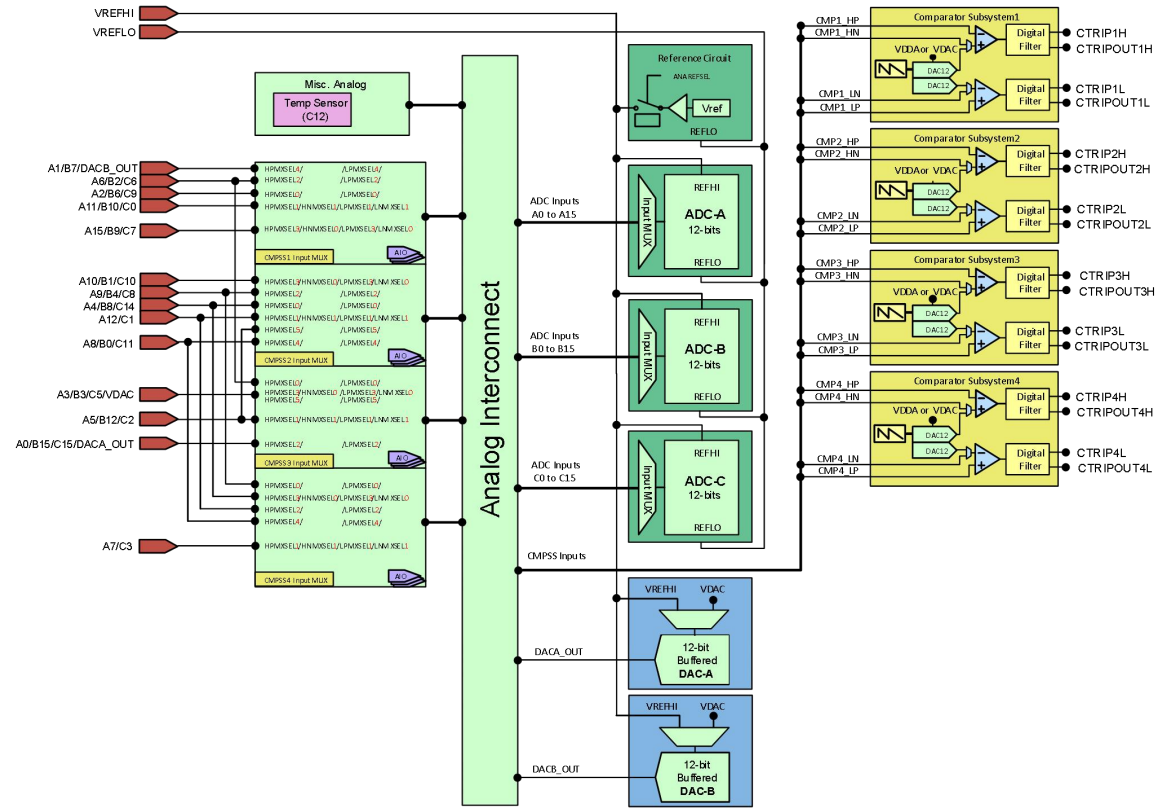
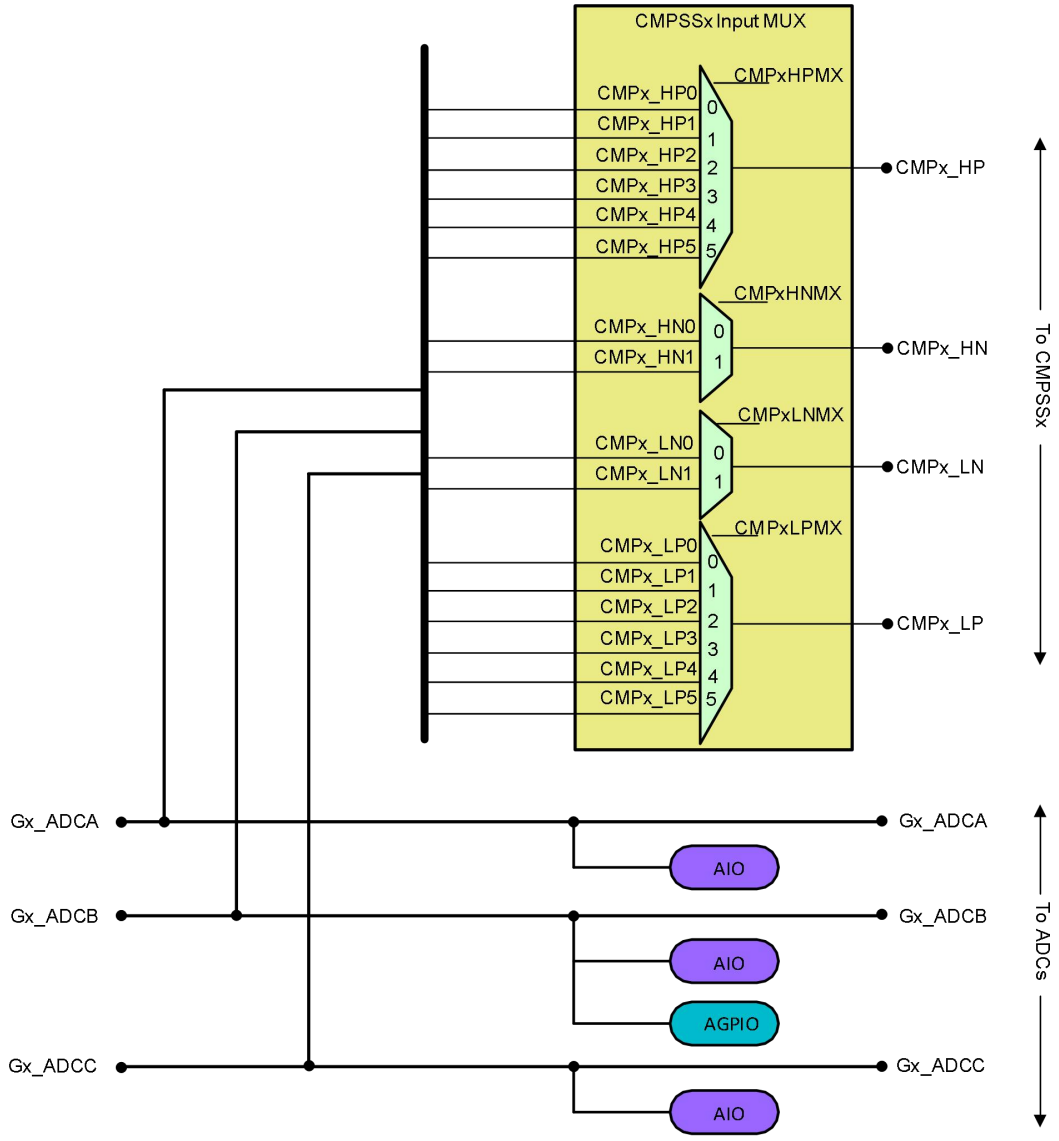


Figure 6- 39 Analog Subsystem Block Diagram (48-Pin LQFP)



A. AIOs support digital input mode only.
AIO 仅支持数字输入模式。

Figure 6- 40 Analog Group Connections

6.13.1 Analog Pins and Internal Connections
模拟引脚和内部连接

Table 6-8 Analog Pins and Internal Connections

Pin Name	Package Pin				ADC			Comparator Subsystem (MUX)				AIO Input
	100QP	80QP	64QP	48QP	A	B	C	High Positive	High Negative	Low Positive	Low Negative	
VREFHI	24, 25	20	16	12								
VREFLO	26, 27	21	17	13	A13	B13	C13					
Analog Group 1								**CMP1**				
A6	14	10	6	4(1)	A6	-	-	CMP1(HPMXSEL=2)		CMP1(LPMXSEL=2)		AIO228
A2/B6/C9	17	13	9	6	A2	B6	C9	CMP1(HPMXSEL=0)		CMP1(LPMXSEL=0)		AIO224
A15	-	14	10	7	A15	-	-	CMP1(HPMXSEL=3)	CMP1(HNMXSEL=0)	CMP1(LPMXSEL=3)	CMP1(LNMXSEL=0)	AIO233
B9/C7	18				-	B9	C7					
A11/B10/C0	20	16	12	8	A11	B10	C0	CMP1(HPMXSEL=1)	CMP1(HNMXSEL=1)	CMP1(LPMXSEL=1)	CMP1(LNMXSEL=1)	AIO237
A1/B7/DACB_OUT	22	18	14	10	A1	B7	-	CMP1(HPMXSEL=4)		CMP1(LPMXSEL=4)		AIO232
Analog Group 2								**CMP2**				
A10/B1/C10	40	29	25	21	A10	B1	C10	CMP2(HPMXSEL=3)	CMP2(HNMXSEL=0)	CMP2(LPMXSEL=3)	CMP2(LNMXSEL=0)	AIO230
Analog Group 3								**CMP3**				
B2/C6	15	11	7	4(1)	-	B2	C6	CMP3(HPMXSEL=0)		CMP3(LPMXSEL=0)		AIO226
B3/VDAC(2)	16	12	8	5	-	B3	-	CMP3(HPMXSEL=3)	CMP3(HNMXSEL=0)	CMP3(LPMXSEL=3)	CMP3(LNMXSEL=0)	AIO242
C5	28				-	-	C5					
A3	-				A3	-	-	CMP3(HPMXSEL=5)		CMP3(LPMXSEL=5)		
	18	-	-	-		-	-					AIO229
A14/B14/C4	19	15	11	-	A14	B14	C4	CMP3(HPMXSEL=4)		CMP3(LPMXSEL=4)		AIO239
A0/B15/C15/DACA_OUT	23	19	15	11	A0	B15	C15	CMP3(HPMXSEL=2)		CMP3(LPMXSEL=2)		AIO231
Analog Group 4								**CMP4**				
A7/C3	31	23	19	15	A7	-	C3	CMP4(HPMXSEL=1)	CMP4(HNMXSEL=1)	CMP4(LPMXSEL=1)	CMP4(LNMXSEL=1)	AIO245
Combined Analog Group 2/3								**CMP2/3**				
A5	35	-	-	-	A5	-	-	CMP2(HPMXSEL=5)		CMP2(LPMXSEL=5)		AIO249
	-					-	-					
B12/C2	21	17	13	9	-	B12	C2	CMP3(HPMXSEL=1)	CMP3(HNMXSEL=1)	CMP3(LPMXSEL=1)	CMP3(LNMXSEL=1)	AIO244
Combined Analog Group 2/4								**CMP2/4**				
A12	28	22	18	14	A12	-	-	CMP2(HPMXSEL=1)	CMP2(HNMXSEL=1)	CMP2(LPMXSEL=1)	CMP2(LNMXSEL=1)	AIO238
C1	29				-	-	C1	CMP4(HPMXSEL=2)		CMP4(LPMXSEL=2)		AIO248

Table 6-8. Analog Pins and Internal Connections (continued)

Pin Name	Package Pin				ADC			Comparator Subsystem (MUX)				AIO Input
	100QP	80QP	64QP	48QP	A	B	C	High Positive	High Negative	Low Positive	Low Negative	
A8	37	-	-	-	A8	-	-	CMP4(HPMXSEL=4)		CMP4(LPMXSEL=4)		AIO240
	-	24	20	16		-	-					AIO241
B0/C11	-				-	B0	C11	CMP2(HPMXSEL=4)		CMP2(LPMXSEL=4)		
	41	-	-	-	-							AIO253
A4/B8	36	27	23	19	A4	B8	-	CMP2(HPMXSEL=0)		CMP2(LPMXSEL=0)		AIO225
C14	-				-	-	C14	CMP4(HPMXSEL=3)	CMP4(HNMXSEL=0)	CMP4(LPMXSEL=3)	CMP4(LNMXSEL=0)	
	42	-	-	-	-	-						AIO247
A9	38	28	24	20	A9	-	-	CMP2(HPMXSEL=2)		CMP2(LPMXSEL=2)		AIO227
B4/C8	39				-	B4	C8	CMP4(HPMXSEL=0)		CMP4(LPMXSEL=0)		AIO236
Other Analog												
B5	32	-	-	-	-	B5	-	CMP1(HPMXSEL=5)		CMP1(LPMXSEL=5)		AIO252
B5/GPIO20(3)	48	33	-	-	-		-					GPIO20
B11	30	-	-	-	-	B11	-	CMP4(HPMXSEL=5)		CMP4(LPMXSEL=5)		AIO251
B11/GPIO21(3)	49	34	-	-	-		-					GPIO21
温度传感器(4)	-	-	-	-	-	-	C12					

(1) A6 and C6 is double bonded as pin # 4.
A6 和C6 键合为引脚 4。

(2) Optional external reference voltage for on-chip COMPDACs/GPDACs. There is an internal capacitance to VSSA on this pin whether used for ADC input or COMPDAC/GPDAC reference. If used as a VDAC reference, place at least a 1-µF capacitor on this pin.
片上COMPDAC/GPDAC 的可选外部基准电压。无论用于 ADC 输入还是 COMPDAC/GPDAC 基准，该引脚上都有一个连接到 VSSA 的内部电容。如果用作 VDAC 基准，请在该引脚上至少放置一个 1µF 电容器。

(3) The GPIOs on these analog pins support full digital input and output functionality and are referred to as AGPIOs. By default, the AGPIOs are unconnected; that is, the analog and digital functions are both disabled. For configuration details, see the *Digital Inputs and Outputs on ADC Pins (AGPIOs)* section.
这些模拟引脚上的 GPIO 支持完整的数字输入和输出功能，称为AGPIO。默认情况下，AGPIO 处于未连接状态；也就是说，模拟和数字功能都被禁用。有关配置详细信息，请参阅*ADC 引脚上的数字输入和输出(AGPIO)*一节。

(4) Internal connection only; does not come to a device pin.
仅限内部连接；不连接到器件引脚。

Table 6-9 Analog Signal Descriptions

Signal Name	Description
AIOx	Digital input on ADC pin
GPIOx	Digital input/output pin with ADC functionality
Ax	ADC A Input
Bx	ADC B Input
Cx	ADC C Input
CMPx_DACH	Comparator subsystem high DAC output
CMPx_DACL	Comparator subsystem low DAC output
CMPx_HNy	Comparator subsystem high comparator negative input
CMPx_HPy	Comparator subsystem high comparator positive input
CMPx_LNy	Comparator subsystem low comparator negative input
CMPx_LPy	Comparator subsystem low comparator positive input
DACx_OUT	Buffered DAC Output
TempSensor	Internal temperature sensor
VDAC	Optional external reference voltage for on-chip COMPDACs. This pin has a higher capacitance compared to the other analog pins. *See the Per-Channel Parasitic Capacitance* table for details. This capacitance is present whether the pin is being used for ADC input or COMPDAC/GPDAC reference and cannot be disabled. If this pin is being used as a reference for the on-chip COMPDAC/GPDACs, place at least a 1-µF capacitor on this pin.

6.13.2 Analog-to-Digital Converter (ADC)
模数转换器(ADC)

The ADC module described here is a successive approximation (SAR) style ADC with resolution of 12 bits. This section refers to the analog circuits of the converter as the “core,” and includes the channel-select MUX, the sample-and-hold (S/H) circuit, the successive approximation circuits, voltage reference circuits, and other analog support circuits. The digital circuits of the converter are referred to as the “wrapper” and include logic for programmable conversions, result registers, interfaces to analog circuits, interfaces to the peripheral buses, post-processing circuits, and interfaces to other on-chip modules.
Each ADC module consists of a single sample-and-hold (S/H) circuit. The ADC module is designed to be duplicated multiple times on the same chip, allowing simultaneous sampling or independent operation of multiple ADCs.

此处描述的 ADC 模块是分辨率为 12 位的逐次逼近(SAR)型ADC。本节将转换器的模拟电路称为“内核”，包括通道选择 MUX、采样保持(S/H)电路、逐次逼近电路、电压基准电路和其他模拟支持电路。转换器的数字电路被称为“包装器”，包括用于可编程转换的逻辑、结果寄存器、模拟电路接口、外设总线接口、后处理电路以及其他片上模块接口。

每个 ADC 模块都包含一个采样保持(S/H)电路。ADC 模块被设计成在同一个芯片上重复多次，从而实现多个 ADC 的同步采样或独立运行。

Each ADC has the following features:

- Resolution of 12 bits
- Ratiometric external reference set by VREFHI/VREFLO
- Selectable internal reference of 2.5 V or 3.3 V
- Single-ended signaling
- Input multiplexer with up to 16 channels
- 16 configurable SOCs
- 16 individually addressable result registers
- Multiple trigger sources
  - S/W: software immediate start
  - All ePWMs: ADCSOC A or B
  - GPIO XINT2
  - CPU Timers 0/1/2
  - ADCINT1/2
- Four flexible PIE interrupts
- Burst-mode triggering option
- Four post-processing blocks, each with:
  - Saturating offset calibration
  - Error from setpoint calculation
  - High, low, and zero-crossing compare, with interrupt and ePWM trip capability
  - Trigger-to-sample delay capture

Note

Not every channel may be pinned out from all ADCs. See the Pin Configuration and Functions section to determine which channels are available.

每个ADC 具有以下特性：

- 分辨率：12 位
- 由 VREFHI/VREFLO 设定的比例式外部基准
- 2.5V 或 3.3V 的可选内部基准电压

- 单端信号指示
- 多达 16 个通道的输入多路复用器
- 16 个可配置 SOC
- 16 个可单独寻址的结果寄存器
- 多个触发源
- S/W：软件立即启动
- 所有 ePWM：ADCSOCA 或 B
- GPIOXINT2
- CPU 定时器 0/1/2
- ADCINT1/2
- 四个灵活的 PIE 中断
- 突发模式触发选项
- 四个后处理块，每块具有：
  - 饱和偏移量校准
  - 设定点计算的误差
  - 具有中断和 ePWM 跳变功能的高电平、低电平和过零比较
  - 触发至采样延迟采集

备注

并非每个通道都可以从所有 ADC 输出引脚。请参阅“引脚配置和功能”部分以确定哪些通道可用

The block diagram for the ADC core and ADC wrapper are shown in Figure 6-41.
ADC 内核和ADC 包装器的方框图如图 6-41 所示。

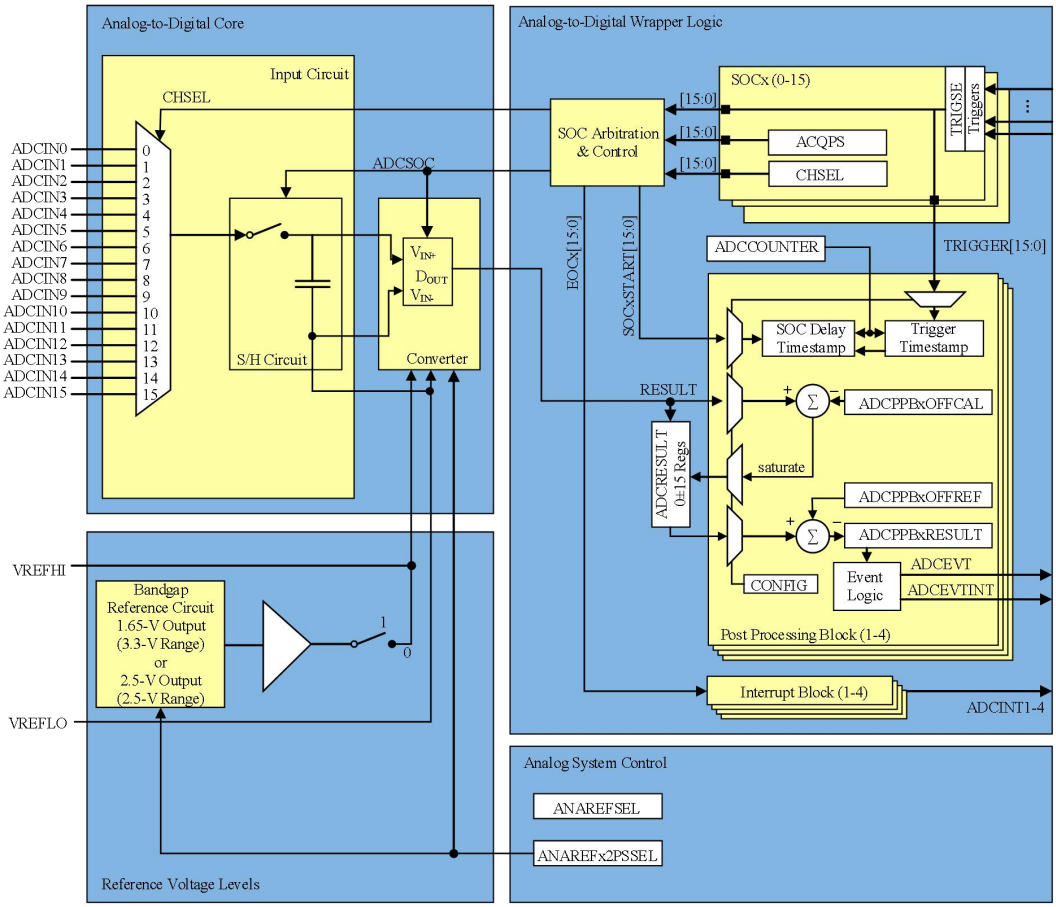


Figure 6-41 ADC Module Block Diagram

6.13.2.1 ADC ConfigurabilityADC 可配置性

Some ADC configurations are individually controlled by the SOCs, while others are globally controlled per ADC module. Table 6-10 summarizes the basic ADC options and their level of configurability.

一些ADC 配置由 SOC 单独控制，而其他配置则由每个 ADC 模块全局控制。表6-10 汇总了基本的 ADC 选项及其可配置性级别。

Table 6-10 ADC Options and Configuration Levels

OPTIONS	CONFIGURABILITY
Clock	Per module(1)
Resolution	Not configurable (12-bit resolution only)
Signal mode	Not configurable (single-ended signal mode only)
Reference voltage source	Either external or internal for all modules
Trigger source	Per SOC(1)
Converted channel	Per SOC
Acquisition window duration	Per SOC(1)
EOC location	Per module
Burst mode	Per module(1)

(1) Writing these values differently to different ADC modules could cause the ADCs to operate asynchronously.
将这些值以不同方式写入不同 ADC 模块可能会导致 ADC 异步工作。

6.13.2.1.1 Signal Mode
信号模式

The ADC supports single-ended signaling. The input voltage to the converter is sampled through a single pin (ADCINx), referenced to VREFLO.

ADC 支持单端信号模式。以 VREFLO 为基准通过单个引脚(ADCINx)对转换器的输入电压进行采样。

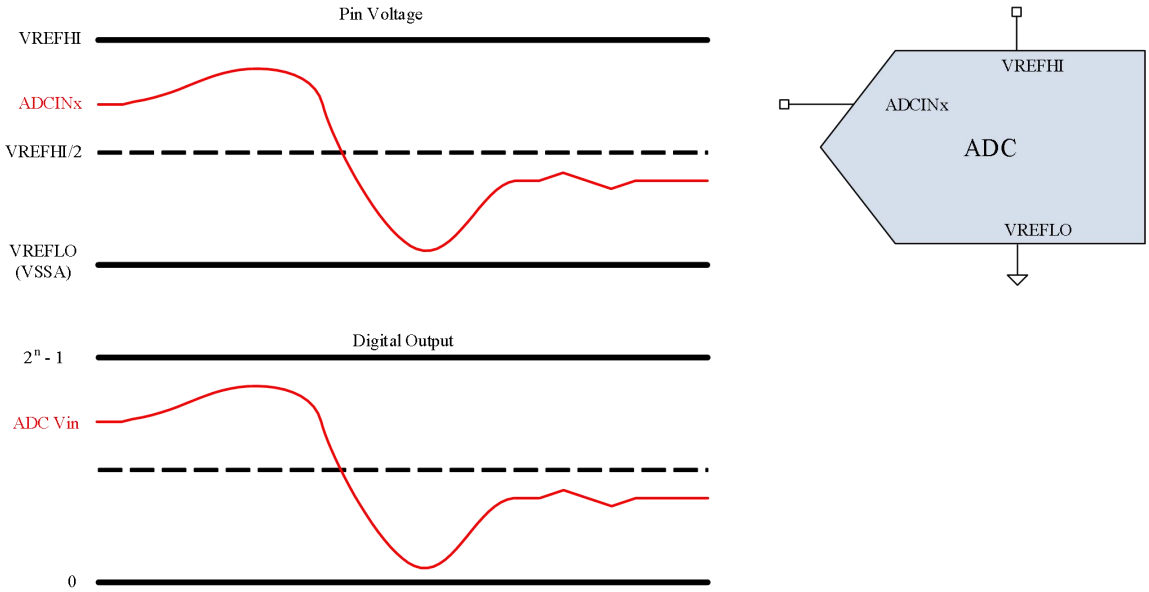


Figure 6- 42 Single-ended Signaling Mode

6.13.2.2 ADC 电气数据和时序
ADC Electrical Data and Timing

Note

The ADC inputs should be kept below VDDA + 0.3 V during operation. If an ADC input exceeds this level, the V_{REF} internal to the device may be disturbed, which can impact results for other ADC inputs using the same V_{REF}.

工作过程中，ADC 输入应保持低于 VDDA+0.3V。如果 ADC 输入超过此电平，器件内部的 VREF 可能会受到干扰，这可能会影响使用相同 VREF 的其他 ADC 输入的结果。

Note

The VREFHI pin must be kept below VDDA + 0.3 V to ensure proper functional operation. If the VREFHI pin exceeds this level, a blocking circuit may activate, and the internal value of VREFHI may float to 0 V internally, giving improper ADC conversion.

VREFHI 引脚必须保持低于 VDDA+0.3V，以确保正常工作。如果 VREFHI 引脚超过此电平，可能会激活阻塞电路，并且 VREFHI 的内部值可能会在内部浮动至 0V，从而导致 ADC 转换不正确。

6.13.2.2.1 ADC Operating Conditions
ADC 运行条件

over operating free-air temperature range (unless otherwise noted)

在自然通风条件下的工作温度范围内测得（除非另有说明）

PARAMETER	TEST CONDITIONS	MIN	TYP	MAX	UNIT
ADCCLK (derived from PERx.SYSCLK)		5		60	MHz
Sample rate	120MHzSYSCLK			4	MSPS
	120MHzSYSCLK (AGPIO Pin)			3.75	MSPS
Sample window duration (set by ACQPS and PERx.SYSCLK)(1)	With 50 Ω or less R_s	75			ns
	With 50 Ω or less R_s (AGPIO Pin)	90			ns
VREFHI	External Reference	2.4	2.5 或3.0	VDDA	V
VREFHI(2)	Internal Reference = 3.3V Range		1.65		V
	Internal Reference = 2.5V Range		2.5		V
VREFLO		VSSA		VSSA	V
VREFHI-VREFLO		2.4		VDDA	V
Conversion range	Internal Reference = 3.3 V Range	0		3.3	V
	Internal Reference = 2.5 V Range	0		2.5	V
	External Reference	VREFLO		VREFHI	V

(1) The sample window must also be at least as long as 1 ADCCLK cycle for correct ADC operation.
采样窗口还必须至少达到 1 个ADCCLK 周期的长度，才能确保 ADC 正确运行。

(2) In internal reference mode, the reference voltage is driven out of the VREFHI pin by the device. The user should not drive a voltage into the pin in this mode.
在内部基准模式下，基准电压由器件从 VREFHI 引脚驱动。在此模式下，用户不应将电压驱动到引脚中。

6.13.2.2.2 ADC Characteristics
ADC 特性

over operating free-air temperature range (unless otherwise noted)
在自然通风条件下的工作温度范围内测得（除非另有说明）

PARAMETER	TEST CONDITIONS	MIN	TYP	MAX	UNIT
General					
ADCCLK Conversion Cycles	120MHzSYSCLK	10.1		11	ADCCLK
Power Up Time	External Reference mode			500	µs
	Internal Reference mode			5500	µs
	Internal Reference mode, when switching between 2.5-V range and 3.3-V range.			5000	µs
VREFHI input current(1)			73.5		µA
Internal Reference Capacitor Value(2)		2.2			µF
External Reference Capacitor Value(2)		2.2			µF
DC Characteristics					
Gain Error	Internal reference	–45		45	LSB
	External reference	–15	–9	5	
Offset Error		–5	±2	5	LSB
Channel-to-Channel Gain Error(4)			10		LSB
Channel-to-Channel Offset Error(4)			2		LSB
ADC-to-ADC Gain Error(5)	Identical VREFHI and VREFLO for all ADCs		4		LSB
ADC-to-ADC Offset Error(5)	Identical VREFHI and VREFLO for all ADCs		2		LSB
DNL Error		–1	±0.5	2	LSB
INL Error		–4	±2	4	LSB
ADC-to-ADC Isolation	VREFHI = 2.5 V, synchronous ADCs	–1		1	LSB
AC Characteristics					
SNR(3)	External VREFHI/Internal VREFHI = 2.5 V, fin = 100 kHz, SYSCLK from X1		67.9		dB
	Internal VREFHI = 1.65 V (0 to 3.3 V range), fin = 100 kHz, SYSCLK from X1		67.6		dB
	External/Internal VREFHI, fin = 100 kHz, SYSCLK from INTOSC		60.1		
THD(3)	External VREFHI/Internal VREFHI = 2.5 V, fin = 100 kHz, SYSCLK from X1		73.4		dB
	Internal VREFHI = 1.65 V (0 to 3.3 V range), fin = 100 kHz, SYSCLK from X1		71.2		dB
SFDR(3)	External/Internal VREFHI , fin = 100 kHz		76.7		dB
SINAD(3)	External VREFHI/Internal VREFHI = 2.5V, fin = 100 kHz, SYSCLK from X1		66.8		dB
	Internal VREFHI = 1.65 V (0 to 3.3 V range), fin = 100 kHz, SYSCLK from X1		66		dB
	External/Internal VREFHI, fin = 100 kHz, SYSCLK from INTOSC		60		
ENOB(3)	External VREFHI/Internal VREFHI = 2.5 V, fin = 100 kHz, SYSCLK from X1, single and synchronous ADCs		10.8		位
	Internal VREFHI = 1.65 V (0 to 3.3 V range), fin = 100 kHz, SYSCLK from X1, single and synchronous ADCs		10.7		

PARAMETER	TEST CONDITIONS	MIN	TYP	MAX	UNIT
	Any VREF mode, fin = 100 kHz, SYSCLK from X1, asynchronous ADCs		Not Supported		
PSRR	VDD = 1.1-V DC + 100mV DC up to Sine at 1 kHz		60		dB
	VDD = 1.1-V DC + 100 mV DC up to Sine at 300 kHz		57		
	VDDA = 3.3-V DC + 200 mV DC up to Sine at 1 kHz		60		
	VDDA = 3.3-V DC + 200 mV Sine at 900 kHz		57		

(1) Load current on VREFHI increases when ADC input is greater than VDDA. This causes inaccurate conversions.
当ADC 输入大于VDDA 时，VREFHI 上的负载电流会增加。这会导致转换不准确。

(2) A ceramic capacitor with package size of 0805 or smaller is preferred. Up to ±20% tolerance is acceptable.
最好使用封装尺寸为 0805 或更小的陶瓷电容器。可接受高达±20%的容差。

(3) IO activity is minimized on pins adjacent to ADC input and VREFHI pins as part of best practices to reduce capacitive coupling and crosstalk.
作为减少电容耦合和串扰的最佳实践的一部分，与ADC 输入和VREFHI 引脚相邻的引脚上的 IO 活动已尽可能减少。

(4) Variation across all channels belonging to the same ADC module.
同一ADC 模块的所有通道之间的差异。

(5) Worst case variation compared to other ADC modules.
与其他ADC 模块相比的最坏情况变化。

6.13.2.2.3 ADC Input Model
ADC 输入模型

The ADC input characteristics are given by Table 6-11 and Figure 6-43.
表 6-11 和图 6-43 给出了 ADC 输入特性。

Table 6-11 Input Model Parameters

	DESCRIPTION	REFERENCE MODE	VALUE
C_p	Parasitic input capacitance	All	See Table 6-12 to Table 6-15
R_{on}	Sampling switch resistance	External Reference, 2.5-V Internal Reference	500 Ω
		3.3-V Internal Reference	860 Ω
C_h	Sampling capacitor	External Reference, 2.5-V Internal Reference	12.5 pF
		3.3-V Internal Reference	7.5 pF
R_s	Nominal source impedance	All	50 Ω

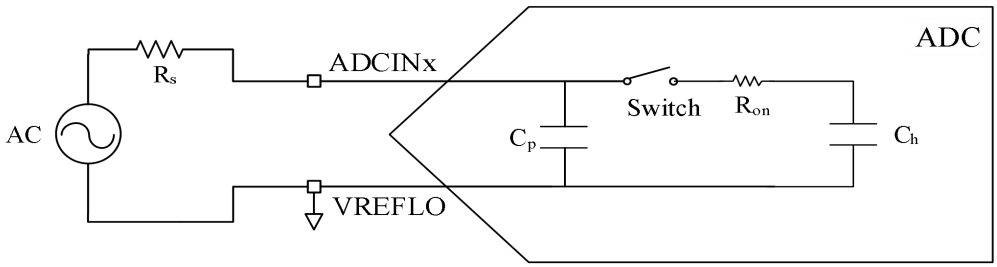


Figure 6-43 Input Model

This input model should be used with actual signal source impedance to determine the acquisition window duration.
应将此输入模型与实际信号源阻抗配合使用，以确定采集窗口持续时间。

Table 6-12 Per-Channel Parasitic Capacitance for 100-Pin LQFP

ADC CHANNEL	C_p(pF)	
	COMPARATOR DISABLED	COMPARATOR ENABLED
A0/B15/C15/DACA_OUT	9.1	11.6
A1/B7/DACB_OUT	7.4	9.9
A2/B6/C9	4.1	6.6
A3	3.3	5.8
A4/B8	3.8	6.3
A5	3.5	6
A6	3.2	5.7
A7/C3	3.8	6.3
A8	4.1	6.6
A9	3.1	5.6
A10/B1/C10	4.7	7.2
A11/B11/C0	4	6.5
A12	3.4	5.9
A14/B14/C4	3.8	6.3
B0/C11	4.1	6.6
B2/C6	3.9	6.4
B3/VDAC	75	77.5
B4/C8	3.8	6.3
B5	3.5	6
B9/C7	3.3	5.8
B11	3	5.5
B12/C2	3.6	6.1
C1	3	5.5
C5	3.6	6.1
C14	4.2	6.7
AGPIO_B5	3.2	5.7
AGPIO_B11	3.1	5.6

Table 6-13 Per-Channel Parasitic Capacitance for 80-Pin LQFP

ADC CHANNEL	C_p(pF)	
	COMPARATOR DISABLED	COMPARATOR ENABLED
A0/B15/C15/DACA_OUT	9.1	11.6
A1/B7/DACB_OUT	7.4	9.9
A2/B6/C9	4.1	6.6
A3/B3/C5/VDAC	81.9	89.4
A4/B8/C14	8	13
A5/B12/C2	7.1	12.1
A6	3.2	5.7
A7/C3	3.8	6.3
A8/B0/C11	8.2	13.2
A9/B4/C8	6.9	11.9
A10/B1/C10	4.7	7.2
A11/B11/C0	4	6.5
A12/C1	6.4	11.4

ADC CHANNEL	C_p(pF)	
	COMPARATOR DISABLED	COMPARATOR ENABLED
A14/B14/C4	3.8	6.3
A15/B9/C7	7.1	12.1
B2/C6	3.9	6.4
AGPIO_B5	3.2	5.7
AGPIO_B11	3.1	5.6

Table 6-14 Per-Channel Parasitic Capacitance for 64-Pin LQFP

ADC CHANNEL	C_p(pF)	
	COMPARATOR DISABLED	COMPARATOR ENABLED
A0/B15/C15/DACA_OUT	9.1	11.6
A1/B7/DACB_OUT	7.4	9.9
A2/B6/C9	4.1	6.6
A3/B3/C5/VDAC	81.9	89.4
A4/B8/C14	8	13
A5/B12/C2	7.1	12.1
A6	3.2	5.7
A7/C3	3.8	6.3
A8/B0/C11	8.2	13.2
A9/B4/C8	6.9	11.9
A10/B1/C10	4.7	7.2
A11/B11/C0	4	6.5
A12/C1	6.4	11.4
A14/B14/C4	3.8	6.3
A15/B9/C7	7.1	12.1
B2/C6	3.9	6.4

Table 6-15 Per-Channel Parasitic Capacitance for 48-Pin LQFP

ADC CHANNEL	C_p(pF)	
	COMPARATOR DISABLED	COMPARATOR ENABLED
A0/B15/C15/DACA_OUT	9.1	11.6
A1/B7/DACB_OUT	7.4	9.9
A2/B6/C9	4.1	6.6
A3/B3/C5/VDAC	81.9	89.4
A4/B8/C14	8	13
A5/B12/C2	7.1	12.1
A6/B2/C6	3.2	5.7
A7/C3	3.8	6.3
A8/B0/C11	8.2	13.2
A9/B4/C8	6.9	11.9
A10/B1/C10	4.7	7.2
A11/B11/C0	4	6.5
A12/C1	6.4	11.4
A15/B9/C7	3.8	6.3

6.13.2.2.4 ADC Timing Diagrams
ADC 时序图

Figure 6-44 shows the ADC conversion timings for two SOCs given the following assumptions:

- SOC0 and SOC1 are configured to use the same trigger.
- No other SOCs are converting or pending when the trigger occurs.
- The round-robin pointer is in a state that causes SOC0 to convert first.
- ADCINTSEL is configured to set an ADCINT flag upon end of conversion for SOC0 (whether this flag propagates through to the CPU to cause an interrupt is determined by the configurations in the PIE module).

Table 6-16 lists the descriptions of the ADC timing parameters. Table 6-17 lists the ADC timings.

图 6-44 显示了在下列假设下两个 SOC 的 ADC 转换时序：

- SOC0 和 SOC1 配置为使用相同的触发器。
- 触发发生时，没有其他 SOC 正在转换或挂起。
- 轮循指针处于使 SOC0 首先转换的状态。
- ADCINTSEL 配置为在 SOC0 的转换结束时设置一个 ADCINT 标志（该标志是否传播到 CPU 以引起中断由 PIE 模块中的配置决定）。

表 6-16 列出了 ADC 时序参数的说明。表 6-17 列出了 ADC 时序。

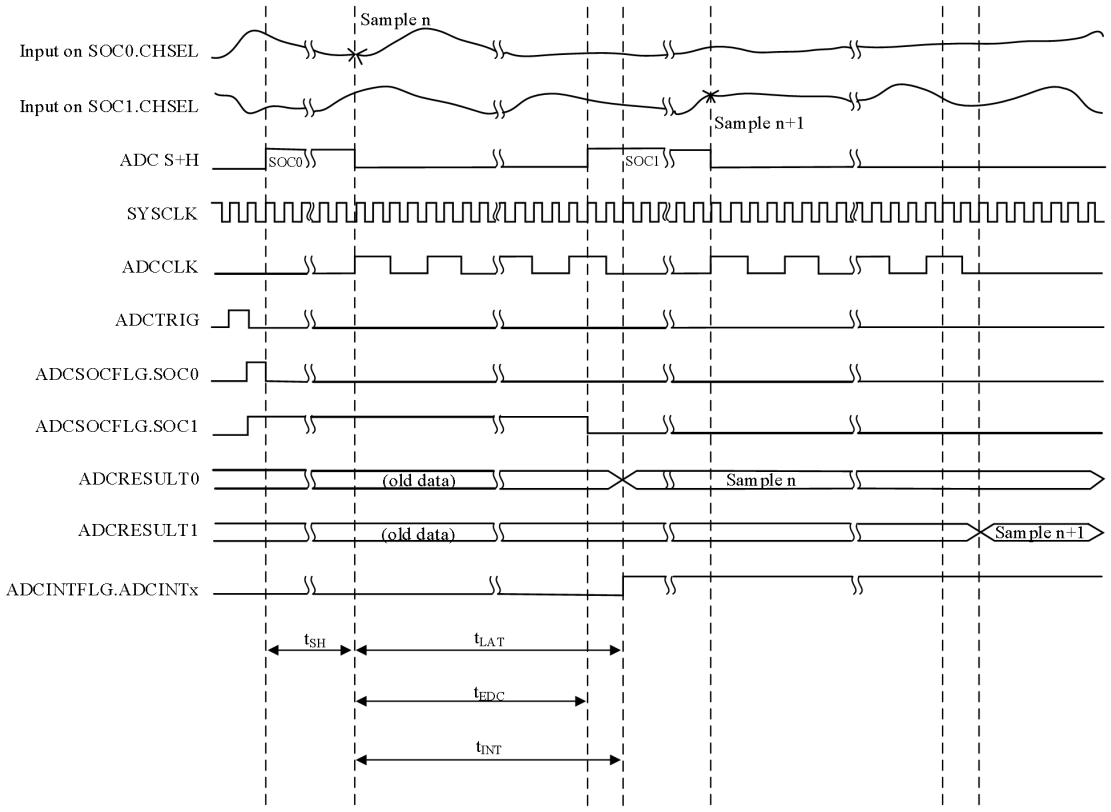


Figure 6-44 ADC Timings

Table 6-16 ADC Timing Parameters

PARAMETER	DESCRIPTION
t_{SH}	The duration of the S+H window. At the end of this window, the value on the S+H capacitor becomes the voltage to be converted into a digital value. The duration is given by (ACQPS + 1) SYSCLK cycles. ACQPS can be configured individually for each SOC, so t_{SH} will not necessarily be the same for different SOCs. **Note:** The value on the S+H capacitor will be captured approximately 5 ns before the end of the S+H window regardless of device clock settings.
t_{LAT}	The time from the end of the S+H window until the ADC results latch in the ADCRESULTx register. If the ADCRESULTx register is read before this time, the previous conversion results will be returned.
t_{EOC}	The time from the end of the S+H window until the S+H window for the next ADC conversion can begin. The subsequent sample can start before the conversion results are latched.
t_{INT}	The time from the end of the S+H window until an ADCINT flag is set (if configured). If the INTPULSEPOS bit in the ADCCTL1 register is set, t_{INT} will coincide with the conversion results being latched into the result register. If the INTPULSEPOS bit is 0, t_{INT} will coincide with the end of the S+H window. If t_{INT} triggers a read of the ADC result register (directly through DMA or indirectly by triggering an ISR that reads the result), care must be taken to ensure the read occurs after the results latch (otherwise, the previous results will be read). If the INTPULSEPOS bit is 0, and the OFFSET field in the ADCINTCYCLE register is not 0, then there will be a delay of OFFSET SYSCLK cycles before the ADCINT flag is set. This delay can be used to enter the ISR or trigger the DMA at exactly the time the sample is ready.

Table 6-17 ADC Timings

ADCCLK PRESCALE		SYSCLK CYCLES				ADCCLK CYCLES
ADCCTL2 [PRESCALE]	RATIO ADCCLK:SYSCLK	t_{EOC}	t^{LAT}	$t_{INT(EARLY)}$ (1)	$t_{INT(LATE)}$	t_{EOC}
0	1	11	13	1	11	11
2	2	21	23	1	21	10.5
4	3	31	34	1	31	10.3
6	4	41	44	1	41	10.3
8	5	51	55	1	51	10.2
10	6	61	65	1	61	10.2
12	7	71	76	1	71	10.1
14	8	81	86	1	81	10.1

(1) By default, t_{INT} occurs one SYSCLK cycle after the S+H window if INTPULSEPOS is 0. This can be changed by writing to the OFFSET field in the ADCINTCYCLE register.
默认情况下，如果INTPULSEPOS 为0，则t_{INT} 在S+H 窗口后的一个 SYSCLK 周期内发生。这可以通过写入 ADCINTCYCLE 寄存器的OFFSET 域来改变。

6.13.3 Temperature Sensor
温度传感器

6.13.3.1 Temperature Sensor Electrical Data and Timing
温度传感器电气数据和时序

The temperature sensor can be used to measure the device junction temperature. The temperature sensor is sampled through an internal connection to the ADC and translated into a temperature through software. When sampling the temperature sensor, the ADC must meet the acquisition time in the Temperature Sensor Characteristics table.

温度传感器可用于测量器件结温。温度传感器通过与 ADC 的内部连接进行采样，并通过软件转换为温度。在对温度传感器进行采样时，ADC 必须满足“温度传感器特性”表中的采集时间要求。

6.13.3.1.1 Temperature Sensor Characteristics
温度传感器特征

over recommended operating conditions (unless otherwise noted)
在建议运行条件下测得（除非另有说明）

PARAMETER		TEST CONDITION	MIN	TYP	MAX	UNIT
T_{acc}	Temperature Accuracy	External reference		±15		℃
$t_{startup}$	Start-up time (TSNSCTL[ENABLE] to sampling temperature sensor)			500		μs
t_{acq}	ADC acquisition time		450			ns

6.13.4 Comparator Subsystem (CMPSS)
比较器子系统(CMPSS)

The Comparator Subsystem (CMPSS) consists of analog comparators and supporting circuits that are useful for power applications such as peak current mode control, switched-mode power, power factor correction, voltage trip monitoring, and so forth.

The comparator subsystem is built around a number of modules. Each subsystem contains two comparators, two reference 12-bit DACs, and two digital filters. The subsystem also includes one ramp generator. Comparators are denoted "H" or "L" within each module where “H” and “L” represent high and low, respectively. Each comparator generates a digital output which indicates whether the voltage on the positive input is greater than the voltage on the negative input. The positive input of the comparator is driven from an external pin. The negative input can be driven by an external pin or by the programmable reference 12-bit DAC. Each comparator output passes through a programmable digital filter that can remove spurious trip signals. An unfiltered output is also available if filtering is not required. A ramp generator circuit is optionally available to control the reference 12-bit DAC value for the high comparator in the subsystem.

比较器子系统(CMPSS)由模拟比较器和支持电路组成，这些电路对于峰值电流模式控制、开关模式电源、功率因数校正、电压跳闸监控等电源应用非常有用。

比较器子系统基于多个模块构建而成。每个子系统包含两个比较器、两个基准 12 位DAC 和两个数字滤波器。该子系统还包括一个斜坡发生器。比较器在每个模块中用“H”或“L”表示，其中“H”代表高，“L”代表低。每个比较器都会生成一个数字输出，指示正输入端的电压是否大于负输入端的电压。比较器的正输入由一个 外 部 引 脚 驱 动 ，负输入可由外部引脚或可编程基准 12 位DAC 驱动。每个比较器输出都会通过一个可编程的数字滤波器，该滤波器可以去除伪跳变信号。如果不需要滤波，也可以使用未滤波的输出。一个斜坡发生器电路可用于控制该子系统中高比较器的基准 12 位DAC 值。

Each CMPSS includes:

- Two analog comparators
- Two programmable reference 12-bit DACs
- One ramp generator
- Ability to synchronize submodules with EPWMSYNCPER
- Ability to extend clear signal with EPWMBLANK
- Ability to synchronize output with SYSCLK
- Ability to latch output
- Ability to invert output
- Option to use hysteresis on the input
- Option for negative input of comparator to be driven by an external signal or by the reference DAC
- Option to choose between VDDA or VDAC to be the DAC reference voltage

每个CMPSS 包含：

- 两个模拟比较器
- 两个可编程基准 12 位 DAC
- 一个斜坡发生器
- 能够将各个子模块与 EPWMSYNCPER 同步
- 能够通过 EPWMBLANK 扩展清除信号
- 能够将输出与 SYSCLK 同步

- 能够锁存输出
- 能够反转输出
- 可选择在输入端使用迟滞
- 可选择通过外部信号或基准 DAC 驱动比较器的负输入
- 可选择 VDDA 或 VDAC 作为 DAC 基准电压

6.13.4.1 CMPSS Connectivity Diagram
CMPSS 连接图

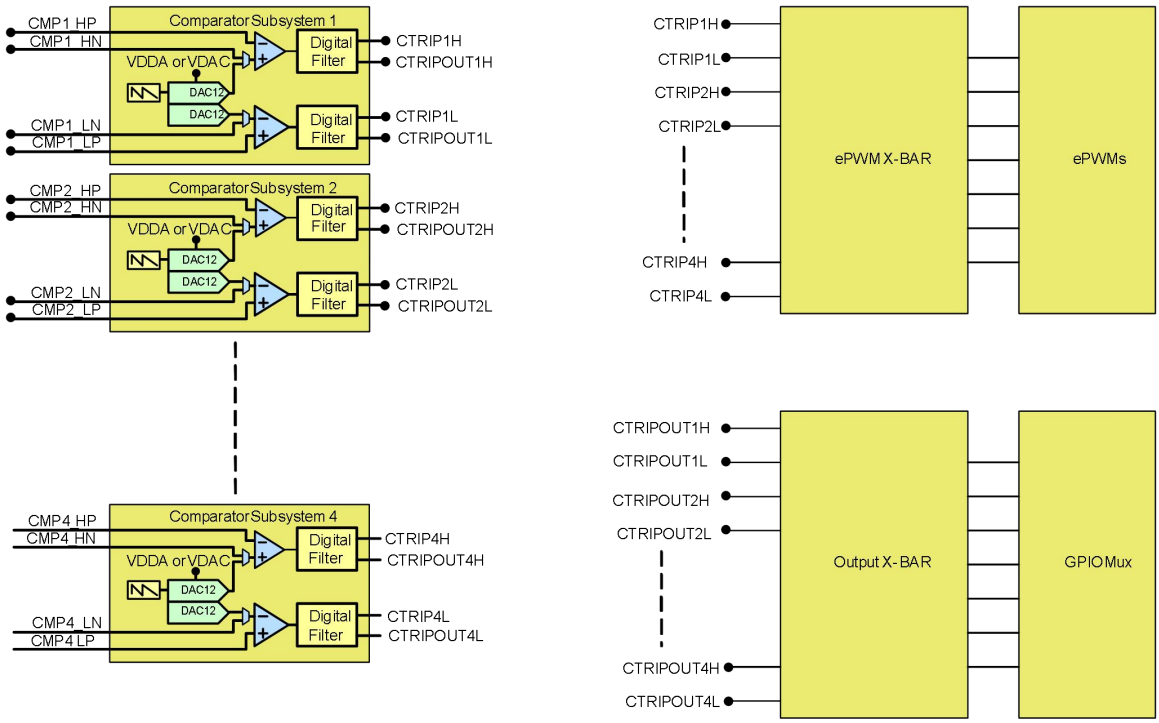


Figure 6- 45 CMPSS Connectivity

6.13.4.2 Block Diagram 框图

The block diagram for the CMPSS is shown in Figure 6- 46.

- CTRIPx(x= "H" or "L") signals are connected to the ePWM X-BAR for ePWM trip response.
- CTRIPxOUTx(x= "H" or "L") signals are connected to the Output X-BAR for external signaling.

CMPSS 框图如图 6-46 所示。

- CTRIPx（x="H"或"L"）信号连接到 ePWMX-BAR，用于 ePWM 跳闸响应。
- CTRIPxOUTx（x="H"或"L"）信号连接到输出 X-BAR，用于外部信号。

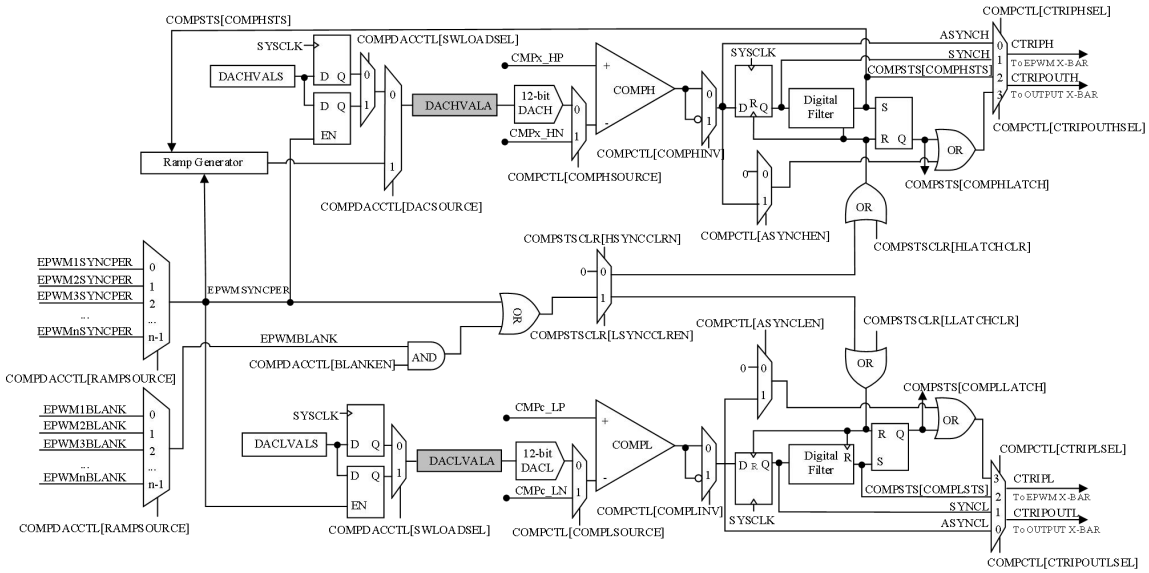


Figure 6- 46 CMPSS Module Block Diagram

6.13.4.3 CMPSS Electrical Data and Timing
CMPSS 电气数据和时序

6.13.4.3.1 Comparator Electrical Characteristics
比较器电气特性

over recommended operating conditions (unless otherwise noted)
在建议运行条件下测得（除非另有说明）

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
TPU	Power-up time				500	µs
Comparator input (CMPINxx) range			0		VDDA	V
Input referred offset error		Low common mode, inverting input set to 50mV	–20		20	mV
Hysteresis(1)		1x	4	12	20	LSB
		2x	17	24	33	
		3x	25	36	50	
		4x	30	48	67	
Response time (delay from CMPINx input pin change to GPIO output pin through either ePWM X-BAR or Output X-BAR)		Step response		21	60	ns
		Ramp response (1.65V/µs)		26		
		Ramp response (8.25mV/µs)		30		ns
PSRR	Power Supply Rejection Ratio	Up to 250 kHz		46		dB
CMRR	Common Mode Rejection Ratio		40			dB

(1) The CMPSS DAC is used as the reference to determine how much hysteresis to apply. Therefore, hysteresis will scale with the CMPSS DAC reference voltage. Hysteresis is available for all comparator input source configurations.
CMPSSDAC 用作确定应用多少迟滞的基准。因此，迟滞将随 CMPSSDAC 基准电压而变化。迟滞适用于所有比较器输入源配置。

CMPSS Comparator Input Referred Offset and Hysteresis
CMPSS 比较器以输入为基准的偏移量和迟滞

Note

The CMPSS inputs must be kept below VDDA + 0.3 V to ensure proper functional operation. If a CMPSS input exceeds this level, an internal blocking circuit isolates the internal comparator from the external pin until the external pin voltage returns below VDDA + 0.3 V. During this time, the internal comparator input is floating and can decay below VDDA within approximately 0.5 µs. After this time, the comparator could begin to output an incorrect result depending on the value of the other comparator input.

CMPSS 输入必须保持低于 VDDA+0.3V，以确保正常工作。如果 CMPSS 输入超过此电平，内部阻塞电路将内部比较器与外部引脚隔离，直至外部引脚电压返回到 VDDA+0.3V 以下。在此期间，内部比较器输入将处于悬空状态，并能在大约 0.5µs 内衰减至 VDDA 以下。在此之后，比较器可能会开始输出不正确的结果，具体取决于其他比较器输入的值。

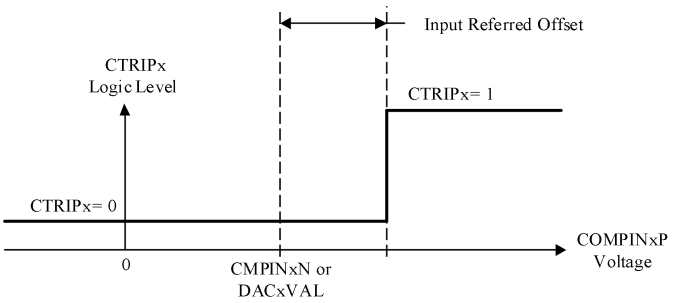


Figure 6- 47 CMPSS Comparator Input Referred Offset

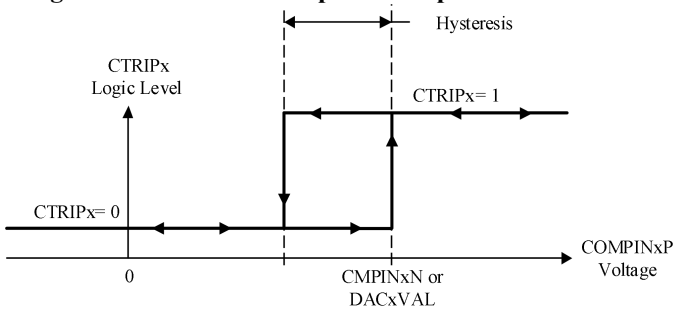


Figure 6- 48 CMPSS Comparator Hysteresis

6.13.4.3.2 CMPSS DAC Static Electrical Characteristics
CMPSSDAC 静态电气特性

over recommended operating conditions (unless otherwise noted)
在建议运行条件下测得（除非另有说明）

PARAMETER	TEST CONDITIONS	MIN	TYP	MAX	UNIT
CMPSS DAC output range	Internal reference	0		VDDA	V
	External reference	0		VDAC(4)	
Static offset error(1)		–25		25	mV
Static gain error(1)		–2		2	% of FSR
Static DNL	Endpoint corrected	–1		4	LSB
Static INL	Endpoint corrected	–16		16	LSB
Settling time	Settling to 1LSB after full-scale output change			1	µs
Resolution			12		bits
CMPSS DAC output disturbance(2)	Error induced by comparator trip or CMPSS DAC code change within the same CMPSS module	–100		100	LSB
CMPSS DAC disturbance time(2)				200	ns
VDAC reference voltage	When VDAC is reference	2.4	2.5/3.0	VDDA	V
VDAC load(3)	When VDAC is reference	6	8	10	kΩ,

(1) Includes comparator input referred errors.
包含以比较器输入为基准的误差。
(2) Disturbance error may be present on the CMPSS DAC output for a certain amount of time after a comparator trip.
在比较器跳闸后的一段时间内，CMPSSDAC 输出可能会出现干扰误差。
(3) Per active CMPSS module.
每个有源CMPSS 模块。
(4) The maximum output voltage is VDDA when VDAC > VDDA.
当VDAC>VDDA 时，最大输出电压为 VDDA。

6.13.4.3.3 CMPSS Illustrative Graphs
CMPSS 示意图

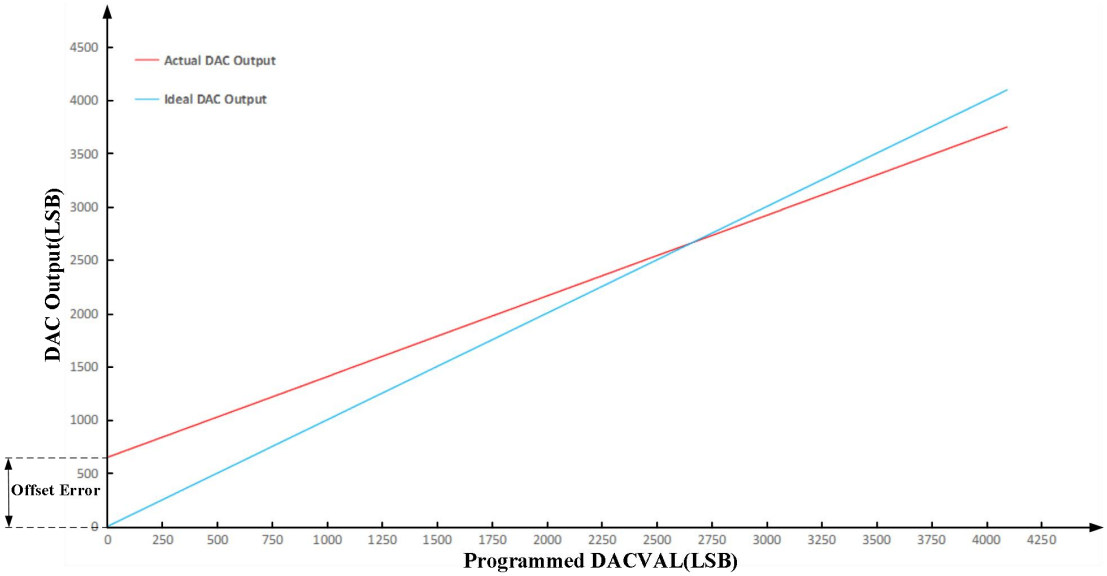


Figure 6- 49 CMPSS DAC Static Offset

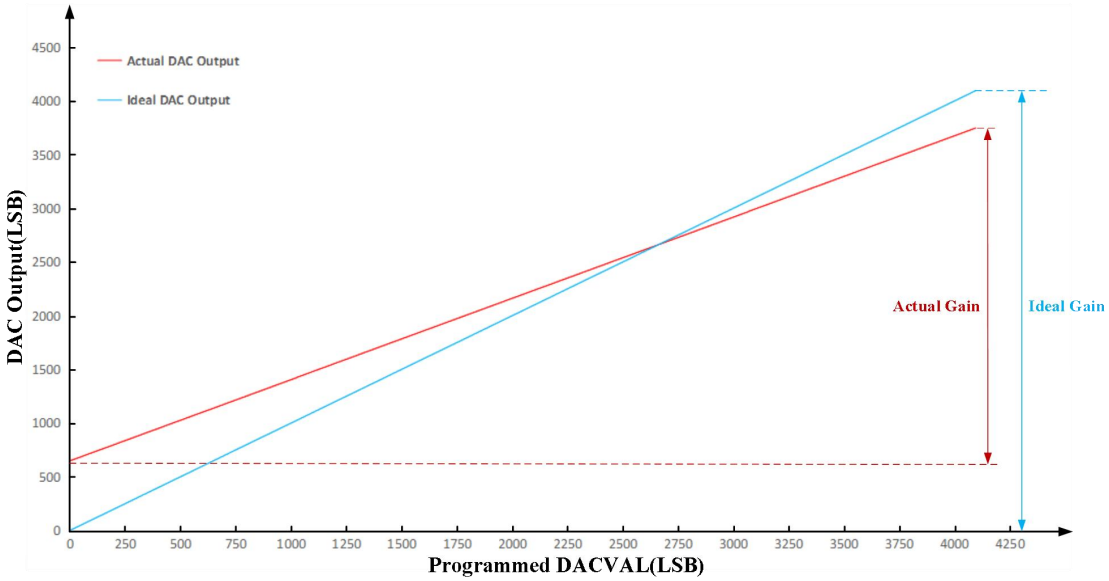


Figure 6- 50 CMPSS DAC Static Gain

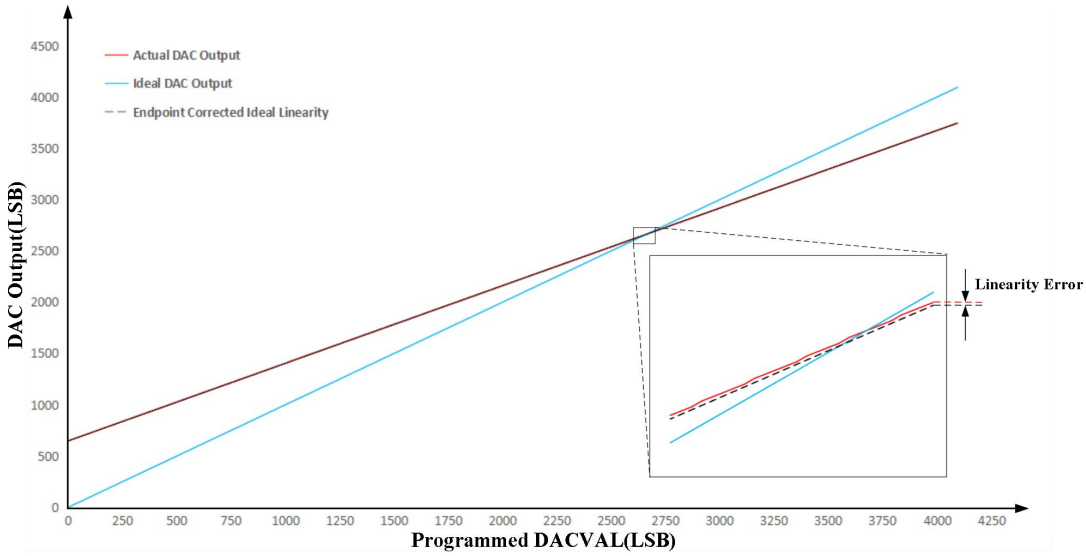


Figure 6- 51 CMPSS DAC Static Linearity

6.13.4.3.4 CMPSS DAC Dynamic Error
CMPSS DAC 动态误差

When using the ramp generator to control the internal DAC, the step size can vary based on the application need. Since the step size of the DAC is less than a full scale transition, the settling time is improved from the electrical specification listed in the *CMPSS DAC Static Electrical Characteristics* table. The equation below and Figure 6- 52 can give guidance on the expected voltage error from ideal based on different RAMPxDECVALA values.

当使用斜坡发生器控制内部 DAC 时，阶跃大小可以根据应用需求而变化。由于 DAC 的阶跃大小小于满量程转换，因此，稳定时间比 CMPSS DAC 静态电气特性表中列出的电气规格有所改善。下面的公式和图 6-52 可以根据不同的 RAMPxDECVALA 值，提供有关与理想值之间预期电压误差的指导。

$$\mathrm{DYNAMICERROR} = \mathrm{m} \times \mathrm{RAMPxDECVALA} + \mathrm{b}$$

Table 6-18 DAC Max Dynamic Error Terms

EQUATION PARAMETER	MIN (LSB)	MAX (LSB)
m	0.10	0.18
b	3.7	5.6

Note

Above error terms are based on the max SYSCLK of the target device. If operating below the max SYSCLK then the "m" error term should be scaled accordingly.

上述误差项基于目标器件的最大 SYSCLK。如果在最大 SYSCLK 以下运行，则“m”误差项应相应调整。

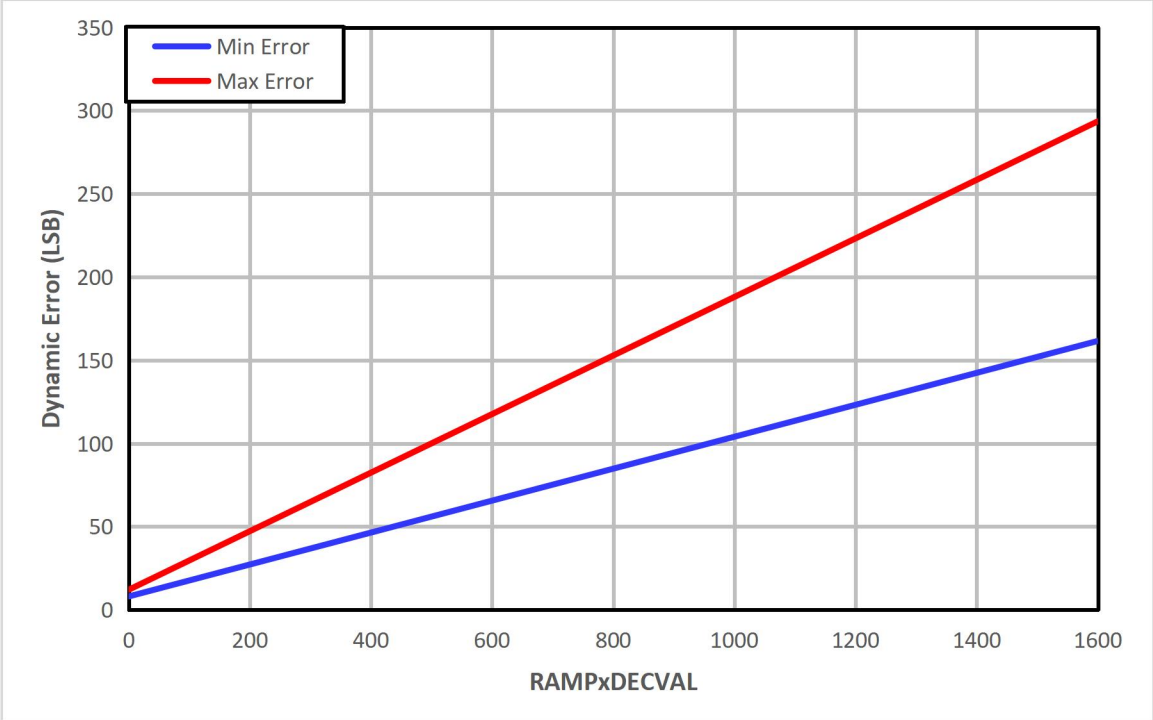


Figure 6- 52 CMPSS DAC Dynamic Error

6.13.5 Buffered Digital-to-Analog Converter (DAC)
缓冲数模转换器(DAC)

The buffered DAC module consists of an internal 12-bit DAC and an analog output buffer that can drive an external load. For driving even higher loads than typical, a trade-off can be made between load size and output voltage swing. For the load conditions of the buffered DAC, see the Buffered DAC Electrical Data and Timing section. The buffered DAC is a general-purpose DAC that can be used to generate a DC voltage or AC waveforms such as sine waves, square waves, triangle waves and so forth. Software writes to the DAC value register can take effect immediately or can be synchronized with EPWMSYNCO events.
Each buffered DAC has the following features:

- 12-bit resolution
- Selectable reference voltage source
- x1 and x2 gain modes when using internal VREFHI
- Ability to synchronize with EPWMSYNCPER

缓冲 DAC 模块由一个内部 12 位DAC 和一个可以驱动外部负载的模拟输出缓冲器组成。为了驱动比典型值更高的负载，可以在负载大小和输出电压摆幅之间进行权衡。对于缓冲 DAC 的负载条件，请参阅“缓冲 DAC 电气数据和时序”部分。缓冲 DAC 是一种通用 DAC，可用于生成直流电压或交流波形，例如正弦波、方波、三角波等。软件写入 DAC 值寄存器可立即生效，也可以与 EPWMSYNCO 事件同步。

每个缓冲 DAC 具有以下特性：

- 12 位分辨率
- 可选择的基准电压源
- 使用内部 VREFHI 时的 X1 和 x2 增益模式
- 能够与 EPWMSYNCPER 同步

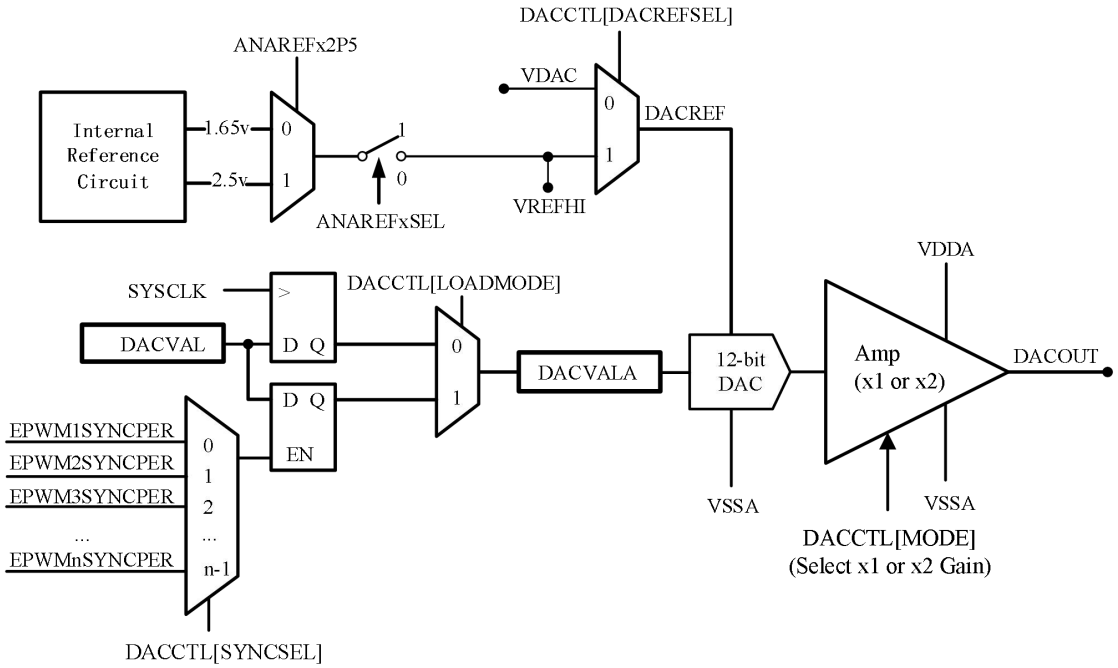


Figure 6- 53 DAC Module Block Diagram

6.13.5.1 Buffered DAC Electrical Data and Timing
缓冲 DAC 电气数据和时序

6.13.5.1.1 Buffered DAC Operating Conditions
缓冲 DAC 运行条件

over recommended operating conditions (unless otherwise noted)(1)
在建议运行条件下测得（除非另有说明）(1)

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
R_L	Resistive Load(2)		5			kΩ
C_L	Capacitive Load				100	pF
VOUT	Valid Output Voltage Range(3)	R_L = 5 kΩ	0.3		VDDA–0.3	V
		R_L = 1 kΩ	0.6		VDDA–0.6	V
Reference Voltage(4)		VDAC or VREFHI	2.4	2.5 or 3.0	VDDA	V

(1) Typical values are measured with VREFHI = 3.3 V and VREFLO = 0 V, unless otherwise noted. Minimum and maximum values are tested or characterized with VREFHI = 2.5 V and VREFLO = 0 V.
典型值是在 VREFHI=3.3V 和VREFLO=0V 时测得的，除非另外注明。在 VREFHI=2.5V 和VREFLO=0V 条件下对最小值和最大值进行测试或表征。

(2) DAC can drive a minimum resistive load of 1 kΩ, but the output range will be limited.
DAC 可以驱动最小 1kΩ的阻性负载，但输出范围会受到限制。

(3) This is the linear output range of the DAC. The DAC can generate voltages outside this range, but the output voltage will not be linear due to the buffer.
这是DAC 的线性输出范围。DAC 可以产生此范围以外的电压，但由于缓冲器的原因，输出电压将不呈线性。

(4) For best PSRR performance, VDAC or VREFHI should be less than VDDA.
为了获得卓越 PSRR 性能，VDAC 或VREFHI 应小于VDDA。

6.13.5.1.2 Buffered DAC Electrical Characteristics
缓冲 DAC 电气特性

over recommended operating conditions (unless otherwise noted)(1)
在建议运行条件下测得（除非另有说明）(1)

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
General 通用						
Resolution				12		bits
Load Regulation			–1		1	mV/V
Glitch Energy				1.5		V-ns
Voltage Output Settling Time Full-Scale		Settling to 2 LSBs after 0.3V- to-3V transition			2	µs
Voltage Output Settling Time 1/4th Full-Scale		Settling to 2 LSBs after 0.3V- to-0.75V transition			1.6	µs
Voltage Output Slew Rate		Slew rate from 0.3V- to-3V transition	2.8		4.5	V/µs
Load Transient Settling Time		5-kΩ Load			328	ns
		1-kΩ Load			557	ns
Reference Input Resistance(2)		VDAC or VREFHI	160	200	240	kΩ
TPU	Power Up Time	External Reference mode			500	µs
		Internal Reference mode			5000	µs
DC Characteristics 直流特性						
Offset	Offset Error	Midpoint	–10		10	mV

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
Gain	Gain Error(3)		–2.5		2.5	% of FSR
DNL	Differential Non Linearity(4)	Endpoint corrected	–1	±0.4	1	LSB
INL	Integral Non Linearity	Endpoint corrected	–5	±2	5	LSB
AC Characteristics 交流特性						
Output Noise		Integrated noise from 100 Hz to 100 kHz		600		µVrms
		Noise density at 10 kHz		800		nVrms/√Hz
SNR	Signal to Noise Ratio	1kHz，200KSPS		64		dB
THD	Total Harmonic Distortion	1kHz，200KSPS		–64.2		dB
SFDR	Spurious Free Dynamic Range	1kHz，200KSPS		66		dB
SINAD	Signal to Noise and Distortion Ratio	1kHz，200KSPS		61.7		dB
PSRR	Power Supply Rejection Ratio(5)	DC		70		dB
		100kHz		30		dB

(1) Typical values are measured with VREFHI = 3.3 V and VREFLO = 0 V, unless otherwise noted. Minimum and maximum values are tested or characterized with VREFHI = 2.5 V and VREFLO = 0 V.
典型值是在 VREFHI=3.3V 和VREFLO=0V 时测得的，除非另外注明。在 VREFHI=2.5V 和VREFLO=0V 条件下对最小值和最大值进行测试或表征。

(2) Per active Buffered DAC module.
每个有源缓冲 DAC 模块。

(3) Gain error is calculated for linear output range.
增益误差是在线性输出范围内计算得出。

(4) The DAC output is monotonic.
DAC 输出是单调输出。

(5) VREFHI=3.2V, VDDA=3.3VDC+100mV Sine。

6.14 Control Peripherals 控制外设

6.14.1 Enhanced Pulse Width Modulator (ePWM) 增强型脉宽调制器(ePWM)

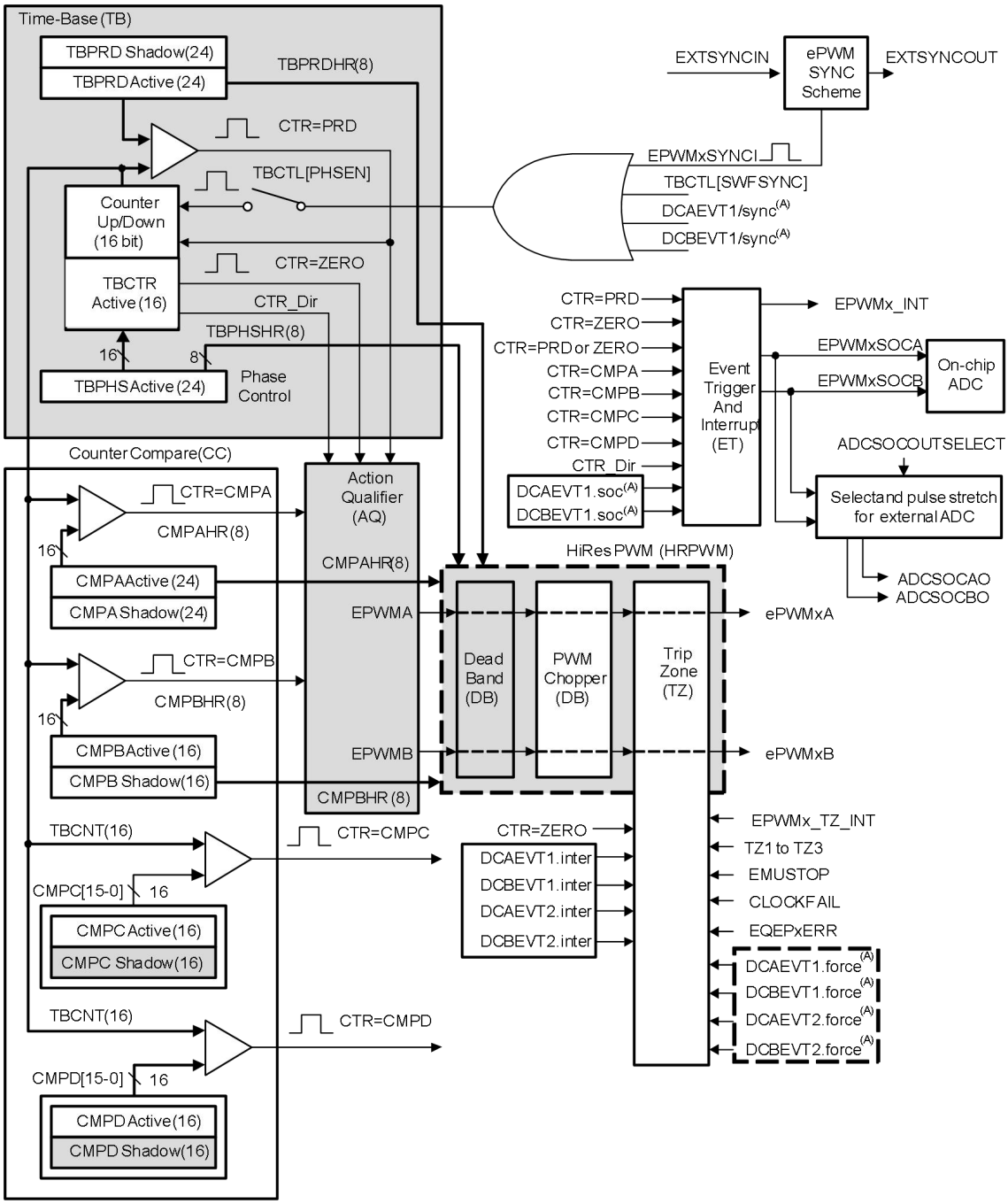
The ePWM peripheral is a key element in controlling many of the power electronic systems found in both commercial and industrial equipment. The ePWM type-4 module is able to generate complex pulse width waveforms with minimal CPU overhead by building the peripheral up from smaller modules with separate resources that can operate together to form a system. Some of the highlights of the ePWM type-4 module include complex waveform generation, dead-band generation, a flexible synchronization scheme, advanced trip-zone functionality, and global register reload capabilities.
The ePWM and eCAP synchronization scheme on the device provides flexibility in partitioning the ePWM and eCAP modules and allows localized synchronization within the modules.
Figure 6- 54 shows the ePWM module. Figure 6- 55 shows the ePWM trip input connectivity.

ePWM 外设是控制商业和工业设备中的许多电力电子系统的关键元件。通过从具有独立资源（这些独立资源可以一起运行形成一个系统）的较小模块构建外设，ePWMType-4 模块能够以最小的 CPU 开销生成复杂的脉冲宽度波形。ePWM Type-4 模块的一些亮点包括复杂波形生成、死区生成、灵活的同步方案、高级跳变区功能和全局寄存器重载功能。

借助该器件上的 ePWM 和eCAP 同步方案，可灵活地划分 ePWM 和eCAP 模块，并可在这些模

块内进行局部同步。

图 6-54 展示了 ePWM 模块。图 6-55 显示了 ePWM 跳变输入连接。



A. These events are generated by the ePWM digital compare (DC) submodule based on the levels of the TRIPIN inputs.
这些事件由 ePWM 数字比较(DC)子模块根据 TRIPIN 输入电平生成。

Figure 6- 54 ePWM Submodules and Critical Internal Signal Interconnects

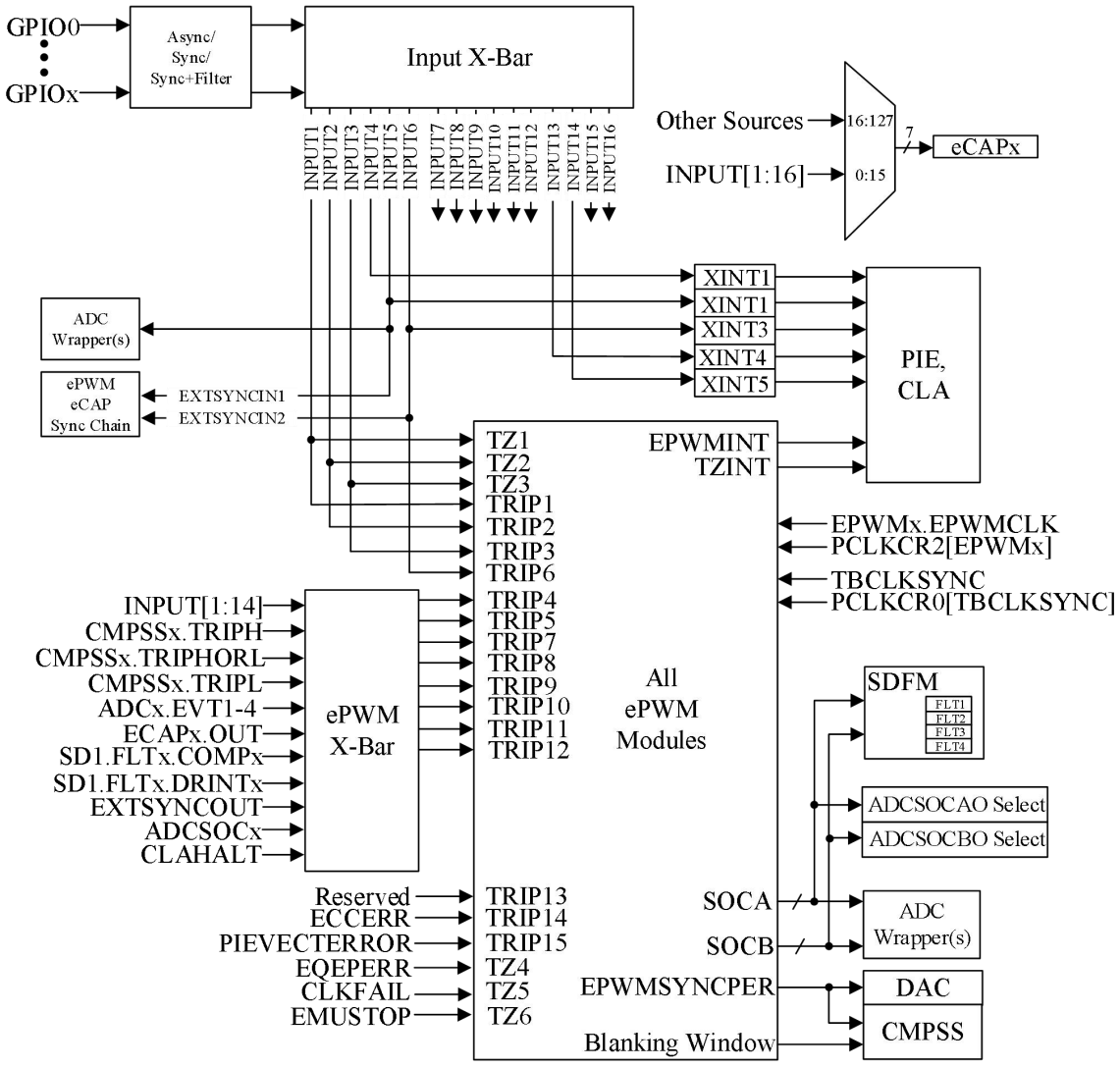


Figure 6- 55 ePWM Trip Input Connectivity

6.14.1.1 ePWM Electrical Data and Timing
ePWM 电气数据和时序

For an explanation of the input qualifier parameters, see the General-Purpose Input Timing Requirements table.

有关输入滤波参数的说明，请参阅“通用输入时序要求”表。

6.14.1.1.1 ePWM Timing Requirements
ePWM 时序要求

PARAMETER[1]			MIN	MAX	UNIT
$t_{w(SYNCIN)}$	Sync input pulse width	Asynchronous	$2t_{c(EPWMCLK)}$		cycles
		Synchronous	$2t_{c(EPWMCLK)}$		
		With input qualifier	$1t_{c(EPWMCLK)}+t_{w(IQSW)}$		

6.14.1.1.2 ePWM Switching Characteristics
ePWM 开关特性

over recommended operating conditions (unless otherwise noted)

在建议运行条件下测得（除非另有说明）

PARAMETER[1]		MIN	MAX	UNIT
$t_{w(PWM)}$	Pulse duration, PWMx output high/low	20		ns
$t_{w(SYNCOUT)}$	Sync output pulse width	$8t_{c(SYSCLK)}$		cycles
$t_{d(TZ-PWM)}$	Delay time, trip input active to PWM forced high Delay time, trip input active to PWM forced low Delay time, trip input active to PWM Hi-Z		25	ns

(1) 20-pF load on pin.

引脚上20pF 负载。

6.14.1.1.3 Trip-Zone Input Timing
跳闸区输入时序

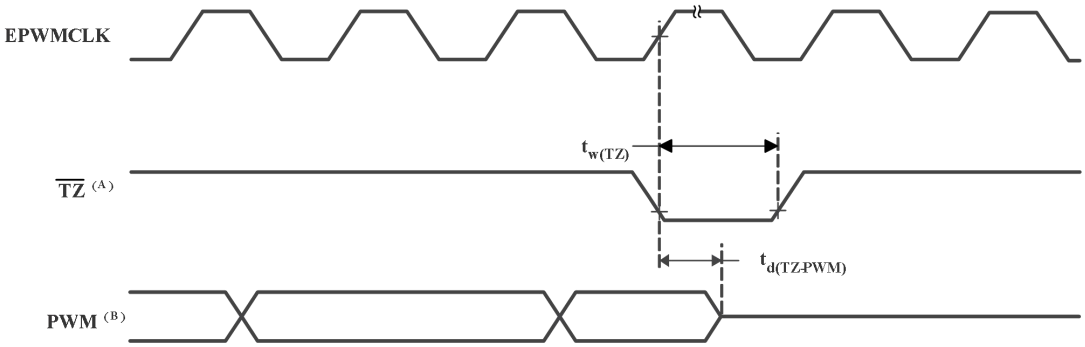
For an explanation of the input qualifier parameters, see the General-Purpose Input Timing Requirements table.

有关输入滤波参数的说明，请参阅“通用输入时序要求”表。

6.14.1.1.3.1 Trip-Zone Input Timing Requirements
跳闸区域输入时序要求

PARAMETER[1]			MIN	MAX	UNIT
$t_{w(TZ)}$	Pulse duration, $\overline{TZx}$ input low	Asynchronous	$1t_{c(EPWMCLK)}$		cycles
		Synchronous	$2t_{c(EPWMCLK)}$		cycles
		With input qualifier	$1t_{c(EPWMCLK)}+t_{w(IQSW)}$		cycles

6.14.1.1.3.2 PWM Hi-Z Characteristics Timing Diagram
PWM 高阻态特征时序图



A. $\overline{TZ}$: $\overline{TZ1}$, $\overline{TZ2}$, $\overline{TZ3}$, TRIP1-TRIP12

$\overline{TZ}$: $\overline{TZ1}$、$\overline{TZ2}$、$\overline{TZ3}$、TRIP1 至TRIP12

B. PWM refers to all the PWM pins in the device. The state of the PWM pins after $\overline{TZ}$ is taken high depends on the PWM recovery software.

PWM 是指器件内的所有 PWM 引脚。$\overline{TZ}$置于高电平后 PWM 引脚的状态取决于 PWM 恢复软件。

Figure 6- 56 PWM Hi-Z Characteristics

6.14.2 High-Resolution Pulse Width Modulator (HRPWM)
高分辨率脉宽调制器(HRPWM)

The HRPWM combines multiple delay lines in a single module and a simplified calibration system by using a dedicated calibration delay line. For each ePWM module, there are two HR outputs:

- HR Duty and Deadband control on Channel A
- HR Duty and Deadband control on Channel B

The HRPWM module offers PWM resolution (time granularity) that is significantly better than what can be achieved using conventionally derived digital PWM methods. The key points for the HRPWM module are:

- Significantly extends the time resolution capabilities of conventionally derived digital PWM
- This capability can be used in both single edge (duty cycle and phase-shift control) as well as dual edge control for frequency/period modulation.
- Finer time granularity control or edge positioning is controlled through extensions to the Compare A, B, phase, period and deadband registers of the ePWM module.

通过使用专用的校准延迟电路，HRPWM 在单个模块和简化的校准系统内结合了多条延迟电路。对于每个 ePWM 模块，都有两个 HR 输出：

- 通道 A 上的 HR 占空比和死区控制
- 通道 B 上的 HR 占空比和死区控制

HRPWM 模块提供 PWM 分辨率（时间粒度），此分辨率明显优于使用传统数字 PWM 方法所能达到的分辨率。HRPWM 模块的关键点为：

- 大大扩展了传统导出数字 PWM 的时间分辨率能力
- 此功能可用于单边沿（占空比和相移控制）以及双边沿控制，以实现频率/周期调制。
- 通过对 ePWM 模块的比较 A、B、相位、周期和死区寄存器的扩展来控制更加精细的时间粒度控制或边沿定位。

6.14.2.1 HRPWM Electrical Data and Timing
HRPWM 电气数据和时序

6.14.2.1.1 High-Resolution PWM Characteristics
高分辨率 PWM 特征

PARAMETER	MIN	TYP	MAX	UNIT
Micro Edge Positioning (MEP) step size(1)		150	310	ps

(1) The MEP step size will be largest at high temperature and minimum voltage on V_{DD}. MEP step size will increase with higher temperature and lower voltage and decrease with lower temperature and higher voltage. Applications that use the HRPWM feature should use MEP Scale Factor Optimizer (SFO) estimation software functions.
MEP 步长在高温和 VDD 上电压最小时最大。MEP 步长将随温度的升高和电压的下降而增加，并随温度的下降和电压的升高而减小。使用 HRPWM 特性的应用应该使用 MEP 比例因子优化器(SFO)估计软件功能。SFO 功能有助于在 HRPWM 运行时动态估计每个 SYSCLK 周期的 MEP 步数。

6.14.3 External ADC Start-of-Conversion Electrical Data and Timing
外部 ADC 转换启动电气数据和时序

6.14.3.1 External ADC Start-of-Conversion Switching Characteristics
外部 ADC 转换启动开关特性

over recommended operating conditions (unless otherwise noted)
在建议运行条件下测得（除非另有说明）

PARAMETER		MIN	MAX	UNIT
$t_{w(ADCSOCL)}$	Pulse duration, $\overline{ADCSOCxO}$ low	$32t_{c(SYSCLK)}$		cycles

6.14.3.2 $\overline{ADCSOCAO}$ or $\overline{ADCSOCAO}$ Timing Diagram
$\overline{ADCSOCAO}$或$\overline{ADCSOCAO}$时序图

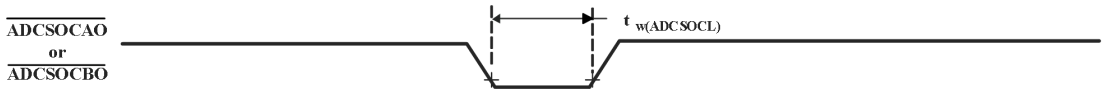


Figure 6- 57 $\overline{ADCSOCAO}$ or $\overline{ADCSOCBO}$ Timing

6.14.4 Enhanced Capture (eCAP)
增强型捕获(eCAP)

The eCAP module can be used in systems where accurate timing of external events is important. eCAP/HRCAP on this device is Type-2.
Applications for eCAP include:

- Speed measurements of rotating machinery (for example, toothed sprockets sensed through Hall sensors)
- Elapsed time measurements between position sensor pulses
- Period and duty cycle measurements of pulse train signals
- Decoding current or voltage amplitude derived from duty cycle encoded current/voltage sensors

eCAP 模块可用于外部事件的准确计时非常重要的系统中。该器件上的/HRCAP 为Type-2。

eCAP 的应用包含：

- 旋转机械的速度测量（例如，通过霍尔传感器感应齿状链轮）
- 位置传感器脉冲之间的持续时间测量
- 脉冲序列信号的周期和占空比测量
- 解码来自占空比编码电流/电压传感器的电流或电压振幅

The eCAP module includes the following features:

- 4-event time-stamp registers (each 32 bits)
- Edge-polarity selection for up to four sequenced time-stamp capture events
- Interrupt on either of the four events
- Single shot capture of up to four event timestamps
- Continuous mode capture of timestamps in a four-deep circular buffer
- Absolute time-stamp capture
- Difference (Delta) mode time-stamp capture
- All of the above resources dedicated to a single input pin
- When not used in capture mode, the eCAP module can be configured as a single-channel PWM output (APWM).

eCAP 模块包括以下特性：

- 4 事件时间戳寄存器（每个 32 位）
- 边缘极性选择，最多选择四个序列时间戳采集事件
- 对 4 个事件中的任何一个事件进行中断
- 单次采集多达 4 个事件时间戳
- 在四深循环缓冲器中连续模式采集时间戳
- 绝对时间戳采集
- 差分(Δ)模式时间戳采集
- 所有上述资源都专用于单个输入引脚
- 当未用于采集模式时，eCAP 模块可配置为单通道 PWM 输出(APWM)。

The capture functionality of the Type-1 eCAP is enhanced from the Type-0 eCAP with the following added features:

- Event filter reset bit
  - Writing a 1 to ECCTL2[CTRFILTRESET] will clear the event filter, the modulo counter, and any pending interrupts flags. Resetting the bit is useful for initialization and debug.

- Modulo counter status bits
  - The modulo counter (ECCTL2 [MODCTRSTS]) indicates which capture register will be loaded next. In the Type-0 eCAP, it was not possible to know current state of modulo counter.
- DMA trigger source
  - eCAPxDMA is added as a DMA trigger. CEVT[1–4] can be configured as the source for eCAPxDMA.
- Input multiplexer
  - ECCTL0 [INPUTSEL] selects one of 128 input signals.
- EALLOW protection
  - EALLOW protection is added to critical registers. To maintain software compatibility with the Type-0 eCAP, configure DEV_CFG_REGS.ECAPTYPE to make these registers unprotected.

Type-1 eCAP 的捕获功能通过 Type-0eCAP 得到增强，增加了以下特性：

- 事件过滤器复位位
  - 向 ECCTL2[CTRFILTRESET]写入 1 将清零事件滤波器、模计数器和任何挂起的中断标志。复位该位对于初始化和调试很有用。
- 模数计数器状态位。
  - 模数计数器(ECCTL2[MODCTRSTS])指示接下来将加载哪个捕获寄存器。在 Type-0eCAP 中，无法知道模数计数器的当前状态。
- DMA 触发源
  - eCAPxDMA 被添加为一个 DMA 触发器。CEVT[1–4]可以配置为 eCAPxDMA 的源。
- 输入多路复用器
  - ECCTL0[INPUTSEL]选择 128 个输入信号之一。
- EALLOW 保护
  - EALLOW 保护已添加到关键寄存器。为了维持与 Type-0eCAP 的软件兼容性，请配置 DEV_CFG_REGS.ECAPTYPE 以使这些寄存器不受保护。

The capture functionality of the Type-2 eCAP is enhanced from the Type-1 eCAP with the following added features:

- ECAPxSYNCINSEL register
  - The ECAPSxYNCINSEL register is added for each eCAP to select an external SYNCIN. Every eCAP can have a separate SYNCIN signal.

Type-2 eCAP 的捕获功能通过 Type-1eCAP 得到增强，增加了以下特性：

- ECAPxSYNCINSEL 寄存器
  - 为每个 eCAP 添加了 ECAPSxYNCINSEL 寄存器以选择外部 SYNCIN。每个 eCAP 可以有一个单独的 SYNCIN 信号。

The eCAP inputs connect to any GPIO input through the Input X-BAR. The APWM outputs connect to GPIO pins through the Output X-BAR to OUTPUTx positions in the GPIO mux. See the *GPIO Input X-BAR* section and the *GPIO Output X-BAR, CLB X-BAR, CLB Output X-BAR, and ePWM X-BAR* section.

The eCAP module is clocked by PERx.SYSCLK.

The clock enable bits (ECAP1–ECAP3) in the PCLKCR3 register turn off the eCAP module

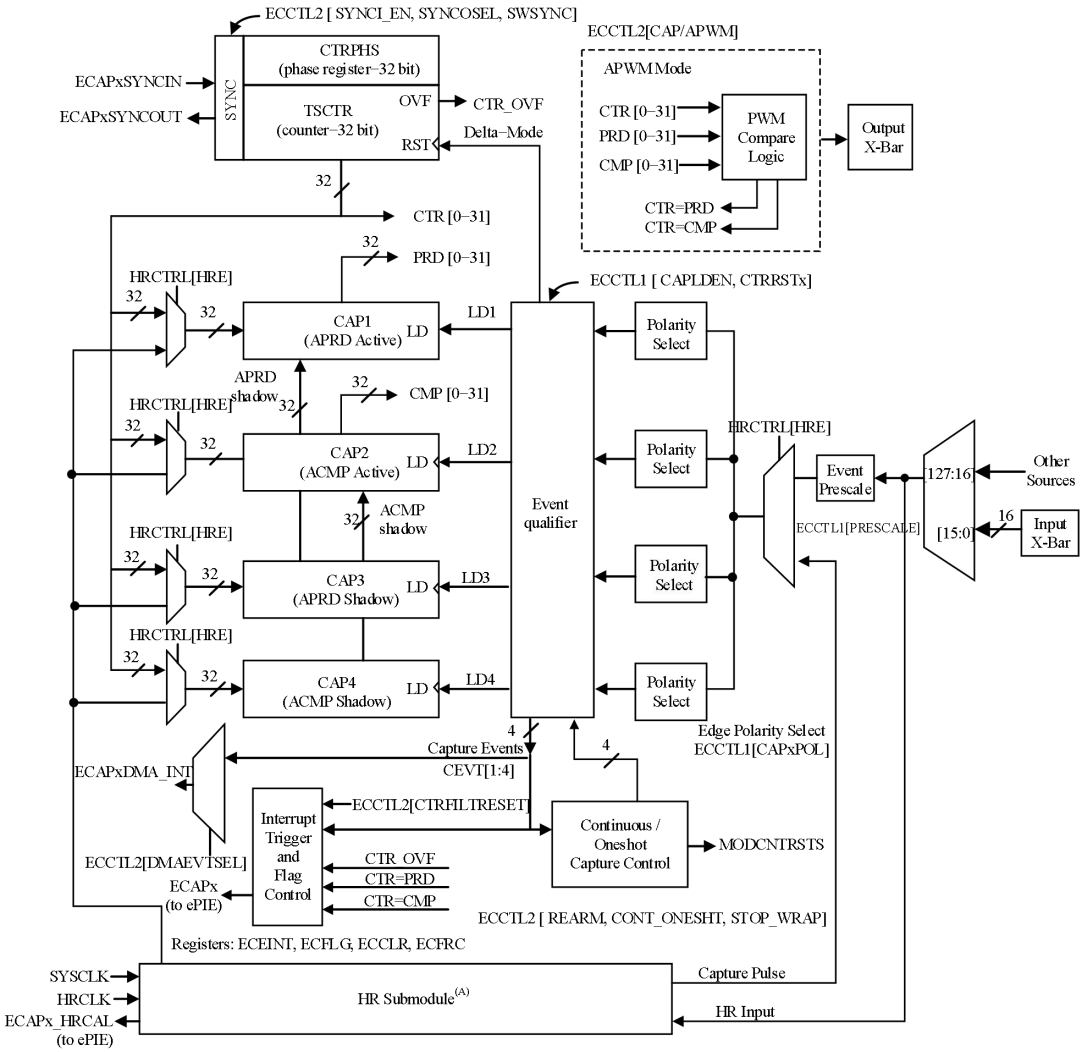
individually (for low-power operation). Upon reset, ECAP1ENCLK is set to low, indicating that the peripheral clock is off.

eCAP 输入通过输入 X-BAR 连接到任何 GPIO 输入。APWM 输出通过 GPIO 多路复用器中的输出 X-BAR 到 OUTPUTx 位置连接到 GPIO 引脚。请参阅 GPIO 输入 X-BAR 部分和 GPIO 输出 X-BAR、CLBX-BAR、CLB 输出 X-BAR 和 ePWMX-BAR 部分。

eCAP 模块由 PERx.SYSCLK 计时。

PCLKCR3 寄存器中的时钟使能位(ECAP1–ECAP3)可单独关闭 eCAP 模块(以实现低功耗运行)。复位时，ECAP1ENCLK 设置为低电平，表明外设时钟已关闭。

6.14.4.1 eCAP and HRCAP Block Diagram
eCAP 和 HRCAP 框图



A. The HRCAP submodule is not available on all eCAP modules; in this case, the high-resolution muxes and hardware are not implemented.
HRCAP 子模块并非在所有 eCAP 模块上都可用；在这种情况下，不会使用高分辨率多路复用器和硬件。

Figure 6- 58 eCAP and HRCAP Block Diagram

6.14.4.2 eCAP Synchronization
eCAP 同步

The eCAP modules can be synchronized with each other by selecting a common SYNCIN source. SYNCIN source for eCAP can be either software sync-in or external sync-in. The external sync-in signal can come from EPWM, eCAP, or X-Bar. The SYNC signal is defined by the selection in the ECAPxSYNCINSEL[SEL] bit for ECAPx as shown in Figure 6- 59.

通过选择一个公共的 SYNCIN 源，eCAP 模块可以互相同步。eCAP 的 SYNCIN 源可以是软件同步输入或外部同步输入。外部同步输入信号可来自 EPWM、eCAP 或 X-Bar。如图 6-59 所示，SYNC 信号由 ECAPx 的 ECAPxSYNCINSEL[SEL]位中的选择定义。

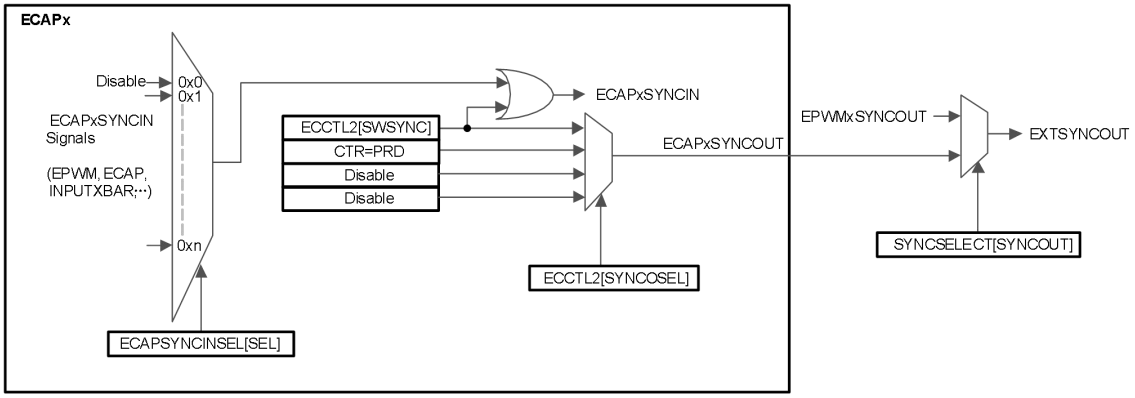


Figure 6- 59 eCAP Synchronization Scheme

6.14.4.3 eCAP Electrical Data and Timing
eCAP 电气数据和时序

6.14.4.3.1 eCAP Timing Requirements
eCAP 时序要求

PARAMETER			MIN	TYP	MAX	UNIT
$t_{w(CAP)}$	Capture input pulse width	Asynchronous	$2t_{c(SYSCLK)}$			ns
		Synchronous	$2t_{c(SYSCLK)}$			
		With input qualifier	$1t_{c(SYSCLK)}+t_{w_(IQSW)}$			

6.14.4.3.2 eCAP Switching Characteristics
eCAP 开关特性

over recommended operating conditions (unless otherwise noted)
在建议运行条件下测得（除非另有说明）

PARAMETER		MIN	TYP	MAX	UNIT
$t_{w(APWM)}$	Pulse duration, APWMx output high/low	20			ns

6.14.5 High-Resolution Capture (HRCAP)
高分辨率捕获(HRCAP)

The eCAP3 module can be configured as high-resolution capture (HRCAP) submodules. The HRCAP submodule measures the difference, in time, between pulses asynchronously to the system clock. This submodule is new to the eCAP Type 1 module, and features many enhancements over the Type 0 HRCAP module.
Applications for the HRCAP include:

- Capacitive touch applications
- High-resolution period and duty-cycle measurements of pulse train cycles
- Instantaneous speed measurements
- Instantaneous frequency measurements
- Voltage measurements across an isolation boundary
- Distance/sonar measurement and scanning
- Flow measurements

eCAP3 模块可以配置为高分辨率捕获(HRCAP)子模块。HRCAP 子模块可以测量与系统时钟异步的脉冲之间的时间差。该子模块是 eCAPType-1 模块新增的子模块，与 Type-0HRCAP 模块相比具有许多增强功能。

HRCAP 的应用包括：

- 电容式触控应用
- 脉冲序列周期的高分辨率周期和占空比测量
- 瞬时速度测量
- 瞬时频率测量
- 在一个隔离边界上的电压测量
- 距离/声纳测量和扫描
- 流量测量

The HRCAP submodule includes the following features:

- Pulse-width capture in either non-high-resolution or high-resolution modes
- Absolute mode pulse-width capture
- Continuous or "one-shot" capture
- Capture on either falling or rising edge
- Continuous mode capture of pulse widths in 4-deep buffer
- Hardware calibration logic for precision high-resolution capture
- All of the resources in this list are available on any pin using the Input X-BAR.

HRCAP 子模块包含以下特性：

- 在非高分辨率或高分辨率模式下进行脉宽捕获
- 绝对模式脉宽捕获
- 连续或“一次性”捕获
- 在下降沿或上升沿捕获
- 4 深度缓冲器中脉冲宽度的持续模式捕获
- 通过硬件校准逻辑实现精密高分辨率捕获
- 使用输入 X-BAR 的任何引脚上均可使用此列表中的所有资源。

The HRCAP submodule includes one high-resolution capture channel in addition to a calibration block. The calibration block allows the HRCAP submodule to be continually recalibrated, at a set interval, with no “down time”. Because the HRCAP submodule now uses the same hardware as its respective eCAP, if the HRCAP is used, the corresponding eCAP will be unavailable.
Each high-resolution-capable channel has the following independent key resources.

- All hardware of the respective eCAP
- High-resolution calibration logic
- Dedicated calibration interrupt

HRCAP 子模块包含一个高分辨率捕获通道以及一个校准块。校准块允许 HRCAP 子模块在设定的时间间隔内持续重新校准，不存在“中断时间”。由于 HRCAP 子模块现在使用与其相应 eCAP 相同的硬件，因此如果使用 HRCAP，则相应的 eCAP 将不可用。

每个支持高分辨率的通道都具有以下独立的关键资源。

- 相应 eCAP 的所有硬件
- 高分辨率校准逻辑
- 专用校准中断

6.14.5.1 eCAP and HRCAP Block Diagram
eCAP 和 HRCAP 方框图

For the HRCAP Block Diagram, see the eCAP and HRCAP Block Diagram in the *Enhanced Capture (eCAP)* section.

有关 HRCAP 方框图，请参阅*增强型捕获(eCAP)*一节中的 eCAP 和 HRCAP 方框图。

6.14.5.2 HRCAP Electrical Data and Timing
HRCAP 电气数据和时序

6.14.5.2.1 HRCAP Switching Characteristics
HRCAP 开关特性

over recommended operating conditions (unless otherwise noted)
在建议运行条件下测得（除非另有说明）

PARAMETER	TEST CONDITIONS	MIN	TYP	MAX	UNIT
Input pulse width		110			ns
Accuracy(1)(2)(3)(4)	Measurement length ≤ 5 µs		±390	540	ps
	Measurement length > 5 µs		±450	1450	ps
Standard deviation			See HRCAPStandard Deviation Characteristics figure		
Resolution			300		ps

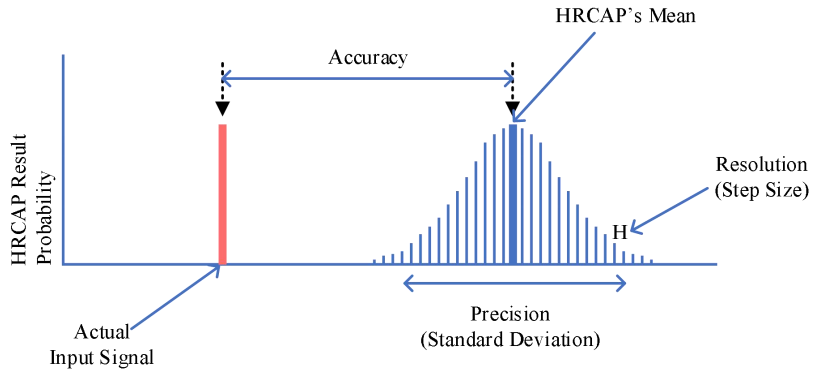
(1) Value obtained using an oscillator of 100 PPM, oscillator accuracy directly affects the HRCAP accuracy.
使用100PPM 的振荡器获得的值，振荡器精度直接影响 HRCAP 精度。

(2) Measurement is completed using rising-rising or falling-falling edges
使用上升-上升沿或下降-下降沿完成测量。

(3) Opposite polarity edges will have an additional inaccuracy due to the difference between V_{IH} and V_{IL}. This effect is dependent on the signal's slew rate.
由于V_{IH}和V_{IL}之间的差异，极性相反的边沿将进一步降低精度。这种影响取决于信号的压摆率。

(4) Accuracy only applies to time-converted measurements.
精度仅适用于经过时间转换的测量。

6.14.5.2.2 HRCAP Figure and Graph
HRCAP 图表



A. The HRCAP has some variation in performance, this results in a probability distribution which is described using the following terms:
- Accuracy: The time difference between the input signal and the mean of the HRCAP's distribution.
- Precision: The width of the HRCAP's distribution, this is given as a standard deviation.
- Resolution: The minimum measurable increment.

HRCAP 在性能上有一些变化，其概率分布可以使用以下术语描述：
- 误差（Accuracy）：输入信号与 HRCAP 分布均值之间的时间差。
- 精度（Precision）：HRCAP 分布的宽度，以标准偏差的形式给出。
- 分辨率（Resolution）：最小可测量增量。

Figure 6- 60 HRCAP Accuracy Precision and Resolution

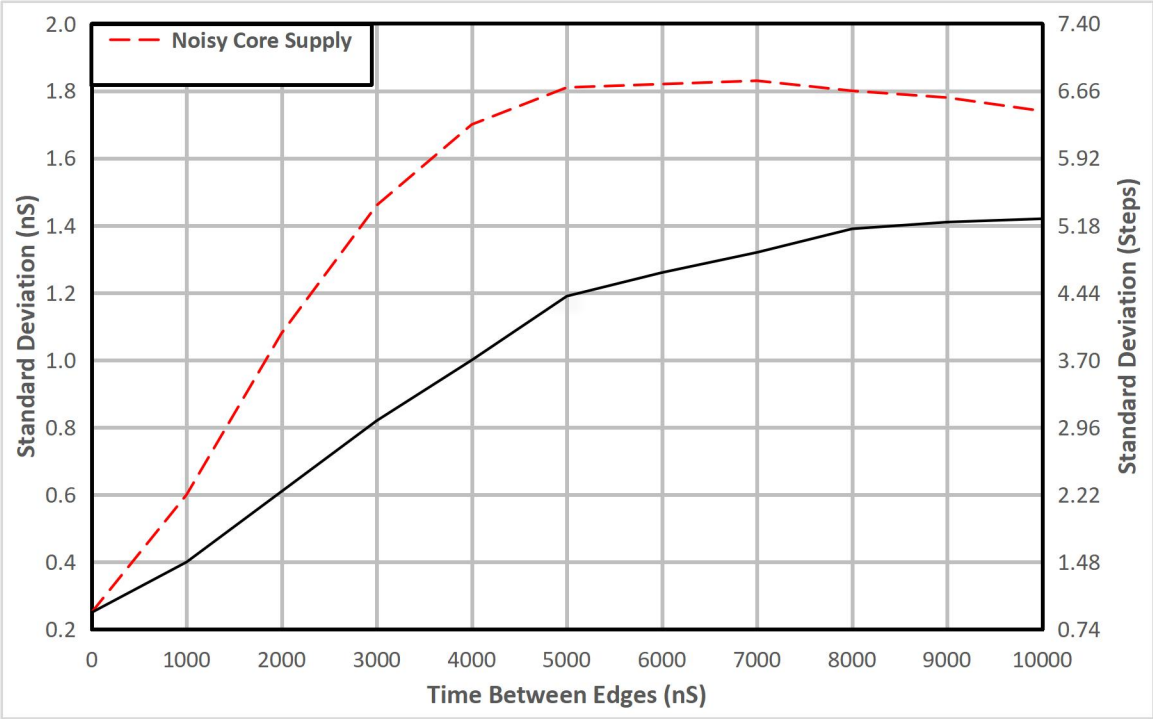


Figure 6- 61 HRCAP Standard Deviation Characteristics

6.14.6 Enhanced Quadrature Encoder Pulse (eQEP)
增强型正交编码器脉冲(eQEP)

The eQEP module on this device is Type-2. The eQEP interfaces directly with linear or rotary incremental encoders to obtain position, direction, and speed information from rotating machines used in high-performance motion and position control systems.
The eQEP peripheral contains the following major functional units (see Figure 6- 62):

- Programmable input qualification for each pin (part of the GPIO MUX)
- Quadrature decoder unit (QDU)
- Position counter and control unit for position measurement (PCCU)
- Quadrature edge-capture unit for low-speed measurement (QCAP)
- Unit time base for speed/frequency measurement (UTIME)
- Watchdog timer for detecting stalls (QWDOG)
- Quadrature Mode Adapter (QMA)

该器件上的 eQEP 模块为 Type-2。eQEP 直接与线性或旋转增量编码器相连，以便从高性能运动和位置控制系统中使用的旋转机器中获得位置、方向和速度信息。

该 eQEP 外设包含以下主要功能单元（请参阅图 6-62）：

- 针对每个引脚的可编程输入鉴定（GPIOMUX 的一部分）
- 正交解码器单元(QDU)
- 用于位置测量的位置计数器和控制单元(PCCU)
- 用于低速测量的正交边沿捕获单元(QCAP)
- 用于速度/频率测量的单位时基(UTIME)
- 用于检测失速的看门狗定时器(QWDOG)
- 正交模式适配器(QMA)

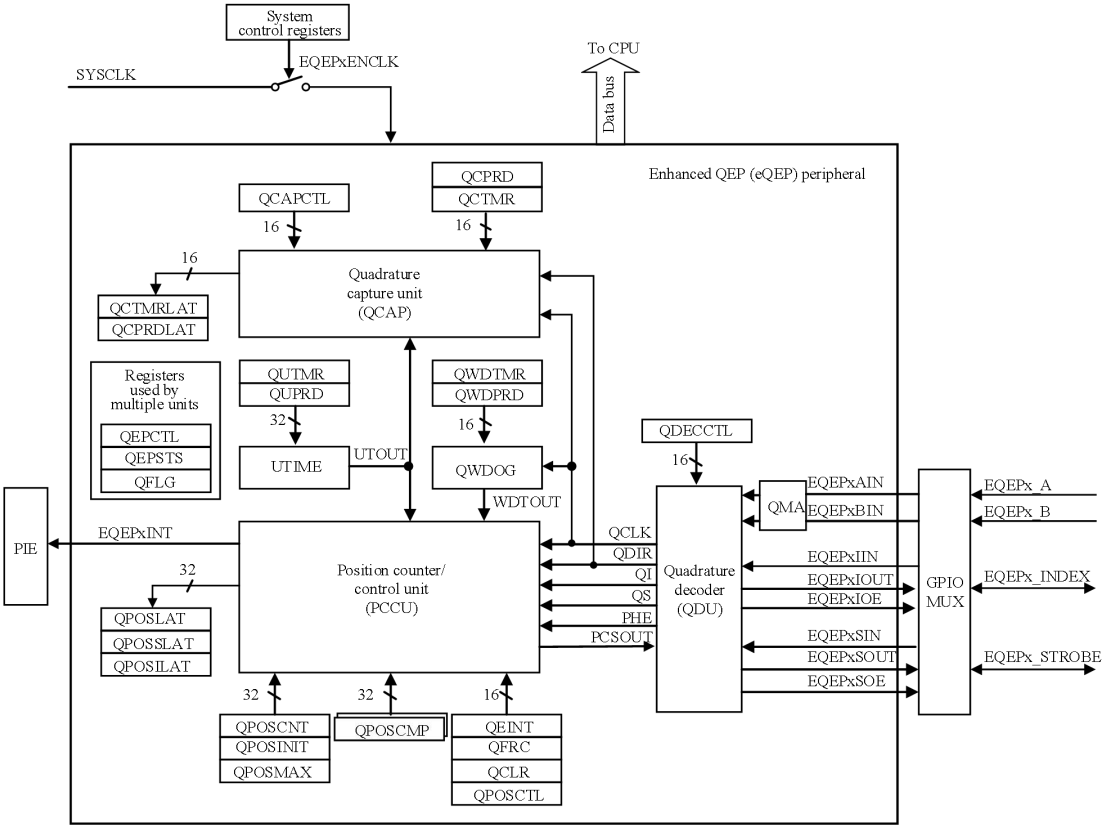


Figure 6- 62 eQEP Block Diagram

6.14.6.1 eQEP Electrical Data and Timing
eQEP 电气数据和时序

For an explanation of the input qualifier parameters, see the General-Purpose Input Timing Requirements table.

有关输入滤波参数的说明，请参阅“通用输入时序要求”表。

6.14.6.1.1 eQEP Timing Requirements
eQEP 时序要求

PARAMETER			MIN	MAX	UNIT
$t_{w(QEPP)}$	QEP input period	Synchronous(1)	$4t_{c(SYSCLK)}$		cycles
		Synchronous with input qualifier	$4[1t_{c(SYSCLK)}+t_{w(IQSW)}]$		
$t_{w(INDEXH)}$	QEP Index Input High time	Synchronous(1)	$2t_{c(SYSCLK)}$		cycles
		Synchronous with input qualifier	$2t_{c(SYSCLK)}+t_{w(IQSW)}$		cycles
$t_{w(INDEXL)}$	QEP Index Input Low time	Synchronous(1)	$2t_{c(SYSCLK)}$		cycles
		Synchronous with input qualifier	$2t_{c(SYSCLK)}+t_{w(IQSW)}$		cycles
$t_{w(STROBH)}$	QEP Strobe High time	Synchronous(1)	$2t_{c(SYSCLK)}$		cycles
		Synchronous with input qualifier	$2t_{c(SYSCLK)}+t_{w(IQSW)}$		cycles
$t_{w(STROBL)}$	QEP Strobe Input Low time	Synchronous(1)	$2t_{c(SYSCLK)}$		cycles
		Synchronous with input qualifier	$2t_{c(SYSCLK)}+t_{w(IQSW)}$		

(1) GPIOGPxQSELn 异步模式不得用于 eQEP 模块输入引脚。

6.14.6.1.2 eQEP Switching Characteristics
eQEP 开关特性

over recommended operating conditions (unless otherwise noted)

在建议运行条件下测得（除非另有说明）

PARAMETER		MIN	MAX	UNIT
$t_{d(CNTR)xin}$	Delay time, external clock to counter increment		$5t_{c(SYSCLK)}$	cycles
$t_{d(PCS-OUT)QEP}$	Delay time, QEP input edge to position compare sync output		$7t_{c(SYSCLK)}$	cycles

6.14.7 Σ-Δ滤波器模块(SDFM)

SDFM features include:

- Eight external pins per SDFM module
  - Four sigma-delta data input pins per SDFM module (SD-Dx, where x = 1 to 4)
  - Four sigma-delta clock input pins per SDFM module (SD-Cx, where x = 1 to 4)
- Different configurable modulator clock modes supported:
  - Mode 0: Modulator clock rate equals the modulator data rate.
- Four independent, configurable secondary filter (comparator) units per SDFM module:
  - Four different filter type selection (Sinc1/Sinc2/SincFast/Sinc3) options available
  - Ability to detect over-value condition, under-value condition, and Threshold-crossing conditions
    1. Two independent Higher Threshold comparators (used to detect over-value condition)
    2. Two independent Lower Threshold comparators (used to detect under-value condition)
    3. One independent Threshold-Crossing comparator (used to measure duty cycle/frequency with eCAP)
  - OSR value for comparator filter unit (COSR) programmable from 1 to 32
- Four independent configurable primary filter (data filter) units per SDFM module:
  - Four different filter type selection (Sinc1/Sinc2/SincFast/Sinc3) options available
  - OSR value for data filter unit (DOSR) programmable from 1 to 256
  - Ability to enable or disable (or both) individual filter module
  - Ability to synchronize all four independent filters of an SDFM module by using the Master Filter Enable (MFE) bit or by using PWM signals
- Data filter output can be represented in either 16 bits or 32 bits.
- Data filter unit has a programmable mode FIFO to reduce interrupt overhead. The FIFO has the following features:
  - The primary filter (data filter) has a 16-deep x 32-bit FIFO.
  - The FIFO can interrupt the CPU after programmable number of data-ready events.
  - FIFO Wait-for-Sync feature: Ability to ignore data-ready events until the PWM synchronization signal (SDSYNC) is received. Once the SDSYNC event is received, the FIFO is populated on every data-ready event.
  - Data filter output can be represented in either 16 bits or 32 bits.
- PWMx.SOCA/SOCB can be configured to serve as SDSYNC source on a per-data-filter-channel basis.
- PWMs can be used to generate a modulator clock for sigma-delta modulators.
- Configurable Input Qualification available for both SD-Cx and SD-Dx
- Ability to use one filter channel clock (SD-C1) to provide clock to other filter clock channels.
- Configurable digital filter available on comparator filter events to blankout comparator events caused by spurious noise

SDFM 的特性包含：

- 每个 SDFM 模块八个外部引脚
  - 每个 SDFM 模块有四个Σ-Δ数据输入引脚（SD-Dx，其中 x=1 至 4）
  - 每个 SDFM 模块有四个Σ-Δ时钟输入引脚（SD-Cx，其中 x=1 至 4）
- 支持不同的可配置调制器时钟模式：
  - 模式 0：调制器时钟速率等于调制器数据速率。
- 每个 SDFM 模块有四个独立的可配置次级滤波器（比较器）单元：
  - 提供四个不同的滤波器类型选择(Sinc1/Sinc2/SincFast/Sinc3)选项
  - 能够检测超值条件、低值条件和超限条件
    1. 两个独立的上限阈值比较器（用于检测超值条件）

2. 两个独立的下限阈值比较器（用于检测低值条件）
3. 一个独立的超限比较器（用于通过 eCAP 测量占空比/频率）

- 比较器滤波器单元的 OSR 值(COSR)可编程为 1 至 32

- 每个 SDFM 模块有四个独立的可配置初级滤波器（数据滤波器）单元：
  - 提供四个不同的滤波器类型选择(Sinc1/Sinc2/SincFast/Sinc3)选项
  - 数据滤波器单元的 OSR 值(DOSR)可编程为 1 至 256
  - 能够启用或禁用独立的滤波器模块（或全部两个模块）
  - 能够使用主滤波器使能(MFE)位或 PWM 信号来同步 SDFM 模块的所有四个独立滤波器
- 数据滤波器输出可以用 16 位或 32 位表示。
- 数据滤波器单元具有可编程模式 FIFO 来减少中断开销。该 FIFO 具有以下特性：
  - 初级滤波器（数据滤波器）具有一个 16 深 x32 位 FIFO。
  - FIFO 可在达到可编程数量的数据就绪事件后中断 CPU。
  - FIFO 等待同步功能：能够忽略数据就绪事件，直至接收到 PWM 同步信号(SDSYNC)。一旦接收到 SDSYNC 事件，就会在每个数据就绪事件时填充 FIFO。
  - 数据滤波器输出可以用 16 位或 32 位表示。
- 可根据每个数据滤波器通道将 PWMx.SOCA/SOCB 配置为 SDSYNC 源。
- 可使用 PWM 为Σ-Δ调制器生成调制器时钟。
- SD-Cx 和 SD-Dx 均可配置输入滤波
- 能够使用一个滤波器通道时钟(SD-C1)为其他滤波器时钟通道提供时钟。
- 在发生比较器滤波器事件时可以使用可配置的数字滤波器来清除杂散噪声引起的比较器事件

Figure 6- 63 shows the SDFM module block diagram.
图 6-63 展示了 SDFM 模块方框图。

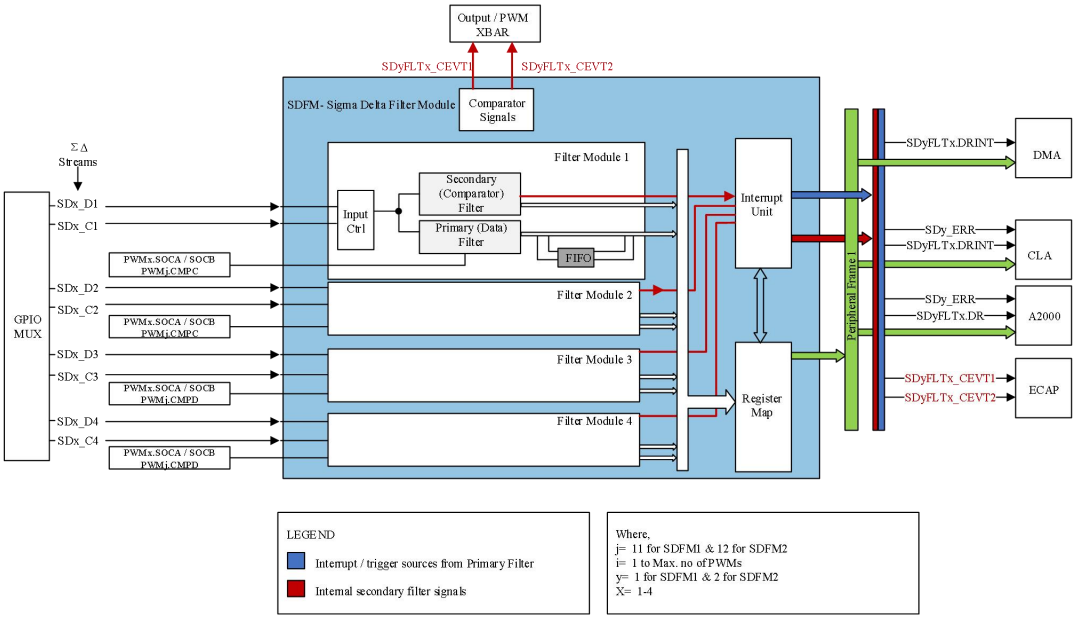


Figure 6- 63 Sigma Delta Filter Module (SDFM) Block Diagram

6.14.7.1 SDFM Electrical Data and Timing
SDFM 电气数据和时序

WARNING

Special precautions should be taken on both SD-Cx and SD-Dx signals to ensure a clean and noise-free signal that meets SDFM timing requirements. Precautions such as series termination resistors for ringing noise due to any impedance mismatch of clock driver and spacing of traces from other noisy signals are recommended.
应该对 SD-Cx 和SD-Dx 信号采取特殊的预防措施，确保信号干净且无干扰，满足 SDFM 时序要求。建议采取的预防措施包括对时钟驱动器的任何阻抗不匹配而导致的振铃噪声采用串联终端电阻，以及将布线与其他噪声信号隔离开来，等等。

Note

The SDFM SD-Cx and SD-Dx signals, when synchronized to PLLRAWCLK, provide protection against SDFM module corruption due to occasional random noise glitches that may result in a false comparator trip and filter output. However, the signals do not provide protection against persistent violations of the above timing requirements. Timing violations will result in data corruption proportional to the number of bits which violate the requirements.
SDFMSD-Cx 和 SD-Dx 信号与 PLLRAWCLK 同步后可防止由于偶尔出现的随机噪声干扰(这些干扰可能造成比较器跳闸和滤波器输出错误）而导致的 SDFM 模块损坏。但是，这些信号对持续违反上述时序要求的情况不提供保护。时序违规将损坏与违反要求的数据位数成正比的数据。

6.14.7.1.1 SDFM Timing Requirements When Using Asynchronous GPIO - ASYNC - Option
使用异步 GPIO(ASYNC)选项时的 SDFM 时序要求

		MIN	MAX	UNIT
Mode 0				
$t_{c(SDC)M0}$	Cycle time, SDx_Cy	$4*t_{c(PLLRAWCLK)}$	256 *SYSCLK period	ns
$t_{w(SDDHL)M0}$	Pulse duration, SDx_Dy (high / Low)	$2*t_{c(PLLRAWCLK)}$		ns
$t_{su(SDDV-SDCH)M0}$	Setup time, SDx_Dy valid before SDx_Cy goes high	$1*t_{c(PLLRAWCLK)}+3$		ns
$t_{h(SDCH-SDD)M0}$	Hold time, SDx_Dy wait after SDx_Cy goes high	$1*t_{c(PLLRAWCLK)}+3$		ns

6.15 Communications Peripherals
通信外设

6.15.1 Controller Area Network (CAN)
控制器局域网(CAN)

Note

The CAN module uses the IP known as *DCAN*. This document uses the names *CAN* and *DCAN* interchangeably to reference this peripheral.

CAN 模块使用称为 DCAN 的 IP。本文档交替使用名称 CAN 和 DCAN 来引用此外设。

The CAN module implements the following features:

- Complies with ISO11898-1 (Bosch® CAN protocol specification 2.0 A and B)
- Bit rates up to 1 Mbps
- Multiple clock sources
- 32 message objects (mailboxes), each with the following properties:
  - Configurable as receive or transmit
  - Configurable with standard (11-bit) or extended (29-bit) identifier
  - Supports programmable identifier receive mask
  - Supports data and remote frames
  - Holds 0 to 8 bytes of data
  - Parity-checked configuration and data RAM
- Individual identifier mask for each message object
- Programmable FIFO mode for message objects
- Programmable loopback modes for self-test operation
- Suspend mode for debug support
- Software module reset
- Automatic bus on after bus-off state by a programmable 32-bit timer
- Two interrupt lines
- DMA support

CAN 模块实现下列特性：

- 符合 ISO11898-1（Bosch®CAN 协议规范 2.0A 和 B）
- 最高 1Mbps 的比特率
- 多个时钟源
- 32 个消息对象（邮箱），每个对象具有以下属性：
  - 可配置为接收或者发送

  - 可配置标准（11 位）或扩展（29 位）标识符
  - 支持可编程标识符接收掩码
  - 支持数据和远程帧
  - 保留 0 到 8 个字节的数据
  - 奇偶校验配置和数据 RAM
- 每个消息对象的单独标识符掩码
- 消息对象的可编程 FIFO 模式
- 用于自检操作的可编程回环模式
- 调试支持的挂起模式
- 软件模块复位
- 由一个可编程 32 位定时器实现在总线关闭状态后自动开启总线
- 2 条中断线路
- DMA 支持

Note

For a CAN bit clock of 100 MHz, the smallest bit rate possible is 3.90625Kbps.

对于 100MHz 的 CAN 位时钟，最小比特率可以为 3.90625Kbps。

Note

The accuracy of the on-chip zero-pin oscillator is in the INTOSC Characteristics table. Depending on parameters such as the CAN bit timing settings, bit rate, bus length, and propagation delay, the accuracy of this oscillator may not meet the requirements of the CAN protocol. In this situation, an external clock source must be used.

片上零引脚振荡器的精度可在“INTOSC 特性”表中找到。根据相关参数（如 CAN 位时序设置、比特率、总线长度和传播延迟），此振荡器的精度可能不符合 CAN 协议的要求。在这种情况下，必须使用外部时钟源。

Figure 6-64 shows the CAN block diagram.
图 6-64 所示为 CAN 功能框图。

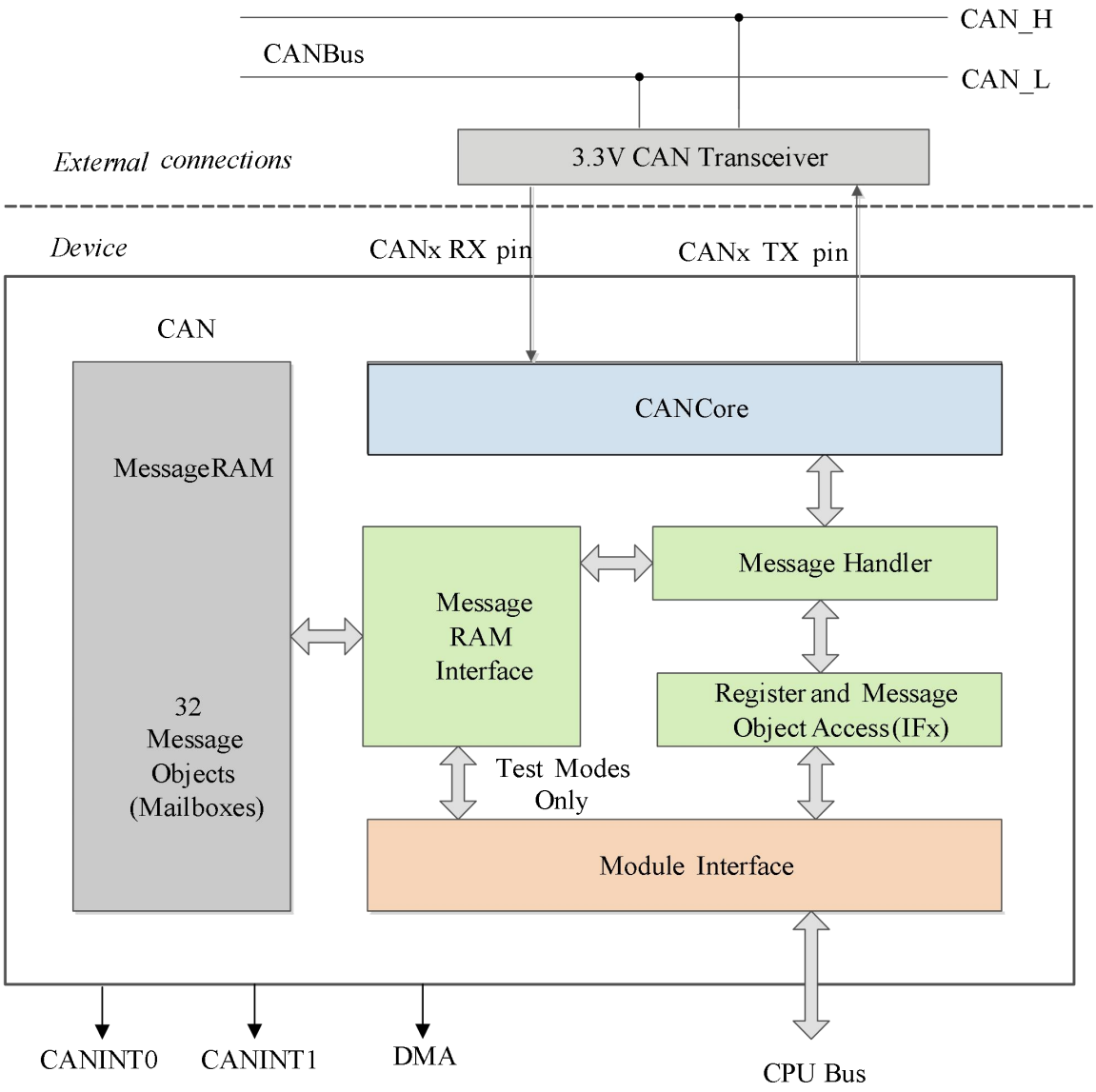


Figure 6-64 CAN Block Diagram

6.15.2 Modular Controller Area Network (MCAN)
模块化控制器局域网(MCAN)

The Controller Area Network (CAN) is a serial communications protocol that efficiently supports distributed real-time control with a high level of reliability. CAN has high immunity to electrical interference and the ability to detect various type of errors. In CAN, many short messages are broadcast to the entire network, which provides data consistency in every node of the system.
The MCAN module supports both classic CAN and CAN FD (CAN with flexible data-rate) protocols. The CAN FD feature allows higher throughput and increased payload per data frame. Classic CAN and CAN FD devices may coexist on the same network without any conflict provided that partial network transceivers, which can detect and ignore CAN FD without generating bus errors, are used by the classic CAN devices. The MCAN module is compliant to ISO 11898-1:2015.

控制器局域网(CAN)是一种串行通信协议，用于有效地为具有高可靠性的分布式实时控制提供支持。CAN 具有较高的抗电气干扰能力，并且能够检测各种类型的错误。在 CAN 中，许多较短的信息会广播到整个网络，从而在系统的每个节点中提供数据一致性。

MCAN 模块支持经典 CAN 和CANFD（具有灵活数据速率的 CAN）协议。CANFD 特性可实现更高的吞吐量和增加每个数据帧的有效负载。经典 CAN 和 CANFD 器件可以在同一网络上共存而不会发生任何冲突，前提是经典 CAN 器件使用部分网络收发器，其中该收发器可以检测和忽略 CANFD，而不产生总线错误。MCAN 模块符合 ISO11898-1:2015 标准。

Note

The availability of the CAN FD feature is dependent on the device's part number. Refer to the device data manual for more information.
CANFD 特性的可用性取决于器件的器件型号。更多信息，请参阅器件数据手册。

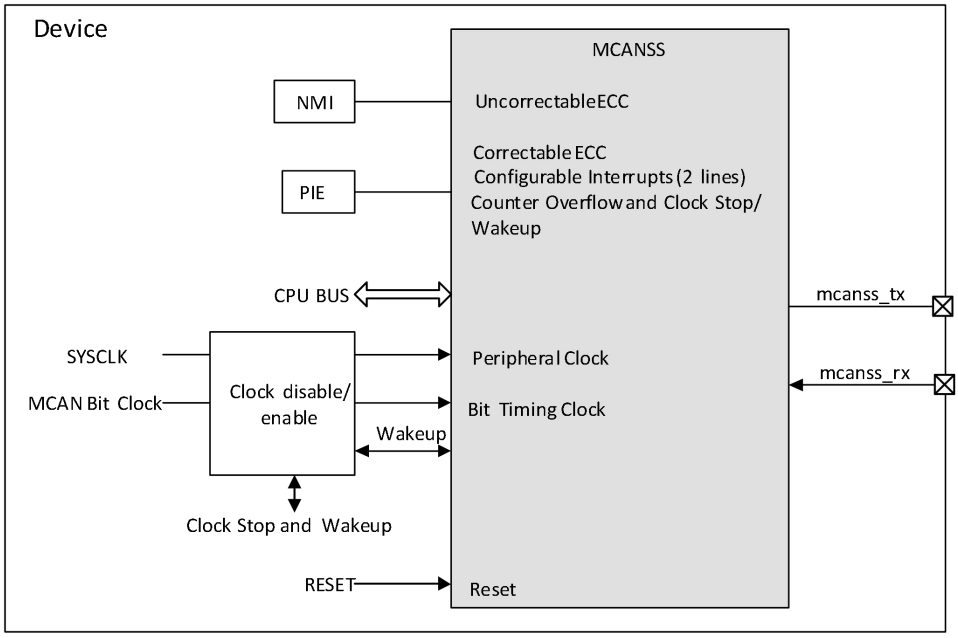


Figure 6- 65 MCAN Module Overview

The MCAN module implements the following features:

- Conforms with CAN Protocol 2.0 A, B and ISO 11898-1:2015
- Full CAN FD support (up to 64 data bytes)
- AUTOSAR and SAE J1939 support
- Up to 32 dedicated transmit buffers
- Configurable transmit FIFO, up to 32 elements
- Configurable transmit queue, up to 32 elements
- Configurable transmit Event FIFO, up to 32 elements
- Up to 64 dedicated receive buffers
- Two configurable receive FIFOs, up to 64 elements each
- Up to 128 filter elements
- Loop-back mode for self-test
- Maskable interrupt (two configurable interrupt lines, correctable ECC, counter overflow and clock stop/ wakeup)
- Non-maskable interrupt (uncorrectable ECC)
- Two clock domains (CAN clock/host clock)
- ECC check for Message RAM
- Clock stop and wakeup support
- Timestamp counter

Non-supported features:

- Host bus firewall
- Clock calibration
- Debug over CAN

MCAN 模块可实现以下特性：

- 符合 CAN 协议 2.0A、B 和 ISO11898-1:2015 标准
- 完全支持 CANFD（最多 64 个数据字节）
- 支持 AUTOSAR 和 SAEJ1939
- 多达 32 个专用发送缓冲器
- 可配置的发送 FIFO，最多 32 个元素
- 可配置的发送队列，最多 32 个元素
- 可配置的发送事件 FIFO，最多 32 个元素
- 多达 64 个专用接收缓冲器
- 两个可配置的接收 FIFO，每个 FIFO 最多 64 个元素
- 多达 128 个滤波器元素
- 用于自检的回环模式
- 可屏蔽中断（两条可配置的中断线路、可纠正的 ECC、计数器溢出和时钟停止/唤醒）
- 不可屏蔽中断（不可纠正的 ECC）
- 两个时钟域（CAN 时钟/主机时钟）
- 针对消息 RAM 的 ECC 检查
- 支持时钟停止和唤醒
- 时间戳计数器不支持的特性：
- 主机总线防火墙
- 时钟校准
- 通过 CAN 进行调试

6.15.3 Inter-Integrated Circuit (I2C)
内部集成电路(I2C)

The I2C module has the following features:

- Compliance with the NXP Semiconductors I^2C-bus specification (version 2.1):
  - Support for 8-bit format transfers
  - 7-bit and 10-bit addressing modes
  - General call
  - START byte mode
  - Support for multiple master-transmitters and slave-receivers
  - Support for multiple slave-transmitters and master-receivers
  - Combined master transmit/receive and receive/transmit mode
  - Data transfer rate from 10Kbps up to 400Kbps (Fast-mode)
- Supports voltage thresholds compatible to:
  - SMBus 2.0 and below
  - PMBus 1.2 and below
- One 16-byte receive FIFO and one 16-byte transmit FIFO
- Supports two ePIE interrupts
  - I2Cx interrupt – Any of the below conditions can be configured to generate an I2Cx interrupt:
    - Transmit Ready
    - Receive Ready
    - Register-Access Ready
    - No-Acknowledgment
    - Arbitration-Lost
    - Stop Condition Detected
    - Addressed-as-Slave
  - I2Cx_FIFO interrupts:
    - Transmit FIFO interrupt
    - Receive FIFO interrupt
- Module enable and disable capability
- Free data format mode

Figure 6- 66 shows how the I2C peripheral module interfaces within the device.

I2C 模块具有以下特性：

- 符合 NXPSemiconductorI2C 总线规范（版本 2.1）：
  - 支持 8 位格式传输
  - 7 位和 10 位寻址模式
  - 常规调用
  - START 字节模式
  - 支持多个主发送器和从接收器
  - 支持多个从发送器和主接收器
  - 组合主器件发送/接收和接收/发送模式
  - 数据传输速率从 10kbps 到高达 400Kbps（快速模式）
- 支持与以下兼容的电压阈值：
  - SMBus2.0 及更低版本
  - PMBus1.2 及更低版本
- 一个 16 字节接收 FIFO 和一个 16 字节发送 FIFO
- 支持两个 ePIE 中断

- I2Cx 中断-可以配置以下任何条件来生成 I2Cx 中断：
  - 发送就绪
  - 接收就绪
  - 寄存器访问就绪
  - 无确认
  - 仲裁丢失
  - 检测到停止条件
  - 被寻址为从器件
- I2Cx_FIFO 中断：
  - 发送 FIFO 中断
  - 接收 FIFO 中断
- 模块启用和禁用能力
- 自由数据格式模式

图 6-66 显示了 I2C 外设模块如何在器件内连接。

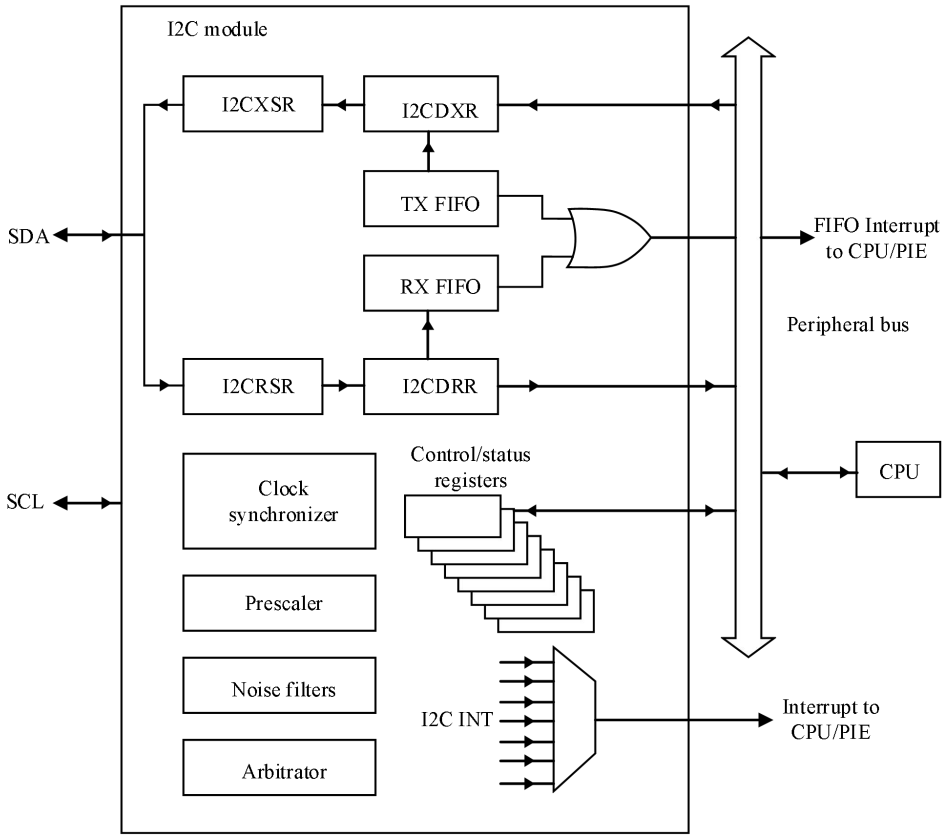


Figure 6- 66 I2C Peripheral Module Interfaces

6.15.3.1 I2C Electrical Data and Timing
I2C 电气数据和时序

Note

To meet all of the I2C protocol timing specifications, the I2C module clock must be configured in the range from 7 MHz to 12 MHz.

为了满足所有的 I2C 协议时序规范，I2C 模块时钟必须配置为 7MHz 至 12MHz 范围内的值。

6.15.3.1.1 I2C Timing Requirements
I2C 时序要求

NO.	PARAMETER		MIN	MAX	UNIT
Standard mode 标准模式					
T0	f_{mod}	I2C module frequency	7	12	MHz
T1	$t_{h(SDA-SCL)START}$	Hold time, START condition, SCL fall delay after SDA fall	4.0		µs
T2	$t_{su(SCL-SDA)START}$	Setup time, Repeated START, SCL rise before SDA fall delay	4.0		µs
T3	$t_{h(SCL-DAT)}$	Hold time, data after SCL fall	0		µs
T4	$t_{su(DAT-SCL)}$	Setup time, data before SCL rise	250		ns
T5	$t_{r(SDA)}$	Rise time, SDA		1000	ns
T6	$t_{r(SCL)}$	Rise time, SCL		1000	ns
T7	$t_{f(SDA)}$	Fall time, SDA		300	ns
T8	$t_{f(SCL)}$	Fall time, SCL		300	ns
T9	$t_{su(SCL-SDA)STOP}$	Setup time, STOP condition, SCL rise before SDA rise delay	4.0		µs
T10	$t_{w(SP)}$	Pulse duration of spikes that will be suppressed by filter	0	50	ns
T11	C_b	capacitance load on each bus line		400	pF
Fast mode 快速模式					
T0	f_{mod}	I2C module frequency	7	12	MHz
T1	$t_{h(SDA-SCL)START}$	Hold time, START condition, SCL fall delay after SDA fall	0.6		µs
T2	$t_{su(SCL-SDA)START}$	Setup time, Repeated START, SCL rise before SDA fall delay	0.6		µs
T3	$t_{h(SCL-DAT)}$	Hold time, data after SCL fall	0		µs
T4	$t_{su(DAT-SCL)}$	Setup time, data before SCL rise	100		ns
T5	$t_{r(SDA)}$	Rise time, SDA	20	300	ns
T6	$t_{r(SCL)}$	Rise time, SCL	20	300	ns
T7	$t_{f(SDA)}$	Fall time, SDA	11.4	300	ns
T8	$t_{f(SCL)}$	Fall time, SCL	11.4	300	ns
T9	$t_{su(SCL-SDA)STOP}$	Setup time, STOP condition, SCL rise before SDA rise delay	0.6		µs
T10	$t_{w(SP)}$	Pulse duration of spikes that will be suppressed by filter	0	50	ns
T11	C_b	capacitance load on each bus line		400	pF

6.15.3.1.2 I2C Switching Characteristics
I2C 开关特征

over recommended operating conditions (unless otherwise noted)
在推荐的工作条件下（除非另有说明）

NO.	PARAMETER		TEST CONDITIONS	MIN	MAX	UNIT
Standard mode 标准模式						
S1	f_{SCL}	SCL clock frequency		0	100	kHz
S2	T_{SCL}	SCL clock period		10		μs
S3	$t_{w(SCLL)}$	Pulse duration, SCL clock low		4.7		μs
S4	$t_{w(SCLH)}$	Pulse duration, SCL clock high		4.0		μs
S5	t_{BUF}	Bus free time between STOP and START conditions		4.7		μs
S6	$t_{v(SCL-DAT)}$	Valid time, data after SCL fall			3.45	μs
S7	$t_{v(SCL-ACK)}$	Valid time, Acknowledge after SCL fall			3.45	μs
S8	I_I	Input current on pins	$0.1V_{bus}<V_i<0.9V_{bus}$	–10	10	μA
Fast mode 快速模式						
S1	f_{SCL}	SCL clock frequency		0	400	kHz
S2	T_{SCL}	SCL clock period		2.5		μs
S3	$t_{w(SCLL)}$	Pulse duration, SCL clock low		1.3		μs
S4	$t_{w(SCLH)}$	Pulse duration, SCL clock high		0.6		μs
S5	t_{BUF}	Bus free time between STOP and START conditions		1.3		μs
S6	$t_{v(SCL-DAT)}$	Valid time, data after SCL fall			0.9	μs
S7	$t_{v(SCL-ACK)}$	Valid time, Acknowledge after SCL fall			0.9	μs
S8	I_I	Input current on pins	$0.1V_{bus}<V_i<0.9V_{bus}$	–10	10	μA

6.15.3.1.3 I2C Timing Diagram
I2C 时序图

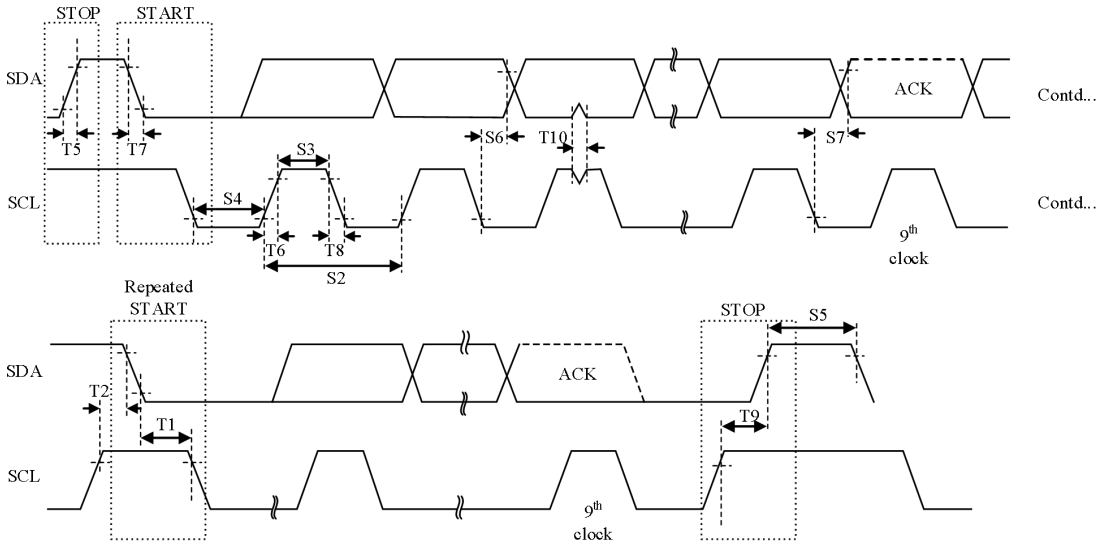


Figure 6- 67 I2C Timing Diagram

6.15.4 Power Management Bus (PMBus) Interface
电源管理总线(PMBus)接口

The PMBus module has the following features:

- Compliance with the SMI Forum PMBus Specification (Part I v1.0 and Part II v1.1)
- Supports voltage thresholds compatible to:
  - PMBus 1.2 and below
  - SMBus 2.0 and below
- Support for master and slave modes
- Support for I2C mode
- Support for two speeds:
  - Standard Mode: Up to 100 kHz
  - Fast Mode: 400 kHz
- Packet error checking
- CONTROL and ALERT signals
- Clock high and low time-outs
- Four-byte transmit and receive buffers
- One maskable interrupt, which can be generated by several conditions:
  - Receive data ready
  - Transmit buffer empty
  - Slave address received
  - End of message
  - ALERT input asserted
  - Clock low time-out
  - Clock high time-out
  - Bus free

PMBus 模块具有以下特性：

- 符合 SMIForumPMBus 规范（第 I 部分 v1.0 和第 II 部分 v1.1）
- 支持与以下兼容的电压阈值：
  - PMBus1.2 及更低版本
  - SMBus2.0 及更低版本
- 支持主模式和从模式
- 支持 I2C 模式
- 支持两种速度：
  - 标准模式：高达 100kHz
  - 快速模式：400kHz
- 数据包错误检查
- 控制和警报信号
- 时钟高电平和低电平超时
- 四字节发送和接收缓冲器
- 一个可屏蔽中断，可由以下几个条件生成：
  - 接收数据就绪
  - 发送缓冲器为空
  - 接收到从器件地址
  - 消息结束

- 警报输入被置为有效
- 时钟低超时
- 时钟高电平超时
- 总线空闲

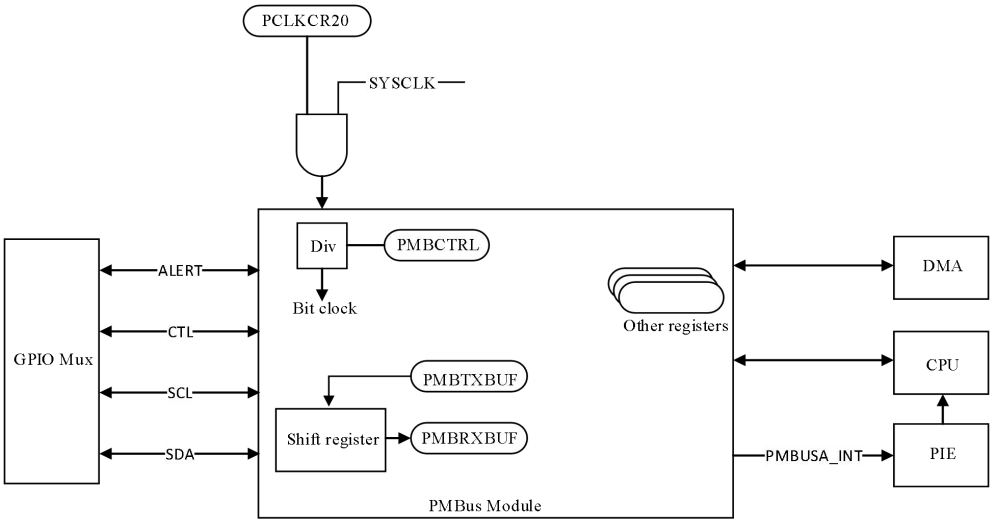


Figure 6- 68 PMBus Block Diagram

6.15.4.1 PMBus Electrical Data and Timing
PMBus 电气数据和时序

6.15.4.1.1 PMBus Electrical Characteristics
PMBus 电气特性

over recommended operating conditions (unless otherwise noted)
在建议运行条件下测得（除非另有说明）

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
V_{IL}	Valid low-level input voltage				0.8	V
V_{IH}	Valid high-level input voltage		2.1		VDDIO	V
V_{OL}	Low-level output voltage	At I_{pullup}=4mA			0.4	V
I_{OL}	Low-level output current	V_{OL}≤0.4V	4			mA
t_{SP}	Pulse width of spikes that must be suppressed by the input filter		0		50	ns
I_i	Input leakage current on each pin	0.1Vbus<Vi<0.9Vbus	–10		10	µA
C_i	Capacitance on each pin				10	pF

6.15.4.1.2 PMBus Fast Mode Switching Characteristics
PMBus 快速模式开关特性

over recommended operating conditions (unless otherwise noted)
在建议运行条件下测得（除非另有说明）

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
f_{FSM_CLK}	FSM_CLK clock frequency		SYSCLK/32		10	MHz
f_{SCL}	SCL clock frequency		10		400	kHz
t_{BUF}	Bus free time between STOP and START conditions		1.3			µs
$t_{HD;STA}$	START condition hold time -- SDA fall to SCL fall delay		0.6			µs
$t_{SU;STA}$	Repeated START setup time -- SCL rise to SDA fall delay		0.6			µs
$t_{SU;STO}$	STOP condition setup time -- SCL rise to SDA rise delay		0.6			µs
$t_{HD;DAT}$	Data hold time after SCL fall		300			ns
$t_{SU;DAT}$	Data setup time before SCL rise		100			ns
$t_{Timeou}t$	Clock low time-out		25		35	ms
t_{LOW}	Low period of the SCL clock		1.3			µs
t_{HIGH}	High period of the SCL clock		0.6		50	µs
$t_{LOW;SEXT}$	Cumulative clock low extend time (slave device)	From START to STOP			25	ms
$t_{LOW;MEXT}$	Cumulative clock low extend time (master device)	Within each byte			10	ms
t_r	Rise time of SDA and SCL	5% to 95%	20		300	ns
t_f	Fall time of SDA and SCL	95% to 5%	20		300	ns

6.15.4.1.3 PMBus Standard Mode Switching Characteristics
PMBus 标准模式开关特性

over recommended operating conditions (unless otherwise noted)
在建议运行条件下测得（除非另有说明）

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
f_{FSM_CLK}	FSM_CLK clock frequency		SYSCLK/32		10	MHz
f_{SCL}	SCL clock frequency		10		100	kHz
t_{BUF}	Bus free time between STOP and START conditions		4.7			µs
$t_{HD;STA}$	START condition hold time -- SDA fall to SCL fall delay		4			µs
$t_{SU;STA}$	Repeated START setup time -- SCL rise to SDA fall delay		4.7			µs
$t_{SU;STO}$	STOP condition setup time -- SCL rise to SDA rise delay		4			µs
$t_{HD;DAT}$	Data hold time after SCL fall		300			ns
$t_{SU;DAT}$	Data setup time before SCL rise		250			ns
$t_{Timeout}$	Clock low time-out		25		35	ms
t_{LOW}	Low period of the SCL clock		4.7			µs
t_{HIGH}	High period of the SCL clock		4		50	µs
$t_{LOW;SEXT}$	Cumulative clock low extend time (slave device)	From START to STOP			25	ms
$t_{LOW;MEXT}$	Cumulative clock low extend time (master device)	Within each byte			10	ms
t_r	Rise time of SDA and SCL				1000	ns
t_f	Fall time of SDA and SCL				300	ns

6.15.5 Serial Communications Interface (SCI)
串行通信接口(SCI)

The SCI is a 2-wire asynchronous serial port, commonly known as a UART. The SCI module supports digital communications between the CPU and other asynchronous peripherals that use the standard non-return-to-zero (NRZ) format.
The SCI receiver and transmitter each have a 16-level-deep FIFO for reducing servicing overhead, and each has its own separate enable and interrupt bits. Both can be operated independently for half-duplex communication, or simultaneously for full-duplex communication. To specify data integrity, the SCI checks received data for break detection, parity, overrun, and framing errors. The bit rate is programmable to different speeds through a 16-bit baud-select register.

SCI 是一种双线制异步串行端口，通常称为 UART。SCI 模块支持 CPU 与其他异步外设之间使用标准非归零码（NRZ)格式的数字通信。

SCI 发送器和接收器都有一个用于减少服务开销的 16 级深度 FIFO，且具有各自独立的使能位和中断位。两者都能独立进行半双工通信，或同时进行全双工通信。为了指定数据完整性，SCI 检查接收到的数据是否存在中断检测、奇偶校验、超限和成帧错误。比特率通过 16 位波特选择寄存器可编程为不同的速度。

Features of the SCI module include:

- Two external pins:
  - SCITXD: SCI transmit-output pin
  - SCIRXD: SCI receive-input pin
  - Baud rate programmable to 64K different rates
- Data-word format
  - 1 start bit
  - Data-word length programmable from 1 to 8 bits
  - Optional even/odd/no parity bit
  - 1 or 2 stop bits
- Four error-detection flags: parity, overrun, framing, and break detection
- Two wake-up multiprocessor modes: idle-line and address bit
- Half- or full-duplex operation
- Double-buffered receive and transmit functions
- Transmitter and receiver operations can be accomplished through interrupt-driven or polled algorithms with status flags.
  - Transmitter: TXRDY flag (transmitter-buffer register is ready to receive another character) and TX EMPTY flag (transmitter-shift register is empty)
  - Receiver: RXRDY flag (receiver-buffer register is ready to receive another character), BRKDT flag (break condition occurred), and RX ERROR flag (monitoring four interrupt conditions)
- Separate enable bits for transmitter and receiver interrupts (except BRKDT)
- NRZ format
- Auto baud-detect hardware logic
- 16-level transmit and receive FIFO

SCI 模块的特性包括：

- 两个外部引脚：
  - SCITXD：SCI 发送-输出引脚
  - SCIRXD：SCI 接收-输入引脚
  - 波特率可编程为 64K 不同速率
- 数据字格式
  - 1 个起始位

  - 数据字长度可在 1 至 8 位之间编程
  - 可选偶数/奇数/无奇偶校验位
  - 1 个或 2 个停止位
- 四个错误检测标志：奇偶、溢满、帧格式和中断检测
- 两种唤醒多处理器模式：空闲线和地址位
- 半双工或全双工操作
- 双缓冲接收和发送功能
- 发送器和接收器操作可通过带有状态标志的中断驱动或轮询算法来完成。
  - 发送器：TXRDY 标志 (发送器缓冲寄存器已准备好接收另一个字符) 和 TXEMPTY 标志（发送器移位寄存器为空）
  - 接收器：RXRDY 标志（接收器缓冲寄存器已准备好接收另一个字符）、BRKDT 标志（发生了中断条件）和 RXERROR 标志（监测四个中断条件）
- 发送器和接收器中断的独立使能位（BRKDT 除外）
- NRZ 格式
- 自动波特检测硬件逻辑
- 16 级发送和接收 FIFO

Note

All registers in this module are 8-bit registers. When a register is accessed, the register data is in the lower byte (bits 7–0), and the upper byte (bits 15–8) is read as zeros. Writing to the upper byte has no effect.

此模块中的所有寄存器均为 8 位寄存器。当寄存器被访问时，寄存器数据位于低位字节（位 7-0），高位字节（位 15-8）读取为零。对高字节进行写入无效。

Figure 6- 69 shows the SCI block diagram.

图 6-69 显示了 SCI 模块框图。

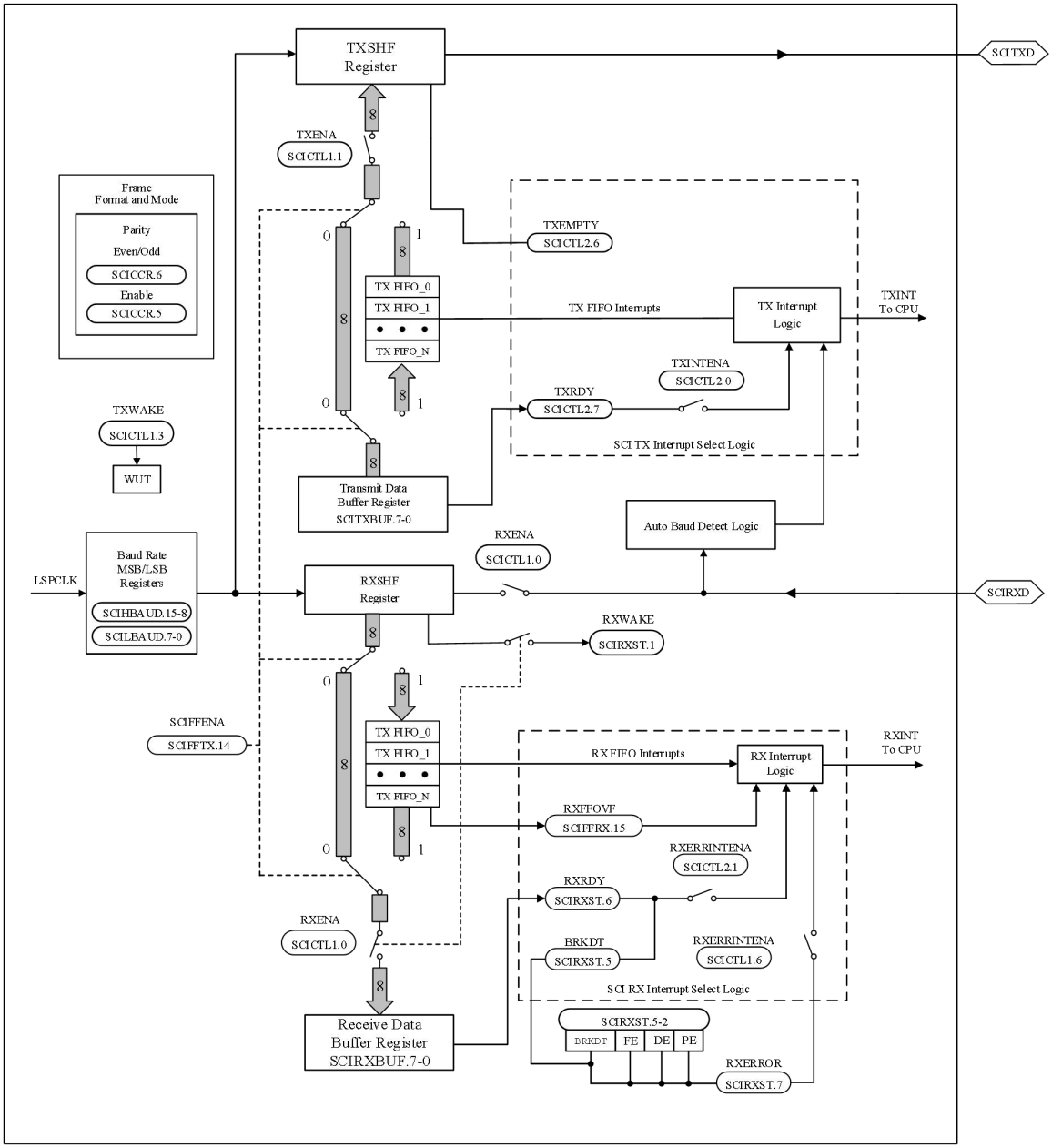


Figure 6- 69 SCI Block Diagram

6.15.6 Serial Peripheral Interface (SPI)
串行外设接口(SPI)

The serial peripheral interface (SPI) is a high-speed synchronous serial input and output (I/O) port that allows a serial bit stream of programmed length (1 to 16 bits) to be shifted into and out of the device at a programmed bit- transfer rate. The SPI is normally used for communications between the MCU controller and external peripherals or another controller. Typical applications include external I/O or peripheral expansion through devices such as shift registers, display drivers, and analog-to-digital converters (ADCs). Multidevice communications are supported by the master or slave operation of the SPI. The port supports a 16-level, receive and transmit FIFO for reducing CPU servicing overhead.

串行外设接口(SPI)是一种高速同步串行输入和输出(I/O)端口，其允许以编程的位传输速率将编程长度（1 至16 位）的串行位流移入和移出器件。SPI 通常用于 MCU 控制器与外部外设或另一控制器之间的通信。典型应用包括外部 I/O 或者通过诸如移位寄存器、显示驱动器和模数转换器(ADC)等器件进行外设扩展。SPI 的主/从工作模式支持多器件通信。该端口支持 16 级接收和发送 FIFO，以减少 CPU 服务开销。

The SPI module features include:

- SPISOMI: SPI slave-output/master-input pin
- SPISIMO: SPI slave-input/master-output pin
- $\overline{\text{SPISTE}}$: SPI slave transmit-enable pin
- SPICLK: SPI serial-clock pin
- Two operational modes: Master and Slave
- Baud rate: 125 different programmable rates. The maximum baud rate that can be employed is limited by the maximum speed of the I/O buffers used on the SPI pins.
- Data word length: 1 to 16 data bits
- Four clocking schemes (controlled by clock polarity and clock phase bits) include:
  - Falling edge without phase delay: SPICLK active-high. SPI transmits data on the falling edge of the SPICLK signal and receives data on the rising edge of the SPICLK signal.
  - Falling edge with phase delay: SPICLK active-high. SPI transmits data one half-cycle ahead of the falling edge of the SPICLK signal and receives data on the falling edge of the SPICLK signal.
  - Rising edge without phase delay: SPICLK inactive-low. SPI transmits data on the rising edge of the SPICLK signal and receives data on the falling edge of the SPICLK signal.
  - Rising edge with phase delay: SPICLK inactive-low. SPI transmits data one half-cycle ahead of the rising edge of the SPICLK signal and receives data on the rising edge of the SPICLK signal.
- Simultaneous receive and transmit operation (transmit function can be disabled in software)
- Transmitter and receiver operations are accomplished through either interrupt-driven or polled algorithm
- 16-level transmit/receive FIFO
- DMA support
- High-speed mode
- Delayed transmit control
- 3-wire SPI mode
- $\overline{\text{SPISTE}}$ inversion for digital audio interface receive mode on devices with two SPI modules

SPI 模块的特性包括：

- SPISOMI：SPI 从器件输出/主器件输入引脚
- SPISIMO：SPI 从器件输入/主器件输出引脚
- $\overline{\text{SPISTE}}$：SPI 从器件发送使能引脚
- SPICLK：SPI 串行时钟引脚

- 两种工作模式：主器件和从器件
- 波特率：125 个不同的可编程速率。可采用的最大波特率受限于 SPI 引脚上使用的 I/O 缓冲器的最大速度。
- 数据字长度：1 至 16 数据位
- 四种计时方案（由时钟极性和时钟相位的位控制）包含：
  - 无相位延迟的下降沿：SPICLK 高电平有效。SPI 在 SPICLK 信号的下降沿上发送数据，在 SPICLK 信号的上升沿上接收数据。
  - 有相位延迟的下降沿：SPICLK 高电平有效。SPI 在 SPICLK 信号下降沿提前半个周期发送数据，在 SPICLK 信号的下降沿上接收数据。
  - 无相位延迟的上升沿：SPICLK 低电平无效。SPI 在 SPICLK 信号的上升沿上发送数据，在 SPICLK 信号的下降沿上接收数据。
  - 有相位延迟的上升沿：SPICLK 低电平无效。SPI 在 SPICLK 信号上升沿的半个周期之前发送数据，而在 SPICLK 信号的上升沿上接收数据。
- 同时接收和发送操作（可在软件中禁用发送功能）
- 发送器和接收器操作通过中断驱动或轮询算法完成
- 16 级发送/接收 FIFO
- DMA 支持
- 高速模式
- 延迟的发送控制
- 3 线 SPI 模式
- 在带有两个 SPI 模块的器件上实现数字音频接口接收模式的$\overline{\text{SPISTE}}$反转

Figure 6- 70 shows the SPI CPU interfaces.

图 6-70 所示为 SPI CPU 接口。

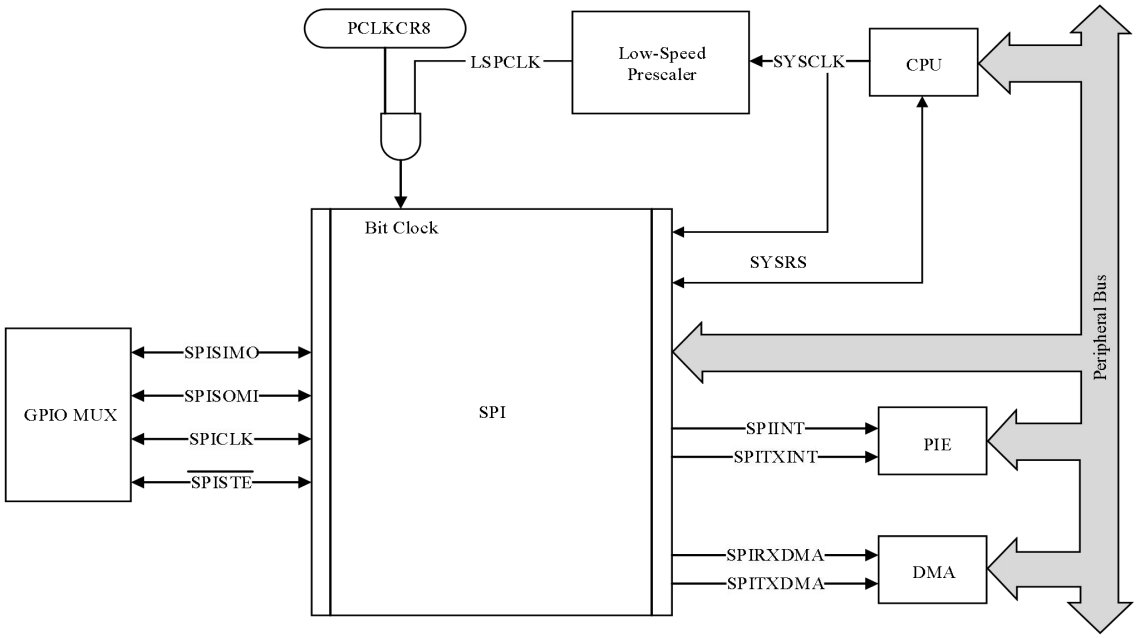


Figure 6- 70 SPI CPU Interface

6.15.6.1 SPI Master Mode Timings
SPI 主模式时序

The following section contains the SPI Master Mode Timings.
下一节介绍了 SPI 主模式时序。

Note

All timing parameters for SPI High-Speed Mode assume a load capacitance of 5 pF on SPICLK, SPISIMO, and SPISOMI.
SPI 高速模式的所有时序参数都假设 SPICLK、SPISIMO 和 SPISOMI 上的负载电容为 5pF。

6.15.6.1.1 SPI Master Mode Timing Requirements
SPI 主模式时序要求

NO.			(BRR+1)(1)	MIN	MAX	UNIT
High-Speed Mode 高速模式						
8	$t_{su(SOMI)M}$	Setup time, SPISOMI valid before SPICLK	Even, Odd	1		ns
9	$t_{h(SOMI)M}$	Hold time, SPISOMI valid after SPICLK	Even, Odd	6.5		ns
Normal Mode 正常模式						
8	$t_{su(SOMI)M}$	Setup time, SPISOMI valid before SPICLK	Even, Odd	15		ns
9	$t_{h(SOMI)M}$	Hold time, SPISOMI valid after SPICLK	Even, Odd	0		ns

(1) The (BRR + 1) condition is Even when (SPIBRR + 1) is even or SPIBRR is 0 or 2. It is Odd when (SPIBRR + 1) is odd and SPIBRR is greater than 3.
当(SPIBRR+1)为偶数或 SPIBRR 为0 或2 时，（BRR+1)条件为偶数。当(SPIBRR+1)为奇数且 SPIBRR 大于3 时，（BRR+1)条件为奇数。

6.15.6.1.2 SPI Master Mode Switching Characteristics - Clock Phase 0
SPI 主模式开关特性-时钟相位 0

over recommended operating conditions (unless otherwise noted)
在推荐的工作条件下（除非另有说明）

NO.	PARAMETER(1)(2)		(BRR+1)(3)	MIN	MAX	UNIT
General 通用						
1	$t_{c(SPC)M}$	Cycle time, SPICLK	Even	$4t_{c(LSPCLK)}$	$128t_{c(LSPCLK)}$	ns
			Odd	$5t_{c(LSPCLK)}$	$127t_{c(LSPCLK)}$	
2	$t_{w(SPC1)M}$	Pulse duration, SPICLK, first pulse	Even	$0.5t_{c(SPC)M}-1$	$0.5t_{c(SPC)M}+1$	ns
			Odd	$0.5t_{c(SPC)M}+0.5t_{c(LSPCLK)}-1$	$0.5t_{c(SPC)M}+0.5t_{c(LSPCLK)}+1$	
3	$t_{w(SPC2)M}$	Pulse duration, SPICLK, second pulse	Even	$0.5t_{c(SPC)M}-1$	$0.5t_{c(SPC)M}+1$	ns
			Odd	$0.5t_{c(SPC)M}-0.5t_{c(LSPCLK)}-1$	$0.5t_{c(SPC)M}-0.5t_{c(LSPCLK)}+1$	
23	$t_{d(SPC)M}$	Delay time, $\overline{SPISTE}$ active to SPICLK	Even	$1.5t_{c(SPC)M}-3t_{c(SYSCLK)}-3$	$1.5t_{c(SPC)M}-3t_{c(SYSCLK)}+3$	ns
			Odd	$1.5t_{c(SPC)M}-4t_{c(SYSCLK)}-3$	$1.5t_{c(SPC)M}-4t_{c(SYSCLK)}+3$	
24	$t_{v(STE)M}$	Valid time, SPICLK to $\overline{SPISTE}$ inactive	Even	$0.5t_{c(SPC)M}-3$	$0.5t_{c(SPC)M}+3$	ns
			Odd	$0.5t_{c(SPC)M}-0.5t_{c(LSPCLK)}-3$	$0.5t_{c(SPC)M}-0.5t_{c(LSPCLK)}+3$	
High-Speed Mode 高速模式						
4	$t_{d(SIMO)M}$	Delay time, SPICLK to SPISIMO valid	Even, Odd		1	ns
5	$t_{v(SIMO)M}$	Valid time, SPISIMO valid after SPICLK	Even	$0.5t_{c(SPC)M}-3$		ns
			Odd	$0.5t_{c(SPC)M}-0.5t_{c(LSPCLK)}-3$		
Normal Mode 正常模式						
4	$t_{d(SIMO)M}$	Delay time, SPICLK to SPISIMO valid	Even, Odd		2	ns
5	$t_{v(SIMO)M}$	Valid time, SPISIMO valid after SPICLK	Even	$0.5t_{c(SPC)M}-3$		ns
			Odd	$0.5t_{c(SPC)M}-0.5t_{c(LSPCLK)}-3$		

(1) 10-pF load on pin for High-Speed Mode.
高速模式下引脚上的负载为 10pF。

(2) 20-pF load on pin for Normal Mode.
正常模式下引脚上的负载为 20pF。

(3) The (BRR + 1) condition is Even when (SPIBRR + 1) is even or SPIBRR is 0 or 2. It is Odd when (SPIBRR + 1) is odd and SPIBRR is greater than 3.
当(SPIBRR+1)为偶数或 SPIBRR 为 0 或 2 时，(BRR+1)条件为偶数。当(SPIBRR+1)为奇数且 SPIBRR 大于 3 时，(BRR+1)条件为奇数。

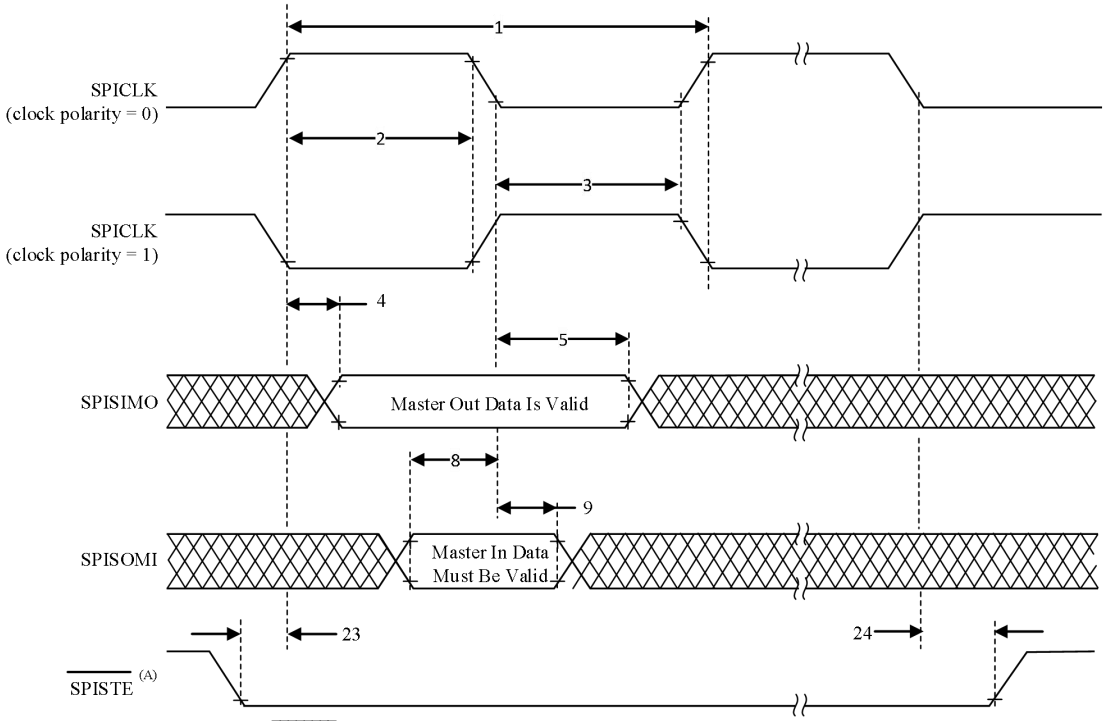
6.15.6.1.3 SPI Master Mode Switching Characteristics - Clock Phase 1
SPI 主模式开关特性-时钟相位 1

over recommended operating conditions (unless otherwise noted)
在推荐的工作条件下（除非另有说明）

NO.	PARAMETER(1)(2)		(BRR+1)	MIN	MAX	UNIT
General 通用						
1	$t_{c(SPC)M}$	Cycle time, SPICLK	Even	$4t_{c(LSPCLK)}$	$128t_{c(LSPCLK)}$	ns
			Odd	$5t_{c(LSPCLK)}$	$127t_{c(LSPCLK)}$	
2	$t_{w(SPCH)M}$	Pulse duration, SPICLK, first pulse	Even	$0.5t_{c(SPC)M}-1$	$0.5t_{c(SPC)M}+1$	ns
			Odd	$0.5t_{c(SPC)M}-0.5t_{c(LSPCLK)}-1$	$0.5t_{c(SPC)M}-0.5t_{c(LSPCLK)}+1$	
3	$t_{w(SPC2)M}$	Pulse duration, SPICLK, second pulse	Even	$0.5t_{c(SPC)M}-1$	$0.5t_{c(SPC)M}+1$	ns
			Odd	$0.5t_{c(SPC)M}+0.5t_{c(LSPCLK)}-1$	$0.5t_{c(SPC)M}+0.5t_{c(LSPCLK)}+1$	
23	$t_{d(SPC)M}$	Delay time, $\overline{SPISTE}$ valid to SPICLK	Even, Odd	$2t_{c(SPC)M}-3t_{c(SYSCLK)}-3$	$2t_{c(SPC)M}-3t_{c(SYSCLK)}+3$	ns
24	$t_{d(STE)M}$	Delay time, SPICLK to $\overline{SPISTE}$ invalid	Even	-3	3	ns
			Odd	-3	3	
High-Speed Mode 高速模式						
4	$t_{d(SIMO)M}$	Delay time, SPISIMO valid to SPICLK	Even	$0.5t_{c(SPC)M}-2$		ns
			Odd	$0.5t_{c(SPC)M}+0.5t_{c(LSPCLK)}-2$		
5	$t_{v(SIMO)M}$	Valid time, SPISIMO valid after SPICLK	Even	$0.5t_{c(SPC)M}-3$		ns
			Odd	$0.5t_{c(SPC)M}-0.5t_{c(LSPCLK)}-3$		
Normal Mode 正常模式						
4	$t_{d(SIMO)M}$	Delay time, SPISIMO valid to SPICLK	Even	$0.5t_{c(SPC)M}-2$		ns
			Odd	$0.5t_{c(SPC)M}+0.5t_{c(LSPCLK)}-2$		
5	$t_{v(SIMO)M}$	Valid time, SPISIMO valid after SPICLK	Even	$0.5t_{c(SPC)}M-3$		ns
			Odd	$0.5t_{c(SPC)M}-0.5t_{c(LSPCLK)}-3$		

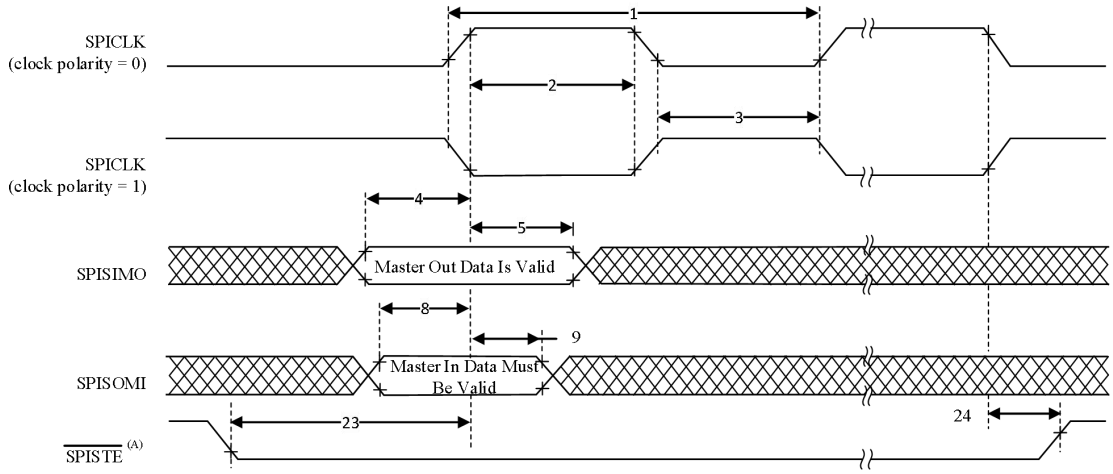
(1) 10-pF load on pin for High-Speed Mode.
高速模式下引脚上的负载为 10pF。
(2) 20-pF load on pin for Normal Mode.
正常模式下引脚上的负载为 20pF。

6.15.6.1.4 SPI Master Mode Timing Diagrams
SPI 主模式时序图



A. On the trailing end of the word, SPISTE will go inactive except between back-to-back transmit words in both FIFO and non-FIFO modes.
除了在FIFO 和非FIFO 模式下的背对背传输字之间的情况外，在字的尾端，SPISTE将变为停止状态。

Figure 6- 71 SPI Master Mode External Timing (Clock Phase = 0)



A. On the trailing end of the word, SPISTE will go inactive except between back-to-back transmit words in both FIFO and non-FIFO modes.
除了在FIFO 和非FIFO 模式下的背对背传输字之间的情况外，在字的尾端，SPISTE将变为停止状态。

Figure 6- 72 SPI Master Mode External Timing (Clock Phase = 1)

6.15.6.2 SPI Slave Mode Timings
SPI 从模式时序

The following section contains the SPI Slave Mode Timings.
下一节介绍了 SPI 从模式时序。

6.15.6.2.1 SPI Slave Mode Timing Requirements
SPI 从模式时序要求

NO.			MIN	MAX	UNIT
12	$t_{c(SPC)S}$	Cycle time, SPICLK	$4t_{c(SYSCLK)}$		ns
13	$t_{w(SPC1)S}$	Pulse duration, SPICLK, first pulse	$2t_{c(SYSCLK)}-1$		ns
14	$t_{w(SPC2)S}$	Pulse duration, SPICLK, second pulse	$2t_{c(SYSCLK)}-1$		ns
19	$t_{su(SIMO)S}$	Setup time, SPISIMO valid before SPICLK	$1.5t_{c(SYSCLK)}$		ns
20	$t_{h(SIMO)S}$	Hold time, SPISIMO valid after SPICLK	$1.5t_{c(SYSCLK)}$		ns
25	$t_{su(STE)S}$	Setup time, $\overline{SPISTE}$ valid before SPICLK (Clock Phase = 0)	$2t_{c(SYSCLK)}+15$		ns
		Setup time, $\overline{SPISTE}$ valid before SPICLK (Clock Phase = 1)	$2t_{c(SYSCLK)}+15$		ns
26	$t_{h(STE)S}$	Hold time, $\overline{SPISTE}$ invalid after SPICLK	$1.5t_{c(SYSCLK)}$		ns

6.15.6.2.2 SPI Slave Mode Switching Characteristics
SPI 从模式开关特性

over recommended operating conditions (unless otherwise noted)
在推荐的工作条件下（除非另有说明）

NO.	PARAMETER(1)		MIN	MAX	UNIT
15	$t_{d(SOMI)S}$	Delay time, SPICLK to SPISOMI valid		16.5	ns
16	$t_{v(SOMI)S}$	Valid time, SPISOMI valid after SPICLK	0		ns

(1) 20-pF load on pin.
引脚上20pF 负载。

6.15.6.2.3 SPI Slave Mode Timing Diagrams
SPI 从模式时序图

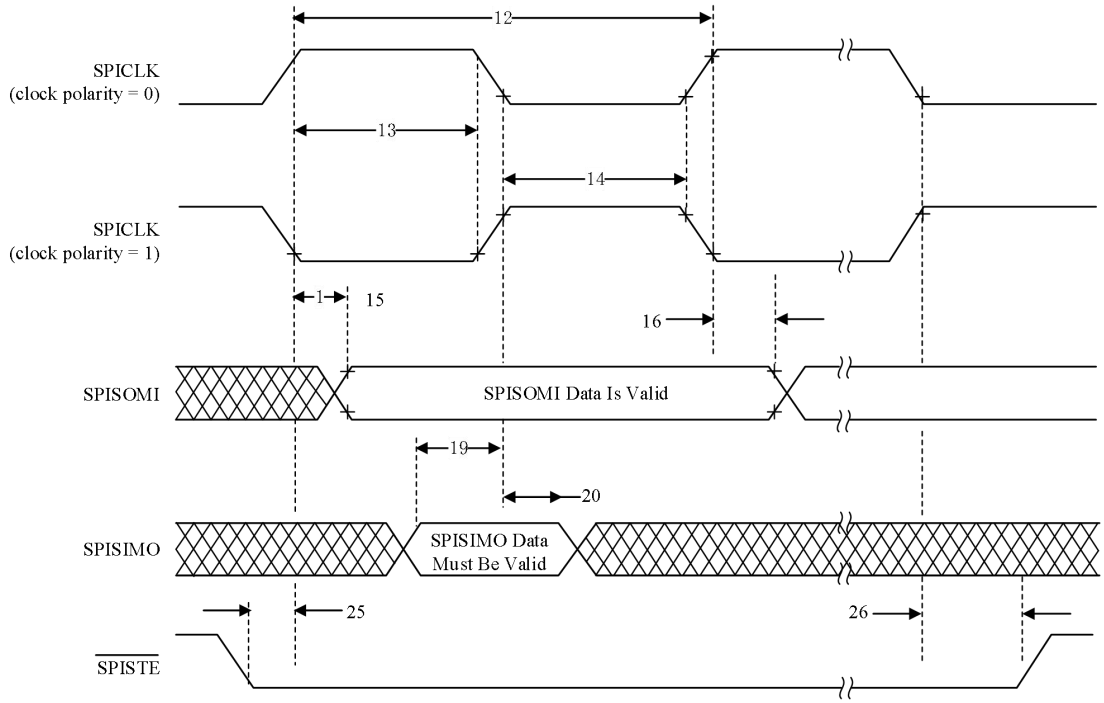


Figure 6- 73 SPI Slave Mode External Timing (Clock Phase = 0)

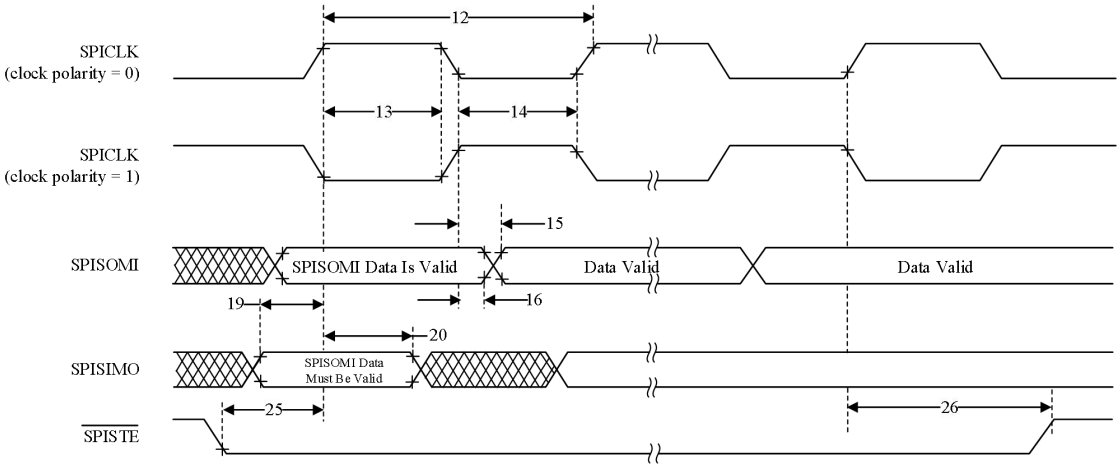


Figure 6- 74 SPI Slave Mode External Timing (Clock Phase = 1)

6.15.7 Local Interconnect Network (LIN)
本地互连网络(LIN)

This device contains one Local Interconnect Network (LIN) module. The LIN module adheres to the LIN 2.1 standard as defined by the *LIN Specification Package Revision 2.1*. The LIN is a low-cost serial interface designed for applications where the CAN protocol may be too expensive to implement, such as small subnetworks for cabin comfort functions like interior lighting or window control in an automotive application.
The LIN standard is based on the SCI (UART) serial data link format. The communication concept is single- master and multiple-slave with a message identification for multicast transmission between any network nodes.
The LIN module can be programmed to work either as an SCI or as a LIN as the core of the module is an SCI. The hardware features of the SCI are augmented to achieve LIN compatibility. The SCI module is a universal asynchronous receiver-transmitter (UART) that implements the standard non-return-to-zero format.
Though the registers are common for LIN and SCI, the register descriptions have notes to identify the register/bit usage in different modes. Because of this, code written for this module cannot be directly ported to the stand- alone SCI module and vice versa.

此器件包含一个本地互连网络(LIN)模块。LIN 模块遵循 *LIN 规范包修订版2.1* 定义的 LIN2.1 标准。LIN 是一种低成本串行接口，专为 CAN 协议实施成本太高的应用而设计，例如用于汽车应用中车内照明或车窗控制等舒适功能的小型子网。

LIN 标准基于 SCI(UART)串行数据连接格式。该接口的通信概念是在任何网络节点之间实现具有消息标识的单主/多从式多播传输。

LIN 模块可以编程作为一个 SCI 或作为一个 LIN 运行，因为此模块的内核为 SCI。SCI 的硬件特性得到增强以实现 LIN 兼容性。SCI 模块是一个通用异步收发器(UART)，可实现标准的非归零格式。

虽然LIN 和SCI 的寄存器是通用的，但是寄存器说明中有相应注释指出不同模式下的寄存器/位用法。因此，为该模块编写的代码无法直接移植到独立 SCI 模块，反之亦然。

The LIN module has the following features:

- Compatibility with LIN 1.3, 2.0 and 2.1 protocols
- Configurable baud rate up to 20 kbps (as per LIN 2.1 protocol)
- Two external pins: LINRX and LINTX
- Multibuffered receive and transmit units
- Identification masks for message filtering
- Automatic master header generation
  - Programmable synchronization break field
  - Synchronization field
  - Identifier field
- Slave automatic synchronization
  - Synchronization break detection
  - Optional baud rate update
  - Synchronization validation
- 2^{31} programmable transmission rates with 7 fractional bits
- Wakeup on LINRX dominant level from transceiver
- Automatic wakeup support
  - Wakeup signal generation
  - Expiration times on wakeup signals
- Automatic bus idle detection
- Error detection
  - Bit error
  - Bus error

  - No-response error
  - Checksum error
  - Synchronization field error
  - Parity error
- Capability to use direct memory access (DMA) for transmit and receive data
- Two interrupt lines with priority encoding for:
  - Receive
  - Transmit
  - ID, error, and status
- Support for LIN 2.0 checksum
- Enhanced synchronizer finite state machine (FSM) support for frame processing
- Enhanced handling of extended frames
- Enhanced baud rate generator
- Update wakeup/go to sleep

LIN 模块具有以下特性：

- 与 LIN1.3、2.0 和 2.1 协议兼容
- 波特率最高可配置为 20kbps（根据 LIN2.1 协议）
- 两个外部引脚：LINRX 和 LINTX
- 多缓冲接收和发送单元
- 针对信息过滤的识别掩码
- 自动主器件头文件生成
  - 可编程同步中断域
  - 同步域
  - 标识符域
- 从器件自动同步
  - 同步中断检测
  - 可选波特率更新
  - 同步验证
- 使用 7 个分数位支持 231 个可编程传输速率
- 从收发器在 LINRX 主级别上唤醒
- 自动唤醒支持
  - 唤醒信号生成
  - 唤醒信号超期时间
- 自动总线闲置检测
- 错误检测
  - 位错误
  - 总线错误
  - 无响应错误
  - 校验和错误
  - 同步域错误
  - 奇偶校验错误

- 能够使用直接存储器访问(DMA)发送和接收数据
- 两个中断线路带有以下项目的优先级编码：
  - 接收
  - 发送
  - ID、错误和状态
- 支持 LIN2.0 校验和
- 增强型同步器有限状态机(FSM)支持帧处理
- 增强对扩展帧的处理能力
- 增强型波特率发生器
- 更新唤醒/进入睡眠模式

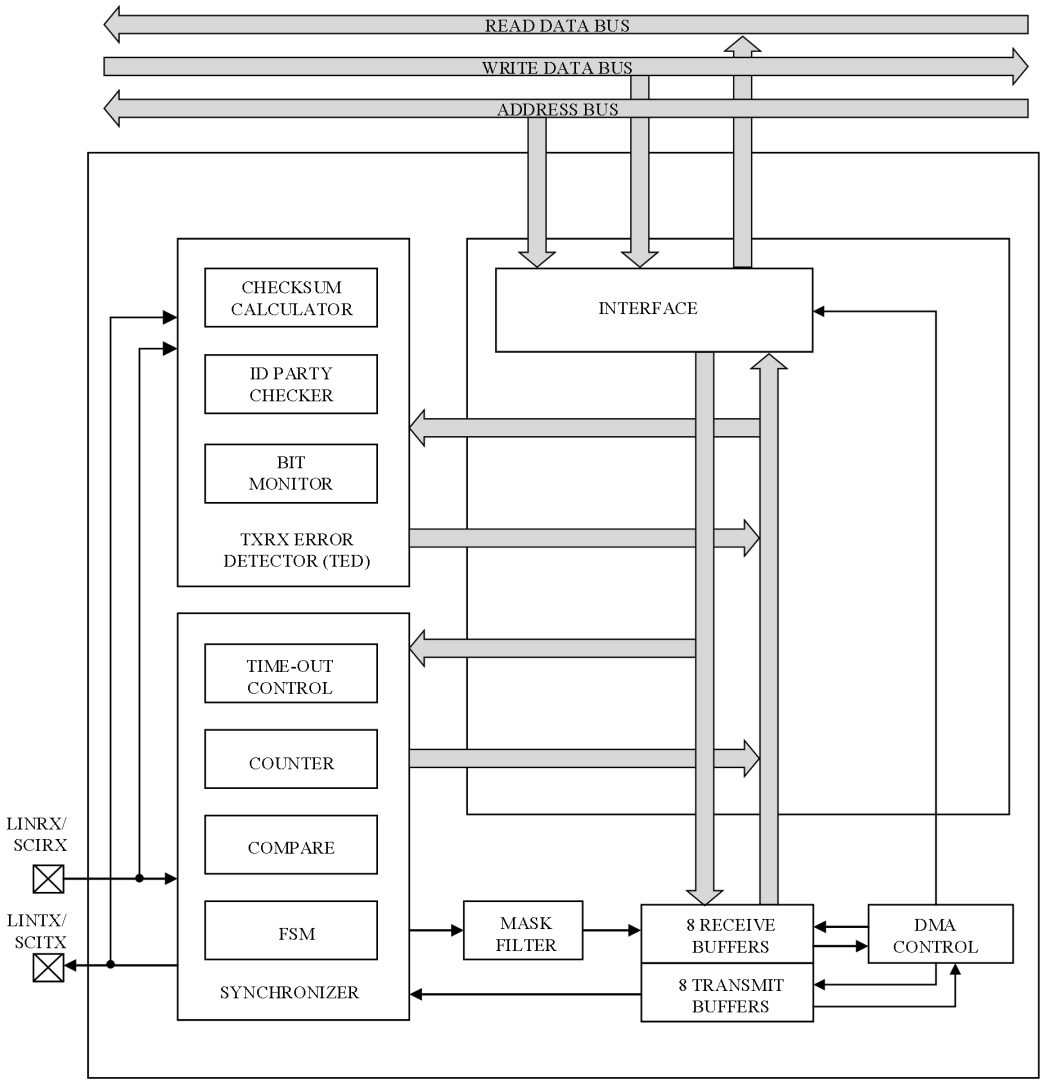


Figure 6- 75 LIN Block Diagram

6.15.8 Fast Serial Interface (FSI)
快速串行接口(FSI)

The Fast Serial Interface (FSI) module is a serial communication peripheral capable of reliable and robust high-speed communications. The FSI is designed to ensure data robustness across many system conditions such as chip-to-chip as well as board-to-board across an isolation barrier. Payload integrity checks such as CRC, start- and end-of-frame patterns, and user-defined tags, are encoded before transmit and then verified after receipt without additional CPU interaction. Line breaks can be detected using periodic transmissions, all managed and monitored by hardware. The FSI is also tightly integrated with other control peripherals on the device. To ensure that the latest sensor data or control parameters are available, frames can be transmitted on every control loop period. An integrated skew-compensation block has been added on the receiver to handle skew that may occur between the clock and data signals due to a variety of factors, including trace-length mismatch and skews induced by an isolation chip. With embedded data robustness checks, data-link integrity checks, skew compensation, and integration with control peripherals, the FSI can enable high-speed, robust communication in any system. These and many other features of the FSI follow.

快速串行接口(FSI)是一种能够进行可靠而稳健的高速通信的串行通信外设。FSI 旨在确保在芯片到芯片之间以及跨越隔离栅的板到板之间等诸多系统条件下实现数据稳健性。CRC、帧起始和结束模式以及用户定义的标签等有效载荷完整性检查在发送前进行编码，然后在接收后进行验证，无需额外的 CPU 交互。可以使用定期发送的方式检测线路中断，所有这些均由硬件管理和监控。FSI 还与器件上的其他控制外设紧密集成。为了确保提供最新的传感器数据或控制参数，可以在每个控制环路周期发送帧。为了处理由于各种因素而在时钟和数据信号之间可能出现的偏移（包括布线长度不匹配和隔离芯片引起的偏移），接收器上添加了一个集成的偏移补偿块。借助嵌入式数据稳健性检查、数据链路完整性检查、偏移补偿以及与控制外设的集成，FSI 可以在任何系统中实现高速、稳健的通信。FSI 除了这些特性，还有许多其他特性，如下所示。

The FSI module includes the following features:

- Independent transmitter and receiver cores
- Source-synchronous transmission
- Dual data rate (DDR)
- One or two data lines
- Programmable data length
- Skew adjustment block to compensate for board and system delay mismatches
- Frame error detection
- Programmable frame tagging for message filtering
- Hardware ping to detect line breaks during communication (ping watchdog)
- Two interrupts per FSI core
- Externally triggered frame generation
- Hardware- or software-calculated CRC
- Embedded ECC computation module
- Register write protection
- DMA support
- SPI compatibility mode (limited features available)

FSI 模块包含以下特性：

- 独立的发送器和接收器内核
- 源同步发送
- 双数据速率(DDR)
- 一条或两条数据线路
- 可编程数据长度

- 偏移调整块可针对电路板和系统延迟的不匹配部分进行补偿
- 帧错误检测
- 通过可编程的帧标记进行消息过滤
- 通过硬件 ping 检测通信期间的线路中断（ping 看门狗）
- 每个 FSI 内核对应两个中断
- 外部触发帧生成
- 由硬件或软件计算 CRC
- 嵌入式 ECC 计算模块
- 寄存器写保护
- DMA 支持
- SPI 兼容模式（可用功能受限）

Operating the FSI at maximum speed (60 MHz) at dual data rate (120Mbps) may require the integrated skew compensation block to be configured according to the specific operating conditions on a case-by-case basis. The FSI consists of independent transmitter (FSITX) and receiver (FSIRX) cores. The FSITX and FSIRX cores are configured and operated independently. The features available on the FSITX and FSIRX are described in the *FSI Transmitter* section and the *FSI Receiver* section, respectively.

为了让 FSI 在双数据速率(120Mbps)下以最大速度(60MHz)运行，可能需要根据具体运行条件逐例配置集成的偏差补偿块。

FSI 包含独立发送器(FSITX)和接收器(FSIRX)内核。FSITX 和 FSIRX 内核是独立配置和运行的。FSI 发送器部分和 FSI 接收器部分分别介绍了 FSITX 和 FSIRX 上可用的功能。

6.15.8.1 FSI Transmitter
FSI 变送器

The FSI transmitter module handles the framing of data, CRC generation, signal generation of TXCLK, TXD0, and TXD1, as well as interrupt generation. The operation of the transmitter core is controlled and configured through programmable control registers. The transmitter control registers let the CPU program, control, and monitor the operation of the FSI transmitter. The transmit data buffer is accessible by the CPU and the DMA.

FSI 变送器模块会处理数据组帧、CRC 生成、TXCLK、TXD0 和 TXD1 的信号生成以及中断生成。变送器内核的运行通过可编程控制寄存器进行控制和配置。变送器控制寄存器可让 CPU 对 FSI 变送器的运行进行编程、控制和监控。CPU 和 DMA 均可访问发送数据缓冲器。

The transmitter has the following features:

- Automated ping frame generation
- Externally triggered ping frames
- Externally triggered data frames
- Software-configurable frame lengths
- 16-word data buffer
- Data buffer underrun and overrun detection
- Hardware-generated CRC on data bits
- Software ECC calculation on select data
- DMA support

变送器具有以下特性：

- 自动生成 ping 帧
- 外部触发 ping 帧
- 外部触发数据帧
- 可通过软件配置帧长度
- 16 字数据缓冲器
- 数据缓冲器欠运转和溢出检测
- 硬件生成数据位 CRC
- 使用软件对选定数据进行 ECC 计算
- DMA 支持

Figure 6- 76 shows the FSITX CPU interface. Figure 6- 77 shows the high-level block diagram of the FSITX. Not all data paths and internal connections are shown. This diagram provides a high-level overview of the internal modules present in the FSITX.

图 6-76 所示为 FSITXCPU 接口。图 6-77 所示为 FSITX 的简要方框图。图中并未显示所有数据路径和内部连接。此图提供了 FSITX 中存在的内部模块的简要概览。

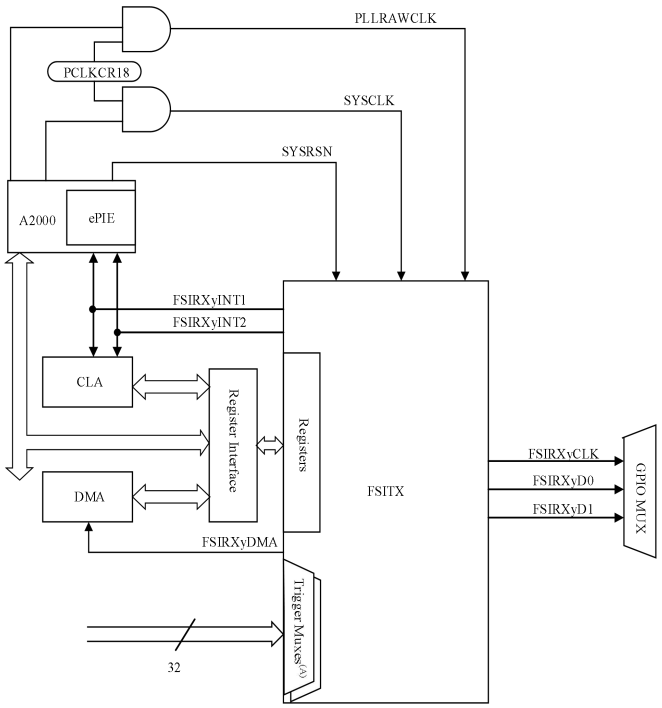


Figure 6- 76 FSITX CPU Interface

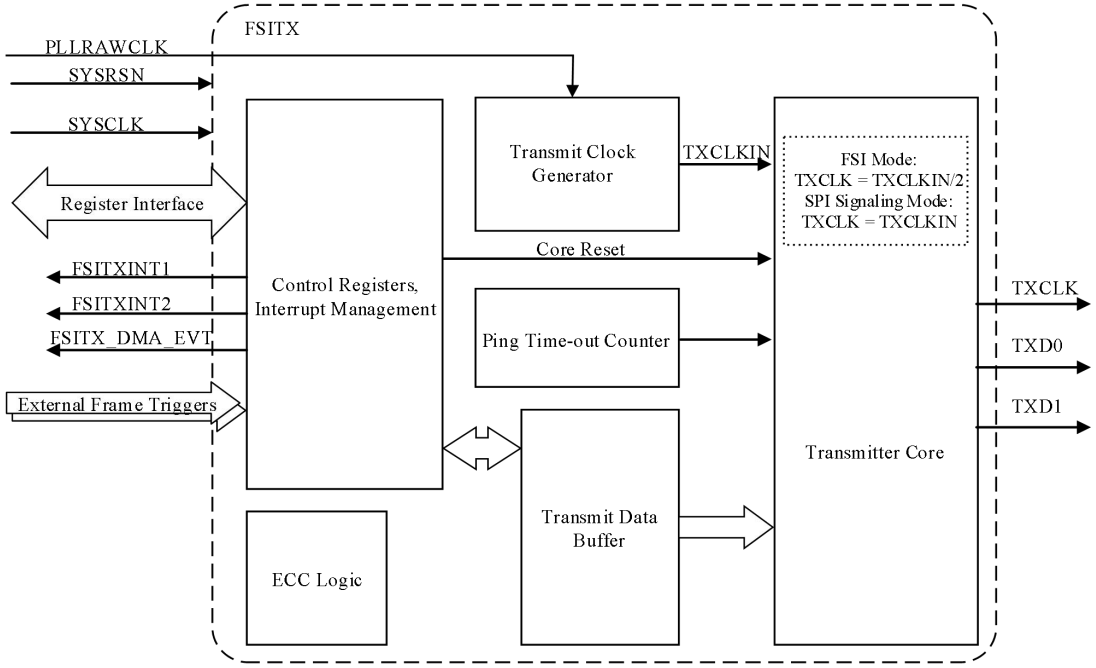


Figure 6- 77 FSITX Block Diagram

6.15.8.1.1 FSITX Electrical Data and Timing
FSITX 电气数据和时序

6.15.8.1.1.1 FSITX Switching Characteristics
FSITX 开关特性

over operating free-air temperature range (unless otherwise noted)
在自然通风条件下的工作温度范围内测得（除非另有说明）

NO.	PARAMETER(1)		MIN	MAX	UNIT
1	$t_{c(TXCLK)}$	Cycle time, TXCLK	16.67		ns
2	$t_{w(TXCLK)}$	Pulse width, TXCLK low or TXCLK high	$(0.5t_{c(TXCLK)})–1$	$(0.5t_{c(TXCLK)})+1$	ns
3	$t_{d(TXCLK–TXD)}$	Delay time, TXCLK rising or falling toTXD valid	$(0.25t_{c(TXCLK)})–2$	$(0.25t_{c(TXCLK)})+2$	ns
4	$t_{d(TXCLK)}$	TXCLK delay compensation at TX_DLYLINE_CTRL[TXCLK_DLY]=31	9.95	30	ns
5	$t_{d(TXD0)}$	TXD0 delay compensation at TX_DLYLINE_CTRL[TXD0_DLY]=31	9.95	30	ns
6	$t_{d(TXD1)}$	TXD1 delay compensation at TX_DLYLINE_CTRL[TXD1_DLY]=31	9.95	30	ns
7	$t_{d(DELAY_ELEMENT)}$	Incremental delay of each delay line element for TXCLK, TXD0, and TXD1	0.3	1	ns
TDM1	$t_{skew(TDM_CLK-TDM_Dx)}$	Delay skew introduced between TXCLK- TDM_CLK delay and TXDx-TDM_Dx delays	-2.5	2.5	ns

(1) 10-pF load on pin.
引脚上的10pF 负载。

6.15.8.1.1.2 FSITX Timings
FSITX 时序

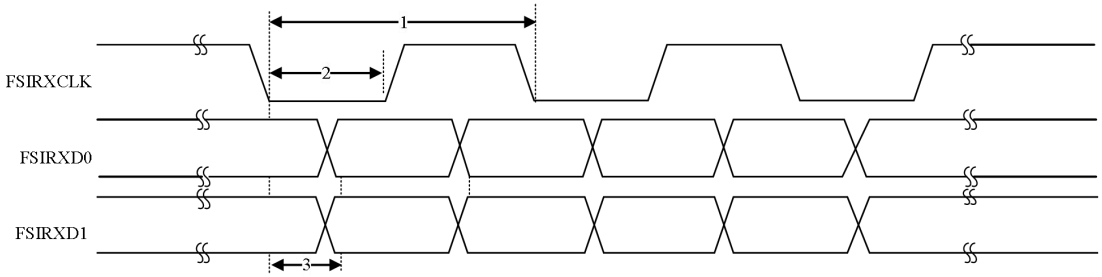


Figure 6- 78 FSITX Timings

6.15.8.2 FSI Receiver
FSI 接收器

The receiver module interfaces to the FSI clock (RXCLK), and data lines (RXD0 and RXD1) after they pass through an optional programmable delay line. The receiver core handles the data framing, CRC computation, and frame-related error checking. The receiver bit clock and state machine are run by the RXCLK input, which is asynchronous to the device system clock.
The receiver control registers let the CPU program, control, and monitor the operation of the FSIRX. The receive data buffer is accessible by the CPU, HIC, and the DMA.

接收器模块在通过可选的可编程延迟线路后连接到 FSI 时钟(RXCLK)和数据线路（RXD0 和 RXD1）。接收器内核会处理数据组帧、CRC 计算和与帧相关的错误检查。接收器位时钟和状态机由与器件系统时钟异步的 RXCLK 输入运行。

接收器控制寄存器可让 CPU 对FSIRX 的运行进行编程、控制和监控。CPU、HIC 和DMA 均可访问接收数据缓冲器。

The receiver core has the following features:

- 16-word data buffer
- Multiple supported frame types
- Ping frame watchdog
- Frame watchdog
- CRC calculation and comparison in hardware
- ECC detection
- Programmable delay line control on incoming signals
- DMA support
- SPI compatibility mode

接收器内核具有以下特性：

- 16 字数据缓冲器
- 支持多种帧类型
- Ping 帧看门狗
- 帧看门狗
- 在硬件中进行 CRC 计算和比较
- ECC 检测
- 针对输入信号的可编程延迟线路控制
- DMA 支持

- SPI 兼容模式

Figure 6-79 shows the FSIRX CPU interface. Figure 6-80 provides a high-level overview of the internal modules present in the FSIRX. Not all data paths and internal connections are shown.

图6-79 所示为 FSIRXCPU 接口。图6-80 提供了 FSIRX 中存在的内部模块的简要概览。图中并未显示所有数据路径和内部连接。

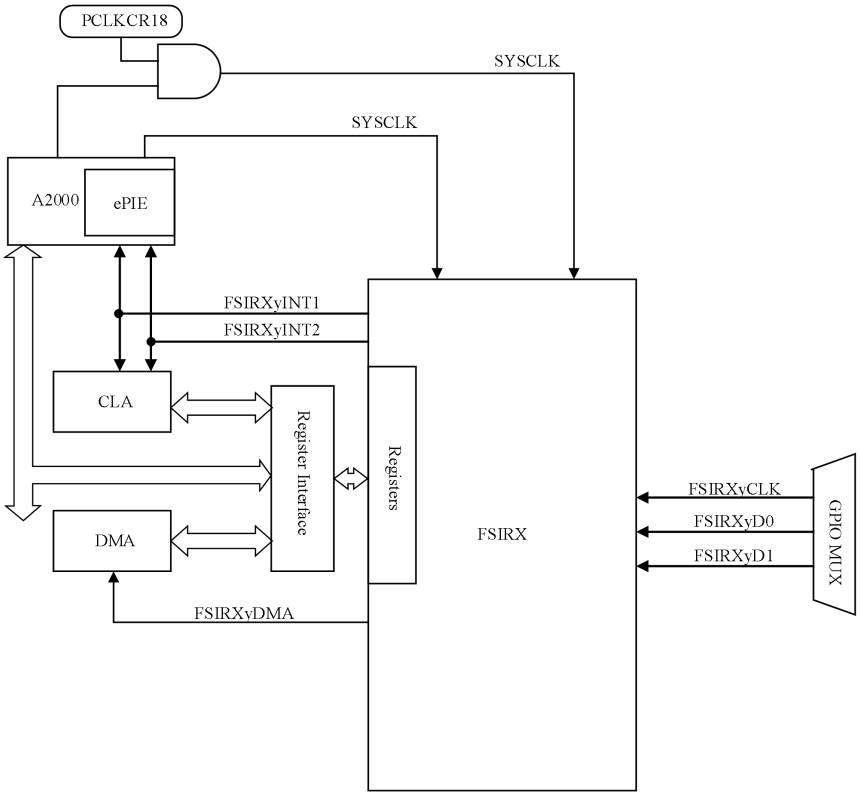


Figure 6-79 FSIRX CPU Interface

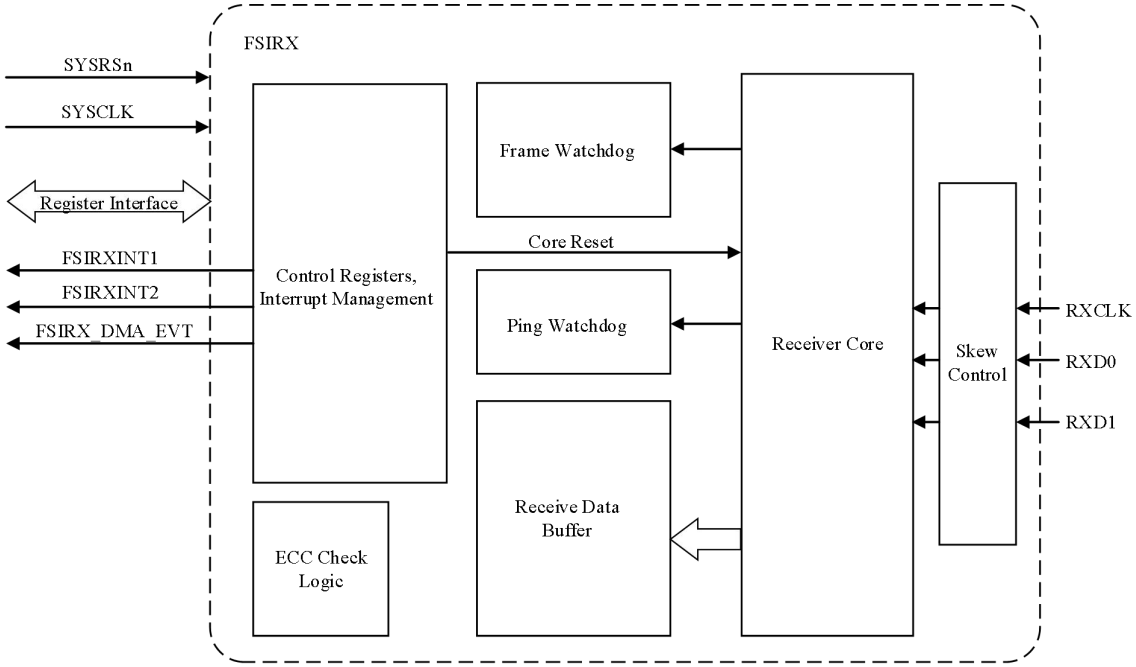


Figure 6- 80 FSIRX Block Diagram

6.15.8.2.1 FSIRX Electrical Data and Timing
FSIRX 电气数据和时序

6.15.8.2.1.1 FSIRX Timing Requirements
FSIRX 时序要求

NO.	PARAMETER[1]		MIN	MAX	UNIT
1	$t_{c(RXCLK)}$	Cycle time, RXCLK	16.67		ns
2	$t_{w(RXCLK)}$	Pulse width, RXCLK low or RXCLK high.	$0.35t_{c(RXCLK)}$	$0.65t_{c(RXCLK)}$	ns
3	$t_{su(RXCLK–RXD)}$	Setup time with respect to RXCLK, applies to both edges of the clock	1.7		ns
4	$t_{h(RXCLK–RXD)}$	Hold time with respect to RXCLK, applies to both edges of the clock	2		ns

6.15.8.2.1.2 FSIRX Switching Characteristics
FSIRX 开关特性

NO.	PARAMETER[1]		MIN	MAX	UNIT
1	$t_{d(RXCLK)}$	RXCLK delay compensation at RX_DLYLINE_CTRL[RXCLK_DLY]=31	10	30	ns
2	$t_{d(RXD0)}$	RXD0 delay compensation at RX_DLYLINE_CTRL[RXD0_DLY]=31	10	30	ns
3	$t_{d(RXD1)}$	RXD1 delay compensation at RX_DLYLINE_CTRL[RXD1_DLY]=31	10	30	ns
4	$t_{d(DELAY_ELEMENT)}$	Incremental delay of each delay line element for RXCLK, RXD0, and RXD1	0.3	1	ns
TDM1	$t_{skew(TDM_CLK-TDM_Dx)}$	Delay skew introduced between RXCLK-TDM_CLK delay and RXDx-TDM_Dx delays	-3	3	ns

(1) 10-pF load on pin.
引脚上的10pF 负载。

6.15.8.2.1.3 FSIRX Timings
FSIRX 时序

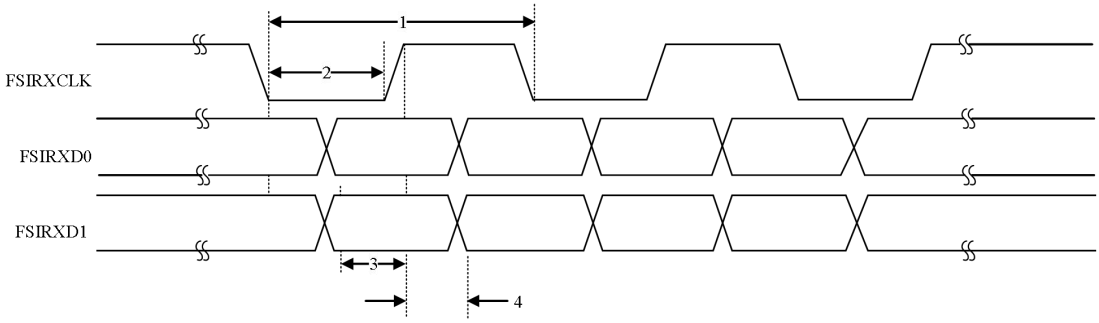


Figure 6-81 FSIRX Timings

6.15.8.3 FSI SPI Compatibility Mode
FSI SPI 兼容模式

The FSI supports a SPI compatibility mode to enable communication with programmable SPI devices. In this mode, the FSI transmits its data in the same manner as a SPI in a single clock configuration mode. While the FSI is able to physically interface with a SPI in this mode, the external device must be able to encode and decode an FSI frame to communicate successfully. This is because the FSI transmits all SPI frame phases with the exception of the preamble and postamble. The FSI provides the same data validation and frame checking as if it was in standard FSI mode, allowing for more robust communication without consuming CPU cycles. The external SPI is required to send all relevant information and can access standard FSI features such as the ping frame watchdog on the FSIRX, frame tagging, or custom CRC values. The list of features of SPI compatibility mode follows:

- Data will transmit on rising edge and receive on falling edge of the clock.
- Only 16-bit word size is supported.
- TXD1 will be driven like an active-low chip-select signal. The signal will be low for the duration of the full frame transmission.
- No receiver chip-select input is required. RXD1 is not used. Data is shifted into the receiver on every active clock edge.
- No preamble or postamble clocks will be transmitted. All signals return to the idle state after the frame phase is finished.
- It is not possible to transmit in the SPI slave configuration because the FSI TXCLK cannot take an external clock source.

FSI 支持 SPI 兼容模式以实现与可编程 SPI 器件的通信。在此模式下，FSI 发送数据的方式与 SPI 在单时钟配置模式下发送数据的方式相同。虽然 FSI 在此模式下能够通过物理方式连接 SPI，但外部器件必须能够对 FSI 帧进行编码和解码才能成功进行通信。这是因为 FSI 会发送除前同步码和后同步码之外的所有 SPI 帧相位。FSI 提供与标准 FSI 模式相同的数据验证和帧检查功能，从而在不占用 CPU 周期的情况下实现更稳健的通信。为此，需要使用外部 SPI 来发送所有相关信息，并可用于访问标准 FSI 功能，例如 FSIRX 上的 ping 帧看门狗、帧标记或自定义 CRC 值。SPI 兼容模式的特性如下：

- 将在时钟的上升沿发送数据，并在下降沿接收数据。
- 仅支持 16 位字大小。
- TXD1 将像低电平有效片选信号一样被驱动。信号在整个帧发送期间将处于低电平。
- 无需接收器片选输入。不使用 RXD1。数据在每个有效时钟边沿转移到接收器中。
- 不会发送前同步码或后同步码时钟。帧相位结束后，所有信号都恢复到空闲状态。
- 由于FSITXCLK 不能采用外部时钟源，因此无法在 SPI 从器件配置下进行发送。

6.15.8.3.1 FSITX SPI Signaling Mode Electrical Data and Timing
FSITX SPI 信令模式电气数据和时序

Special timings are not required for the FSIRX in SPI signaling mode. FSIRX timings listed in the FSIRX Timing Requirements table are applicable in SPI compatibility mode. Setup and Hold times are only valid on the falling edge of FSIRXCLK because this is the active edge in SPI signaling mode.

在 SPI 信令模式下，FSIRX 不需要特殊时序。“FSIRX 时序要求”表中列出的 FSIRX 时序适用于 SPI 兼容模式。设置和保持时间仅在 FSIRXCLK 的下降沿有效，因为这是 SPI 信令模式中的有效边沿。

6.15.8.3.1.1 FSITX SPI Signaling Mode Switching Characteristics
FSITXSPI 信令模式开关特性

over operating free-air temperature range (unless otherwise noted)
在自然通风条件下的工作温度范围内测得（除非另有说明）

NO.	PARAMETER(1)		MIN	MAX	UNIT
1	$t_{c(TXCLK)}$	Cycle time, TXCLK	16.67		ns
2	$t_{w(TXCLK)}$	Pulse width, TXCLK low or TXCLK high	$(0.5t_{c(TXCLK)})–1$	$(0.5t_{c(TXCLK)})+1$	ns
3	$t_{d(TXCLKH–TXD0)}$	Delay time, TXD0 valid after TXCLK high		3	ns
4	$t_{d(TXD1-TXCLK)}$	Delay time, TXCLK high after TXD1 low	$t_{w(TXCLK)}–3$		ns
5	$t_{d(TXCLK-TXD1)}$	Delay time, TXD1 high after TXCLK low	$t_{w(TXCLK)}$		ns

(1) 10-pF load on pin
引脚上的 10pF 负载

6.15.8.3.1.2 FSITX SPI Signaling Mode Timings
FSITXSPI 信令模式时序

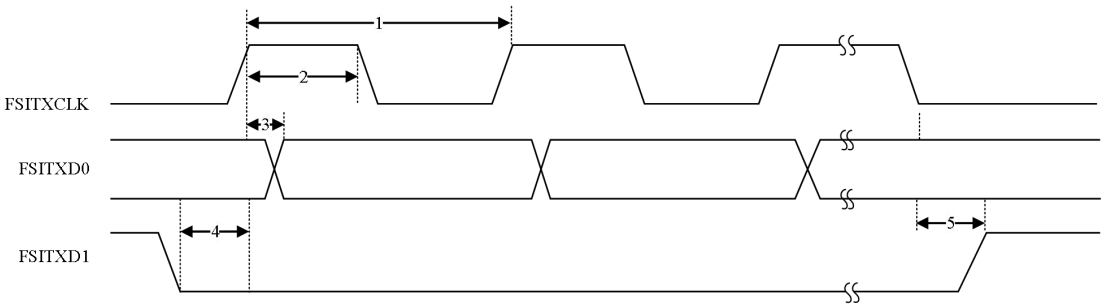


Figure 6- 82 FSITX SPI Signaling Mode Timings

6.15.9 Host Interface Controller (HIC)
主机接口控制器(HIC)

The HIC module allows an external host controller (master) to directly access resources of the device (slave) by emulating the ASRAM protocol. It has two modes of operation: direct access and mailbox access. In direct access mode, device resources is written to and read from directly by the external host. In mailbox access mode, external host and device write to and read from a buffer and notify each other when the buffer write/read is complete. For security reasons, the HIC has to be enabled by the device before the external host can access it.
Features of the HIC include:

- Configurable I/O data lines of 8 bits and 16 bits
- Direct and mailbox access modes
- 8 address lines and 8 configurable base addresses for a total of 2048 possible addressable regions
- Two 64-byte buffers for external host and device when using mailbox access mode
- Interrupt generation on buffer full/empty
- High throughput
- Trigger HIC activity from other peripherals
- Error indicators to the system or interface
- Commit feature that blocks writes to configuration registers

HIC 模块允许外部主机控制器（主器件）通过仿真 ASRAM 协议直接存取器件（从器件）的资源。该模块有两种运行模式：直接访问和邮箱访问。在直接访问模式下，器件资源直接写入到外部主机并由外部主机进行读取。在邮箱访问模式下，外部主机和器件对缓冲区进行写入和读取，并在缓冲区写入/读取完成时相互通知。出于安全考虑，必须先由器件启用 HIC，然后外部主机才能访问 HIC。

HIC 的特性包括：

- 8 位和16 位可配置 I/O 数据线
- 直接访问模式和邮箱访问模式
- 8 条地址线和 8 个可配置的基址，总共2048 个可能的可寻址区域
- 使用邮箱访问模式时有两个用于外部主机和器件的 64 字节缓冲器
- 缓冲器满/空时产生中断
- 高吞吐量
- 从其他外设触发 HIC 活动
- 系统或接口的错误指示器
- 提交功能可阻止写入配置寄存器

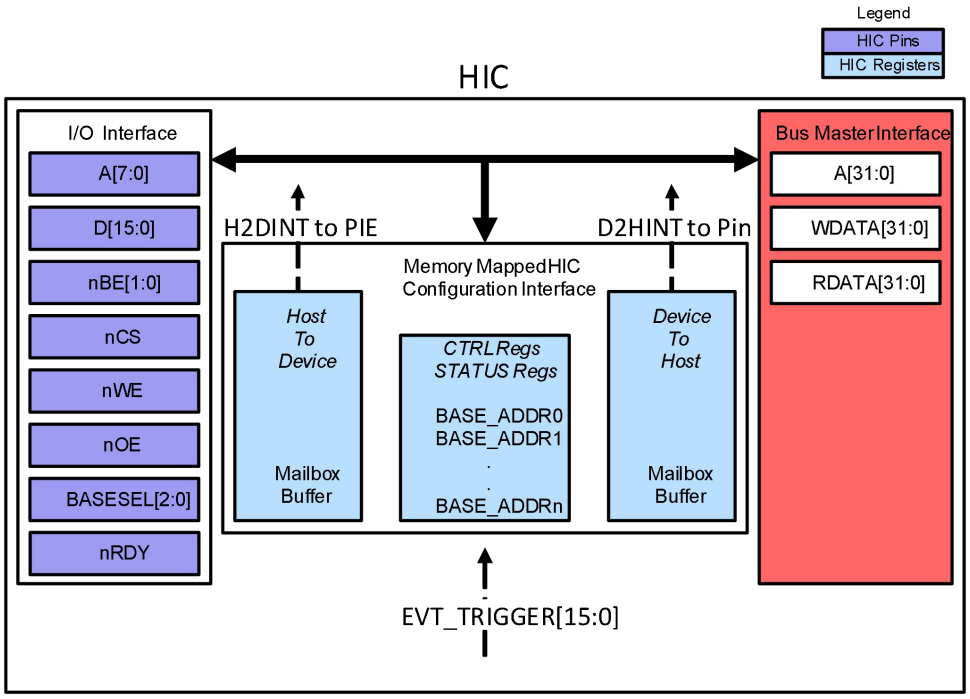


Figure 6- 83 HIC Block Diagram

6.15.9.1 HIC Electrical Data and Timing
HIC 电气数据和时序

6.15.9.1.1 HIC Timing Requirements
HIC 时序要求

over operating free-air temperature range (unless otherwise noted)
在自然通风条件下的工作温度范围内测得（除非另有说明）

		MIN	MAX	UNIT
Read/Write Parameters with nOE and nWE pins - Dual Read/Write pins				
$t_{su(ABBV-OEV)}$	Setup time, A/BASESEL/nBE before nOE active	0		ns
$t_{su(ABBV-WEV)}$	Setup time, A/BASESEL/nBE before nWE active	0		ns
$t_{su(CSV-OEV)}$	Setup time, nCS active before nOE active	$0.5t_{c(SYSCLK)}$		ns
$t_{su(CSV-WEV)}$	Setup time, nCS active before nWE active	$0.5t_{c(SYSCLK)}$		ns
$t_{h(ABBV-OEIV)}$	Hold time, A/BASESEL/nBE/nCS after nOE inactive	6		ns
$t_{h(ABBV-WEIV)}$	Hold time, A/BASESEL/nBE/nCS after nWE inactive	6		ns
$t_{w(OEV)}$	Active pulse width of nOE (Read)(1)	$4t_{c(SYSCLK)}$		ns
$t_{w(WEV)}$	Active pulse width of nWE (Write)	$4t_{c(SYSCLK)}$		ns
$t_{w(CSIV)}$	Inactive pulse width of nCS(2)	$3t_{c(SYSCLK)}$		ns
$t_{w(OEIV)}$	Inactive Read pulse width of nOE(2)	$3t_{c(SYSCLK)}$		ns
$t_{w(WEIV)}$	Inactive Write pulse width of nWE(2)	$3t_{c(SYSCLK)}$		ns
$t_{su(DV-WEV)}$	Setup time, D before nWE active	0		ns
$t_{h(DV-WEIV)}$	Hold time, D after nWE inactive	6		ns
Read/Write Parameters with RnW pin - Single Read/Write pin				
$t_{su(ABBV-CSV)}$	Setup time, A/BASESEL/nBE before nCS active	0		ns
$t_{su(RNWV-CSV)}$	Setup time, RnW before nCS active	$0.5t_{c(SYSCLK)}$		ns
$t_{h(ABBV-CSIV)}$	Hold time, A/BASESEL/nBE/RnW after nCS inactive	5		ns
$t_{w(CSV_RD)}$	Active pulse width of nCS for read operation(1)	$4t_{c(SYSCLK)}$		ns
$t_{w(CSV_WR)}$	Active pulse width of nCS for write operation	$4t_{c(SYSCLK)}$		ns
$t_{w(CSIV)}$	Inactive pulse width of nCS(2)	$3t_{c(SYSCLK)}$		ns
$t_{w(RNWIV)}$	Inactive pulse width of RnW(2)	$3t_{c(SYSCLK)}$		ns
$t_{su(DV-CSV)}$	Setup time, D before nCS active	0		ns
$t_{h(DV-CSIV)}$	Hold time, D after nCS inactive	5		ns

(1) For accesses to the device region, additional 2 SYSCLK cycles are required.
要访问器件区域，需要额外的 2 个SYSCLK 周期。

(2) For accesses to the device region with nRDY pin, additional SYSCLK cycle is required.
要使用nRDY 引脚访问器件区域，需要额外的SYSCLK 周期。

6.15.9.1.2 HIC Switching Characteristics
HIC 开关特征

over operating free-air temperature range (unless otherwise noted)
在自然通风条件下的工作温度范围内测得（除非另有说明）

PARAMETER(1)		MIN	MAX	UNIT
Read/Write Parameters with nOE and nWE pins				
$t_{d(OEV-DV)}$	Output data delay time : nOE to D output valid (2)	$3t_{c(SYSCLK)}$	$4t_{c(SYSCLK)}+15$	ns

PARAMETER[1]		MIN	MAX	UNIT
$t_{d(OEIV-DIV)}$	Output data hold time : nOE invalid to D output invalid (tri-state)	$1t_{c(SYSCLK)}$	$2t_{c(SYSCLK)}$+15	ns
$t_{d(OEV-RDYV)}$	Read Ready delay time : nOE to nRDY output valid	0	17	ns
$t_{d(WEV-RDYV)}$	Write Ready delay time : nWE to nRDY output valid	0	12	ns
$t_{d(RDYV-DV)}$	Ready to Data delay time : nRDY output valid to D output valid	-3	45	ns
$t_{w(RDYACT)}$	Active pulse width of nRDY output	$2t_{c(SYSCLK)}$		ns
Read/Write Parameters with RnW pin				
$t_{d(CSV-DV)}$	Output delay time : nCS active to D output valid (2)	$3t_{c(SYSCLK)}$	$4t_{c(SYSCLK)}$+14	ns
$t_{d(CSIV-DIV)}$	Output hold time : nCS inactive to D output invalid (tri-state)	$1t_{c(SYSCLK)}$	$2t_{c(SYSCLK)}$+14	ns
$t_{d(CSV-RDYV)}$	Output delay time : nCS to nRDY output valid	0	12	ns
$t_{d(RDYV-DV)}$	Ready to Data delay time : nRDY output valid to D output valid	-3	45	ns
$t_{w(RDYACT)}$	Active pulse width of nRDY output	$2t_{c(SYSCLK)}$		ns

(1) 10-pF load on pin.
引脚上的 10pF 负载。

(2) Applicable to mailbox accesses only. Direct memory map (Device) accesses are qualified with nRDY pin.
仅适用于邮箱访问。直接存储器映射（器件）访问由 nRDY 引脚滤波。

6.15.9.1.3 HIC Timing Diagrams
HIC 时序图

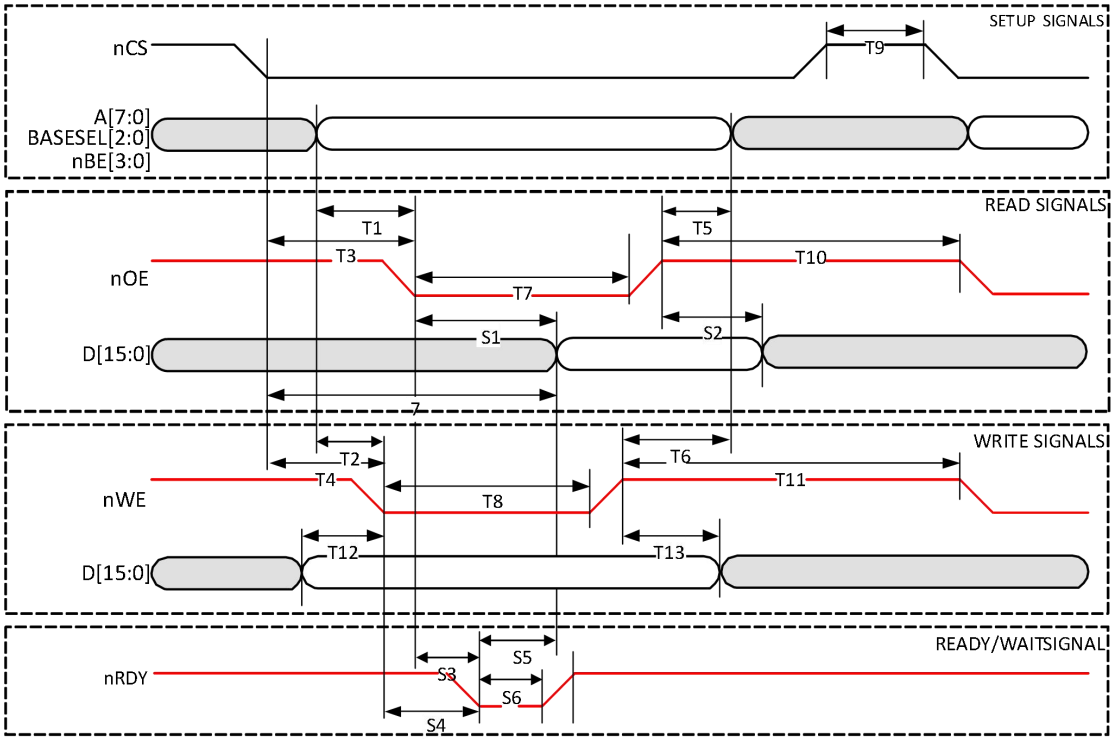


Figure 6- 84 Read/Write Operation With nOE and nWE Pins

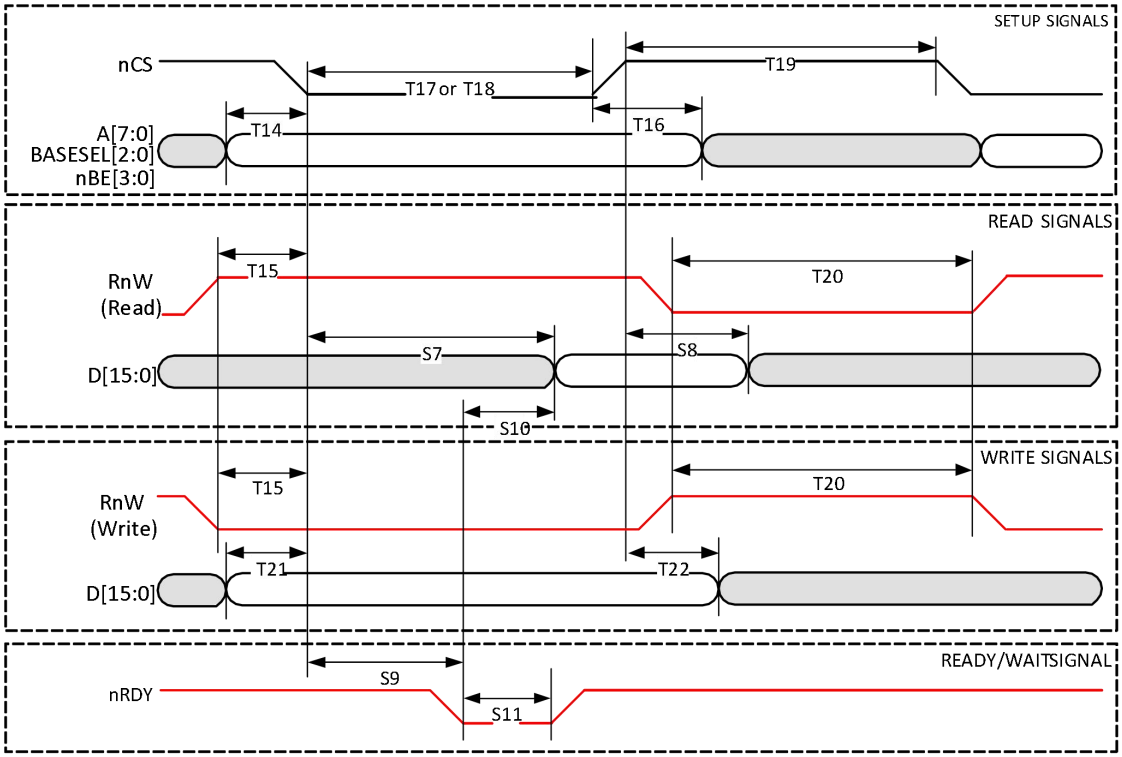


Figure 6- 85 Read/Write Operation With RnW Pin

7 Detailed Description
详细描述

7.1 Overview
概述

A2000 32-bit Real-Time microcontrollers are optimized for processing, sensing, and actuation to improve closed-loop performance in real-time control applications such as industrial motor drives; solar inverters and digital power; electrical vehicles and transportation; motor control; and sensing and signal processing.

The AVP32F003X is a powerful 32-bit floating-point microcontroller unit (MCU) that lets designers incorporate crucial control peripherals, differentiated analog, and nonvolatile memory on a single device.

The real-time control subsystem is based on 32-bit A2000 CPU, which provides 120 MHz of signal processing performance. The A2000 CPU is further boosted by the FPU, new TMU extended instruction set, which enables fast execution of algorithms with trigonometric operations commonly found in transforms and torque loop calculations; and the VCRC extended instruction set, which reduces the latency for complex math operations commonly found in encoded applications.

The CLA allows significant offloading of common tasks from the main A2000 CPU. The CLA is an independent 32-bit floating-point math accelerator that executes in parallel with the CPU. Additionally, the CLA has its own dedicated memory resources and it can directly access the key peripherals that are required in a typical control system. Support of a subset of ANSI C is standard, as are key features like hardware breakpoints and hardware task-switching.

The AVP32F003X supports up to 384KB (192KW) of flash memory divided into three 128KB (64KW) banks, which enable programming and execution in parallel. Up to 69KB (34.5KW) of on-chip SRAM is also available to supplement the flash memory.

The Live Firmware Update hardware enhancements on AVP32F003X allow fast context switching from the old firmware to the new firmware to minimize application downtime when updating the device firmware.

High-performance analog blocks are integrated on the AVP32F003X real-time MCU to further enable system consolidation. Three separate 12-bit ADCs provide precise and efficient management of multiple analog signals, which ultimately boosts system throughput. Four analog comparator modules provide continuous monitoring of input voltage levels for trip conditions.

The A2000 devices contain industry-leading control peripherals with frequency-independent ePWM/ HRPWM and eCAP allow for a best-in-class level of control to the system.

Connectivity is supported through various industry-standard communication ports (such as SPI, SCI, I2C, PMBus, LIN, CAN and CAN FD) and offers multiple muxing options for optimal signal placement in a variety of applications. The Host Interface Controller (HIC) is a high throughput interface that allows an external host to access resources of the AVP32F003X. Additionally, in an industry first, the FSI enables high-speed, robust communication to complement the rich set of peripherals that are embedded in the device.

A specially enabled device variant, AVP32F003X, allows access to the Configurable Logic Block (CLB)

for additional interfacing features. See Table 4-1 for more information.

The Embedded Real-Time Analysis and Diagnostic (ERAD) module enhances the debug and system analysis capabilities of the device by providing additional hardware breakpoints and counters for profiling.

A2000 32 位实时微控制器针对处理、感应和驱动进行了优化，旨在提高实时控制应用（如工业电机驱动器、光伏逆变器和数字电源、电动汽车和运输、电机控制以及感应和信号处理）的闭环性能。

AVP32F003X 是一系列功能强大的 32 位浮点微控制器单元（MCU），可让设计人员在单个器件上集成关键的控制外设、差分模拟和非易失性存储器。

实时控制子系统基于 32 位 A2000 CPU，可提供 120MHz 的信号处理性能。A2000CPU 的性能通过新的 TMU 扩展指令集（即FPU）和VCRC 扩展指令集得到了进一步提升；TMU 扩展指令集可快速执行包含变换和扭矩环路计算中常见三角运算的算法，VCRC 扩展指令集可降低编码应用中常见复杂数学运算的延迟。

CLA 允许从主 A2000 CPU 上大量卸载常见任务。CLA 是一款与 CPU 并行执行的独立 32 位浮点数学加速器。此外，CLA 自带专用存储资源，它可以直接访问典型控制系统中所需的关键外设。与硬件断点和硬件任务切换等主要特性一样，ANSIC 子集支持是标准配置。

AVP32F003X 支持高达 384KB(192KW)的闪存，这些闪存分为三个 128KB(64KW)存储体，支持并行编程和执行。高达 69KB(34.5KW)的片上 SRAM 也可用于补充闪存。

AVP32F003X 上的实时固件更新硬件增强允许从旧固件到新固件的快速上下文切换，以尽可能减少更新器件固件时的应用停机时间。

AVP32F003X 实时 MCU 上集成了高性能模拟块，可进一步支持系统整合。三个独立的 12 位 ADC 可准确、高效地管理多个模拟信号，从而最终提高系统吞吐量。四个模拟比较器模块可以针对跳闸情况对输入电压电平进行持续监控。

A2000 器件包含行业领先的控制外设（具有与频率无关的 ePWM/HRPWM 和eCAP），允许对系统进行一流的控制。

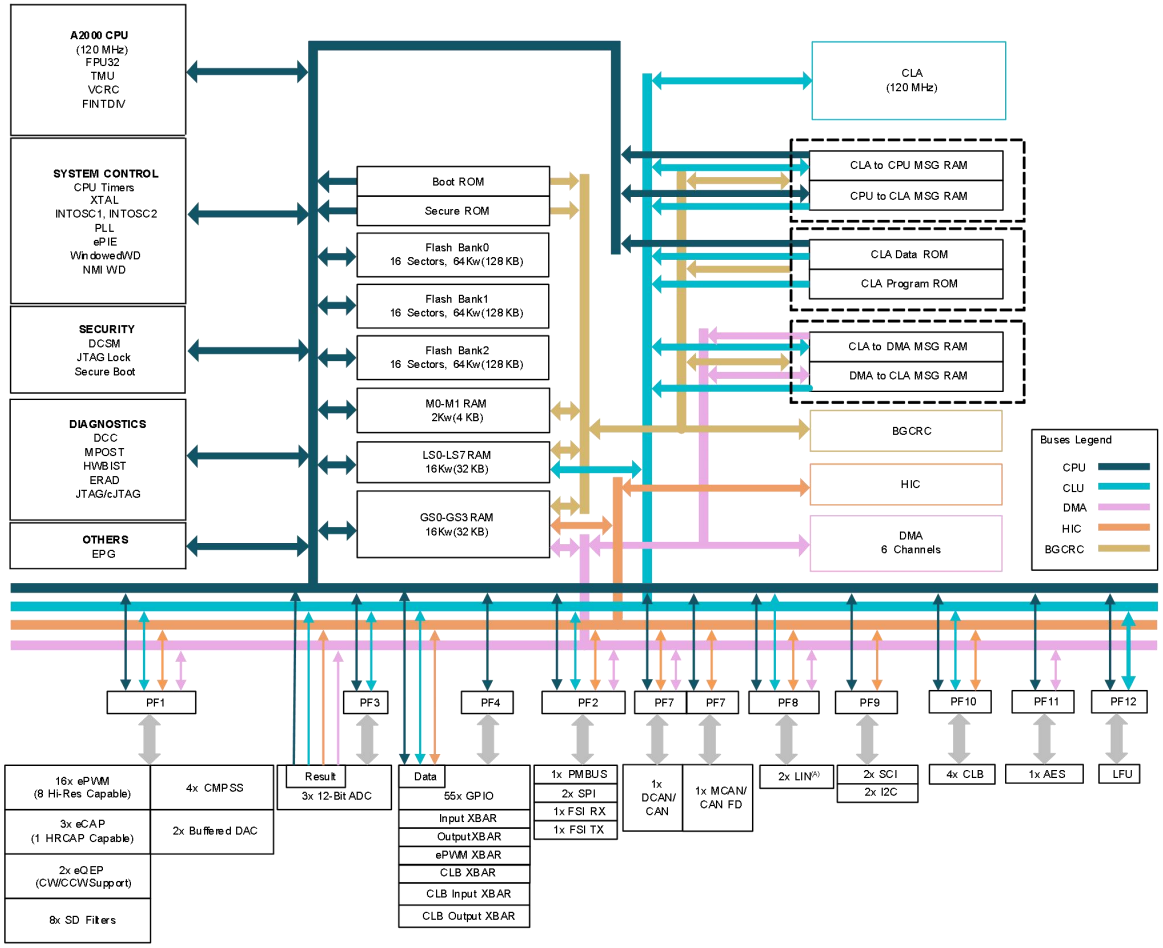
通过各种业界通用通信端口（如SPI、SCI、I2C、PMBus、LIN、CAN 和CANFD）支持连接，并且提供了多个多路复用选项，可在各种应用中实现出色的信号布局。主机接口控制器（HIC）是一个高吞吐量接口，它允许外部主机访问 AVP32F003X 的资源。此外，FSI 率先在业内实现了高速可靠的通信，补充了嵌入该器件的各种外设的功能。

专门实现的器件型号 AVP32F003X 允许访问可配置逻辑块(CLB)以支持额外连接功能。有关更多信息，请参阅表4-1。

嵌入式实时分析和诊断(ERAD)模块通过提供用于分析的附加硬件断点和计数器来增强器件的调试和系统分析功能。

7.2 Functional Block Diagram
功能框图

Figure 7-1 shows the CPU system and associated peripherals.
图 7-1 所示为 CPU 系统及相关外设。



A. The LIN module can also work as an SCI.
LIN 模块还可用作 SCI。

Figure 7-1. Functional Block Diagram

7.3 Memory

7.3.1 Memory Map
内存映射

The Memory Map table describes the memory map.
“存储器映射”表描述了存储器映射。

Table 7-1. Memory Map

MEMORY	SIZE	START ADDRESS	END ADDRESS	HIC ACCESS	DMA ACCESS	CLA ACCESS	ECC/ PARITY	ACCESS PROTECTION	SECURITY
M0RAM	1Kx16	0x00000000	0x000003FF	-	-	-	ECC	Yes	-
M1RAM	1Kx16	0x00000400	0x000007FF	-	-	-	ECC	Yes	-
PieVectTable	512x16	0x00000D00	0x00000EFF	-	-	-	-	-	-
PieVectTable Swap	512x16	0x01000900	0x01000AFF	-	-	-	-	-	-
LS0RAM	2Kx16	0x00008000	0x000087FF	-	-	Yes	ECC	Yes	Yes
LS1RAM	2Kx16	0x00008800	0x00008FFF	-	-	Yes	ECC	Yes	Yes
LS2RAM	2Kx16	0x00009000	0x000097FF	-	-	Yes	ECC	Yes	Yes
LS3RAM	2Kx16	0x00009800	0x00009FFF	-	-	Yes	ECC	Yes	Yes
LS4RAM	2Kx16	0x0000A000	0x0000A7FF	-	-	Yes	ECC	Yes	Yes
LS5RAM	2Kx16	0x0000A800	0x0000AFFF	-	-	Yes	ECC	Yes	Yes
LS6RAM	2Kx16	0x0000B000	0x0000B7FF	-	-	Yes	ECC	Yes	Yes
LS7RAM	2Kx16	0x0000B800	0x0000BFFF	-	-	Yes	ECC	Yes	Yes
GS0RAM	4Kx16	0x0000C000	0x0000CFFF	Yes	Yes	-	ECC	Yes	-
GS1RAM	4Kx16	0x0000D000	0x0000DFFF	Yes	Yes	-	ECC	Yes	-
GS2RAM	4Kx16	0x0000E000	0x0000EFFF	Yes	Yes	-	ECC	Yes	-
GS3RAM	4Kx16	0x0000F000	0x0000FFFF	Yes	Yes	-	ECC	Yes	-
CAN A Message RAM	2Kx16	0x00049000	0x000497FF	Yes	Yes	-	Parity	-	-
MCAN Message RAM	8Kx16	0x00058000	0x00059FFF	Yes	-	-	ECC	-	-
CLA to CPU Message RAM	128x16	0x00001480	0x000014FF	-	-	Yes	ECC	-	-
CPU to CLA Message RAM	128x16	0x00001500	0x0000157F	-	-	Yes	ECC	-	-
CLA to DMA Message RAM	128x16	0x00001680	0x000016FF	-	Yes	Yes	ECC	-	-
DMA to CLA Message RAM	128x16	0x00001700	0x0000177F	-	Yes	Yes	ECC	-	-
ADV OTP(1)	3Kx16	0x00070000	0x00070BFF	-	-	-	ECC	-	Yes(2)
User OTP	3Kx16	0x00078000	0x00078BFF	-	-	-	ECC	-	Yes(2)
Flash	192Kx16	0x00080000	0x000AFFFF	-	-	-	ECC	-	Yes
Secure ROM	24Kx16	0x003F2000	0x003F7FFF	-	-	-	Parity	-	Yes
Boot ROM	32Kx16	0x003F8000	0x003FFFFF	-	-	-	Parity	-	-
Pie Vector Fetch Error (part of Boot ROM)	1x16	0x003FFFBE	0x003FFFBF	-	-	-	Parity	-	-

MEMORY	SIZE	START ADDRESS	END ADDRESS	HIC ACCESS	DMA ACCESS	CLA ACCESS	ECC/ PARITY	ACCESS PROTECTION	SECURITY
Default Vectors (part of Boot ROM)	64x16	0x003FFFC0	0x003FFFFF	-	-	-	Parity	-	-
AVP32F0033									
M0RAM	1Kx16	0x00000000	0x000003FF	-	-	-	ECC	Yes	-
M1RAM	1Kx16	0x00000400	0x000007FF	-	-	-	ECC	Yes	-
PieVectTable	512x16	0x00000D00	0x00000EFF	-	-	-	-	-	-
LS4RAM	2Kx16	0x0000A000	0x0000A7FF	-	-	Yes	ECC	Yes	Yes
LS5RAM	2Kx16	0x0000A800	0x0000AFFF	-	-	Yes	ECC	Yes	Yes
LS6RAM	2Kx16	0x0000B000	0x0000B7FF	-	-	Yes	ECC	Yes	Yes
LS7RAM	2Kx16	0x0000B800	0x0000BFFF	-	-	Yes	ECC	Yes	Yes
GS0RAM	2Kx16	0x0000C000	0x0000C7FF	Yes	Yes	-	ECC	Yes	-
CANA Message RAM	2Kx16	0x00049000	0x000497FF	Yes	Yes	-	Parity	-	-
MCAN Message RAM	8Kx16	0x00058000	0x00059FFF	Yes	-	-	ECC	-	-
ADV OTP(1)	3Kx16	0x00070000	0x00070BFF	-	-	-	ECC	-	Yes(2)
User OTP	3Kx16	0x00078000	0x00078BFF	-	-	-	ECC	-	Yes(2)
Flash	64Kx16	0x00088000	0x00097FFF	-	-	-	ECC	-	Yes
Secure ROM	24Kx16	0x003F2000	0x003F7FFF	-	-	-	Parity	-	Yes
Boot ROM	32Kx16	0x003F8000	0x003FFFFF	-	-	-	Parity	-	-
Pie Vector Fetch Error (part of Boot ROM)	1x16	0x003FFFBE	0x003FFFBF	-	-	-	Parity	-	-
Default Vectors (part of Boot ROM)	64x16	0x003FFFC0	0x003FFFFF	-	-	-	Parity	-	-

(1) ADV OTP is for ADV internal use only.
进芯 OTP 仅供进芯内部使用。
(2) Only a subset is secure.
只有一部分是安全的。

7.3.1.1 Dedicated RAM (Mx RAM)
专用 RAM(MxRAM)

The CPU subsystem has two dedicated ECC-capable RAM blocks: M0 and M1. These memories are small nonsecure blocks that are tightly coupled with the CPU (that is, only the CPU has access to them).

CPU 子系统有两个支持 ECC 功能的专用 RAM 模块：M0 和M1。这些存储器是与 CPU 紧密耦合的小型非安全块（即，只有CPU 可以访问这些存储器）。

7.3.1.2 Local Shared RAM (LSx RAM)
本地共享 RAM(LSxRAM)

Local shared RAMs (LSx RAMs) are accessible to the CPU, CLA, and BGCRC. All LSx RAM blocks have ECC. These memories are secure and have CPU access protection (CPU write/CPU fetch).

CPU、CLA 和BGCRC 可访问本地共享的 RAM(LSxRAM)。所有 LSxRAM 块都具有 ECC。这些存储器都是安全的，且具有 CPU 访问保护（CPU 写入/CPU 获取）。

7.3.1.3 Global Shared RAM (GSx RAM)
全局共享 RAM(GSxRAM)

Global shared RAMs (GSx RAMs) are accessible from the CPU, HIC, BGCRC and DMA. The CPU, HIC, and DMA have full read and write access to these memories. All GSx RAM blocks have ECC. The GSx RAMs have access protection (CPU write/CPU fetch/DMA write/HIC write).

可从 CPU、HIC、BGCRC 和 DMA 访问全局共享的 RAM(GSxRAM)。CPU、HIC 和 DMA 都具有对这些存储器的完全读写访问权限。所有 GSxRAM 块都具有 ECC。GSxRAM 具有访问保护（CPU 写入/CPU 获取/DMA 写入/HIC 写入）。

7.3.1.4 Message RAM
消息 RAM

There are two types of message RAMs on this device that can be used to share between CPU, CLA and DMA. CLA-CPU message RAM shares data between the CLA and CPU while the CLA-DMA message RAM shares data between the CLA and DMA.

此器件上有两种类型的消息 RAM 可用于在 CPU、CLA 和 DMA 之间共享。CLA-CPU 消息 RAM 在 CLA 和 CPU 之间共享数据，而 CLA-DMA 消息 RAM 在 CLA 和 DMA 之间共享数据。

7.3.2 Control Law Accelerator (CLA) Memory Map
控制律加速器(CLA)存储器映射

Table 7-2 shows the CLA data ROM memory map.
表 7-2 显示了 CLA 数据 ROM 存储器映射。

Table 7-2. CLA Data ROM Memory Map

MEMORY	START ADDRESS	END ADDRESS	LENGTH
FFT Tables (Load)	0x01001070	0x0100186F	0x0800
Data (Load)	0x01001870	0x01001FF9	0x078A
Version (Load)	0x01001FFA	0x01001FFF	0x0006
FFT Tables (Run)	0x0000F070	0x0000F86F	0x0800
Data (Run)	0x0000F870	0x0000FFF9	0x078A
Version (Run)	0x0000FFFA	0x0000FFFF	0x0006

7.3.3Flash Memory Map
闪存映射

On the AVP32F003X devices, three flash banks (384KB [192KW]) are available. Code to program the flash should be executed out of RAM, there should not be any kind of access to the flash bank when an erase or program operation is in progress. The Addresses of Flash Sectors table lists the addresses of flash sectors available for each part number.

AVP32F003X 器件上最大有三个闪存存储体(384KB[192KW])。对闪存进行编程的代码应在 RAM 之外执行，在进行擦除或编程操作时，不应以任何形式存取闪存存储体。“闪存扇区地址”表列出了每个型号器件中闪存扇区的地址。

7.3.3.1 Addresses of Flash Sectors
闪存扇区的地址

Table 7-3. Addresses of Flash Sectors

PART NUMBER	SECTOR	ADDRESS			ECC ADDRESS		
		SIZE	START	END	SIZE	START	END
OTP Sectors							
AVP32F003X AVP32F00157P	ADV OTP Bank 0 (Unsecure)	1008x16	0x00070000	0x000703EF	126x16	0x01070000	0x0107007D
	ADV OTP Bank 0 (Secure)	16x16	0x000703F0	0x000703FF	2x16	0x0107007E	0x0107007F
	ADV OTP Bank1	1Kx16	0x00070400	0x000707FF	128x16	0x01070080	0x010700FF
AVP32F003X AVP32F00157P	ADV OTP Bank2	1Kx16	0x00070800	0x00070BFF	128x16	0x01070100	0x0107017F
AVP32F003X AVP32F00157P	User configurable DCSM OTP Bank 0	1Kx16	0x00078000	0x000783FF	128x16	0x01071000	0x0107107F
	User configurable OTP Bank 1	1Kx16	0x00078400	0x000787FF	128x16	0x01071080	0x010710FF
AVP32F003X AVP32F00157P	User configurable OTP Bank 2	1Kx16	0x00078800	0x00078BFF	128x16	0x01071100	0x0107117F
Bank0 Sectors							
AVP32F0039 AVP32F0036 AVP32F00157P	Sector0	4Kx16	0x00080000	0x00080FFF	512x16	0x01080000	0x010801FF
	Sector1	4Kx16	0x00081000	0x00081FFF	512x16	0x01080200	0x010803FF
	Sector2	4Kx16	0x00082000	0x00082FFF	512x16	0x01080400	0x010805FF
	Sector3	4Kx16	0x00083000	0x00083FFF	512x16	0x01080600	0x010807FF
	Sector4	4Kx16	0x00084000	0x00084FFF	512x16	0x01080800	0x010809FF
	Sector5	4Kx16	0x00085000	0x00085FFF	512x16	0x01080A00	0x01080BFF
	Sector6	4Kx16	0x00086000	0x00086FFF	512x16	0x01080C00	0x01080DFF
	Sector7	4Kx16	0x00087000	0x00087FFF	512x16	0x01080E00	0x01080FFF
AVP32F0039 AVP32F0036 AVP32F0033 AVP32F00157P	Sector8	4Kx16	0x00088000	0x00088FFF	512x16	0x01081000	0x010811FF
	Sector9	4Kx16	0x00089000	0x00089FFF	512x16	0x01081200	0x010813FF
	Sector10	4Kx16	0x0008A000	0x0008AFFF	512x16	0x01081400	0x010815FF
	Sector11	4Kx16	0x0008B000	0x0008BFFF	512x16	0x01081600	0x010817FF
	Sector12	4Kx16	0x0008C000	0x0008CFFF	512x16	0x01081800	0x010819FF

PART NUMBER	SECTOR	ADDRESS			ECC ADDRESS		
		SIZE	START	END	SIZE	START	END
	Sector13	4Kx16	0x0008D000	0x0008DFFF	512x16	0x01081A00	0x01081BFF
	Sector14	4Kx16	0x0008E000	0x0008EFFF	512x16	0x01081C00	0x01081DFF
	Sector15	4Kx16	0x0008F000	0x0008FFFF	512x16	0x01081E00	0x01081FFF
Bank1 Sectors							
AVP32F0039 AVP32F0036 AVP32F0033 AVP32F00157P	Sector0	4Kx16	0x00090000	0x00090FFF	512x16	0x01082000	0x010821FF
	Sector1	4Kx16	0x00091000	0x00091FFF	512x16	0x01082200	0x010823FF
	Sector2	4Kx16	0x00092000	0x00092FFF	512x16	0x01082400	0x010825FF
	Sector3	4Kx16	0x00093000	0x00093FFF	512x16	0x01082600	0x010827FF
	Sector4	4Kx16	0x00094000	0x00094FFF	512x16	0x01082800	0x010829FF
	Sector5	4Kx16	0x00095000	0x00095FFF	512x16	0x01082A00	0x01082BFF
	Sector6	4Kx16	0x00096000	0x00096FFF	512x16	0x01082C00	0x01082DFF
	Sector7	4Kx16	0x00097000	0x00097FFF	512x16	0x01082E00	0x01082FFF
AVP32F0039 AVP32F0036 AVP32F00157P	Sector8	4Kx16	0x00098000	0x00098FFF	512x16	0x01083000	0x010831FF
	Sector9	4Kx16	0x00099000	0x00099FFF	512x16	0x01083200	0x010833FF
	Sector10	4Kx16	0x0009A000	0x0009AFFF	512x16	0x01083400	0x010835FF
	Sector11	4Kx16	0x0009B000	0x0009BFFF	512x16	0x01083600	0x010837FF
	Sector12	4Kx16	0x0009C000	0x0009CFFF	512x16	0x01083800	0x010839FF
	Sector13	4Kx16	0x0009D000	0x0009DFFF	512x16	0x01083A00	0x01083BFF
	Sector14	4Kx16	0x0009E000	0x0009EFFF	512x16	0x01083C00	0x01083DFF
	Sector15	4Kx16	0x0009F000	0x0009FFFF	512x16	0x01083E00	0x01083FFF
Bank2 Sectors							
AVP32F0039 AVP32F0036 AVP32F00157P	Sector0	4Kx16	0x000A0000	0x000A0FFF	512x16	0x01084000	0x010841FF
	Sector1	4Kx16	0x000A1000	0x000A1FFF	512x16	0x01084200	0x010843FF
	Sector2	4Kx16	0x000A2000	0x000A2FFF	512x16	0x01084400	0x010845FF
	Sector3	4Kx16	0x000A3000	0x000A3FFF	512x16	0x01084600	0x010847FF
	Sector4	4Kx16	0x000A4000	0x000A4FFF	512x16	0x01084800	0x010849FF
	Sector5	4Kx16	0x000A5000	0x000A5FFF	512x16	0x01084A00	0x01084BFF
	Sector6	4Kx16	0x000A6000	0x000A6FFF	512x16	0x01084C00	0x01084DFF
	Sector7	4Kx16	0x000A7000	0x000A7FFF	512x16	0x01084E00	0x01084FFF
	Sector8	4Kx16	0x000A8000	0x000A8FFF	512x16	0x01085000	0x010851FF
	Sector9	4Kx16	0x000A9000	0x000A9FFF	512x16	0x01085200	0x010853FF
	Sector10	4Kx16	0x000AA000	0x000AAFFF	512x16	0x01085400	0x010855FF
	Sector11	4Kx16	0x000AB000	0x000ABFFF	512x16	0x01085600	0x010857FF
	Sector12	4Kx16	0x000AC000	0x000ACFFF	512x16	0x01085800	0x010859FF
	Sector13	4Kx16	0x000AD000	0x000ADFFF	512x16	0x01085A00	0x01085BFF
	Sector14	4Kx16	0x000AE000	0x000AEFFF	512x16	0x01085C00	0x01085DFF
	Sector15	4Kx16	0x000AF000	0x000AFFFF	512x16	0x01085E00	0x01085FFF

7.3.4Peripheral Registers Memory Map
外设寄存器内存映射

The Peripheral Registers Memory Map table lists the peripheral registers.
“外设寄存器存储器映射”表列出了外设寄存器。

Table 7-4. Peripheral Registers Memory Map

Bit Field Name		DriverLib Name	Base Address	CPU1	DMA	HIC	CLA	Pipeline Protected
Instance	Structure							
Peripheral Frame 0(PF0)								
-	-	M0_RAM_BASE	0x0000_0000	Yes	-	-	-	-
-	-	M1_RAM_BASE	0x0000_0400	Yes	-	-	-	-
AdcaResultRegs	ADC_RESULT_REGS	ADCARESULT_BASE	0x0000_0B00	Yes	Yes	Yes	Yes	-
AdcbResultRegs	ADC_RESULT_REGS	ADCBRESULT_BASE	0x0000_0B20	Yes	Yes	Yes	Yes	-
AdccResultRegs	ADC_RESULT_REGS	ADCCRESULT_BASE	0x0000_0B40	Yes	Yes	Yes	Yes	-
CpuTimer0Regs	CPUTIMER_REGS	CPUTIMER0_BASE	0x0000_0C00	Yes	-	-	-	-
Cla1OnlyRegs	CLA_ONLY_REGS	CLA1_ONLY_BASE	0x0000_0C00	-	-	-	Yes	-
CpuTimer1Regs	CPUTIMER_REGS	CPUTIMER1_BASE	0x0000_0C08	Yes	-	-	-	-
CpuTimer2Regs	CPUTIMER_REGS	CPUTIMER2_BASE	0x0000_0C10	Yes	-	-	-	-
Cla1SoftIntRegs	CLA_SOFTINT_REGS	CLA1_SOFTINT_BASE	0x0000_0CE0	-	-	-	Yes	-
PieCtrlRegs	PIE_CTRL_REGS	PIECTRL_BASE	0x0000_0CE0	Yes	-	-	-	-
PieVectTable	PIE_VECT_TABLE	PIEVECTTABLE_BASE	0x0000_0D00	Yes	-	-	-	-
DmaRegs	DMA_REGS	DMA_BASE	0x0000_1000	Yes	-	-	-	-
Dmach1Regs	DMA_CH_REGS	DMA_CH1_BASE	0x0000_1020	Yes	-	-	-	-
Dmach2Regs	DMA_CH_REGS	DMA_CH2_BASE	0x0000_1040	Yes	-	-	-	-
Dmach3Regs	DMA_CH_REGS	DMA_CH3_BASE	0x0000_1060	Yes	-	-	-	-
Dmach4Regs	DMA_CH_REGS	DMA_CH4_BASE	0x0000_1080	Yes	-	-	-	-
Dmach5Regs	DMA_CH_REGS	DMA_CH5_BASE	0x0000_10A0	Yes	-	-	-	-
Dmach6Regs	DMA_CH_REGS	DMA_CH6_BASE	0x0000_10C0	Yes	-	-	-	-
Cla1Regs	CLA_REGS	CLA1_BASE	0x0000_1400	Yes	-	-	-	-
-	-	LS0_RAM_BASE	0x0000_8000	Yes	-	-	Yes	-
-	-	LS1_RAM_BASE	0x0000_8800	Yes	-	-	Yes	-
-	-	LS2_RAM_BASE	0x0000_9000	Yes	-	-	Yes	-
-	-	LS3_RAM_BASE	0x0000_9800	Yes	-	-	Yes	-
-	-	LS4_RAM_BASE	0x0000_A000	Yes	-	-	Yes	-
-	-	LS5_RAM_BASE	0x0000_A800	Yes	-	-	Yes	-
-	-	LS6_RAM_BASE	0x0000_B000	Yes	-	-	Yes	-
-	-	LS7_RAM_BASE	0x0000_B800	Yes	-	-	Yes	-
-	-	GS0_RAM_BASE	0x0000_C000	Yes	Yes	Yes	-	-
-	-	GS1_RAM_BASE	0x0000_D000	Yes	Yes	Yes	-	-
-	-	GS2_RAM_BASE	0x0000_E000	Yes	Yes	Yes	-	-
-	-	GS3_RAM_BASE	0x0000_F000	Yes	Yes	Yes	-	-
UidRegs	UID_REGS	UID_BASE	0x0007_0200	Yes	-	-	-	-
DcsmZ1OtpRegs	DCSM_Z1_OTP	DCSM_Z1OTP_BASE	0x0007_8000	Yes	-	-	-	-
DcsmZ2OtpRegs	DCSM_Z2_OTP	DCSM_Z2OTP_BASE	0x0007_8200	Yes	-	-	-	-
Peripheral Frame 1 (PF1)								
EPwm1Regs	EPWM_REGS	EPWM1_BASE	0x0000_4000	Yes	Yes	Yes	Yes	Yes
EPwm2Regs	EPWM_REGS	EPWM2_BASE	0x0000_4100	Yes	Yes	Yes	Yes	Yes

Bit Field Name		DriverLib Name	Base Address	CPU1	DMA	HIC	CLA	Pipeline Protected
Instance	Structure							
EPwm3Regs	EPWM_REGS	EPWM3_BASE	0x0000_4200	Yes	Yes	Yes	Yes	Yes
EPwm4Regs	EPWM_REGS	EPWM4_BASE	0x0000_4300	Yes	Yes	Yes	Yes	Yes
EPwm5Regs	EPWM_REGS	EPWM5_BASE	0x0000_4400	Yes	Yes	Yes	Yes	Yes
EPwm6Regs	EPWM_REGS	EPWM6_BASE	0x0000_4500	Yes	Yes	Yes	Yes	Yes
EPwm7Regs	EPWM_REGS	EPWM7_BASE	0x0000_4600	Yes	Yes	Yes	Yes	Yes
EPwm8Regs	EPWM_REGS	EPWM8_BASE	0x0000_4700	Yes	Yes	Yes	Yes	Yes
EQep1Regs	EQEP_REGS	EQEP1_BASE	0x0000_5100	Yes	Yes	Yes	Yes	Yes
EQep2Regs	EQEP_REGS	EQEP2_BASE	0x0000_5140	Yes	Yes	Yes	Yes	Yes
ECap1Regs	ECAP_REGS	ECAP1_BASE	0x0000_5200	Yes	Yes	Yes	Yes	Yes
ECap2Regs	ECAP_REGS	ECAP2_BASE	0x0000_5240	Yes	Yes	Yes	Yes	Yes
ECap3Regs	ECAP_REGS	ECAP3_BASE	0x0000_5280	Yes	Yes	Yes	Yes	Yes
HRCap3Regs	HRCAP_REGS	HRCAP3_BASE	0x0000_52A0	Yes	Yes	Yes	Yes	Yes
DacaRegs	DAC_REGS	DACA_BASE	0x0000_5C00	Yes	Yes	Yes	Yes	Yes
DacbRegs	DAC_REGS	DACB_BASE	0x0000_5C10	Yes	Yes	Yes	Yes	Yes
Cmpss1Regs	CMPSS_REGS	CMPSS1_BASE	0x0000_5C80	Yes	Yes	Yes	Yes	Yes
Cmpss2Regs	CMPSS_REGS	CMPSS2_BASE	0x0000_5CA0	Yes	Yes	Yes	Yes	Yes
Cmpss3Regs	CMPSS_REGS	CMPSS3_BASE	0x0000_5CC0	Yes	Yes	Yes	Yes	Yes
Cmpss4Regs	CMPSS_REGS	CMPSS4_BASE	0x0000_5CE0	Yes	Yes	Yes	Yes	Yes
Sdfm1Regs	SDFM_REGS	SDFM1_BASE	0x0000_5E00	Yes	Yes	Yes	Yes	Yes
Sdfm2Regs	SDFM_REGS	SDFM2_BASE	0x0000_5E80	Yes	Yes	Yes	Yes	Yes
Peripheral Frame 2 (PF2)								
SpiaRegs	SPI_REGS	SPIA_BASE	0x0000_6100	Yes	Yes	Yes	Yes	Yes
SpibRegs	SPI_REGS	SPIB_BASE	0x0000_6110	Yes	Yes	Yes	Yes	Yes
BgcrcCpuRegs	BGCRC_REGS	BGCRC_CPU_BASE	0x0000_6340	Yes	-	-	-	Yes
BgcrcCla1Regs	BGCRC_REGS	BGCRC_CLA1_BASE	0x0000_6380	Yes	-	-	Yes	Yes
PmbusaRegs	PMBUS_REGS	PMBUSA_BASE	0x0000_6400	Yes	Yes	Yes	Yes	Yes
HicRegs	HIC_CFG_REGS	HIC_BASE	0x0000_6500	Yes	Yes	-	-	Yes
FsiTxaRegs	FSI_TX_REGS	FSITXA_BASE	0x0000_6600	Yes	Yes	Yes	Yes	Yes
FsiRxaRegs	FSI_RX_REGS	FSIRXA_BASE	0x0000_6680	Yes	Yes	Yes	Yes	Yes
Peripheral Frame 3 (PF3)								
AdcaRegs	ADC_REGS	ADCA_BASE	0x0000_7400	Yes	-	-	Yes	Yes
AdcbRegs	ADC_REGS	ADCB_BASE	0x0000_7480	Yes	-	-	Yes	Yes
AdccRegs	ADC_REGS	ADCC_BASE	0x0000_7500	Yes	-	-	Yes	Yes
Peripheral Frame 4 (PF4)								
InputXbarRegs	INPUT_XBAR_REGS	INPUTXBAR_BASE	0x0000_7900	Yes	-	-	-	Yes
XbarRegs	XBAR_REGS	XBAR_BASE	0x0000_7920	Yes	-	-	-	Yes
SyncSocRegs	SYNC_SOC_REGS	SYNCSOC_BASE	0x0000_7940	Yes	-	-	-	Yes
ClbInputXbarRegs	INPUT_XBAR_REGS	CLBINPUTXBAR_BASE	0x0000_7960	Yes	-	-	-	Yes
DmaClaSrcSelRegs	DMA_CLA_SRC_SEL_REGS	DMACLASRCSEL_BASE	0x0000_7980	Yes	-	-	-	Yes
EPwmXbarRegs	EPWM_XBAR_REGS	EPWMXBAR_BASE	0x0000_7A00	Yes	-	-	-	Yes
ClbXbarRegs	CLB_XBAR_REGS	CLBXBAR_BASE	0x0000_7A40	Yes	-	-	-	Yes
OutputXbarRegs	OUTPUT_XBAR_REGS	OUTPUTXBAR_BASE	0x0000_7A80	Yes	-	-	-	Yes
ClbOutputXbarRegs	OUTPUT_XBAR_REGS	CLBOUTPUTXBAR_BASE	0x0000_7BC0	Yes	-	-	-	Yes
GpioCtrlRegs	GPIO_CTRL_REGS	GPIOCTRL_BASE	0x0000_7C00	Yes	-	-	-	Yes
GpioDataRegs	GPIO_DATA_REGS	GPIODATA_BASE	0x0000_7F00	Yes	-	-	Yes	Yes

Bit Field Name		DriverLib Name	Base Address	CPU1	DMA	HIC	CLA	Pipeline Protected
Instance	Structure							
GpioDataReadRegs	GPIO_DATA_READ_REGS	GPIODATAREAD_BASE	0x0000_7F80	Yes	-	Yes	Yes	Yes
ClkCfgRegs	CLK_CFG_REGS	CLKCFG_BASE	0x0005_D200	Yes	-	-	-	Yes
CpuSysRegs	CPU_SYS_REGS	CPUSYS_BASE	0x0005_D300	Yes	-	-	-	Yes
SysStatusRegs	SYS_STATUS_REGS	SYSSTAT_BASE	0x0005_D400	Yes	-	-	-	Yes
PeriphAcRegs	PERIPH_AC_REGS	PERIPHAC_BASE	0x0005_D500	Yes	-	-	-	Yes
AnalogSubsysRegs	ANALOG_SUBSYS_REGS	ANALOGSUBSYS_BASE	0x0005_D700	Yes	-	-	-	Yes
Peripheral Frame 5 (PF5)								
DevCfgRegs	DEV_CFG_REGS	DEVCFG_BASE	0x0005_D000	Yes	-	-	-	Yes
EradGlobalRegs	ERAD_GLOBAL_REGS	ERAD_GLOBAL_BASE	0x0005_E800	Yes	-	-	-	Yes
EradHWBP1Regs	ERAD_HWBP_REGS	ERAD_HWBP1_BASE	0x0005_E900	Yes	-	-	-	Yes
EradHWBP2Regs	ERAD_HWBP_REGS	ERAD_HWBP2_BASE	0x0005_E908	Yes	-	-	-	Yes
EradHWBP3Regs	ERAD_HWBP_REGS	ERAD_HWBP3_BASE	0x0005_E910	Yes	-	-	-	Yes
EradHWBP4Regs	ERAD_HWBP_REGS	ERAD_HWBP4_BASE	0x0005_E918	Yes	-	-	-	Yes
EradHWBP5Regs	ERAD_HWBP_REGS	ERAD_HWBP5_BASE	0x0005_E920	Yes	-	-	-	Yes
EradHWBP6Regs	ERAD_HWBP_REGS	ERAD_HWBP6_BASE	0x0005_E928	Yes	-	-	-	Yes
EradHWBP7Regs	ERAD_HWBP_REGS	ERAD_HWBP7_BASE	0x0005_E930	Yes	-	-	-	Yes
EradHWBP8Regs	ERAD_HWBP_REGS	ERAD_HWBP8_BASE	0x0005_E938	Yes	-	-	-	Yes
EradCounter1Regs	ERAD_COUNTER_REGS	ERAD_COUNTER1_BASE	0x0005_E980	Yes	-	-	-	Yes
EradCounter2Regs	ERAD_COUNTER_REGS	ERAD_COUNTER2_BASE	0x0005_E990	Yes	-	-	-	Yes
EradCounter3Regs	ERAD_COUNTER_REGS	ERAD_COUNTER3_BASE	0x0005_E9A0	Yes	-	-	-	Yes
EradCounter4Regs	ERAD_COUNTER_REGS	ERAD_COUNTER4_BASE	0x0005_E9B0	Yes	-	-	-	Yes
EradCRCGlobalRegs	ERAD_CRC_GLOBAL_REGS	ERAD_CRC_GLOBAL_BASE	0x0005_EA00	Yes	-	-	-	Yes
EradCRC1Regs	ERAD_CRC_REGS	ERAD_CRC1_BASE	0x0005_EA10	Yes	-	-	-	Yes
EradCRC2Regs	ERAD_CRC_REGS	ERAD_CRC2_BASE	0x0005_EA20	Yes	-	-	-	Yes
EradCRC3Regs	ERAD_CRC_REGS	ERAD_CRC3_BASE	0x0005_EA30	Yes	-	-	-	Yes
EradCRC4Regs	ERAD_CRC_REGS	ERAD_CRC4_BASE	0x0005_EA40	Yes	-	-	-	Yes
EradCRC5Regs	ERAD_CRC_REGS	ERAD_CRC5_BASE	0x0005_EA50	Yes	-	-	-	Yes
EradCRC6Regs	ERAD_CRC_REGS	ERAD_CRC6_BASE	0x0005_EA60	Yes	-	-	-	Yes
EradCRC7Regs	ERAD_CRC_REGS	ERAD_CRC7_BASE	0x0005_EA70	Yes	-	-	-	Yes
EradCRC8Regs	ERAD_CRC_REGS	ERAD_CRC8_BASE	0x0005_EA80	Yes	-	-	-	Yes
Epg1Regs	EPG_REGS	EPG1_BASE	0x0005_EC00	Yes	-	-	-	Yes
Epg1MuxRegs	EPG_MUX_REGS	EPG1MUX_BASE	0x0005_ECD0	Yes	-	-	-	Yes
DcsmZ1Regs	DCSM_Z1_REGS	DCSM_Z1_BASE	0x0005_F000	Yes	-	-	-	Yes
DcsmZ2Regs	DCSM_Z2_REGS	DCSM_Z2_BASE	0x0005_F080	Yes	-	-	-	Yes

Bit Field Name		DriverLib Name	Base Address	CPU1	DMA	HIC	CLA	Pipeline Protected
Instance	Structure							
DcsmCommonRegs	DCSM_COMMON_REGS	DCSMCOMMON_BASE	0x0005_F0C0	Yes	-	-	-	Yes
MemCfgRegs	MEM_CFG_REGS	MEMCFG_BASE	0x0005_F400	Yes	-	-	-	Yes
AccessProtectionRegs	ACCESS_PROTECTION_REGS	ACCESSPROTECTION_BASE	0x0005_F500	Yes	-	-	-	Yes
MemoryErrorRegs	MEMORY_ERROR_REGS	MEMORYERROR_BASE	0x0005_F540	Yes	-	-	-	Yes
TestErrorRegs	TEST_ERROR_REGS	TESTERROR_BASE	0x0005_F590	Yes	-	-	-	Yes
Flash0CtrlRegs	FLASH_CTRL_REGS	FLASH0CTRL_BASE	0x0005_F800	Yes	-	-	-	Yes
Flash0EccRegs	FLASH_ECC_REGS	FLASH0ECC_BASE	0x0005_FB00	Yes	-	-	-	Yes
Peripheral Frame 7 (PF7)								
CanaRegs	CAN_REGS	CANA_BASE	0x0004_8000	Yes	Yes	Yes	-	Yes
-	-	CANA_MSG_RAM_BASE	0x0004_9000	Yes	Yes	Yes	-	Yes
-	-	MCAN_MSG_RAM_BASE	0x0005_8000	Yes	-	Yes	-	Yes
McanaSsRegs	MCANASS_REGS	MCANASS_BASE	0x0005_C400	Yes	-	Yes	-	Yes
McanaRegs	MCANA_REGS	MCANA_BASE	0x0005_C600	Yes	-	Yes	-	Yes
McanaErrorRegs	MCANA_ERROR_REGS	MCANA_ERROR_BASE	0x0005_C800	Yes	-	Yes	-	Yes
HwbistRegs	HWBIST_REGS	HWBIST_BASE	0x0005_E000	Yes	-	-	-	Yes
MpostRegs	MPOST_REGS	MPOST_BASE	0x0005_E200	Yes	-	-	-	Yes
Dcc0Regs	DCC_REGS	DCC0_BASE	0x0005_E700	Yes	-	-	-	Yes
Dcc1Regs	DCC_REGS	DCC1_BASE	0x0005_E740	Yes	-	-	-	Yes
Peripheral Frame 8 (PF8)								
LinaRegs	LIN_REGS	LINA_BASE	0x0000_6A00	Yes	Yes	Yes	Yes	Yes
LinbRegs	LIN_REGS	LINB_BASE	0x0000_6B00	Yes	Yes	Yes	Yes	Yes
Peripheral Frame 9 (PF9)								
WdRegs	WD_REGS	WD_BASE	0x0000_7000	Yes	-	-	-	Yes
NmiIntruptRegs	NMI_INTRUPT_REGS	NMI_BASE	0x0000_7060	Yes	-	-	-	Yes
XintRegs	XINT_REGS	XINT_BASE	0x0000_7070	Yes	-	-	-	Yes
SciaRegs	SCI_REGS	SCIA_BASE	0x0000_7200	Yes	-	Yes	-	Yes
ScibRegs	SCI_REGS	SCIB_BASE	0x0000_7210	Yes	-	Yes	-	Yes
I2caRegs	I2C_REGS	I2CA_BASE	0x0000_7300	Yes	-	Yes	-	Yes
I2cbRegs	I2C_REGS	I2CB_BASE	0x0000_7340	Yes	-	Yes	-	Yes
Peripheral Frame 10 (PF10)								
Clb1LogicCfgRegs	CLB_LOGIC_CONFIG_REGS	CLB1_LOGICCFG_BASE	0x0000_3000	Yes	-	Yes	Yes	-
Clb1LogicCtrlRegs	CLB_LOGIC_CONTROL_REGS	CLB1_LOGICCTRL_BASE	0x0000_3100	Yes	-	Yes	Yes	-
Clb1DataExchRegs	CLB_DATA_EXCHANGE_REGS	CLB1_DATAEXCH_BASE	0x0000_3180	Yes	-	Yes	Yes	-
Clb2LogicCfgRegs	CLB_LOGIC_CONFIG_REGS	CLB2_LOGICCFG_BASE	0x0000_3400	Yes	-	Yes	Yes	-
Clb2LogicCtrlRegs	CLB_LOGIC_CONTROL_REGS	CLB2_LOGICCTRL_BASE	0x0000_3500	Yes	-	Yes	Yes	-
Clb2DataExchRegs	CLB_DATA_EXCHANGE_REGS	CLB2_DATAEXCH_BASE	0x0000_3580	Yes	-	Yes	Yes	-
Clb3LogicCfgRegs	CLB_LOGIC_CONFIG_REGS	CLB3_LOGICCFG_BASE	0x0000_3800	Yes	-	Yes	Yes	-

Bit Field Name		DriverLib Name	Base Address	CPU1	DMA	HIC	CLA	Pipeline Protected
Instance	Structure							
Clb3LogicCtrlRegs	CLB_LOGIC_CONTROL_REGS	CLB3_LOGICCTRL_BASE	0x0000_3900	Yes	-	Yes	Yes	-
Clb3DataExchRegs	CLB_DATA_EXCHANGE_REGS	CLB3_DATAEXCH_BASE	0x0000_3980	Yes	-	Yes	Yes	-
Clb4LogicCfgRegs	CLB_LOGIC_CONFIG_REGS	CLB4_LOGICCFG_BASE	0x0000_3C00	Yes	-	Yes	Yes	-
Clb4LogicCtrlRegs	CLB_LOGIC_CONTROL_REGS	CLB4_LOGICCTRL_BASE	0x0000_3D00	Yes	-	Yes	Yes	-
Clb4DataExchRegs	CLB_DATA_EXCHANGE_REGS	CLB4_DATAEXCH_BASE	0x0000_3D80	Yes	-	Yes	Yes	-
Peripheral Frame 11 (PF11)								
AesaRegs	AES_IP_REGS	AESA_BASE	0x0004_2000	Yes	Yes	-	-	-
AesaSsRegs	AES_WRAPPER_REGS	AESA_SS_BASE	0x0004_2C00	Yes	Yes	-	-	-
Peripheral Frame 12 (PF12)								
LfuRegs	LFU_REGS	LFU_BASE	0x0000_7FE0	Yes	-	-	Yes	Yes

7.4 Identification
标识

Table 7-5. Device Identification Registers

NAME	ADDRESS	SIZE (x16)	DESCRIPTION
PARTIDH	0x0005D00A	2	Device part identification number AVP32F003X 0x05FF0500 AVP32F00157P 0x05FF0500
REVID	0x0005D00C	2	Silicon revision number 0x00000000
UID_UNIQUE	0x0007 020C	2	Unique identification number. This number is different on each individual device with the same PARTIDH. This unique number can be used as a serial number in the application.

7.5 Bus Architecture – Peripheral Connectivity
总线架构-外设连接

The Peripheral Connectivity table lists a broad view of the peripheral and configuration register accessibility from each bus master.

“外设连接”表列出了每个总线主器件对外设和配置寄存器的可访问性。

Table 7-6. Peripheral Connectivity

PERIPHERAL	DMA	HIC	BGCRC	CLA	A2000
SYSTEM PERIPHERALS 系统外设					
CPU Timers					Yes
ERAD					Yes
GPIO Data		Yes		Yes	Yes
GPIO Pin Mapping and Configuration					Yes
XBAR Configuration					Yes
System Configuration					Yes
AES	Yes				Yes
EPG					Yes
LFU				Yes	Yes
DCC					Yes
MEMORY					
M0/M1			Yes		Yes
LSx			Yes	Yes	Yes
GSx	Yes	Yes	Yes		Yes
ROM			Yes		Yes
FLASH					Yes
CONTROL PERIPHERALS 控制外设					
ePWM/HRPWM	Yes	Yes		Yes	Yes
eCAP/HRCAP	Yes	Yes		Yes	Yes
eQEP(1)	Yes	Yes		Yes	Yes
CLB		Yes		Yes	Yes

PERIPHERAL	DMA	HIC	BGCRC	CLA	A2000
SDFM	Yes	Yes		Yes	Yes
ANALOG PERIPHERALS 模拟外设					
CMPSS(1)	Yes	Yes		Yes	Yes
DAC(1)	Yes	Yes		Yes	Yes
ADC Configuration				Yes	Yes
ADC Results(1)	Yes	Yes		Yes	Yes
COMMUNICATION PERIPHERALS 通信外设					
DCAN	Yes	Yes			Yes
MCAN		Yes			Yes
FSITX/FSIRX	Yes	Yes		Yes	Yes
I2C		Yes			Yes
LIN	Yes	Yes		Yes	Yes
PMBus	Yes	Yes		Yes	Yes
SCI		Yes			Yes
SPI	Yes	Yes		Yes	Yes

(1) These modules are accessible from DMA but cannot trigger a DMA transfer.
这些模块可从 DMA 访问，但不能触发 DMA 传输。

7.6 A2000 Processor
A2000 处理器

The CPU is a 32-bit fixed-point processor. This device draws from the best features of digital signal processing; reduced instruction set computing (RISC); and microcontroller architectures, firmware, and tool sets.
The CPU features include a modified Harvard architecture and circular addressing. The RISC features are single-cycle instruction execution, register-to-register operations, and modified Harvard architecture. The microcontroller features include ease of use through an intuitive instruction set, byte packing and unpacking, and bit manipulation. The modified Harvard architecture of the CPU enables instruction and data fetches to be performed in parallel. The CPU can read instructions and data while it writes data simultaneously to maintain the single-cycle instruction operation across the pipeline. The CPU does this over six separate address/data buses.

CPU是32位定点处理器。该器件借鉴了数字信号处理的最佳特性;精简指令集计算(RISC);以及微控制器架构、固件和工具集。

CPU 的特性包含修改后的 Harvard 架构和循环寻址。RISC 特性是单周期指令执行、寄存器到寄存器操作和修改后的Harvard 架构。微控制器特性包含通过直观的指令集、字节打包和解包以及位操作来实现易用性。CPU 修改后的Harvard 架构使指令和数据获取能够并行执行。CPU 可以读取指令和数据，同时写入数据以在整个流水线中保持单周期指令操作。CPU 通过六条独立的地址/数据总线完成此操作。

7.6.1 Floating-Point Unit (FPU)
浮点单元(FPU)

The A2000 plus floating-point (A2000+FPU) processor extends the capabilities of the A2000 fixed-point CPU by adding registers and instructions to support IEEE single-precision floating-point operations.
Devices with the A2000+FPU include the standard A2000 register set plus an additional set of floating-point unit registers. The additional floating-point unit registers are the following:

- Eight floating-point result registers, RnH (where n = 0–7)
- Floating-point Status Register (STF)
- Repeat Block Register (RB)

All of the floating-point registers, except the RB, are shadowed. This shadowing can be used in high-priority interrupts for fast context save and restore of the floating-point registers.

A2000 加浮点（A2000+FPU）处理器通过增加支持 IEEE 单精度浮点运算的寄存器和指令来扩展 A2000 定点CPU 的功能。

具有A2000+FPU 的器件包含标准 A2000 寄存器集以及一组额外的浮点单元寄存器。额外的浮点单元寄存器如下：

- 八个浮点结果寄存器，RnH（其中 n=0–7）
- 浮点状态寄存器(STF)
- 重复块寄存器(RB)

除 RB 寄存器外，所有浮点寄存器都采用影子化技术。这种影子化可用于高优先级中断，以实现浮点寄存器的快速上下文保存和恢复。

7.6.2Fast Integer Division Unit
快速整数除法单元

The Fast Integer Division (FINTDIV) unit of the A2000 CPU uniquely supports three types of integer division (Truncated, Modulus, Euclidean) of varying data type sizes (16/16, 32/16, 32/32, 64/32, 64/64) in unsigned or signed formats.

- Truncated integer division is naturally supported by C language (/, % operators).
- Modulus and Euclidean divisions are variants that are more efficient for control algorithms and are supported by C intrinsics.

All three types of integer division produce both a quotient and remainder component, are interruptible, and execute in a minimum number of deterministic cycles (10 cycles for a 32/32 division). In addition, the Fast Division capabilities of the A2000 CPU uniquely support fast execution of floating-point 32-bit (in 5 cycles) and 64-bit (in 20 cycles) division.

A2000 CPU 的快速整数除法（FINTDIV）单元独特地支持三种类型的整数除法（截断、模数、欧几里德），这些整数除法具有不同的数据类型大小（16/16、32/16、32/32、64/32、64/64），采用无符号或有符号格式。

- C 语言支持截断整数除法（/、%运算符）。
- 模数除法和欧几里德除法是更高效的控制算法，并受 C 内在函数支持。

所有三种类型的整数除法都会产生商和余数分量，具有可中断特性，并在最小数量的确定性周期内执行（32/32 除法为 10 个周期）。此外，A2000 CPU 的快速除法功能独特地支持浮点 32 位（5 个周期内）和 64 位（20 个周期内）除法的快速执行。

7.6.3Trigonometric Math Unit (TMU)
三角法数学单元(TMU)

The trigonometric math unit (TMU) extends the capabilities of a A2000+FPU by adding instructions and leveraging existing FPU instructions to speed up the execution of common trigonometric and arithmetic operations listed in Table 7-7.

三角函数加速器(TMU)通过增加指令和利用可加速执行常见三角函数和表 7-7 中所列算术运算的现有 FPU 指令来扩展 A2000+FPU 的功能。

Table 7-7. TMU Supported Instructions

Instructions	C Equivalent Operation	Pipeline Cycles
MPY2PIF32RaH，RbH	a=b*2π	2/3
DIV2PIF32RaH，RbH	a=b/2π	2/3
DIVF32RaH，RbH，RcH	a=b/c	5
SQRTF32RaH，RbH	a=sqrt(b)	5
SINPUF32RaH，RbH	a=sin(b*2π)	4
COSPUF32RaH，RbH	a=cos(b*2π)	4
ATANPUF32RaH，RbH	a=atan(b)/2π	4
QUADF32RaH，RbH，RcH，RdH	Operation to assist in calculating ATANPU2	5

No changes have been made to existing instructions, pipeline or memory bus architecture. All TMU instructions use the existing FPU register set (R0H to R7H) to carry out their operations.

对现有指令、流水线或内存总线架构均未做任何更改。所有 TMU 指令都使用现有的 FPU 寄存器集（R0H 至R7H）来执行运算。

7.6.4VCRC Unit
VCRC 单元

Cyclic redundancy check (CRC) algorithms provide a straightforward method for verifying data integrity over large data blocks, communication packets, or code sections. The A2000+VCRC can perform 8-bit, 16-bit, 24-bit, and 32-bit CRCs. For example, the VCRC can compute the CRC for a block length of 10 bytes in 10 cycles. A CRC result register contains the current CRC, which is updated whenever a CRC instruction is executed.

The following are the CRC polynomials used by the CRC calculation logic of the VCRC:

- CRC8 polynomial = 0x07
- CRC16 polynomial 1 = 0x8005
- CRC16 polynomial 2 = 0x1021
- CRC24 polynomial = 0x5d6dcb
- CRC32 polynomial 1 = 0x04c11db7
- CRC32 polynomial 2 = 0x1edc6f41

This module can calculate CRCs for a byte of data in a single cycle. The CRC calculation for CRC8, CRC16, CRC24, and CRC32 is done byte-wise (instead of computing on a complete 16-bit or 32-bit data read by the A2000 core) to match the byte-wise computation requirement mandated by various standards.
The VCRC Unit also allows the user to provide the size (1b-32b) and value of any polynomial to fit custom CRC requirements. The CRC execution time increases to three cycles when using a custom polynomial.

循环冗余校验（CRC）算法提供了一种简单的方法来验证大型数据块、通信数据包或代码段

上的数据完整性。A2000+VCRC 可执行 8 位、16 位、24 位和 32 位 CRC。例如，VCRC 可以在 10 个周期内计算出块长度为 10 字节的 CRC。CRC 结果寄存器包含当前 CRC，每次执行 CRC 指令时，该 CRC 都会更新。

以下是VCRC 的CRC 计算逻辑使用的 CRC 多项式：

- CRC8 多项式=0x07
- CRC16 多项式 1=0x8005
- CRC16 多项式 2=0x1021
- CRC24 多项式=0x5d6dcb
- CRC32 多项式 1=0x04c11db7
- CRC32 多项式 2=0x1edc6f41

该模块可以在单个周期内为一个字节的数据计算 CRC。CRC8、CRC16、CRC24 和CRC32 的 CRC 计算是按字节完成的（而不是计算 A2000 内核读取的完整 16 位或 32 位数据），以便与各种标准规定的按字节计算要求保持一致。VCRC 单元还允许用户提供任何多项式的大小(1b-32b)和值，来满足自定义 CRC 要求。使用自定义多项式时，CRC 执行时间会增加到三个周期。

7.7 Control Law Accelerator (CLA)
控制律加速器(CLA)

The CLA Type-2 is an independent, fully programmable, 32-bit floating-point math processor that brings concurrent control-loop execution to the A2000 family. The low interrupt-latency of the CLA allows it to read ADC samples "just-in-time." This significantly reduces the ADC sample to output delay to enable faster system response and higher MHz control loops. By using the CLA to service time-critical control loops, the main CPU is free to perform other system tasks such as communications and diagnostics.
The control law accelerator extends the capabilities of the A2000 CPU by adding parallel processing. Time-critical control loops serviced by the CLA can achieve low ADC sample to output delay. Thus, the CLA enables faster system response and higher frequency control loops. Using the CLA for time-critical tasks frees up the main CPU to perform other system and communication functions concurrently.

CLAType-2 是一款独立、完全可编程的 32 位浮点数学处理器，为 A2000 系列实现了并发控制环路执行。CLA 的低中断延迟使其能够“及时”读取 ADC 样本。这显著降低了 ADC 采样到输出延迟，从而实现了更快的系统响应和更高的 MHz 控制环路。通过利用 CLA 为时间关键型控制环路提供服务，可腾出主 CPU 来执行其他系统任务，如通信和诊断。

控制律加速器通过添加并行处理来扩展 A2000CPU 的功能。CLA 处理的时间关键控制环路可实现低 ADC 采样输出延迟。因此，CLA 支持更快速的系统响应和更高频率的控制环路。将 CLA 用于时间关键型任务可释放主 CPU 以同时执行其他系统和通信功能。

The following is a list of major features of the CLA:

- Clocked at the same rate as the main CPU (SYSCLKOUT).
- An independent architecture allowing CLA algorithm execution independent of the main A2000 CPU.
  - Complete bus architecture:
    - Program Address Bus (PAB) and Program Data Bus (PDB)
    - Data Read Address Bus (DRAB), Data Read Data Bus (DRDB), Data Write Address Bus (DWAB), and Data Write Data Bus (DWDB)
  - Independent 8-stage pipeline.
  - 16-bit program counter (MPC)
  - Four 32-bit result registers (MR0 to MR3)
  - Two 16-bit auxiliary registers (MAR0, MAR1)
  - Status register (MSTF)
- Instruction set includes:
  - IEEE single-precision (32-bit) floating-point math operations
  - Floating-point math with parallel load or store
  - Floating-point multiply with parallel add or subtract
  - 1/X and 1/sqrt(X) estimations
  - Data type conversions
  - Conditional branch and call
  - Data load/store operations
- The CLA program code can consist of up to eight tasks or interrupt service routines, or seven tasks and a main background task.
  - The start address of each task is specified by the MVECT registers.
  - No limit on task size as long as the tasks fit within the configurable CLA program memory space.
  - One task is serviced at a time until its completion. There is no nesting of tasks.
  - Upon task completion a task-specific interrupt is flagged within the PIE.
  - When a task finishes the next highest-priority pending task is automatically started.
  - The Type-2 CLA can have a main task that runs continuously in the background, while other high-priority events trigger a foreground task.

- Task trigger mechanisms:
  – A2000 CPU through the IACK instruction
  – Task1 to Task8: up to 256 possible trigger sources from peripherals connected to the shared bus on which the CLA assumes secondary ownership.
  – Task8 can be set to be the background task, while Tasks 1 to 7 take peripheral triggers.
- Memory and Shared Peripherals:
  – Two dedicated message RAMs for communication between the CLA and the main CPU.
  – The A2000 CPU can map CLA program and data memory to the main CPU space or CLA space.

以下是CLA 主要特性的列表。

- 时钟速率与主 CPU 一致(SYSCLKOUT)。
- 一个独立的架构使得 CLA 能够独立于主 A2000CPU 之外执行算法。

完整的总线架构:

- 程序地址总线(PAB)和程序数据总线(PDB)
- 数据读取地址总线(DRAB)、数据读取数据总线(DRDB)、数据写入地址总线(DWAB)和数据写入数据总线(DWDB)
  - 独立的 8 级流水线。
  - 16 位程序计数器(MPC)
  - 四个 32 位结果寄存器（MR0 至 MR3）
  - 两个 16 位辅助寄存器（MAR0、MAR1）
  - 状态寄存器(MSTF)
- 指令集包括:
  - IEEE 单精度（32 位）浮点数学运算
  - 涉及并行载入或者存储的浮点数学
  - 涉及并行加法或者减法的浮点乘法
  - 1/X 和 1/sqrt(X)估值
  - 数据类型转换
  - 条件分支指令和调用
  - 数据载入/存储操作
- CLA 程序代码可包含多达 8 个任务或中断处理例程，或者 7 个任务和一个主后台任务。
  - 每一个任务的开始地址由 MVECT 寄存器指定。
  - 只要任务适合可配置的 CLA 程序内存空间，任务大小就没有限制。
  - 每次处理并完成一个任务。无任务嵌套。
  - 任务完成时，在 PIE 内标志一个任务专用中断。
  - 当一个任务结束时，下一个具有最高优先级的等待任务自动开始。
  - Type-2CLA 可以有一个在后台连续运行的主任务，而其他高优先级事件则触发前台任务。
- 任务触发机制:
  - 借助于 IACK 指令的 A2000CPU
  - 任务 1 到任务 8：最多 256 个可能的触发源，来自连接到共享总线的外设，CLA 在

共享总线上承担二级所有权。

- 任务 8 可以设置为后台任务，而任务 1 至 7 采用外设触发。

- 内存和共用外设：
  - 两个专用消息 RAM 用于 CLA 和主 CPU 间的通信。
  - A2000 CPU 能够将 CLA 程序和数据内存映射到主 CPU 空间或者 CLA 空间。

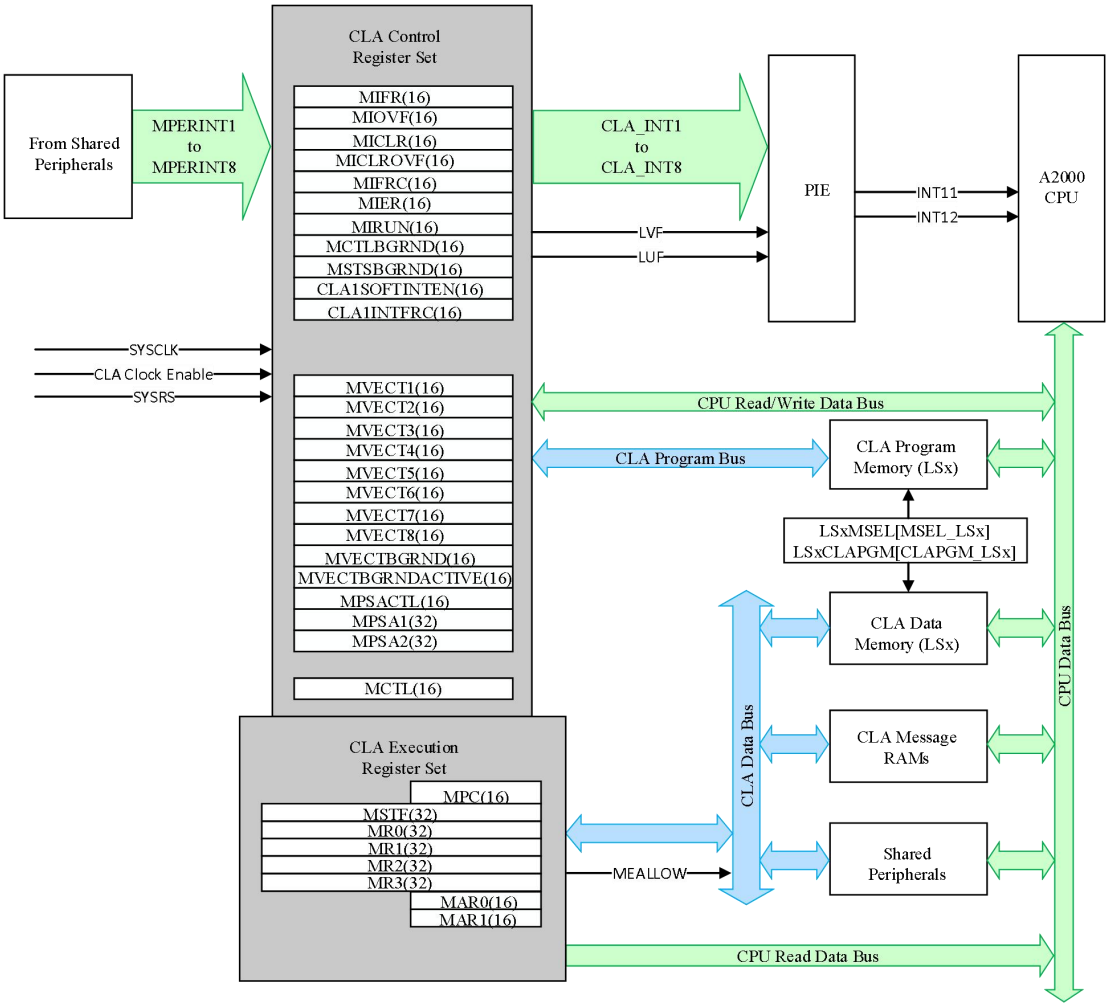


Figure 7-2. CLA Block Diagram

7.8 Embedded Real-Time Analysis and Diagnostic (ERAD)
嵌入式实时分析和诊断(ERAD)

The ERAD module enhances the debug and system-analysis capabilities of the device. The debug and system- analysis enhancements provided by the ERAD module is done outside of the CPU. The ERAD module consists of the Enhanced Bus Comparator units and the System Event Counter units. The Enhanced Bus Comparator units are used to generate hardware breakpoints, hardware watch points, and other output events. The System Event Counter units are used to analyze and profile the system. The ERAD module is accessible by the debugger and by the application software, which significantly increases the debug capabilities of many real-time systems, especially in situations where debuggers are not connected. In the AVP32F003X devices, the ERAD module contains eight Enhanced Bus Comparator units (which increases the number of Hardware breakpoints from two to ten) and four Benchmark System Event Counter units.

ERAD 模块增强了器件的调试和系统分析功能。ERAD 模块提供的调试和系统分析增强功能在 CPU 之外完成。ERAD 模块由增强型总线比较器单元和系统事件计数器单元组成。增强型总线比较器单元用于生成硬件断点、硬件观察点和其他输出事件。系统事件计数器单元用于分析和评测系统。ERAD 模块可由调试器和应用软件访问，这显著提高了许多实时系统的调试功能，尤其是在调试器未连接的情况下。在 AVP32F003X 器件中，ERAD 模块包含八个增强型总线比较器单元（将硬件断点数量从 2 个增加到 10 个）和四个基准测试系统事件计数器单元。

7.9 Background CRC-32 (BGCRC)
后台 CRC-32 (BGCRC)

The Background CRC (BGCRC) module computes a CRC-32 on a configurable block of memory. It accomplishes this by fetching the specified block of memory during idle cycles (when the CPU, HIC, CLA or DMA is not accessing the memory block). The calculated CRC-32 value is compared against a golden CRC-32 value to indicate a pass or fail. In essence, the BGCRC helps identify memory faults and corruption. The BGCRC module has the following features:

- One cycle CRC-32 computation on 32 bits of data
- No CPU bandwidth impact for zero wait state memory
- Minimal CPU bandwidth impact for non-zero wait state memory
- Dual operation modes (CRC-32 mode and scrub mode)
- Watchdog timer to time CRC-32 completion
- Ability to pause and resume CRC-32 computation

后台 CRC（BGCRC）模块在可配置的存储器块上计算 CRC-32。这是通过在 CPU、HIC、CLA 或 DMA 不存取指定的存储块的空闲周期中获取该存储块来实现的。计算出的 CRC-32 值与黄金 CRC-32 值进行比较以指示通过还是未通过。本质上，BGCRC 有助于识别内存故障和损坏。

BGCRC 模块具有以下特性：

- 关于 32 位数据的一个周期 CRC-32 运算
- 对于零等待状态存储器，CPU 带宽不受影响
- 对非零等待状态存储器，CPU 带宽受到的影响极小
- 双工作模式（CRC-32 模式和清理模式）
- 通过看门狗定时器对 CRC-32 完成时间进行计时
- 能够暂停和恢复 CRC-32 计算

7.10 Direct Memory Access (DMA)
直接存储器访问(DMA)

The DMA module provides a hardware method of transferring data between peripherals and/or memory without intervention from the CPU, thereby freeing up bandwidth for other system functions. Additionally, the DMA has the capability to orthogonally rearrange the data as it is transferred as well as “ping-pong” data between buffers. These features are useful for structuring data into blocks for optimal CPU processing. Figure 7-3 shows a device-level block diagram of the DMA.

DMA features include:

- Six channels with independent PIE interrupts
- Peripheral interrupt trigger sources
  - ADC interrupts and EVT signals
  - External Interrupts
  - ePWM SOC signals
  - CPU timers
  - eCAP
  - SPI transmit and receive
  - CAN transmit and receive
  - LIN transmit and receive
- Data sources and destinations:
  - GSx RAM
  - ADC result registers
  - Control peripheral registers (ePWM, eQEP, eCAP)
  - SPI, LIN, CAN, and PMBus registers
- Word Size: 16-bit or 32-bit (SPI limited to 16-bit)
- Throughput: Four cycles per word without arbitration

DMA 模块提供了一种在外设和/或存储器之间传输数据的硬件方法，无需 CPU 干预，从而为其他系统功能释放带宽。此外，DMA 还能够在数据传输时对其进行正交重排，以及在缓冲器之间对数据执行“乒乓”操作。这些特性对于将数据结构化为模块以实现最优 CPU 处理非常有用。图 7-3 为 DMA 的器件级方框图。

DMA 特性包括：

- 六个具有独立 PIE 中断的通道
- 外设中断触发源
  - ADC 中断和 EVT 信号
  - 外部中断
  - ePWMSOC 信号
  - CPU 定时器
  - eCAP
  - SPI 发送和接收
  - CAN 发送和接收
  - LIN 发送和接收
- 数据源和目标：
  - GSxRAM
  - ADC 结果寄存器
  - 控制外设寄存器（ePWM、eQEP、eCAP）

- SPI、LIN、CAN 和 PMBus 寄存器

- 字大小：16 位或 32 位（SPI 限制到 16 位）
- 数据率：每个字四个周期，无需仲裁

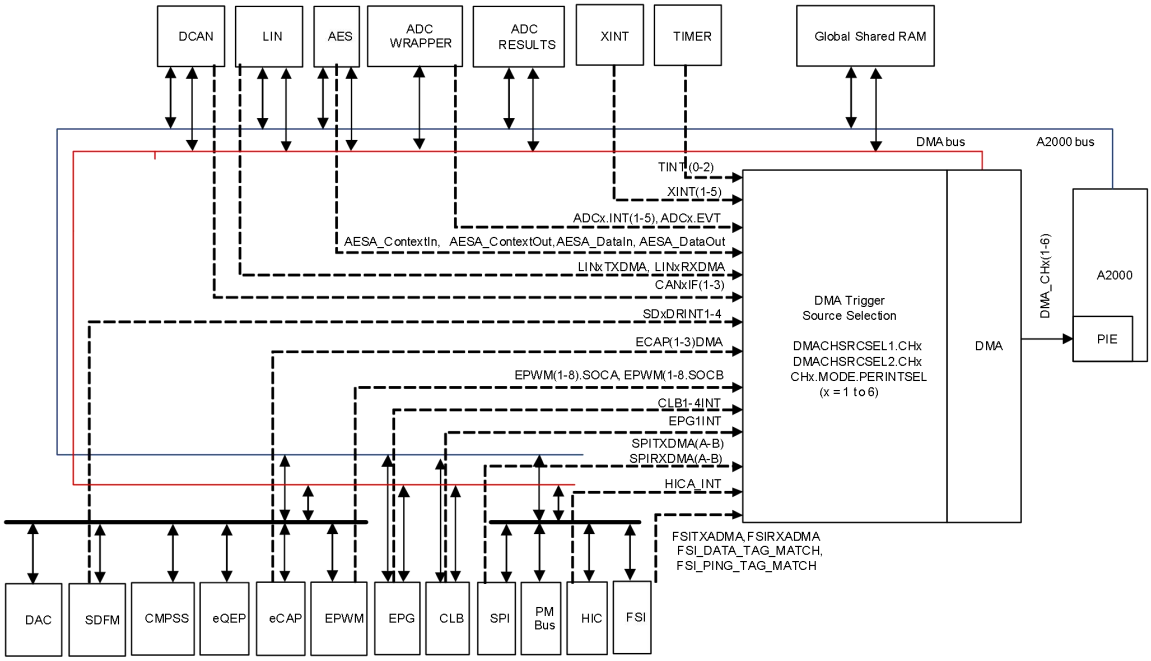


Figure 7-3. DMA Block Diagram

7.11 Device Boot Modes

This section explains the default boot modes, as well as all the available boot modes supported on this device. The boot ROM uses the boot mode select, general-purpose input/output (GPIO) pins to determine the boot mode configuration.
Table 7-8 shows the boot mode options available for selection by the default boot mode select pins. Users have the option to program the device to customize the boot modes selectable in the boot-up table as well as the boot mode select pin GPIOs used.
All the peripheral boot modes that are supported use the first instance of the peripheral module (SCIA, SPIA, I2CA, CANA, and so forth). Whenever these boot modes are referred to in this chapter, such as SCI boot, it is actually referring to the first module instance, which means the SCI boot on the SCIA port. The same applies to the other peripheral boots.
See Section 6.12.2.2.2 and the Power-on Reset figure for tboot-flash, the boot ROM execution time to first instruction fetch in flash.

本节介绍了默认引导模式以及该器件支持的所有可用引导模式。引导 ROM 使用引导模式选择、通用输入/输出（GPIO)引脚来确定引导模式配置。

表 7-8 展示了可供默认引导模式选择引脚选择的引导模式选项。用户可以选择对器件进行编程，以便自定义引导表中可选的引导模式以及使用的引导模式选择引脚 GPIO。

所有支持的外设引导模式都使用外设模块（SCIA、SPIA、I2CA、CANA 等）的第一个实例。凡是本章提到的这些引导模式（例如 SCI 引导）时，实际均指第一个模块实例，如 SCIA 端口上的 SCI 引导。这同样适用于其他外设引导。

请参阅第 6.12.2.2.2 节和“上电复位”图，了解 $t_{boot-flash}$（在闪存中第一次取指令之前的引导 ROM 执行时间）。

Table 7-8. Device Default Boot Modes

BOOT MODE	GPIO24 (DEFAULT BOOT MODE SELECT PIN 1)	GPIO32 (DEFAULT BOOT MODE SELECT PIN 0)
Parallel IO	0	0
SCI / Wait Boot(1)	0	1
CAN	1	0
Flash	1	1

(1) SCI boot mode can be used as a wait boot mode as long as SCI continues to wait for an 'A' or 'a' during the SCI autobaud lock process.
只要SCI 在SCI 自动波特率锁定过程中继续等待“A”或“a”，SCI 引导模式就可用作等待引导模式。

7.11.1 Device Boot Configurations

This section details what boot configurations are available and how to configure them. This device supports from 0 boot mode select pins up to 3 boot mode select pins as well as from 1 configured boot mode up to 8 configured boot modes.
To change and configure the device from the default settings to custom settings for your application, use the following process:

1. Determine all the various ways you want application to be able to boot. (For example: Primary boot option of Flash boot for your main application, secondary boot option of CAN boot for firmware updates, tertiary boot option of SCI boot for debugging, etc)
2. Based on the number of boot modes needed, determine how many boot mode select pins (BMSPs) are required to select between your selected boot modes. (For example: 2 BMSPs are required to select between 3 boot mode options)

3. Assign the required BMSPs to a physical GPIO pin. (For example, BMSP0 to GPIO10, BMSP1 to GPIO51, and BMSP2 left as default which is disabled). Refer to Section 7.11.1.1 for all the details on performing these configurations.
4. Assign the determined boot mode definitions to indexes in your custom boot table that correlate to the decoded value of the BMSPs. For example, BOOTDEF0=Boot to Flash, BOOTDEF1=CAN Boot, BOOTDEF2=SCI Boot; all other BOOTDEFx are left as default/nothing). Refer to Section 7.11.1.2 for all the details on setting up and configuring the custom boot mode table.

Additionally, the Boot Mode Example Use Cases section of the AVP32F003X Real-Time Microcontrollers Technical Reference Manual provides some example use cases on how to configure the BMSPs and custom boot tables.

本节详细介绍了可用的引导配置以及如何配置这些配置。该器件支持 0 引导模式选择引脚到 3 个引导模式选择引脚，以及 1 种配置的引导模式到 8 种配置的引导模式。

要更改器件并将其从默认设置配置为应用程序的自定义设置，请使用以下过程：

1. 确定您希望应用程序能够引导的所有各种方式。（例如：用于主应用程序的闪存引导的主要引导选项、用于固件更新的 CAN 引导的次级引导选项、用于调试的 SCI 引导的三级引导选项等）
2. 根据所需引导模式的数量，确定需要多少个引导模式选择引脚(BMSP)来在所选的引导模式之间进行选择。（例如：需要2 个BMSP 来在3 个引导模式选项之间进行选择）
3. 将所需的 BMSP 分配到物理 GPIO 引脚。（例如，BMSP0 到GPIO10，BMSP1 到GPIO51，BMSP2 保留为默认禁用状态。）有关执行这些配置的所有详细信息，请参阅第7.11.1.1 节。
4. 将确定的引导模式定义分配给自定义引导表中与 BMSP 的解码值相关的索引。例如，BOOTDEF0=引导至闪存，BOOTDEF1=CAN 引导，BOOTDEF2=SCI 引导；所有其他 BOOTDEFx 保留为默认值/无。有关如何设置和配置自定义引导模式表的所有详细信息，请参阅第7.11.1.2 节。

此外，《AVP32F003X 技术参考手册》的“引导模式示例用例”一节提供了一些有关如何配置 BMSP 和自定义引导表的示例用例。

Note

The CAN boot mode turns on the XTAL. Be sure an XTAL is installed in the application before using CAN boot mode.

CAN 引导模式打开 XTAL。在使用 CAN 引导模式之前，请确保在应用程序中安装了 XTAL。

7.11.1.1 Configuring Boot Mode Pins
配置引导模式引脚

This section explains how the boot mode select pins can be customized by the user, by programming the BOOTPIN-CONFIG location (refer to Table 7-9) in the user-configurable dual-zone security module (DCSM) OTP. The location in the DCSM OTP is Z1-OTP-BOOTPIN-CONFIG or Z2-OTP-BOOTPIN-CONFIG. When debugging, EMU-BOOTPIN-CONFIG is the emulation equivalent of Z1-OTP-BOOTPIN-CONFIG/Z2-OTP-BOOTPIN-CONFIG, and can be programmed to experiment with different boot modes without writing to OTP. The device can be programmed to use 0, 1, 2, or 3 boot mode select pins as needed.

本节介绍了用户如何通过在用户可配置的双区域安全模块(DCSM)OTP 中对 BOOTPIN-CONFIG 位置（请参阅表 7-9）进行编程来自定义引导模式选择引脚。DCSMOTP 中的位置是

Z1-OTP-BOOTPIN-CONFIG 或 Z2-OTP-BOOTPIN-CONFIG。调试时，EMU-BOOTPIN-CONFIG 是 Z1-OTP-BOOTPIN-CONFIG/Z2-OTP-BOOTPIN-CONFIG 的仿真等效项，可对其进行编程，以便在不写入 OTP 的情况下使用不同的引导模式进行实验。可根据需要对器件进行编程，以使用 0、1、2 或 3 个引导模式选择引脚。

Note

When using Z2-OTP-BOOTPIN-CONFIG, the configurations programmed in this location will take priority over the configurations in Z1-OTP-BOOTPIN-CONFIG. It is recommended to use Z1-OTP-BOOTPIN-CONFIG first and then if OTP configurations need to be altered, switch to using Z2-OTP-BOOTPIN-CONFIG.

使用 Z2-OTP-BOOTPIN-CONFIG 时，在此位置编程的配置将优先于 Z1-OTP-BOOTPIN-CONFIG 中的配置。建议先使用 Z1-OTP-BOOTPIN-CONFIG，然后如果需要更改 OTP 配置，请改为使用 Z2-OTP-BOOTPIN-CONFIG。

Table 7-9. BOOTPIN-CONFIG Bit Fields

BIT	NAME	DESCRIPTION
31:24	Key	Write 0x5A to these 8-bits to indicate the bits in this register are valid.
23:16	Boot Mode Select Pin 2 (BMSP2)	Refer to BMSP0 description except for BMSP2
15:8	Boot Mode Select Pin 1 (BMSP1)	Refer to BMSP0 description except for BMSP1
7:0	Boot Mode Select Pin 0 (BMSP0)	Set to the GPIO pin to be used during boot (up to 255): – 0x0 = GPIO0 – 0x01 = GPIO1 – and so on Writing 0xFF disables BMSP0 and this pin is no longer used to select the boot mode.

The following GPIOs cannot be used as a BMSP. If selected for a particular BMSP, the boot ROM automatically selects the factory default GPIO (the factory default for BMSP2 is 0xFF, which disables the BMSP).

- GPIO20 and GPIO21
- GPIO36 and GPIO38
- GPIO62 to GPIO223

以下 GPIO 不能用作 BMSP。如果为特定的 BMSP 选择，引导 ROM 会自动选择出厂默认 GPIO（BMSP2 的出厂默认值为 0xFF，这会禁用 BMSP）。

- GPIO20 和GPIO21
- GPIO36 和GPIO38
- GPIO62 至GPIO223

Table 7-10. Standalone Boot Mode Select Pin Decoding

BOOTPIN_CONFIG KEY	BMSP0	BMSP1	BMSP2	REALIZED BOOT MODE
!=0x5A	Don't Care	Don't Care	Don't Care	Boot as defined by the factory default BMSPs
= 0x5A	0xFF	0xFF	0xFF	Boot as defined in the boot table for boot mode 0 (All BMSPs disabled)
	Valid GPIO	0xFF	0xFF	Boot as defined by the value of BMSP0 (BMSP1 and BMSP2 disabled)
	0xFF	Valid GPIO	0xFF	Boot as defined by the value of BMSP1 (BMSP0 and BMSP2 disabled)
	0xFF	0xFF	Valid GPIO	Boot as defined by the value of BMSP2 (BMSP0 and BMSP1 disabled)
	Valid GPIO	Valid GPIO	0xFF	Boot as defined by the values of BMSP0 and BMSP1 (BMSP2 disabled)
	Valid GPIO	0xFF	Valid GPIO	Boot as defined by the values of BMSP0 and BMSP2 (BMSP1 disabled)
	0xFF	Valid GPIO	Valid GPIO	Boot as defined by the values of BMSP1 and BMSP2 (BMSP0 disabled)
	Valid GPIO	Valid GPIO	Valid GPIO	Boot as defined by the values of BMSP0, BMSP1, and BMSP2
	Invalid GPIO	Valid GPIO	Valid GPIO	BMSP0 is reset to the factory default BMSP0 GPIO Boot as defined by the values of BMSP0, BMSP1, and BMSP2
	Valid GPIO	Invalid GPIO	Valid GPIO	BMSP1 is reset to the factory default BMSP1 GPIO Boot as defined by the values of BMSP0, BMSP1, and BMSP2
	Valid GPIO	Valid GPIO	Invalid GPIO	BMSP2 is reset to the factory default state, which is disabled Boot as defined by the values of BMSP0 and BMSP1

Note

When decoding the boot mode, BMSP0 is the least-significant-bit and BMSP2 is the most-significant-bit of the boot table index value. It is recommended when disabling BMSPs to start with disabling BMSP2. For example, in an instance when only using BMSP2 (BMSP1 and BMSP0 are disabled), then only the boot table indexes of 0 and 4 will be selectable. In the instance when using only BMSP0, then the selectable boot table indexes are 0 and 1.

解码引导模式时，BMSP0 是引导表索引值的最低有效位，BMSP2 是最高有效位。建议在禁用 BMSP 时从禁用 BMSP2 开始。例如，在仅使用 BMSP2（禁用 BMSP1 和 BMSP0）的实例中，只能选择引导表索引 0 和 4。在仅使用 BMSP0 的实例中，可选引导表索引 0 和 1。

7.11.1.2 Configuring Boot Mode Table Options
配置引导模式表选项

This section explains how to configure the boot definition table, BOOTDEF, for the device and the associated boot options. The 64-bit location is located in user-configurable DCSM OTP in the Z1-OTP-BOOTDEF-LOW and Z1-OTP-BOOTDEF-HIGH locations. When debugging, EMU-BOOTDEF-LOW and EMU-BOOTDEF-HIGH are the emulation equivalents of Z1-OTP-BOOTDEF-LOW and Z1-OTP-BOOTDEF-HIGH, and can be programmed to experiment with different boot mode options without writing to OTP. The range of customization to the boot definition table depends on how many boot mode select pins (BMSP) are being used. For example, 0 BMSPs equals to 1 table entry, 1 BMSP equals to 2 table entries, 2 BMSPs equals to 4 table entries, and 3 BMSPs equals to 8 table entries.

本节介绍了如何为器件配置引导定义表 BOOTDEF 以及相关的引导选项。64 位位于用户可配置的

DCSMOTP 中的 Z1-OTP-BOOTDEF-LOW 和 Z1-OTP-BOOTDEF-HIGH 位置。调试时，EMU-BOOTDEF-LOW 和 EMU-BOOTDEF-HIGH 功能类似于 Z1-OTP-BOOTDEF-LOW 和 Z1-OTP-BOOTDEF-HIGH，并且可以进行编程，以便在不写入 OTP 的情况下使用不同的引导模式选项进行测试。引导定义表的自定义范围取决于正在使用多少引导模式选择引脚(BMSP)。例如，0 个 BMSP 等于可用 1 个引导模式、1 个 BMSP 等于可用 2 个引导模式、2 个 BMSP 等于可用 4 个引导模式，而 3 个 BMSP 等于可用 8 个引导模式。

Note

The locations Z2-OTP-BOOTDEF-LOW and Z2-OTP-BOOTDEF-HIGH will be used instead of Z1-OTP-BOOTDEF-LOW and Z1-OTP-BOOTDEF-HIGH locations when Z2-OTP-BOOTPIN-CONFIG is configured. Refer to Configuring Boot Mode Pins for more details on BOOTPIN_CONFIG usage.

配置 Z2-OTP-BOOTPIN-CONFIG 时，将使用 Z2-OTP-BOOTDEF-LOW 和 Z2-OTP-BOOTDEF-HIGH 位置，而不是 Z1-OTP-BOOTDEF-LOW 和 Z1-OTP-BOOTDEF-HIGH 位置。更多有关 BOOTPIN_CONFIG 用法的详细信息，请参阅配置引导模式引脚。

Table 7-11. BOOTDEF Bit Fields

BOOTDEF NAME	BYTE POSITION	NAME	DESCRIPTION
BOOT_DEF0	7:0	BOOT_DEF0 Mode/Options	Set the boot for index 0 of the boot table. Different boot modes and their options can include, for example, a boot mode that uses different GPIOs for a specific bootloader or a different flash entry point address. Any unsupported boot mode will cause the device to either go to wait boot or boot to flash. Refer to GPIO Assignments for valid BOOTDEF values to set in the table.
BOOT_DEF1	15:8	BOOT_DEF1 Mode/Options	Refer to BOOT_DEF0 description
BOOT_DEF2	23:16	BOOT_DEF2 Mode/Options	
BOOT_DEF3	31:24	BOOT_DEF3 Mode/Options	
BOOT_DEF4	39:32	BOOT_DEF4 Mode/Options	
BOOT_DEF5	47:40	BOOT_DEF5 Mode/Options	
BOOT_DEF6	55:48	BOOT_DEF6 Mode/Options	
BOOT_DEF7	63:56	BOOT_DEF7 Mode/Options	

7.11.2 GPIO 分配
GPIO Assignments

This section details the GPIOs and boot option values used for boot mode set in the BOOT_DEF memory location located at Z1-OTP-BOOTDEF-LOW/ Z2-OTP-BOOTDEF-LOW and Z1-OTP-BOOTDEF-HIGH/ Z2-OTP-BOOTDEF-HIGH. Refer to Configuring Boot Mode Table Options on how to configure BOOT_DEF. When selecting a boot mode option, make sure to verify that the necessary pins are available in the pin mux options for the specific device package being used.

本节详细介绍了用于在位于 Z1-OTP-BOOTDEF-LOW/Z2-OTP-BOOTDEF-LOW 和 Z1-OTP-BOOTDEF-HIGH/Z2-OTP-BOOTDEF-HIGH 的BOOT_DEF 存储器位置中设置引导模式的 GPIO 和引导选项值。请参阅配置引导模式表选项，了解如何配置 BOOT_DEF。选择引导模式选项时，请确认所用特定器件封装的引脚多路复用器选项中提供了必要的引脚。

Table 7-12. SCI Boot Options

OPTION	BOOTDEF VALUE	SCITXDA GPIO	SCIRXDA GPIO
0 (default)	0x01	GPIO29	GPIO28
1	0x21	GPIO16	GPIO17
2	0x41	GPIO8	GPIO9
3	0x61	GPIO2	GPIO3
4	0x81	GPIO16	GPIO3

Table 7-13. MCAN Boot Options

OPTION	BOOTDEF VALUE	SCITXDA GPIO	SCIRXDA GPIO
0 (default)	0x01	GPIO29	GPIO28
1	0x21	GPIO16	GPIO17
2	0x41	GPIO8	GPIO9
3	0x61	GPIO2	GPIO3
4	0x81	GPIO16	GPIO3

Table 7-14. DCAN Boot Options

OPTION	BOOTDEF VALUE	SCITXDA GPIO	SCIRXDA GPIO
0 (default)	0x01	GPIO29	GPIO28
1	0x21	GPIO16	GPIO17
2	0x41	GPIO8	GPIO9
3	0x61	GPIO2	GPIO3
4	0x81	GPIO16	GPIO3

Table 7-15. I2C Boot Options

OPTION	BOOTDEF VALUE	SCITXDA GPIO	SCIRXDA GPIO
0 (default)	0x01	GPIO29	GPIO28
1	0x21	GPIO16	GPIO17
2	0x41	GPIO8	GPIO9
3	0x61	GPIO2	GPIO3
4	0x81	GPIO16	GPIO3

Table 7-16. RAM Boot Options

OPTION	BOOTDEF VALUE	SCITXDA GPIO	SCIRXDA GPIO
0 (default)	0x01	GPIO29	GPIO28
1	0x21	GPIO16	GPIO17
2	0x41	GPIO8	GPIO9
3	0x61	GPIO2	GPIO3
4	0x81	GPIO16	GPIO3

Table 7-17. Flash Boot Options

OPTION	BOOTDEF VALUE	SCITXDA GPIO	SCIRXDA GPIO
0 (default)	0x01	GPIO29	GPIO28
1	0x21	GPIO16	GPIO17
2	0x41	GPIO8	GPIO9
3	0x61	GPIO2	GPIO3
4	0x81	GPIO16	GPIO3

Table 7-18. LFU Flash Boot Options

OPTION	BOOTDEF VALUE	SCITXDA GPIO	SCIRXDA GPIO
0 (default)	0x01	GPIO29	GPIO28
1	0x21	GPIO16	GPIO17
2	0x41	GPIO8	GPIO9
3	0x61	GPIO2	GPIO3
4	0x81	GPIO16	GPIO3

Table 7-19. Wait Boot Options

OPTION	BOOTDEF VALUE	SCITXDA GPIO	SCIRXDA GPIO
0 (default)	0x01	GPIO29	GPIO28
1	0x21	GPIO16	GPIO17
2	0x41	GPIO8	GPIO9
3	0x61	GPIO2	GPIO3
4	0x81	GPIO16	GPIO3

Table 7-20. SPI Boot Options

OPTION	BOOTDEF VALUE	SPISIMOA	SPISOMIA	SPICLKA	SPISTEA
0	0x06	GPIO2	GPIO1	GPIO3	GPIO5
1	0x26	GPIO16	GPIO1	GPIO3	GPIO0
2	0x46	GPIO8	GPIO10	GPIO9	GPIO11
3	0x66	GPIO8	GPIO17	GPIO9	GPIO11

Table 7-21. Parallel Boot Options

OPTION	BOOTDEF VALUE	D0-D7 GPIO	DSP CONTROL GPIO	HOST CONTROL GPIO
0 (default)	0x01	D0 - GPIO28	GPIO16	GPIO29
		D1 – GPIO1		
		D2 – GPIO2		
		D3 – GPIO3		
		D4 – GPIO4		
		D5 – GPIO5		
		D6 – GPIO6		
		D7 – GPIO7		
1	0x20	D0 – GPIO0	GPIO16	GPIO11
		D1 – GPIO1		
		D2 – GPIO2		
		D3 – GPIO3		
		D4 – GPIO4		
		D5 – GPIO5		
		D6 – GPIO6		
		D7 – GPIO7		

7.12 Security
安全性

Security features are enforced by the Dual Code Security Module (DCSM). The primary layer of defense is securing the boundary of the chip, which should always be enabled. Additionally, the Dual Zone Security feature is available to support code partitioning.
安全功能由双代码安全模块（DCSM）强制执行。主要的防御层是保护芯片边界，该功能应始终启用。此外，还提供了双区域安全功能以支持代码分区。

7.12.1 Securing the Boundary of the Chip
保护芯片边界

The following two features, along with authentication in the firmware update code, should be used to help to prevent unauthorized code from running on the device.
应使用以下两项功能以及固件更新代码中的身份验证来帮助防止未经授权的代码在器件上运行。

7.12.1.1 JTAGLOCK

Enabling the JTAGLOCK feature in the USER OTP disables JTAG access (for example, debug probe) to resources on the device.
在 USER OTP 中启用 JTAGLOCK 功能会禁用对器件资源的 JTAG 访问（例如调试探针）。

7.12.1.2 Zero-pin Boot
零引脚引导

Enabling the Zero-pin Boot option along with Flash Boot in the USER OTP blocks all pin-based external bootloader options (for example, SCI, CAN, Parallel).
在 USER OTP 中启用零引脚引导选项以及闪存引导会阻止所有基于引脚的外部引导加载程序选项（例如 SCI、CAN、并行）。

7.12.2 Dual-Zone Security
双区域安全

The dual-zone security mechanism offers protection for two zones: Zone 1 (Z1) and Zone 2 (Z2). The security implementation for both zones is identical. Each zone has its own dedicated secure resource (OTP memory and secure ROM) and allocated secure resource (LSx RAM and flash sectors).
双区域安全机制为两个区域提供保护：区域 1 (Z1) 和区域 2 (Z2)。这两个区域的安全实现是相同的。每个区域都有自身的专用安全资源（OTP 存储器和安全 ROM）和分配的安全资源（LSx RAM 和闪存扇区）。

Code Security Module Disclaimer

The Code Security Module (CSM) on this device was designed to password protect the data stored in the associated memory (either ROM or Flash) and is warranted by the manufacturer, in accordance with its standard terms and conditions, to conform to manufacturer's published specifications for the warranty period applicable for this device. The manufacturer does not warrant or represent that the CSM cannot compromised or breached or that the data stored in the associated memory cannot be accessed through other means. Moreover, except as set forth above, manufacturer makes no warranties or representations concerning the CSM or operation of this device, including ant implied

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warranties of merchantability or fitness for a particular purpose.
In no event shall manufacturer be liable for any consequential, special, indirect, incidental, or punitive damages, however caused, arising in any way out of your use of the CSM or this device, whether or not manufacturer has been advised of the possibility of such damages. Excluded damages include, but are not limited to loss of data, loss of goodwill, loss of use or interruption of business or the economic loss.
本器件上包含的代码安全模块（CSM）旨在密码保护存储在相关内存（ROM 或 FLASH）中的数据，并由进芯电子根据其标准条款和条件，符合适用的保修规范。
然而，进芯电子不保证或声明 CSM 不能被破坏或破坏存储在相关内存中的数据不能通过其他方式访问。此外，除上文规定外，进芯电子对 CSM 或本器件的操作不作任何保证或陈述，包括任何对适销性或适合于特定用途的隐含保证。
在任何情况下，进芯电子均不对因您使用 CSM 而产生的任何后果性、特殊性、间接性、附带性或惩罚性损害承担责任，无论您是否告知进芯电子此类损害的可能。损害包括且不限于数据损失、商誉损失、使用损失或业务中断等其他经济损失。

7.13 Watchdog
看门狗

The watchdog module is the same as the one on previous A2000 microcontrollers, but with an optional lower limit on the time between software resets of the counter. This windowed countdown is disabled by default, so the watchdog is fully backward-compatible.
The watchdog generates either a reset or an interrupt. It is clocked from the internal oscillator with a selectable frequency divider.
Figure 7-4 shows the various functional blocks within the watchdog module.

该看门狗模块与之前的 A2000 器件上的模块相同，但针对计数器的软件复位之间的时间提供了一个可选的下限。默认情况下会禁用此窗口倒计时，因此该看门狗完全向后兼容。

看门狗生成复位或中断。看门狗使用可选分频器通过内部振荡器计时。图 7-4 显示了看门狗模块内的各种功能块。

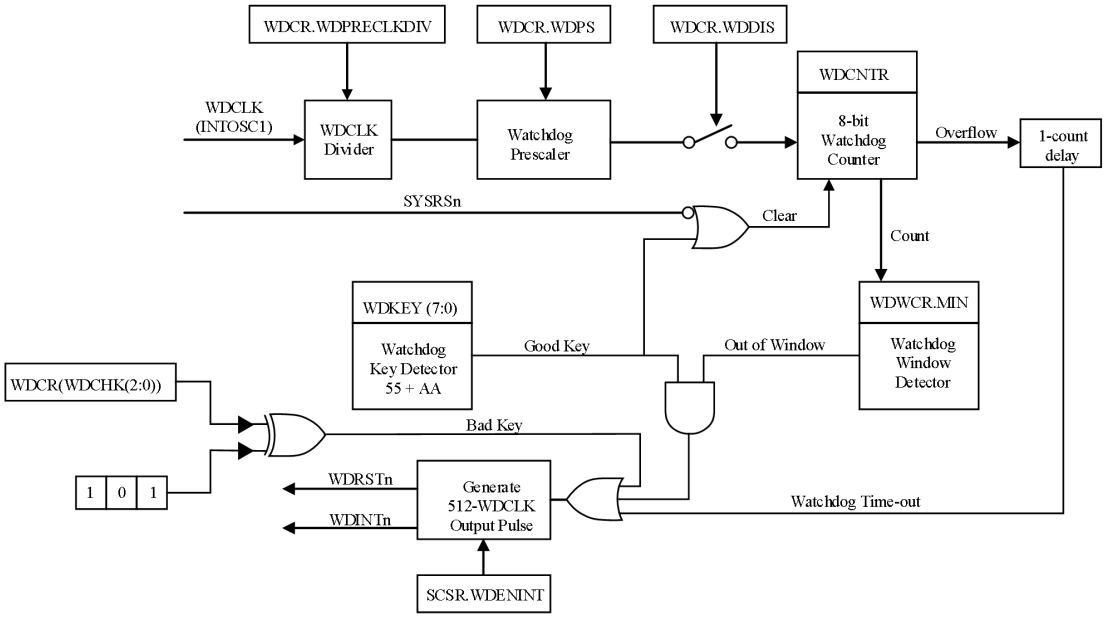


Figure 7-4. Windowed Watchdog

7.14 A2000 Timers
A2000 定时器

CPU-Timers 0, 1, and 2 are identical 32-bit timers with pre-settable periods and with 16-bit clock pre-scaling. The timers have a 32-bit count-down register that generates an interrupt when the counter reaches zero. The counter is decremented at the CPU clock speed divided by the prescale value setting. When the counter reaches zero, it is automatically reloaded with a 32-bit period value.
CPU-Timer 0 is for general use and is connected to the PIE block. CPU-Timer 1 is also for general use and is connected to INT13 of the CPU. CPU-Timer 2 is reserved for ADV-RTOS. It is connected to INT14 of the CPU. If ADV-RTOS is not being used, CPU-Timer 2 is available for general use.
CPU-Timer 2 can be clocked by any one of the following:

- SYSCLK (default)
- Internal zero-pin oscillator 1 (INTOSC1)
- Internal zero-pin oscillator 2 (INTOSC2)
- X1 (XTAL)

CPU 定时器 0，1，和 2 是完全一样的 32 位定时器，具有可预设定周期和 16 位时钟预分频。此定时器具有 32 位递减计数寄存器，该寄存器在计数器达到 0 时生成一中断。计数器以 CPU 时钟速度除以预分频值设置形式递减。当计数器达到 0 时，则自动重新加载 32 位周期值。

CPU 定时器 0 用于普通用途并连接至 PIE 块。CPU 定时器 1 也用于普通用途，并连接至 CPU 的 INT13。CPU 定时器 2 为进芯-RTOS 保留。该定时器连接至 CPU 的 INT14。如果未使用进芯-RTOS，CPU 定时器 2 也可用于普通用途。

CPU 定时器 2 可由下列任一器件计时：

- SYSCLK（默认）
- 内部零引脚振荡器 1 (INTOSC1)
- 内部零引脚振荡器 2 (INTOSC2)
- X1 (XTAL)

7.15 Dual-Clock Comparator (DCC)
双路时钟比较器（DCC）

The DCC module is used for evaluating and monitoring the clock input based on a second clock, which can be a more accurate and reliable version. This instrumentation is used to detect faults in clock source or clock structures, thereby enhancing the system's safety metrics.

DCC 模块用于根据第二个时钟评估和监测时钟输入，第二个时钟可以是更准确和可靠的版本。该仪器用于检测时钟源或时钟结构中的故障，从而增强系统的安全性指标。

7.15.1 Features
特性

The DCC has the following features:

- Allows the application to ensure that a fixed ratio is maintained between frequencies of two clock signals.
- Supports the definition of a programmable tolerance window in terms of the number of reference clock cycles.
- Supports continuous monitoring without requiring application intervention.
- Supports a single-sequence mode for spot measurements.

- Allows the selection of a clock source for each of the counters, resulting in several specific use cases.

DCC 具有以下特性：

- 允许应用确保两个时钟信号的频率之间保持固定的比率。
- 支持根据参考时钟周期数定义可编程公差窗口。
- 支持连续监测，而无需应用干预。
- 支持单序列模式进行点测量。
- 允许为每个计数器选择一个时钟源，从而产生几种特定的用例。

7.15.2 Mapping of DCCx Clock Source Inputs
DCCx 时钟源中断的映射

Table 7-22. DCCx Clock Source0 Table

DCCxCLKSRC0[3:0]	CLOCK NAME
0x0	XTAL/X1
0x1	INTOSC1
0x2	INTOSC2
0x4	TCK
0x5	CPU1.SYSCLK
0x8	AUXCLKIN
0xC	INPUT XBAR (Output16 of input-xbar)
Others	Reserved

Table 7-23. DCCx Clock Source1 Table

	CLOCK NAME
0x0	PLLRAWCLK
0x2	INTOSC1
0x3	INTOSC2
0x6	CPU1.SYSCLK
0x9	Input XBAR (Output15 of the input-xbar)
0xA	AUXCLKIN
0xB	EPWMCLK
0xC	LSPCLK
0xD	ADCCLK
0xE	WDCLK
0xF	CAN0BITCLK
others	Reserved

7.16 Configurable Logic Block (CLB)
可配置逻辑块(CLB)

The A2000 configurable logic block (CLB) is a collection of blocks that can be interconnected using software to implement custom digital logic functions or enhance existing on-chip peripherals. The CLB is able to enhance existing peripherals through a set of crossbar interconnections, which provide a high level of connectivity to existing control peripherals such as enhanced pulse width modulators (ePWM), enhanced capture modules (eCAP), and enhanced quadrature encoder pulse modules (eQEP). The crossbars also allow the CLB to be connected to external GPIO pins. In this way, the CLB can be configured to interact with device peripherals to perform small logical functions such as comparators, or to implement custom serial data exchange protocols. Through the CLB, functions that would otherwise be accomplished using external logic devices can now be implemented inside the MCU.

A2000 可配置逻辑块(CLB)是一组模块的集合，这些模块使用软件进行互连，以实现自定义数字逻辑功能或增强现有的片上外设。CLB 能够通过一组交叉开关互连来增强现有的外设，为现有的控制外设（例如增强型脉宽调制器(ePWM)、增强型采集模块(eCAP)和增强型正交编码器脉冲模块(eQEP)）提供高度连接性。交叉开关还允许将 CLB 连接到外部 GPIO 引脚。通过这种方式，CLB 可以配置为与器件外设交互以执行小型逻辑功能（例如比较器），或实现自定义串行数据交换协议。通过 CLB，原本需要使用外部逻辑器件实现的功能现在可在 MCU 内实现。

The CLB module and its interconnections are shown in Figure 7-5.

CLB 模块及其互连如图 7-5 所示。

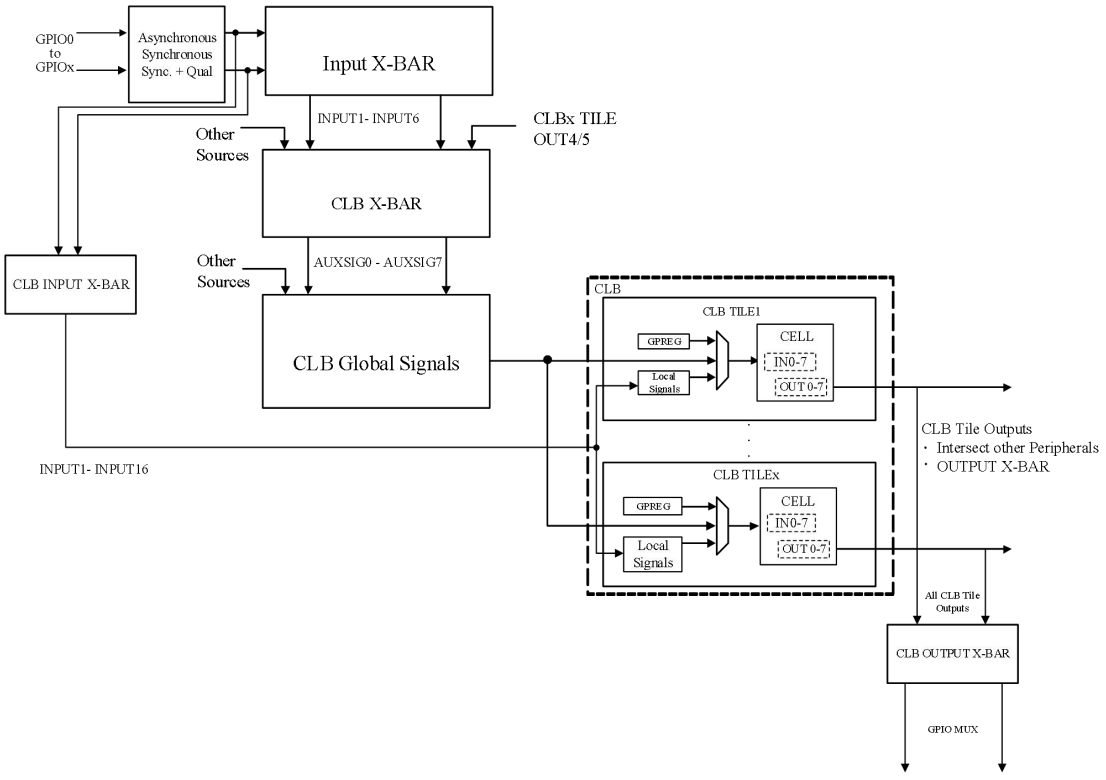


Figure 7-5. GPIO to CLB Tile Connections

8 Device and Documentation Support
器件和文档支持

8.1 Device Nomenclature
器件命名规则

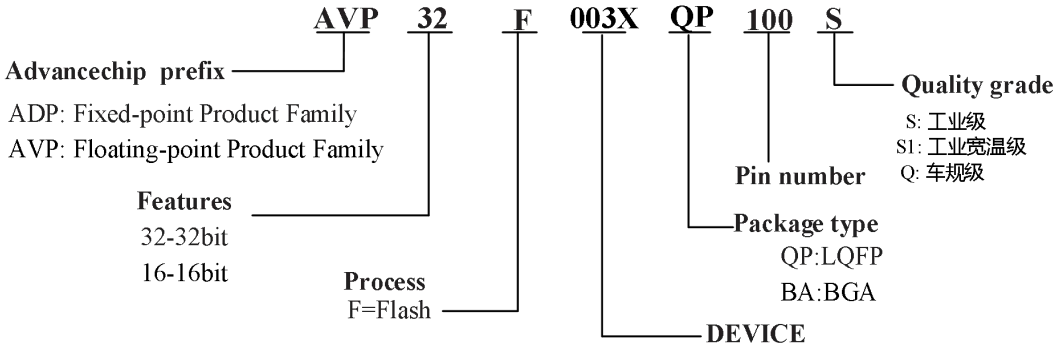


Figure 8-1 Device Nomenclature

8.2 Device Identification
标识

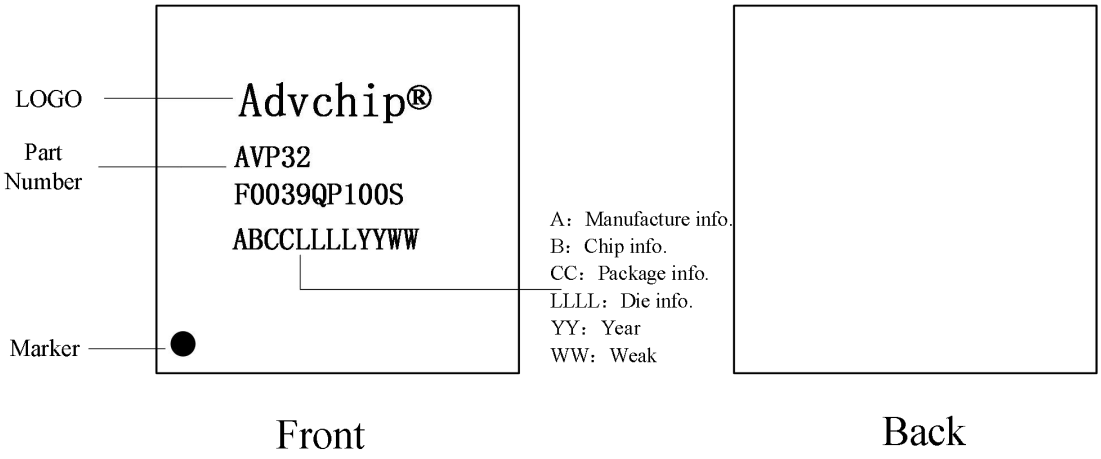


Table 8-2 Device Identification

8.3 Documentation Support
文档支持

The current documentation that describes the processor, related peripherals, and other technical collateral follows.

Errata

AVP32F003X Silicon Errata describes known advisories on silicon and provides workarounds.

Technical Reference Manual

AVP32F003X Technical Reference Manual details the integration, the environment, the functional description,

and the programming models for each peripheral and subsystem in the AVP32F003X DSP.

下面列出了介绍处理器、相关外设以及其他配套技术资料的最新文档。

勘误

《AVP32F003X 器件勘误表》介绍了有关器件的已知勘误项，并给出了应对措施。

技术参考手册

《AVP32F003X 技术参考手册》详述了 AVP32F003X DSP 中每个外设和子系统的集成、环境、功能说明和编程模型。

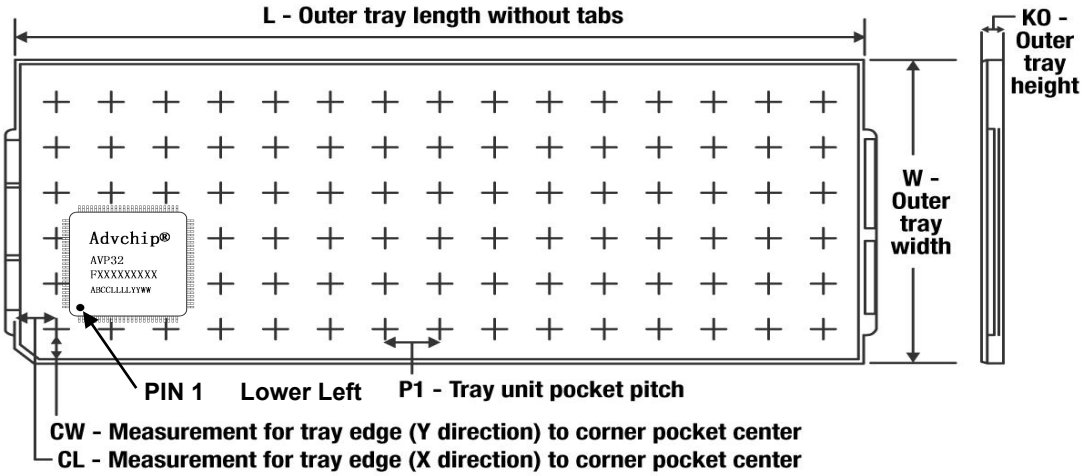
9 Mechanical, Packaging, and Orderable Information
机械、包装及可订购信息

9.1 Orderable Information
可订购信息

9.1.1 Packaging Information
包装信息

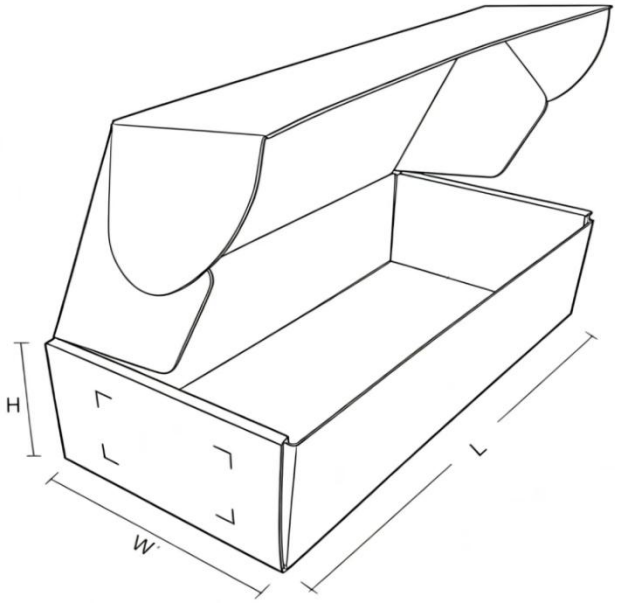
可订购型号 Orderable Device	状态 Status	封装形式 Package Type	引脚数 Pins	包装数量 Package Qty	环保标志 Eco Plan	湿敏等级及回流焊温度 MSL Peak Temp	工作温度 Op Temp (°C)	产品丝印 Device Marking
AVP32F0039QP100S	ACTIVE	LQFP	100	900	ROHS&REACH	Level-3-260C-168 HR	–40°C to 125°C	AVP32F0039QP100S
AVP32F0039QP80S	ACTIVE	LQFP	80	1190	ROHS&REACH	Level-3-260C-168 HR	–40°C to 125°C	AVP32F0039QP80S
AVP32F0039QP64S	ACTIVE	LQFP	64	1600	ROHS&REACH	Level-3-260C-168 HR	–40°C to 125°C	AVP32F0039QP64S
AVP32F0039QP48S	ACTIVE	LQFP	48	2500	ROHS&REACH	Level-3-260C-168 HR	–40°C to 125°C	AVP32F0039QP48S
AVP32F0033QP80S	ACTIVE	LQFP	80	1190	ROHS&REACH	Level-3-260C-168 HR	–40°C to 125°C	AVP32F0033QP80S
AVP32F0039QP100Q	ACTIVE	LQFP	100	900	ROHS&REACH	Level-3-260C-168 HR	–40°C to 125°C	AVP32F0039QP100Q
AVP32F0039QP80Q	ACTIVE	LQFP	80	1190	ROHS&REACH	Level-3-260C-168 HR	–40°C to 125°C	AVP32F0039QP80Q
AVP32F00157PQP80Q	ACTIVE	LQFP	80	1190	ROHS&REACH	Level-3-260C-168 HR	–40°C to 125°C	AVP32F00157PQP80Q
AVP32F0036QP64Q	ACTIVE	LQFP	64	1600	ROHS&REACH	Level-3-260C-168 HR	–40°C to 125°C	AVP32F0036QP64Q
AVP32F0039QP48Q	ACTIVE	LQFP	48	2500	ROHS&REACH	Level-3-260C-168 HR	–40°C to 125°C	AVP32F0039QP48Q
AVP32F0033QP64Q	ACTIVE	LQFP	64	1600	ROHS&REACH	Level-3-260C-168 HR	–40°C to 125°C	AVP32F0033QP64Q

9.1.2Tray Information
托盘信息



Device	Package Type	Package	Pins	SPQ	MPQ	Max temperature (℃)	L (mm)	W (mm)	K0 (mm)	P1 (mm)	CL (mm)	CW (mm)
AVP32F0039QP80S AVP32F0033QP80S AVP32F0039QP80Q AVP32F00157PQP80Q	LQFP	12*12	80	119 （7*17）	1190 Tray （10+1）	150	315	135.9	7.62	17.9	14.3	13.95
AVP32F0039QP64S AVP32F0036QP64Q AVP32F0033QP64Q	LQFP	10*10	64	160 （8*20）	1600 Tray （10+1）	150	315	135.9	7.62	15.2	13.1	13
AVP32F0039QP48S AVP32F0039QP48Q	LQFP	7*7	48	250 （10*25）	2500 Tray （10+1）	150	315	135.9	7.62	12.2	11.1	11.25
AVP32F0039QP100S AVP32F0039QP100Q	LQFP	14*14	100	90 (6*15)	900 Tray （10+1）	150	315	135.9	7.62	20.3	15.4	15.45

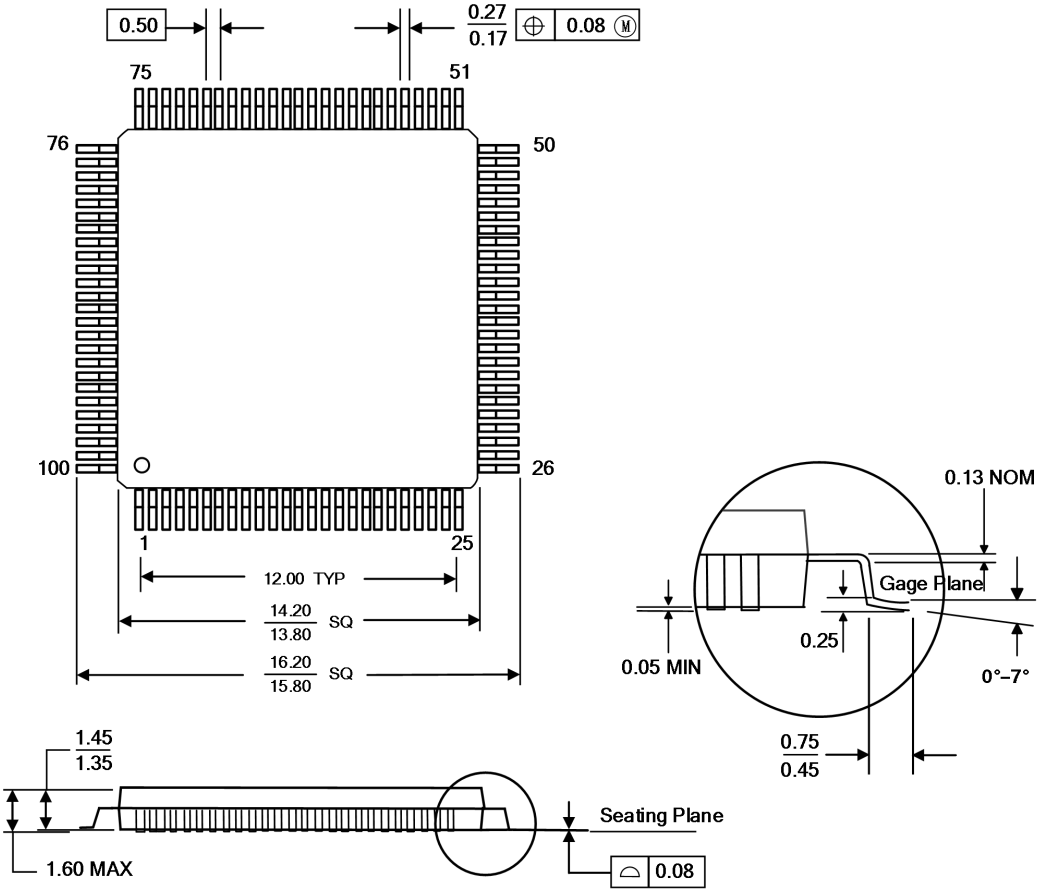
9.1.3BOX Dimensions
包装盒规格



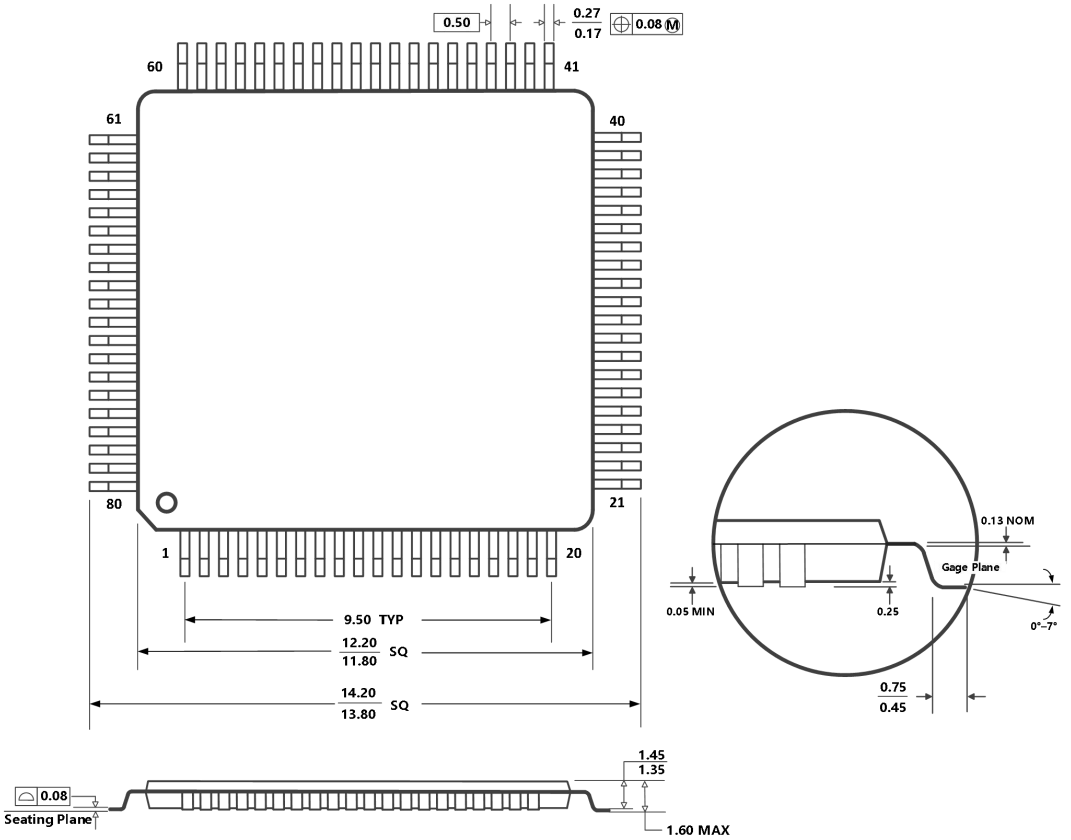
Device	Package Type	Package Drawing	Pins	SPQ	L (mm)	W (mm)	H (mm)
AVP32F0039QP100S	LQFP	Tray	100	900	370	160	90
AVP32F0039QP80S	LQFP	Tray	80	1190	370	160	90
AVP32F0033QP80S	LQFP	Tray	80	1190	370	160	90
AVP32F0039QP64S	LQFP	Tray	64	1600	370	160	90
AVP32F0039QP48S	LQFP	Tray	48	2500	370	160	90
AVP32F0039QP100Q	LQFP	Tray	100	900	370	160	90
AVP32F0039QP80Q	LQFP	Tray	80	1190	370	160	90
AVP32F00157PQP80Q	LQFP	Tray	80	1190	370	160	90
AVP32F0036QP64Q	LQFP	Tray	64	1600	370	160	90
AVP32F0033OP64Q	LQFP	Tray	64	1600	370	160	90
AVP32F0039QP48Q	LQFP	Tray	48	2500	370	160	90

9.2 Packaging Information
封装信息

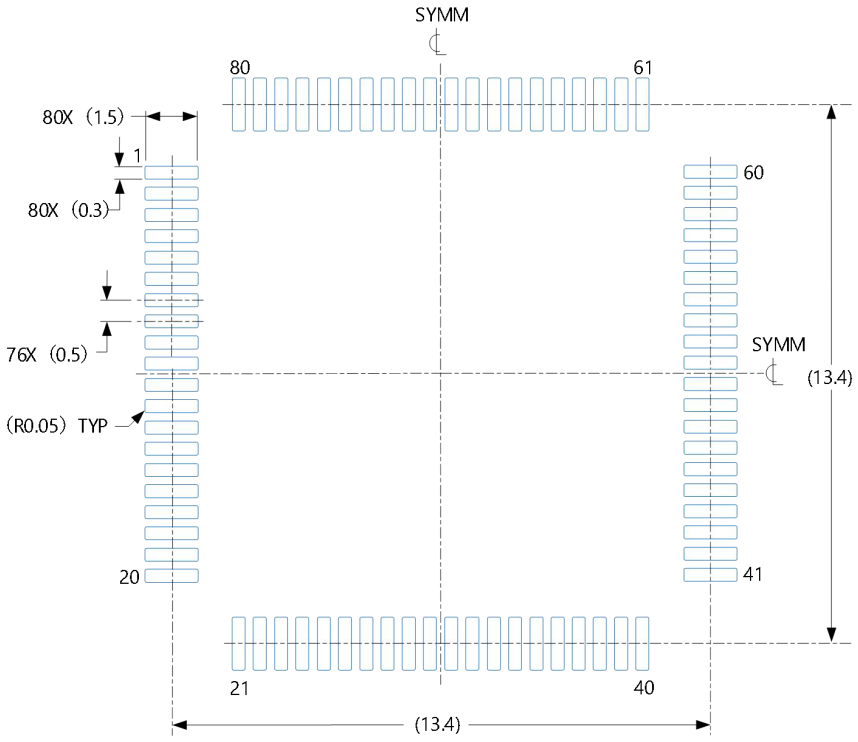
LQFP100 Unit: mm



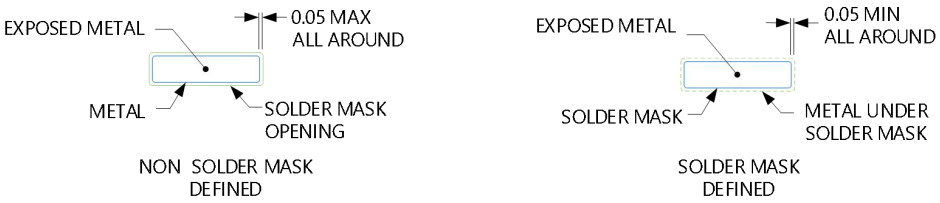
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LQFP80 Unit: mm

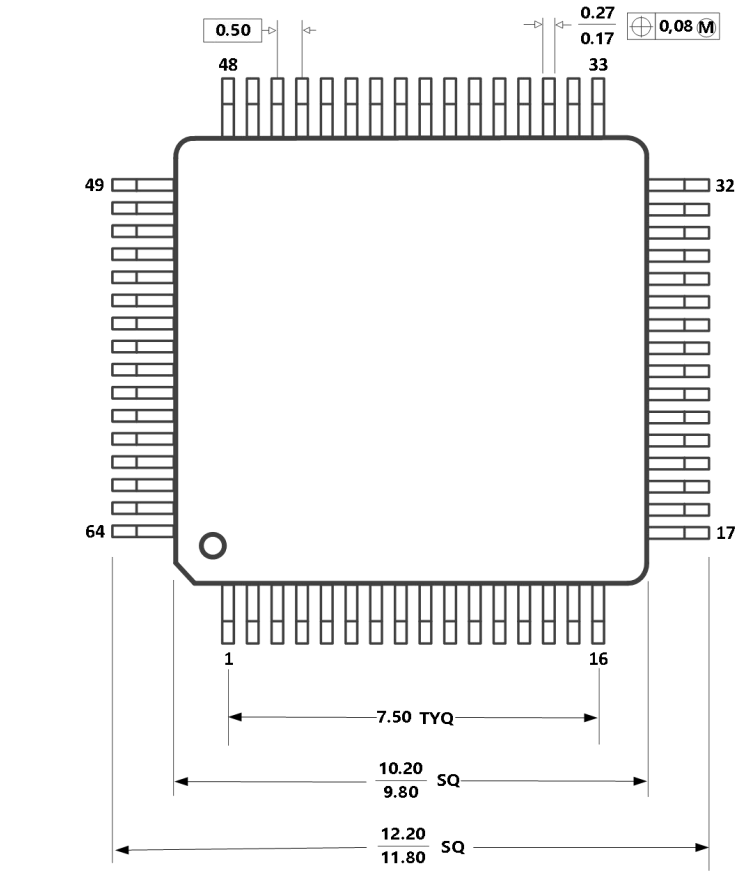


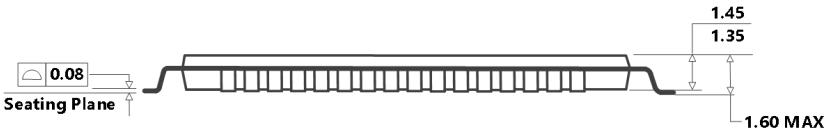
LAND PATTERN EXAMPLE
EXPOSED METAL SHOWN
SCALE:6X

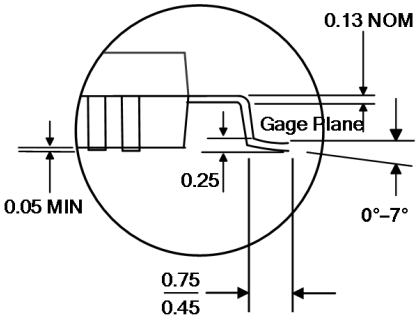


SOLDER MASK DETAILS

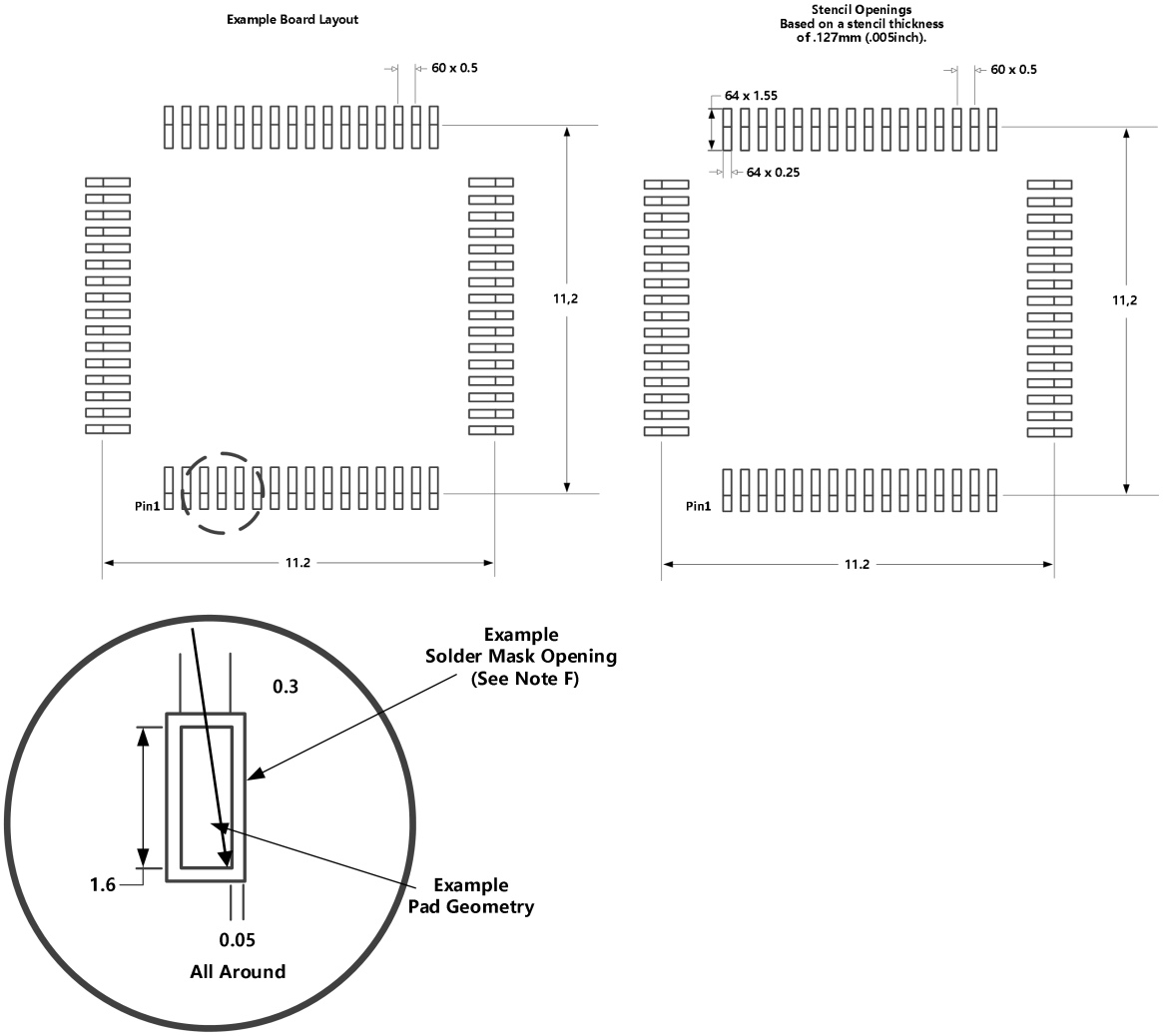
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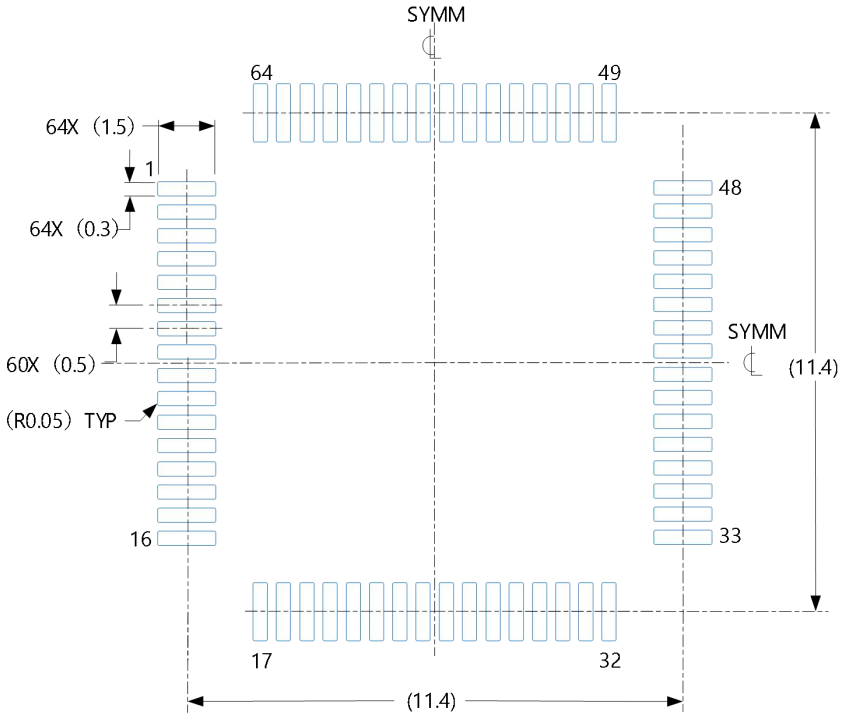




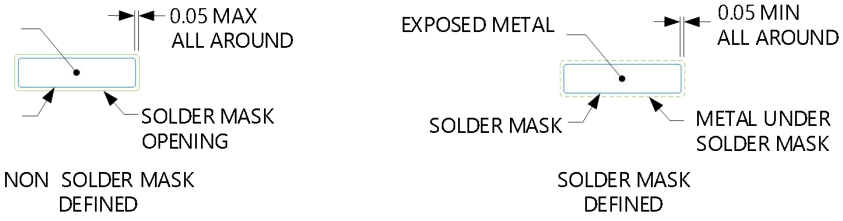
LQFP64 Unit: mm



LQFP64 Unit: mm

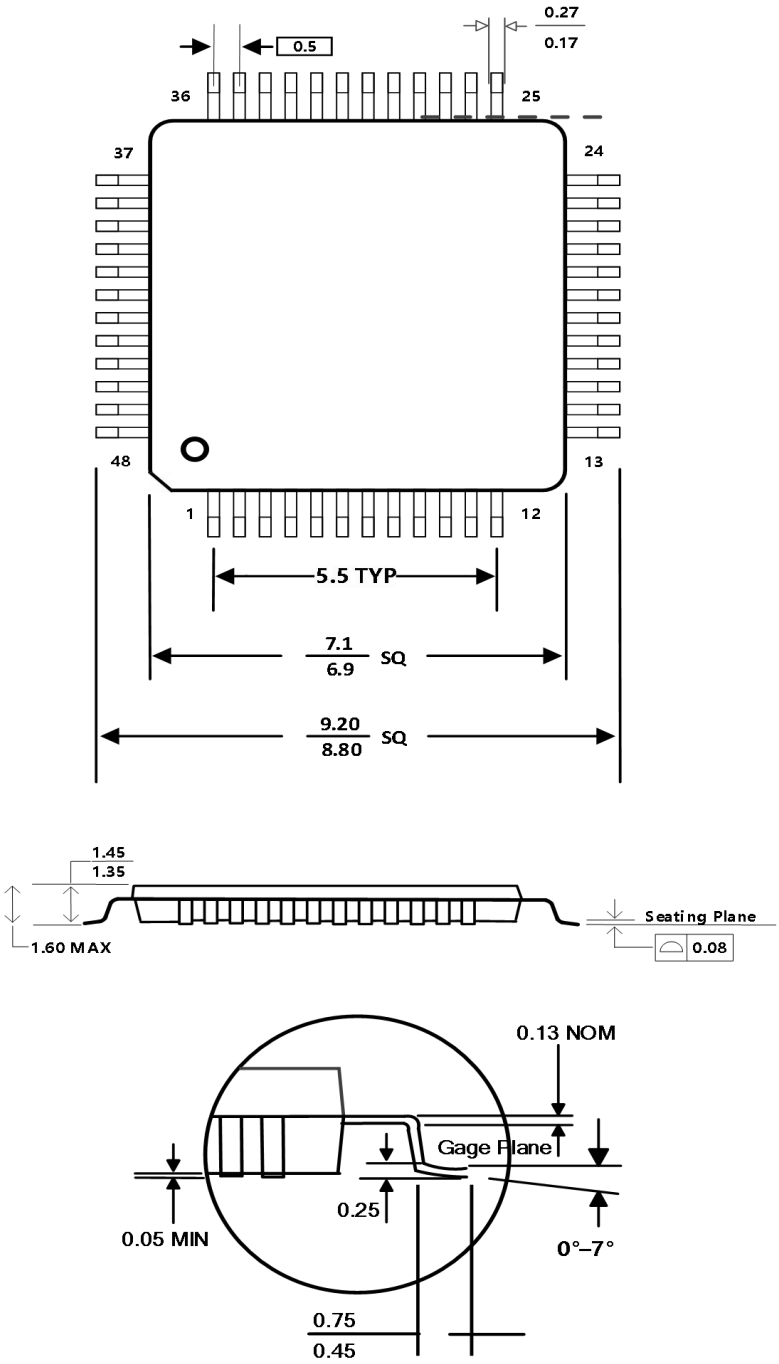


LAND PATTERN EXAMPLE
EXPOSED METAL SHOWN
SCALE:8X

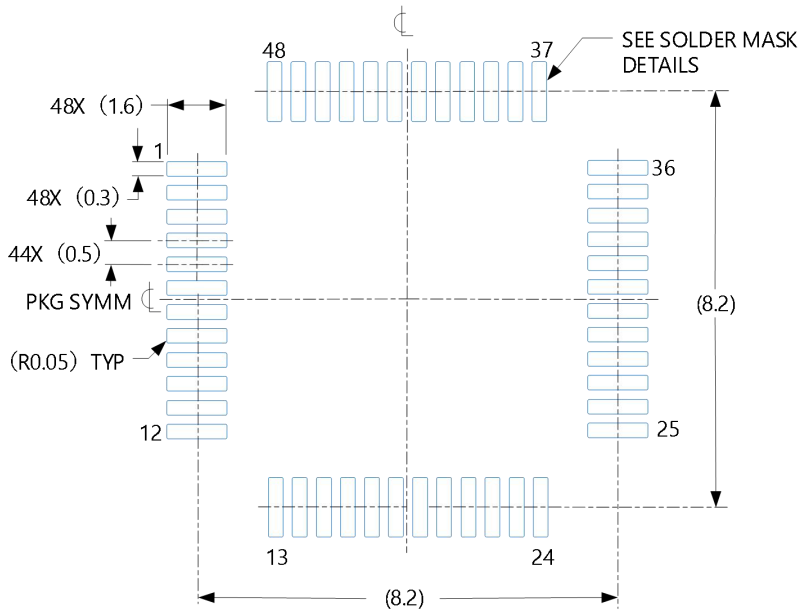


SOLDER MASK DETAILS

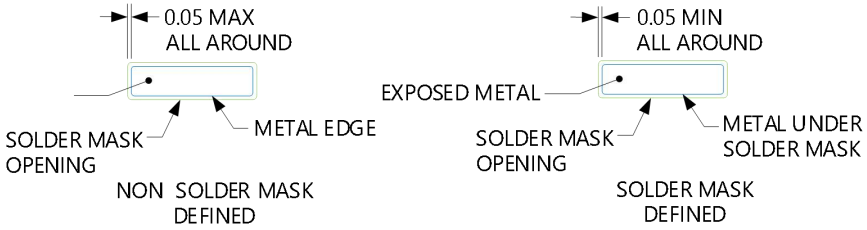
LQFP48 Unit: mm



LQFP48 Unit: mm



LAND PATTERN EXAMPLE
EXPOSED METAL SHOWN
SCALE 10.000



SOLDER MASK DETAILS

10 Applications, Implementation, and Layout
应用、实现及布局

10.1 Applications and Implementation
应用和实施

Note

Information in the following applications sections is not part of the ADVCHIP component specification, and ADVCHIP does not warrant its accuracy or completeness. ADVCHIP's customers are responsible for determining suitability of components for their purposes, as well as validating and testing their design implementation to confirm system functionality.

以下应用部分中的信息不属于 ADVCHIP 器件规格的范围，ADVCHIP 不担保其准确性和完整性。ADVCHIP 的客户应负责确定器件是否适用于其应用。客户应验证并测试其设计，以确保系统功能。

10.2 Key Device Features
器件主要特性

Table 10-1 Key Device Features
表 10-1：关键器件特性

模块 MODULE	特性 FEATURE	系统效果 SYSTEM BENEFIT
PROCESSING 处理器		
Real-time control CPUs 实时控制处理器	Up to 120 MIPS A2000: 120 MIPS CLA: 120MIPs Flash: Up to 384KB RAM : Up to 69KB 32-bit Floating-Point Unit (FPU32) Trigonometric Math Unit (TMU) CRC engine and instructions (VCRC) 高达 120MIPS A2000：120MIPS CLA：120MIPs 闪存：高达 384KB RAM：高达 69KB 32 位浮点单元 (FPU32) 三角法数学单元 (TMU) CRC 引擎和指令 (VCRC)	Provides 120 MHz of signal-processing performance for floating- or fixed-point code running from either on-chip flash or SRAM. **CLA:** 32-bit floating point Control Law Accelerator, parallel execution to A2000 CPU **FPU32:** Native hardware support for IEEE-754 single-precision floating- point operations **TMU:** Accelerators used to speed up execution of trigonometric and arithmetic operations for faster computation (such as PLL and DQ transform) optimized for control applications. Helps in achieving faster control loops, resulting in higher efficiency and better component sizing. Special instructions to support nonlinear PID control algorithms **VCRC:** Provides a straightforward method for verifying data integrity over large data blocks, communication packets, or code sections. 为从片载闪存或 SRAM 运行的浮点或定点代码提供 120MHz 的信号处理性能。 **CLA：** 32 位浮点控制律加速器，与 A2000 CPU 并行执行 **FPU32：** 原生硬件支持 IEEE-754 单精度浮点运算 **TMU：** 使用加速器加快三角函数和算术运算执行速度，从而提高控制应用的计算速度（例如 PLL 和DQ 变换）。有助于实现更快的控制环路从而提高效率和优化元件尺寸。 特殊指令支持非线性 PID 控制算法 **VCRC：** 提供了一种简单的方法来验证大型数据块、通信数据包

模块 MODULE	特性 FEATURE	系统效果 SYSTEM BENEFIT
		或代码段上的数据完整性。
SENSING 传感器		
Analog-to-Digital Converter (ADC) (12-bit) 12 位模数转换器	Up to 3 ADC modules 4 MSPS Up to 23 channels 多达 3 个 ADC 模块 4MSPS 高达 23 通道	ADC provides precise and concurrent sampling of all three-phase currents and DC bus with zero jitter. ADC post-processing – On-chip hardware reduces ADC ISR complexity and shortens current loop cycles. More ADCs help in multiphase applications. Provide better effective MSPS (oversampling) and typical ENOB for better control-loop performance. ADC 对全部三相电流和直流总线进行精准并行采样，且具有零抖动。 ADC 后处理——片上硬件将降低 ADC ISR 复杂度并缩短电流环路周期。 ADC 数量的增加适用于多相应用。提供更高的有效 MSPS（过采样）和典型 ENOB 以实现更好的控制环路性能。
Comparator Subsystem (CMPSS) 比较器子系统	**CMPSS** 4 windowed comparator Dual 12-bit DACs DAC ramp generation Digital filters 60-ns detection to trip time Slope compensation 4 个窗口比较器 两个 12 位 DAC DAC 斜坡生成数字滤波器 60ns 触发检测时间 斜率补偿	**System protection without false alarms:** Comparator Subsystem (CMPSS) modules are useful for applications such as peak-current mode control, switched-mode power, power factor correction, and voltage trip monitoring. PWM trip-triggering and removal of unwanted noise are easy with blanking window and filtering features provided with the analog comparator subsystems. Provides better control accuracy. No need for further CPU configuration to control the PWM with the comparator and 12-bit DAC (CMPSS) Enables protection(CMPSS) and control(ADC) using the same pin. 系统保护无误报： 比较器子系统 (CMPSS) 模块适用于峰值电流模式控制、开关模式电源、功率因数校正和电压跳闸监控等应用。 借助模拟比较器子系统提供的消隐窗口和滤波功能，PWM 跳闸触发和消除不必要噪声变得非常容易。 提供更出色的控制精度。无需进一步的 CPU 配置即可通过比较器和 12 位 DAC (CMPSS) 控制 PWM。 使用同一引脚实现保护 (CMPSS) 和控制 (ADC)。
Enhanced Quadrature Encoder Pulse (eQEP) 增强型正交编码脉冲	2 eQEP modules 2 个eQEP 模块	Used for direct interface with a linear or rotary incremental encoder to get position, direction, and speed information from a rotating machine used in a high-performance motion and position-control system. Support CW/CCW encoding. Also can be used in other applications to count input pulses from an external device (such as a sensor). 用于与线性或旋转增量编码器进行直接连接，以便获得高性能运动和位置控制系统中使用的旋转机器的位置、方向和速度信息。 支持 CW/CCW 编码。另外，也可以在其他应用中用于对来自外部器件（例如传感器）的输入脉冲进行计数。

模块 MODULE	特性 FEATURE	系统效果 SYSTEM BENEFIT
Enhanced Capture (eCAP) 增强型捕获	3 eCAP modules Measures elapsed time between events (up to 4 time-stamped events). Connects to any GPIO through the input X-BAR. When not used in capture mode, the eCAP module can be configured as a single- channel PWM output (APWM). 3 个 eCAP 模块 测量事件之间经过的时间（最多 4 个带时间戳的事件）。 通过输入 X-BAR 连接到任何 GPIO。 当未用于采集模式时，eCAP 模块可配置为单通道 PWM 输出 (APWM)。	**Applications for eCAP include:** Speed measurements of rotating machinery (for example, toothed sprockets sensed through Hall sensors) Elapsed time measurements between position sensor pulses Period and duty cycle measurements of pulse train signals Decoding current or voltage amplitude derived from duty-cycle encoded current/voltage sensors eCAP 的应用包含： 旋转机械的速度测量（例如，通过霍尔传感器感应齿状链轮）位置传感器脉冲之间的持续时间测量。 脉冲序列信号的周期和占空比测量。 对来自占空比编码电流/电压传感器的电流或电压幅度进行解码。
ACTUATION 执行模块		
Enhanced Pulse Width Modulation (ePWM) 增强型脉冲宽度调制器	Up to 16 ePWM channels Ability to generate high-side/low-side PWMs with deadband Supports Valley switching (ability to switch PWM output at valley point) and features like blanking window 多达 16 个 ePWM 通道 能够生成具有死区的高侧/低侧 PWM 支持谷底开关（能够在谷点切换 PWM 输出） 以及消隐窗口等特性	Flexible PWM waveform generation with best power topology coverage. Shadowed Dead band itself and shadowed action qualifier enable adaptive PWM generation and protection for improved control accuracy and reduced power loss. Enables improvement in Power Factor (PF) and Total Harmonic Distortion (THD), which is especially relevant in Power Factor Correction (PFC) applications. Improves light load efficiency. 灵活的 PWM 波形生成功能，具有出色的电源拓扑覆盖范围。 影子化死区本身和影子化动作限定器可实现自适应 PWM 生成和保护，从而提高控制精度并降低功率损耗。 可改善功率因数 (PF) 和总谐波失真 (THD)，这在功率因数校正 (PFC) 应用中尤为重要。可提高轻载效率。
	One-shot and global reload feature 一次性和全局重新加载功能	Critical for variable-frequency and multiphase DC-DC applications and helps in attaining high-frequency control loops (>2 MHz). Enables control of interleaved LLC topologies at high frequencies 对于变频和多相直流/直流应用至关重要，有助于实现高频控制环路（>2MHz）。 能够在高频下控制交错式 LLC 拓扑。
	Independent PWM action on a Cycle-by- Cycle (CBC) trip event and an One-Shot Trip (OST) trip event 针对逐周期 (CBC) 跳闸事件和一次性跳闸 (OST) 触发事件进行独立 PWM 操作	Provides cycle-by-cycle protection and complete shutoff of PWM under fault condition. Helps implement multiphase PFC or DC-DC control. 提供逐周期保护并在故障条件下完全关闭 PWM。有助于实现多相 PFC 或直流/直流控制。
	Load on SYNC (support for shadow-to-active load on a SYNC event) 在 SYNC 时加载（支持在发生 SYNC 事件时的“影子到活动”加载）	Enables variable-frequency applications (allows LLC control in power conversion). 支持变频应用（允许在功率转换中进行 LLC 控制）。
	Ability to shut down the PWMs without software intervention (no ISR latency) 无需软件干预即可关闭 PWM (无ISR 延迟)	Fast protection under fault condition 在出现故障时提供快速保护

模块 MODULE	特性 FEATURE	系统效果 SYSTEM BENEFIT
	Delayed Trip Functionality 延迟跳闸功能	Helps implement the deadband with Peak Current Mode Control (PCMC) Phase-Shifted Full Bride (PSFB) DC-DC easily without occupying much CPU resources (even on trigger events based on comparator, trip, or sync-in events). 有助于利用峰值电流模式控制 (PCMC) 相移全桥 (PSFB) 直流/直流转换器轻松实现死区，无需占用大量 CPU 资源（即使发生基于比较器、跳闸或同步输入事件的触发事件时也是如此）。
	Dead band Generator (DB) submodule 死区发生器 (DB) 子模块	Prevents simultaneous ON conditions of High and Low side gates by adding programmable delay to rising (RED) and falling (FED) PWM signal edges. 通过向 PWM 信号上升沿 (RED) 和下降沿 (FED) 添加可编程延迟，防止高侧和低侧栅极同时导通。
	Flexible PWM Phase Relationships and Timer Synchronization 灵活的 PWM 相位关系和计时器同步	Each ePWM module can be synchronized with other ePWM modules or other peripherals. Keeps PWM edges perfectly in synchronization with certain events. Supports flexible ADC scheduling with specific sampling window in synchronization with power device switching. 每个 ePWM 模块都能与其他 ePWM 模块或其他外设同步。可使 PWM 边沿与特定事件完全保持同步。 支持采用特定采样窗口实现灵活的 ADC 调度，与功率器件切换保持同步。
High-Resolution Pulse Width Modulation (HRPWM) 高分辨率脉冲宽度调制器	8 channels with high-resolution capability (150 ps) Provides 150-ps steps for duty cycle, period, Dead band, and phase offsets for 99% greater precision 8 个具有高分辨率功能 (150ps) 的通道 为占空比、周期、死区以及相位偏移提供 150ps 的步长，精度提高 99%	Beneficial for accurate control and enables better-performance high-frequency power conversion. Achieves cleaner waveforms and avoids oscillations/limit cycle at output. 有利于精确控制并实现性能更佳的高频功率转换。实现更干净的波形并避免输出端产生振荡/限制周期。
CONNECTIVITY 连接		
Serial Peripheral Interface (SPI) 串行外设接口	2 high-speed SPI port 两个高速 SPI 端口	Supports 30 MHz 支持 30MHz
Serial Communication Interface (SCI) 串行通信接口	2 SCI (UART) modules 两个 UART 模块	Interfaces with controllers 控制器的内部连接
Local Interconnect Network (LIN) 本地内联网络	2 LIN 两个 LIN	Provides a low-cost solution where the bandwidth and fault tolerance of a Controller Area Network (CAN) are not required Can also be used as simple UART 提供一种低成本解决方案，无需控制器局域网 (CAN) 的带宽和容错能力也可用作简单 UART
Controller Area Network (CAN/ DCAN) 控制域网络	1 DCAN module 一个 DCAN 模块	Provides compatibility with classic CAN modules 能够兼容经典 CAN 模块
Controller Area Network (CAN FD/ MCAN) 控制域网络	1 CAN FD/MCAN module 一个 CANFD/MCAN 模块	CAN FD (flexible data-rate) is an enhancement to the classic CAN protocol. CAN FD facilitates dynamic switching to higher bit rates (>1 Mbps) for the data segment and allows for up to 64 bytes compared to 8 bytes in classic CAN. This is done without having to change the physical layer. This results in a bandwidth gain over traditional CAN. Systems using CAN-FD benefit from faster in-the-field flash updates.

模块 MODULE	特性 FEATURE	系统效果 SYSTEM BENEFIT
		CAN FD（灵活数据速率）是经典 CAN 协议的增强版本。CAN FD 有助于动态切换到数据段的更高比特率（>1Mbps），并允许最多 64 个字节，而经典 CAN 只允许 8 个字节。这是在无需更改物理层的情况下实现的。因此与传统 CAN 相比，该协议具有更高带宽增益。使用 CAN-FD 的系统受益于更快的现场闪存更新。
Inter-Integrated Circuit (I2C) 内部集成电路	2 I2C modules 两个 I2C 模块	Interfaces with external EEPROMs, sensors, or controllers 与外部 EEPROM、传感器或控制器连接
Power- Management Bus (PMBus) 电源管理总线	1 PMBus module Compliance with the SMI Forum PMBus Specification (Part I v1.0 and Part II v1.1) 一个 PMBUS 模块，兼容 SMI	Seamless HW-based host communication 基于硬件的无缝主机通信
OTHER SYSTEM FEATURES 其它系统特性		
Security enhancers 安全扩展	Dual-zone Code Security Module (DCSM) Watchdog Write Protection on Register Missing Clock Detection Logic (MCD) Error Correction Code (ECC) and parity Dual-Clock Comparator (DCC) 双区域代码安全模块（DCSM） 看门狗 寄存器受写保护 丢失时钟检测逻辑（MCD） 纠错码（ECC）和奇偶校验 双时钟比较器（DCC）	**DCSM:** Prevents duplication and reverse-engineering of proprietary code **Watchdog:** Generates reset if CPU gets stuck in endless loops of execution **Write Protection on Registers:** LOCK protection on system configuration registers Protection against spurious CPU writes **MCD:** Automatic clock failure detection **ECC and parity:** Single-bit error correction and double-bit error detection **DCC:** Used to detect faults in clock source **DCSM：**防止对专有代码进行复制和逆向工程 **看门狗：**如果 CPU 陷入无休止的执行循环，则会产生复位 **寄存器受写保护：** 针对系统配置寄存器进行锁定保护防止虚假 CPU 写入 **MCD：**自动时钟故障检测 **ECC 和奇偶校验：**single-bit 纠错和 double-bit 错误检测 **DCC：**用于检测时钟源故障
Crossbars (XBARs) 交叉开关	Provides flexibility to connect device inputs, outputs, and internal resources in a variety of configurations. • Input X-BAR • Output X-BAR • ePWM X-BAR 可灵活连接各种配置中的器件输入、输出和内 部资源。 •输入 X-BAR •输出 X-BAR •ePWM X-BAR	**Enhances hardware design versatility:** **Input X-BAR:** Routes signals from any GPIO to multiple IP blocks within the chip **Output XBAR:** Routes internal signals onto designated GPIO pins **ePWM X-BAR:** Routes internal signals from various IP blocks to EPWM **增强硬件设计的通用性：** **输入X-BAR：**将信号从任一GPIO 路由到芯片内的多个 IP 块 **输出 XBAR：**将内部信号路由到指定的 GPIO 引脚上 **ePWM X-BAR：**将内部信号从多种 IP 块路由到 ePWM

10.3 Application Information
应用信息

10.3.1 Typical Application
典型应用

The Typical Applications section details some applications of this device.

“典型应用”部分详细介绍了该设备的一些应用。

10.3.1.1 Automotive Pump
汽车泵

Fluid or fuel control pumps are typically used in automotive engine management systems based on the type of powertrain required. Depending on the type of system and load, these actuators are in open loop or closed loop, complete with precise control.

All vehicles—internal combustion engine, electric, or hybrid (ICE/EV/HEV)—need various types of pumps (such as fuel pumps; coolant or water pumps; and oil pumps). Although the purpose of each pump is different, the function of the pump is the same: to move fluid, fuel, or oil from one place to another. In the example of a fuel pump, the pump transfers fuel from the fuel tank to the engine chamber for the engine to use. Depending on the function, pumps can be variable-speed pumps or fixed-speed pumps.

The vehicle’s battery provides the current required to run the fuel pump. An electronic control unit (ECU) regulates the output pressure and volume of the gasoline, as well as meters the incoming fuel from the tank. The ECU assists the car in conserving fuel, resulting in improved economy and power.

流体或燃油控制泵通常用于基于所需动力系统类型的汽车发动机管理系统中。根据系统类型和负载情况，这些执行器采用开环或闭环控制，并具备精确控制功能。

所有车辆——内燃机汽车、电动汽车或混合动力汽车（ICE/EV/HEV）——都需要各种类型的泵（如燃油泵、冷却液泵或水泵以及机油泵）。尽管每种泵的用途不同，但泵的功能是相同的：将流体、燃油或机油从一个地方输送到另一个地方。以燃油泵为例，该泵将燃油从油箱输送到发动机燃烧室供发动机使用。根据功能，泵可以是变速泵或定速泵。

车辆的电池为燃油泵的运行提供所需电流。电子控制单元（ECU）调节汽油的输出压力和流量，并计量从油箱流入的燃油。ECU 有助于汽车节省燃油，从而提高经济性和动力性能。

10.3.1.2 Automotive HVAC Compressor
汽车暖通空调压缩机

In a vehicle, the purpose of a conventional HVAC compressor is to cool the cabin. In hybrid and electric vehicles (HEVs and EVs), the compressor system not only cools the cabin but also the battery which powers the vehicle.

In HEV/EVs, the sizing or the absence of a combustion engine requires the introduction of two additional components that play a key role in the HVAC system:

- A brushless DC (BLDC) motor is a type of DC motor that rotates the AC compressor, instead of the

engine.

- A positive temperature coefficient (PTC) heater or alternatively, a heat pump, heats the coolant, rather than the engine.

Automotive HVAC compressor module designs require:

- Minimized number of isolated components.
- Reduced EMI to optimize system performance.
- Comprehensive diagnostics for fault identification.
- High efficiency and sensorless torque control even at low speeds.

在车辆中，传统暖通空调（HVAC）压缩机的作用是为车厢制冷。在混合动力汽车和电动汽车（HEV 和 EV）中，压缩机系统不仅要为车厢制冷，还要为为车辆提供动力的电池制冷。

在混合动力汽车和电动汽车中，由于没有内燃机或内燃机尺寸的原因，需要引入另外两个在暖通空调系统中起关键作用的部件：

- 无刷直流（BLDC）电机是一种直流电机，它替代发动机驱动交流压缩机旋转。
- 正温度系数（PTC）加热器，或者热泵，用于加热冷却液，而非依靠发动机。

汽车暖通空调压缩机模块设计要求如下：

- 减少隔离部件的数量。
- 降低电磁干扰（EMI）以优化系统性能。
- 具备全面的诊断功能以识别故障。
- 即使在低速运行时也能实现高效且无传感器的扭矩控制。

10.3.1.3 On-Board Charger (OBC)
车载充电机（OBC）

In the OBC and High-Voltage DC-DC charger (HV DCDC) markets, the modular-based design and the combo-box-based design are the two primary architectures adopted. The modular approach provides flexibility in manufacturing and after-service; and the combo-box approach seeks to integrate multiple functions into one enclosure for compactness. F28003x targets the modular-based control architecture and cost-sensitive solutions, which require limited controller performance (≤120 MIPS) with the element functional safety of ASIL B (D) for the controller.

An on-board charger consists of two power stages: PFC (AC-DC) power converter and a subsequent DC-DC power converter. Each power stage is controlled with a single MCU.

OBC-charging design requirements are as follows:

- High-performance and fast digital control loops enabling highly efficient power conversion and increased power density.
- Enabling precise control and fast shutdown in an overcurrent scenario by high bandwidth and fast response current sensing.
- Safely and efficiently controlling and protecting the power switch [insulated-gate bipolar

transistor/silicon carbide (IGBT/SiC)].

在车载充电机（OBC）和高压直流-直流充电器（HV DCDC）市场中，基于模块化的设计和基于组合箱的设计是两种主要采用的架构。模块化方法在制造和售后服务方面提供了灵活性；而组合箱方法则力求将多种功能集成到一个外壳中，以实现紧凑性。F28003x 针对基于模块化的控制架构和对成本敏感的解决方案，这些方案需要有限的控制器性能（≤120 MIPS），且控制器需具备 ASIL B（D）级别的功能安全要素。

车载充电机由两个功率级组成：功率因数校正（PFC，交流 - 直流）功率转换器和后续的直流 - 直流功率转换器。每个功率级由单个微控制器（MCU）控制。

车载充电机充电设计要求如下：

- 高性能且快速的数字控制环路，实现高效的功率转换并提高功率密度。
- 通过高带宽和快速响应的电流感应，在过流情况下实现精确控制和快速关断。
- 安全有效地控制和保护功率开关[绝缘栅双极型晶体管/碳化硅（IGBT/SiC）]。

10.3.1.4 Servo Drive Control Module
伺服驱动器控制模块

Servo drives require high precision current and voltage sensing for accurate torque control and often supports interfaces for multiple encoder types along with communication interfaces. This DSP device can be used either as single chip solution for standalone servo drive or can be used in decentralized systems. In the later case, the device functions as the controller which samples all the voltage and current inputs and generates the correct PWM signals for inverter. Each DSP device serves as real-time controller for a target axis, running motor current control loop. Using the Fast Serial Interface (FSI) peripheral, up to 16 axes can be managed with one DSP device. As an outer loop controller, the DSP device executes main axis motor control, controls data exchange with all secondary axis over FSI and communicates with a host or PLC through EtherCAT.

伺服驱动器需要高精度的电流和电压传感，以实现精确的转矩控制，并且通常支持多种编码器类型的接口以及通信接口。这款 DSP 器件既可以作为独立伺服驱动器的单芯片解决方案，也可以用于分散式系统。在后一种情况下，控制器对所有电压和电流输入进行采样，并为逆变器生成正确的 PWM 信号。每个 DSP 器件作为目标轴的实时控制器，运行电机电流控制回路。通过快速串行接口（FSI）外设，一个 DSP 器件最多可以管理 16 个轴。作为外环控制器，DSP 器件执行主轴电机控制，通过 FSI 控制与所有副轴的数据交换，并通过 EtherCAT 与主机或 PLC 进行通信。

10.3.1.5 Solar Micro Inverter
太阳能微型逆变器

A Solar Micro Inverter consists of a DC-AC inverter power stage and one or more Maximum Power Point Tracking (MPPT) DC-DC power stages. Typical switching frequency for the inverter (DC-AC) is between 20kHz-50kHz and for DC-DC side can be in the range 100kHz-200kHz. A variety of power stage topologies can be used to achieve this and the diagram only depicts a typical power stage and the control & communication requirements. A microcontroller has on-chip EPWM, ADC and analog comparator modules

to implement complete digital control of such micro inverter system.

太阳能微型逆变器由一个直流 - 交流逆变器功率级和一个或多个最大功率点跟踪（MPPT）直流 - 直流功率级组成。逆变器（直流 - 交流）的典型开关频率在 20kHz - 50kHz 之间，直流 - 直流侧的开关频率范围可以在 100kHz - 200kHz 之间。可以使用多种功率级拓扑来实现这一点，该图仅描绘了一个典型的功率级以及控制和通信要求。微控制器具有片上增强型脉冲宽度调制（EPWM）、模数转换器（ADC）和模拟比较器模块，可实现对这种微型逆变器系统的完全数字控制。

10.3.1.6 Merchant Telecom Rectifiers
商用电信整流器

Merchant telecom rectifier consists of a power factor correction (PFC) stage and a DC-DC converter stage. The Totem pole PFC is widely used as the PFC stage. For the DC-DC stage, LLC and phase-shifted full bridge (PSFB) are the two most popular topologies. Single-chip and two-chip architecture can be used in merchant telecom rectifier.

The PFC stage draws sine-wave current from the AC mains in phase with the AC voltage, and maintains a steady DC bus voltage (VDC, typically +400 V) across its output. This output voltage is applied to the input of DC-DC stage, which converts it to an isolated low-output voltage Vout (usually 48 V).

商用电信整流器由功率因数校正（PFC）级和 DC - DC 转换器级组成。图腾柱 PFC 被广泛用作 PFC 级。对于 DC - DC 级，LLC 和移相全桥（PSFB）是两种最常用的拓扑结构。商用电信整流器可采用单芯片和双芯片架构。

PFC 级从交流市电中汲取与交流电压同相的正弦波电流，并在其输出端维持稳定的直流母线电压（VDC，通常为 +400 V）。该输出电压施加到 DC - DC 级的输入端 DC， - DC 级将其转换为隔离的低输出电压 Vout（通常为 48 V）。

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